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(54) **ANTI-GDF-15 ANTIBODIES**

(71) Applicant: **AVEO Pharmaceuticals, Inc.**, Boston, MA (US)

(72) Inventors: **Lorena Lerner**, Newton Centre, MA (US); **Sandra Abbott**, Boston, MA (US); **Ailin Bai**, Newton, MA (US); **Ting Chen**, Acton, MA (US); **Maria Isabel Chiu**, Newton Centre, MA (US); **Qing Liu**, Acton, MA (US); **Laura Poling**, Boston, MA (US); **Nianjun Tao**, Brighton, MA (US); **Solly Weiler**, Newton, MA (US); **Zhigang Weng**, Brookline, MA (US); **William M. Winston, JR.**, Marlborough, MA (US); **Jeno Gyuris**, Lincoln, MA (US)

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C12N 5/10	(2006.01)
C12N 15/10	(2006.01)
C12N 15/63	(2006.01)

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CPC **C07K 16/22** (2013.01); **A61K 39/395** (2013.01); **A61K 45/06** (2013.01); **A61P 3/02** (2018.01); **A61K 2039/505** (2013.01); **C07K 2317/24** (2013.01); **C07K 2317/76** (2013.01); **C07K 2317/92** (2013.01); **C07K 2319/30** (2013.01); **C12N 5/10** (2013.01); **C12N 15/10** (2013.01); **C12N 15/63** (2013.01)

(57)

ABSTRACT

Monoclonal antibodies that bind and inhibit the activity of human GDF15 are disclosed. The antibodies can be used to treat body weight loss, including cachexia, associated with the over-expression of human GDF15.

Specification includes a Sequence Listing.

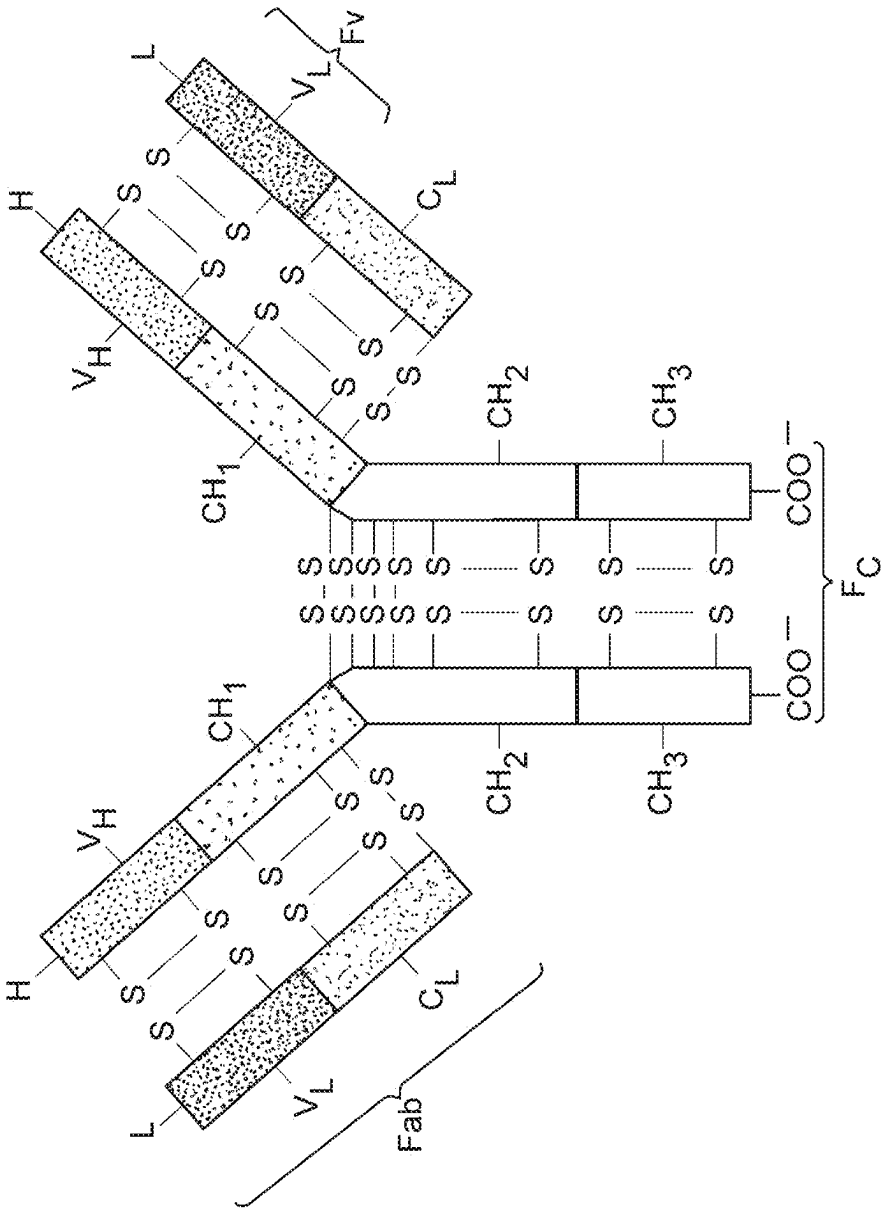


FIG. 1

FIG. 2

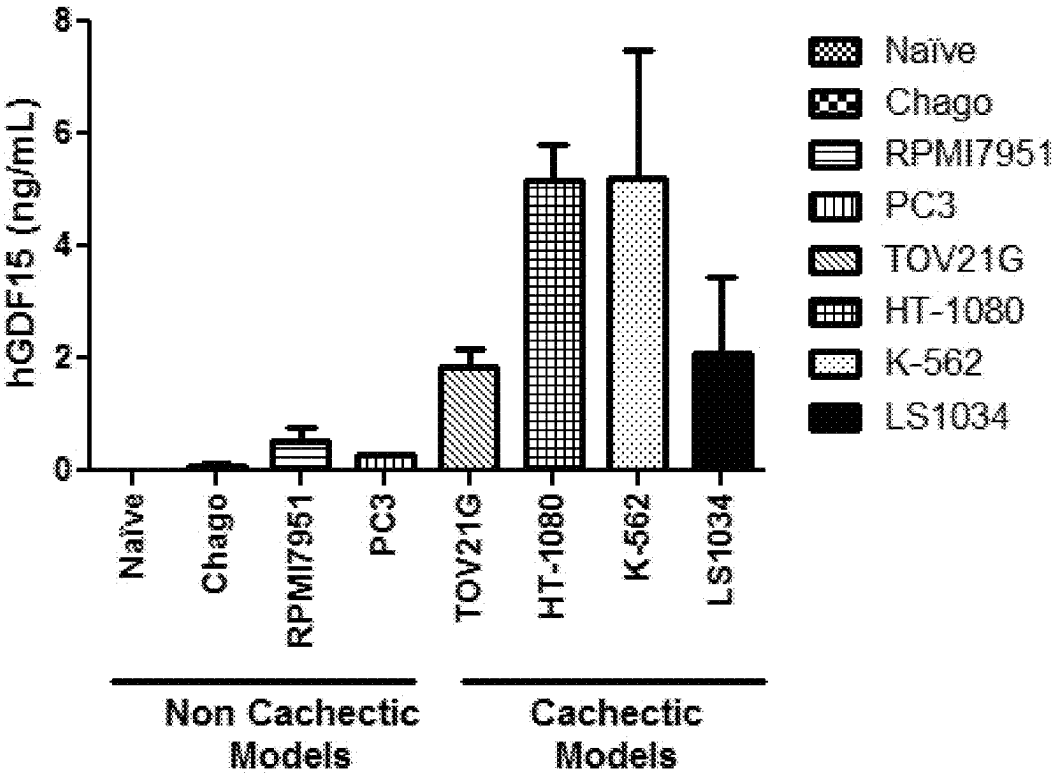


FIG. 3

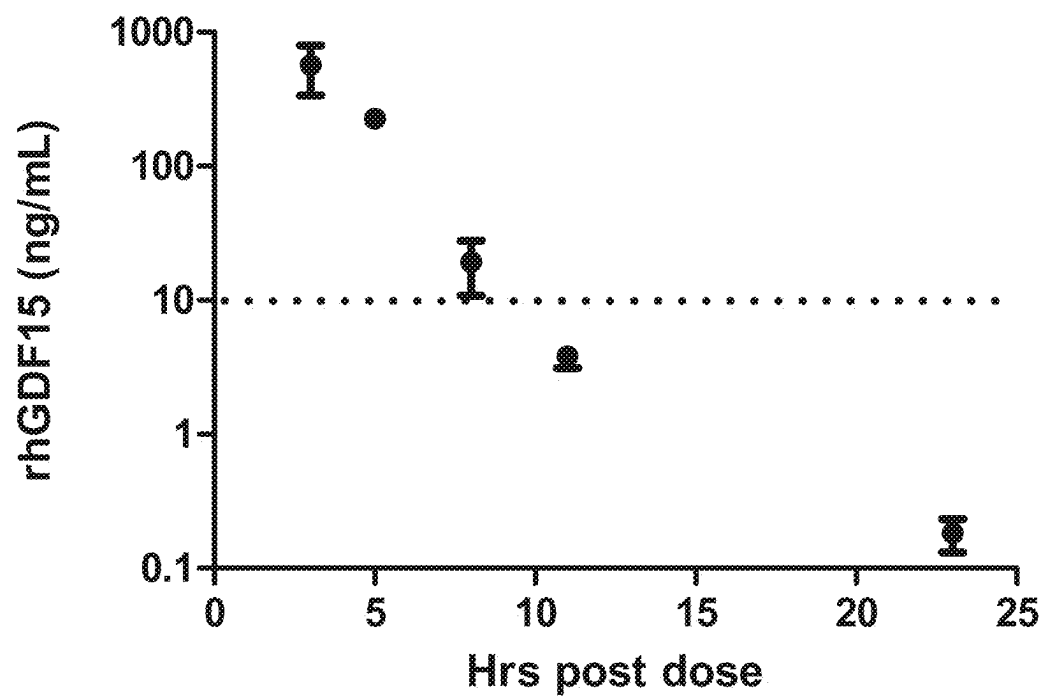


FIG. 4

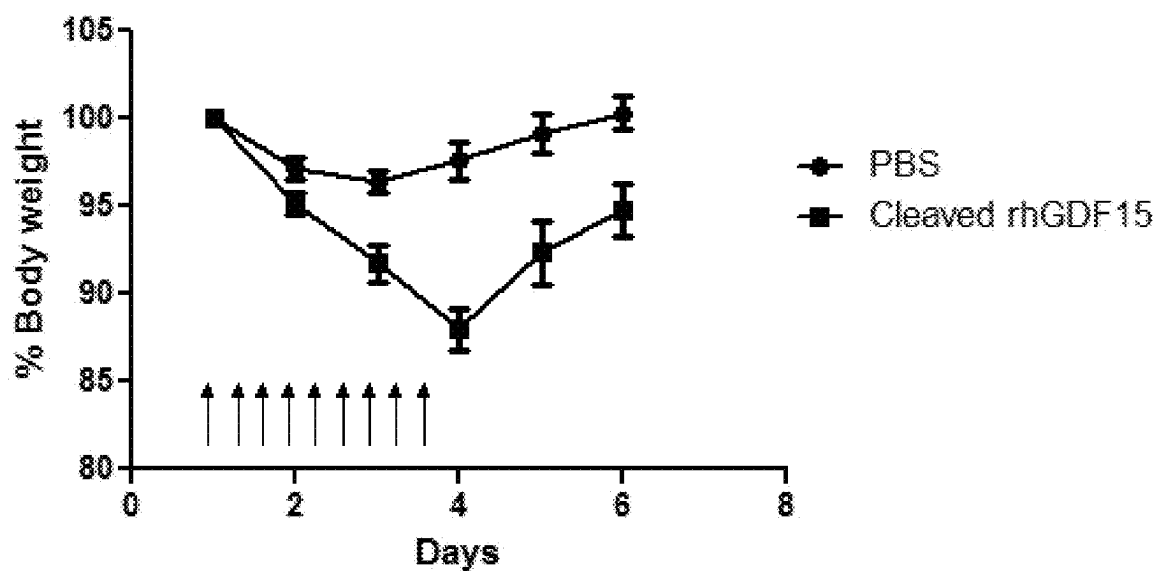


FIG. 5A

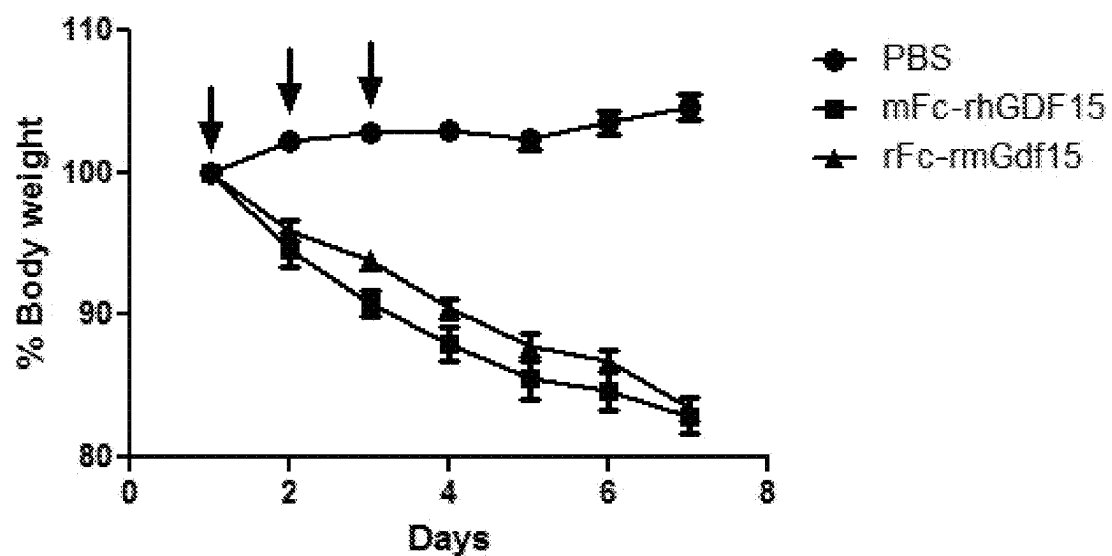


FIG. 5B

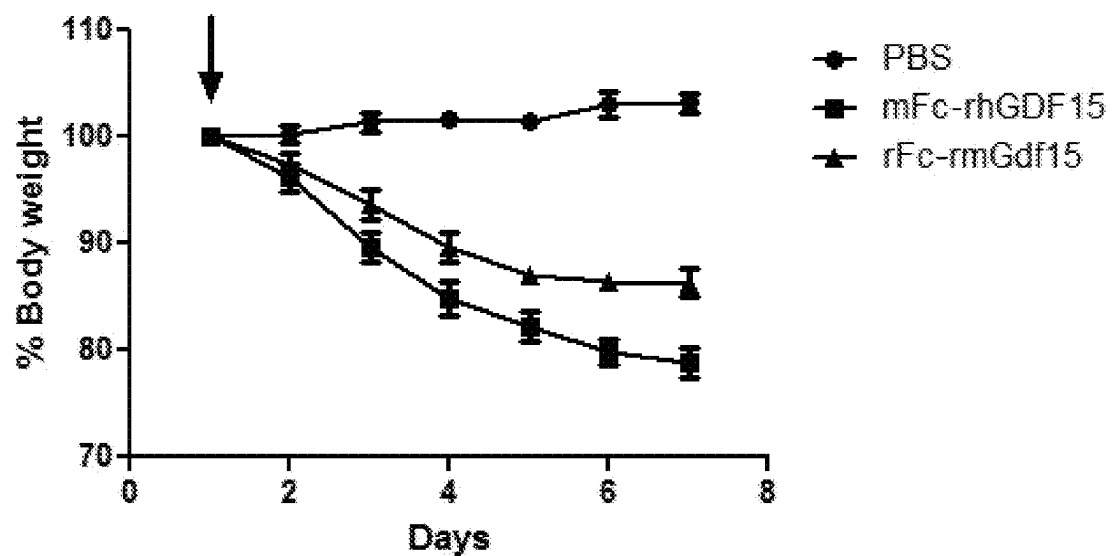


FIG. 6A

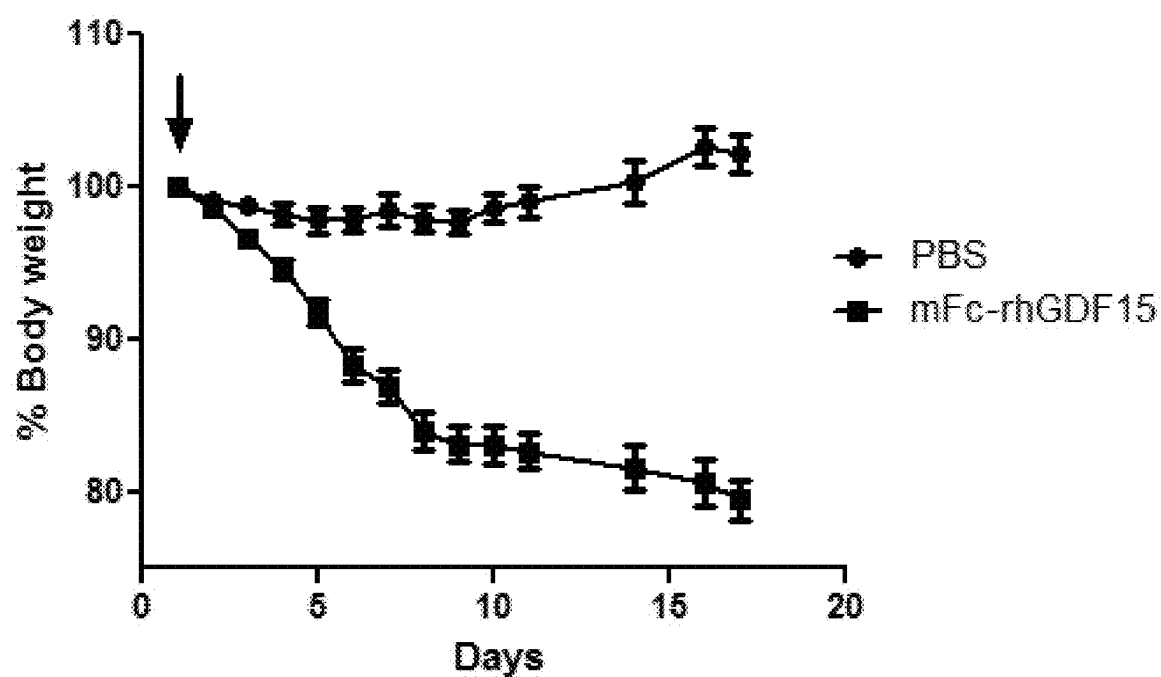


FIG. 6B

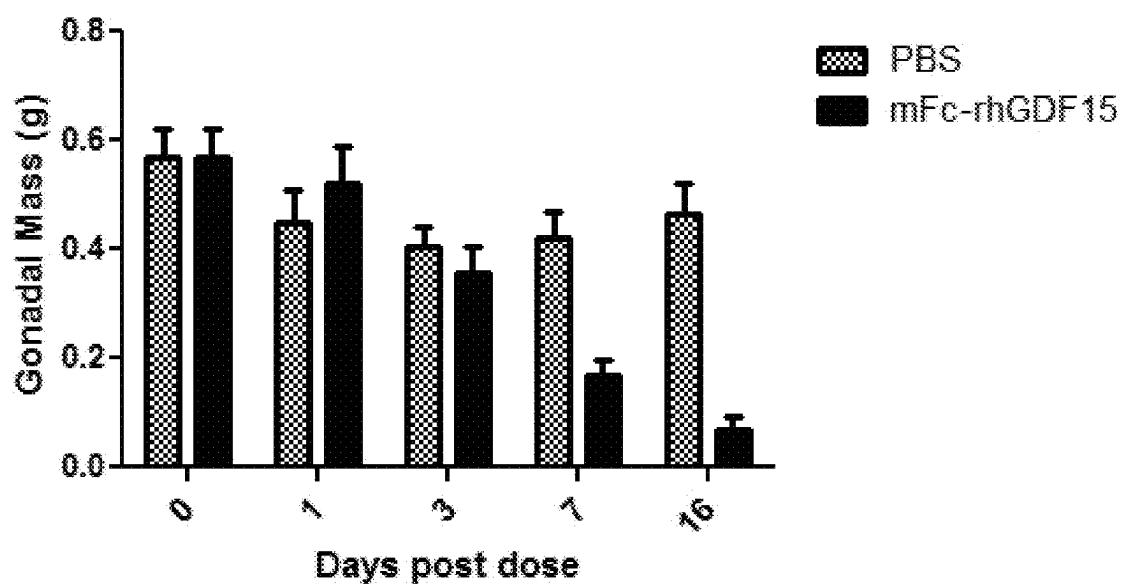


FIG. 6C

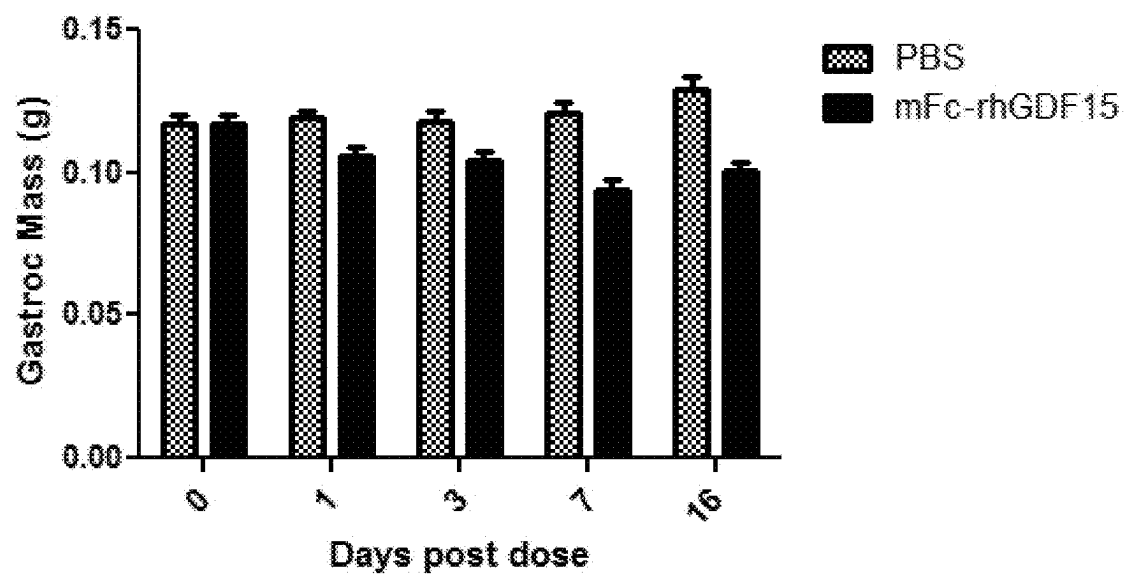


FIG. 6D

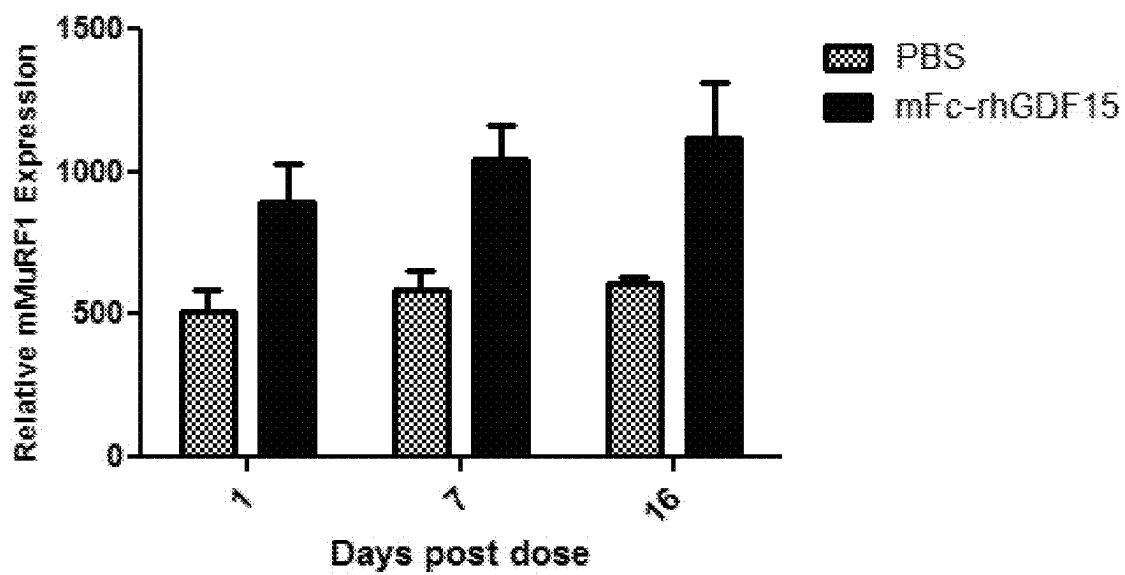


FIG. 6E



FIG. 7

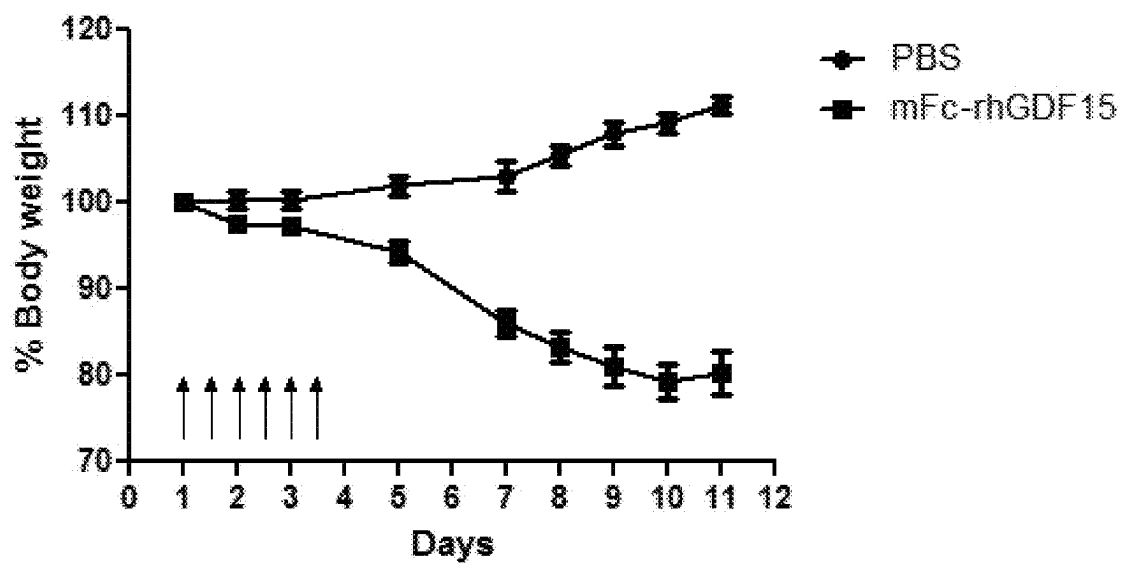


FIG. 8

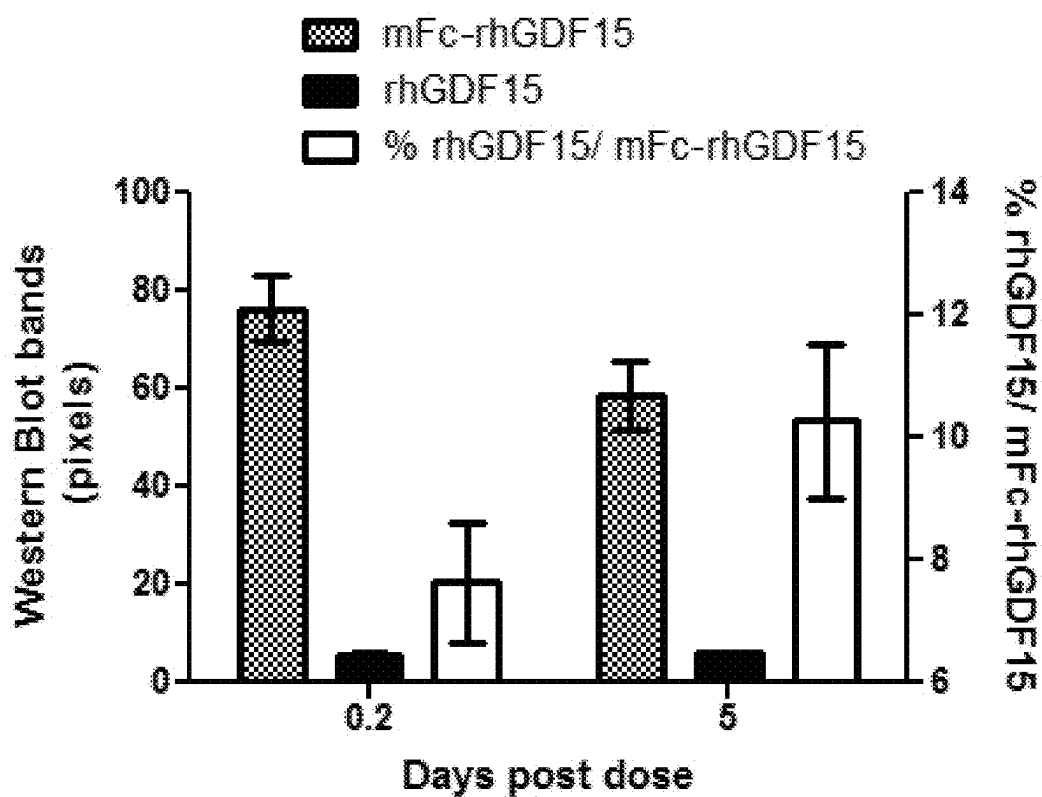


FIG. 9A

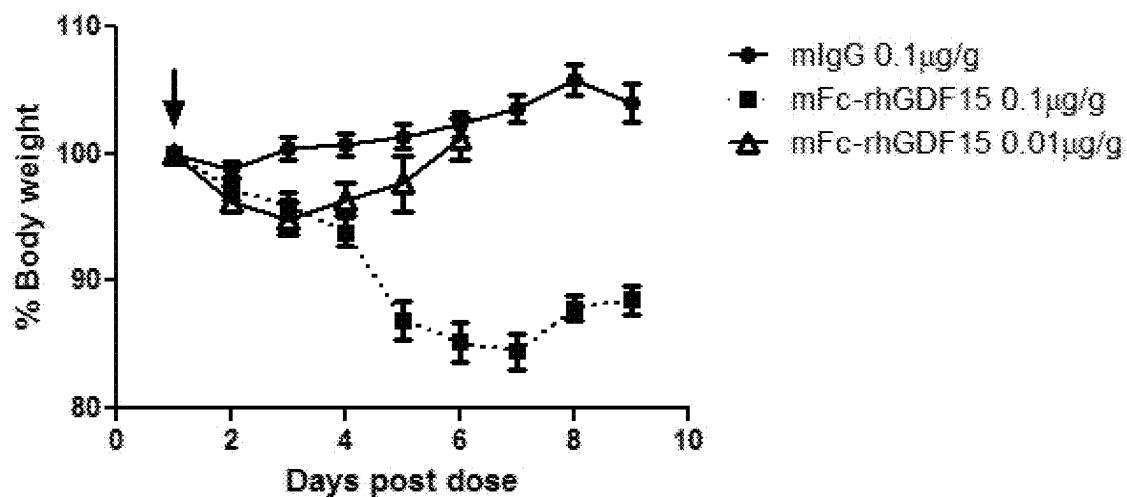
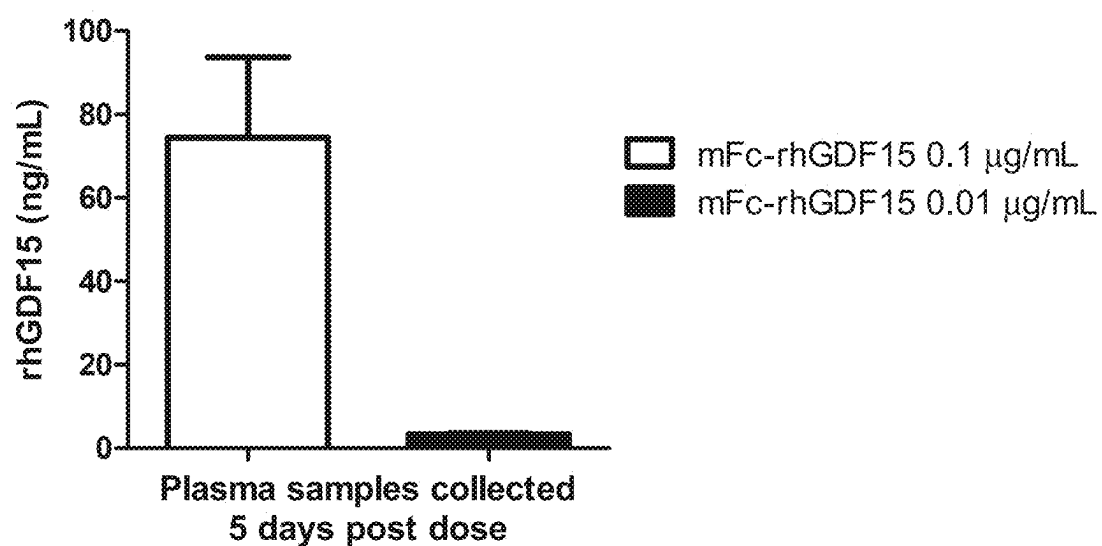


FIG. 9B



Complete Mouse Heavy Chain Variable Region Amino Acid Alignments

Heavy Variable	CDR1							CDR2						
	1	2	3	333	4	5	55	6	7	A				
	0	0	0	555	0	0	22	0	0	AB				
01G06	EVLQQSGPELVKPGASVKIPCKASGYTFTDYNMD--	WVKQSHGKSLEWIG	QINPNNGGIFFNQKFKG	KATLT										
03G05	QVQLQQPGAELVKPGASVKLSCKASGYTFTSYWIH--	WVNQRPGQGLEWIG	DINPSNGRSKYNEKFKN	KATMT										
04F08	QVTLKESGPGILQPSQTLSTLCSFSGFSLSTYGMGVTT	WIRQPSGKGLEWLAH	YIY-WDDDKRYNP	SLKSRLTIS										
06C11	QVTLKESGPGILQPSQTLSTLCSFSGFSLNTYGMGVSV	WIRQPSGKGLEWLAH	YIY-WDDDKRYNP	SLKSRLTIS										
08G01	EVLQQSGPEVVKPGASVKIPCKASGYTFTDYNMD--	WVKQSHGKSLEWIG	EINPNNGGTFYNQKFKG	KATLT										
14F11	QVTLKESGPGILQPSQTLSTLCSFSGFSLSTYGMGVGV	WIRQPSGKGLEWLAH	DIW-WDDDKYNP	SLKSRLTIS										
17B11	QVTLKESGPGILQPSQTLSTLCSFSGFSLSTSGMGVSV	WIRQPSGKGLEWLAH	ND-WDDDKRYKSS	SLKSRLTIS										
										ABC				
	7	8	888	9	1111	1								
	1	0	222	0	0000	1								
			ABC		0000	0								
01G06	VDKSSNTAFMEVRSLSLTSED	TAVVYCAR	EAITTVGAMDYWGQ	TSVTVSS	(SEQ ID NO:40)									
03G05	ADKSSNTAYMQLSSLTSED	SAVVYCAR	EVLDGAM--DYWGQ	TSVTVSS	(SEQ ID NO:42)									
04F08	KDTSNNQVFLKITSV	DATATYCAQ	TGYSNLF--AYWGQ	GLVTVSA	(SEQ ID NO:44)									
06C11	KDASNNRVFLKITSV	DATATYCAQ	RGYDDYW--GYWGQ	GLVTISA	(SEQ ID NO:46)									
08G01	VDKSSSTAYMELRSLTSED	TAVVYCAR	EAITTVGAMDYWGQ	TSVTVSS	(SEQ ID NO:48)									
14F11	KDTSSNEVFLKIAIV	DATATYCAR	RGHYSAM--DYWGQ	TSVTVSS	(SEQ ID NO:50)									
17B11	KDTSRNQVFLKITSV	DATATYCAR	RVGGLEGYFDYWGQ	TTLTVSS	(SEQ ID NO:52)									

FIG. 10

Mouse Heavy Chain CDR Amino Acid Alignments

Heavy Variable	CDR1		CDR2	
01G06	DYNMD--	(SEQ ID NO:1)	QINPNNGGIFFNQKFKG	(SEQ ID NO:7)
03G05	SYWIH--	(SEQ ID NO:2)	DINPSNGRSKYNEKEFN	(SEQ ID NO:8)
04F08	TYGMSGVT	(SEQ ID NO:3)	HIY-WDDDKRYNPSLKS	(SEQ ID NO:9)
06C11	TYGMSGVS	(SEQ ID NO:4)	HIY-WDDDKRYNPSLKS	(SEQ ID NO:9)
08G01	DYNMD--	(SEQ ID NO:1)	EINPNNGGTFYNQKFKG	(SEQ ID NO:10)
14F11	TYGMSGVG	(SEQ ID NO:5)	DIW-WDDDKRYNPSLKS	(SEQ ID NO:11)
17B11	TSGMSGVS	(SEQ ID NO:6)	HND-WDDDKRYKSSLKS	(SEQ ID NO:12)

Heavy Variable	CDR3	
01G06	EAITTVGAMDY	(SEQ ID NO:15)
03G05	EVLDGAM--DY	(SEQ ID NO:16)
04F08	TGYSNLF--AY	(SEQ ID NO:17)
06C11	RGYDDYW--GY	(SEQ ID NO:18)
08G01	EAITTVGAMDY	(SEQ ID NO:15)
14F11	RGHYSAM--DY	(SEQ ID NO:19)
17B11	RVGGLGEGYFDY	(SEQ ID NO:20)

FIG. 11

Complete Mouse Light (Kappa) Chain Variable Region Amino Acid Alignments

Light (Kappa) Variable	CDR1				CDR2			
	1	2	2222	3	4	5	6	7
1	0	0	77777	0	0	0	0	0
ABCD								
01G06	DIQMTQSPASLSASVGETVTITCR	TSSE---	NLHNYLA	WYQQKQKSPQLLVY	DAKTLAD	GVPSRFS	SGSG	TQ
03G05	DIVLTQSPASLAVSLGQRATISCR	ASEVDNYG	ISFMN	WFQQKPGQPPKLLIY	AASNQGS	GVPARFS	SGSG	STD
04F08	DIVMTQSQKFMSTSVGDRVSVTC	KASQ---	NVGTNVA	WYQQKLGQSPKTLIY	SASYRYS	GVPDFRFT	SGSG	STD
06C11	DIVMTQSQKFMSTSVGDRVSVTC	KASQ---	NVGTNVA	WFQQKPGQSPKALIY	SASYRYS	GVPDFRFT	SGSG	STD
08G01	DIQMTQSPASLSASVGETVTITCR	ASG---	NIHNYLA	WYQQKQKSPQLLVY	NAKTLAD	GVPSRFS	SGSG	TQ
14F11	DIVMTQSQKFMSTSVGDRVSVTC	KASQ---	NVGTNVA	WYQQKPGQSPKALIY	SPSYRYS	GVPDFRFT	SGSG	STD
17B11	DIVLTQSPASLAVSLGQRATISCR	ASQSVSTSRFSYMH	WFQQKPGQAPKLLIK	YASNLES	GVPARFS	SGSG	STD	
CDR3								
7	8	9	1					
1	0	0	0					
			0					
01G06	YSLKINSLQPEDFGSYVC	QHFWSSPYT	FGGGTKLEIK	(SEQ ID NO: 76)				
03G05	FSLNIHPMEEDDTAMVFC	QQSKEVPWT	FGGSGKLEIK	(SEQ ID NO: 78)				
04F08	FTLTISNVQSEDLAEYFC	QQYNSYPYT	FGGGTKLEIK	(SEQ ID NO: 80)				
06C11	FILTISNVQSEDLAEYFC	QQYNNYPLT	FGAGTKLELK	(SEQ ID NO: 82)				
08G01	YSLKINSLQPEDFGSYVC	QHFWSSPYT	FGGGTKLEIK	(SEQ ID NO: 84)				
14F11	FTLTISNVQSEDLAEYFC	QQYNSYPHT	FGGGTKLEMK	(SEQ ID NO: 86)				
17B11	FTLNIHPVEGEDTATVYC	QHSWEIPLY	FGGGTKLEIK	(SEQ ID NO: 88)				

FIG. 12

Mouse Light (Kappa) Chain CDR Amino Acid Alignments

Light (Kappa) Variable	CDR1	CDR2
01G06	RTSE----NLHNYLA (SEQ ID NO:21)	DAKTLAD (SEQ ID NO:26)
03G05	RASESVDNYGISFMN (SEQ ID NO:22)	AASNQGS (SEQ ID NO:27)
04F08	KASQ----NVGTNVA (SEQ ID NO:23)	SASYRYS (SEQ ID NO:28)
06C11	KASQ----NVGTNVA (SEQ ID NO:23)	SASYRYS (SEQ ID NO:28)
08G01	RASG----NIHNYLA (SEQ ID NO:24)	NAKTLAD (SEQ ID NO:29)
14F11	KASQ----NVGTNVA (SEQ ID NO:23)	SPSYRYS (SEQ ID NO:30)
17B11	RASQSVSTSRFSYMH (SEQ ID NO:25)	YASNLES (SEQ ID NO:31)

Light (Kappa) Variable	CDR3
01G06	QHFWSPPYT (SEQ ID NO:32)
03G05	QQSKEVPWT (SEQ ID NO:33)
04F08	QQYNSYPYT (SEQ ID NO:34)
06C11	QQYNNYPLT (SEQ ID NO:35)
08G01	QHFWSPPYT (SEQ ID NO:32)
14F11	QQYNSYPHT (SEQ ID NO:36)
17B11	QHSWEIPYT (SEQ ID NO:37)

FIG. 13

FIG. 14

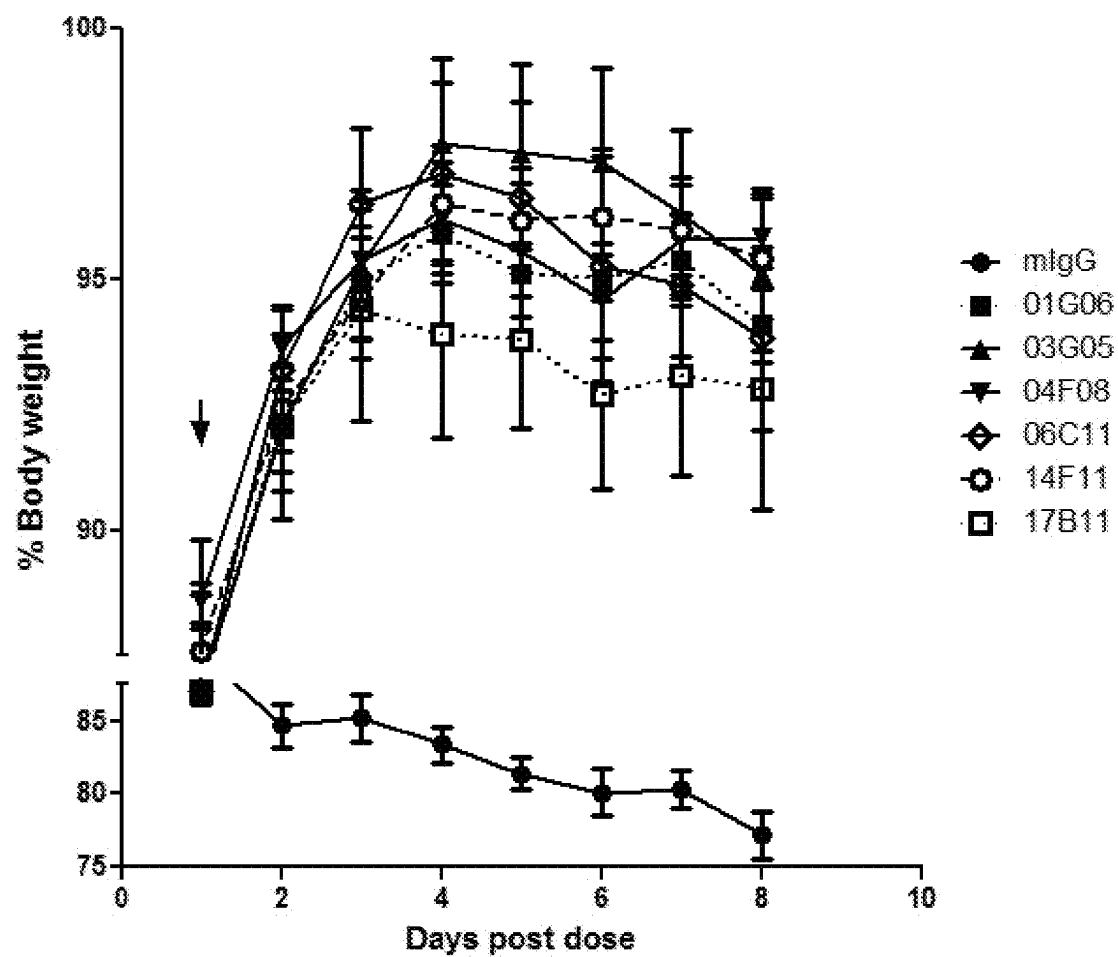


FIG. 15

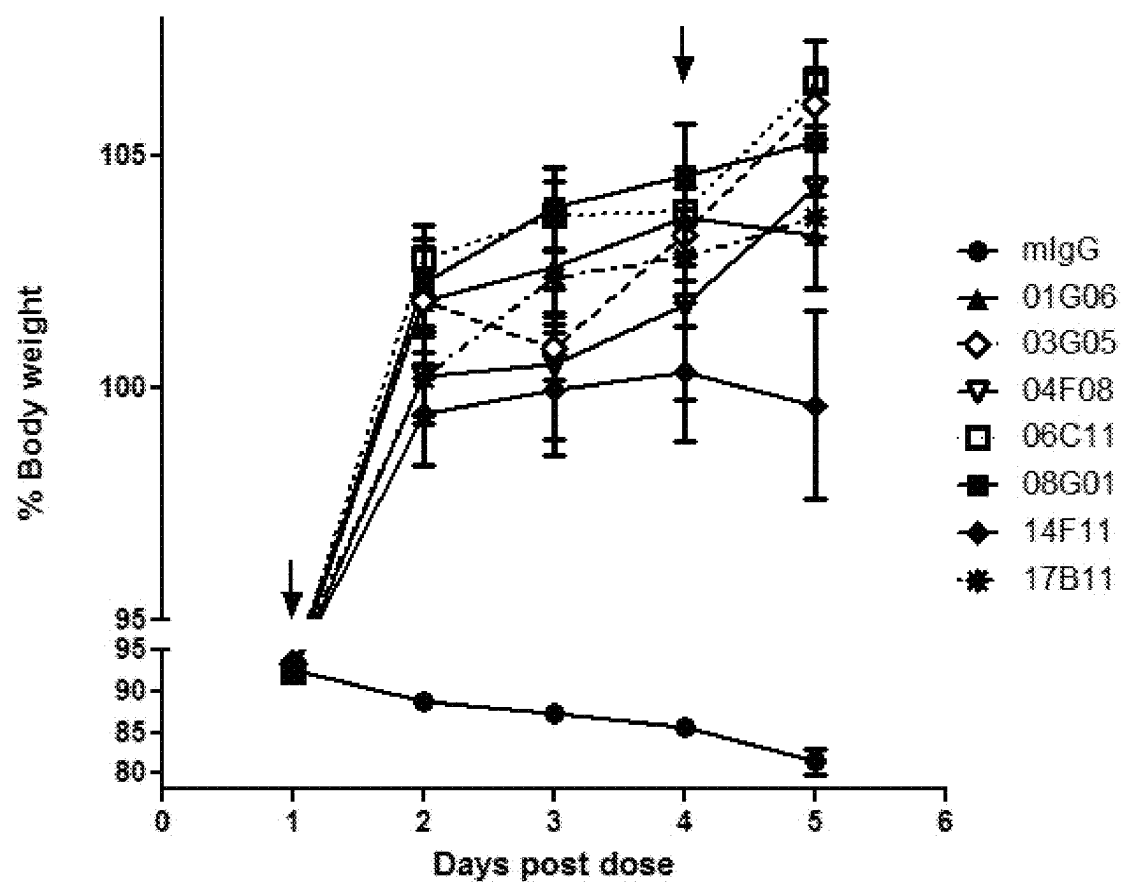


FIG. 16A

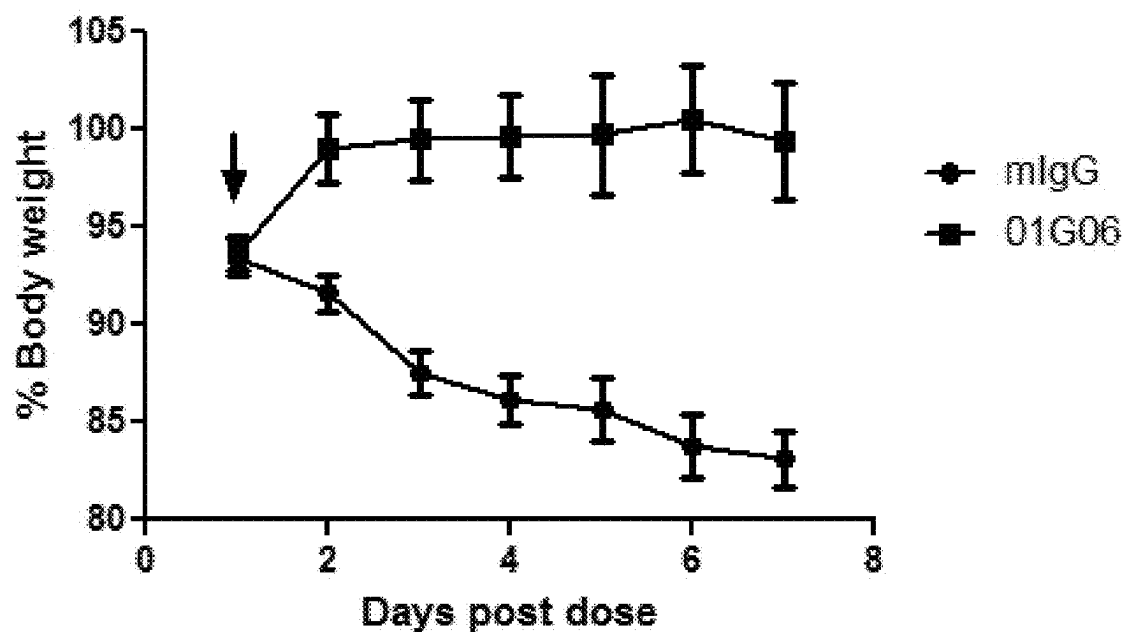


FIG. 16B

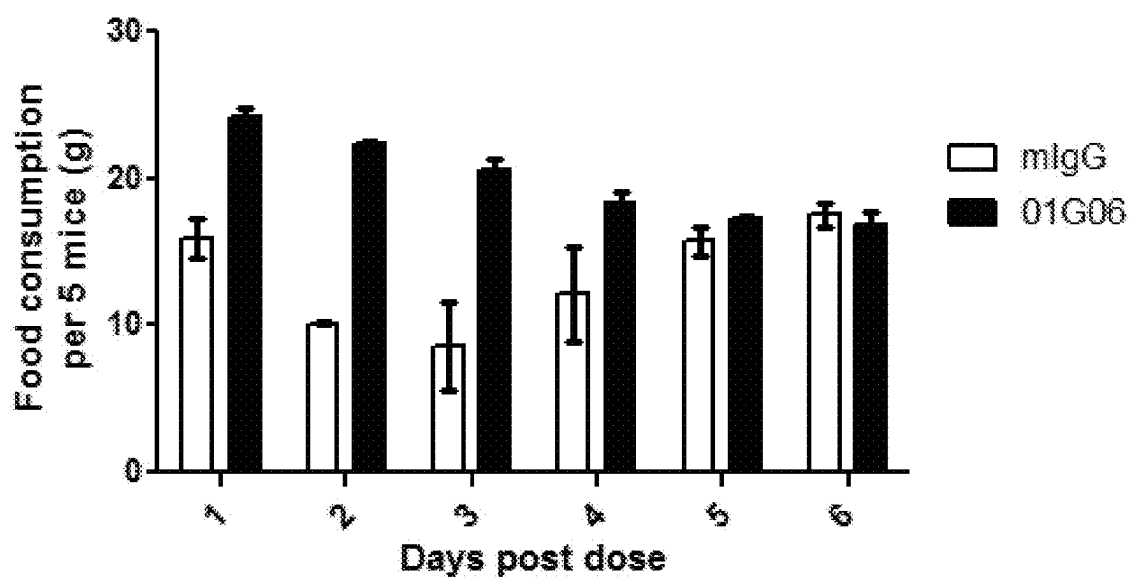


FIG. 16C

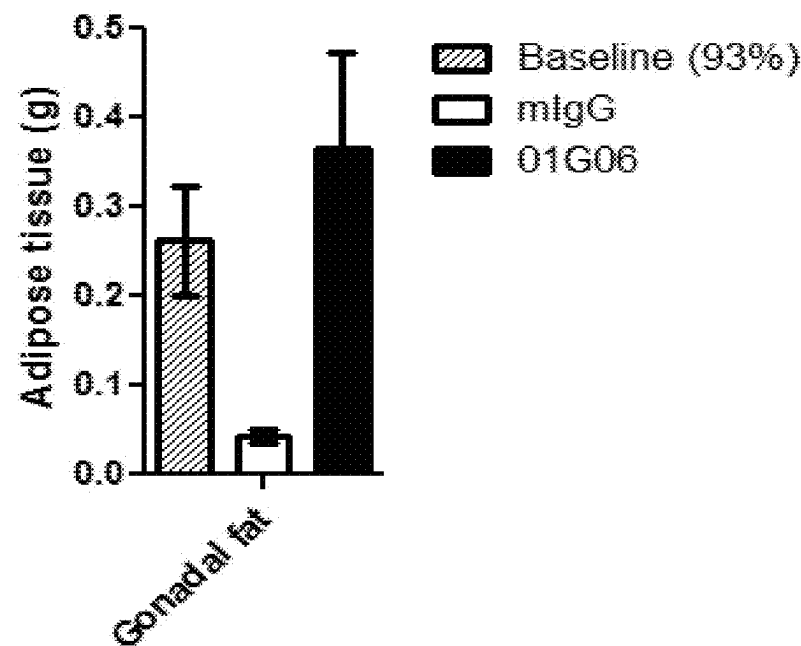


FIG. 16D

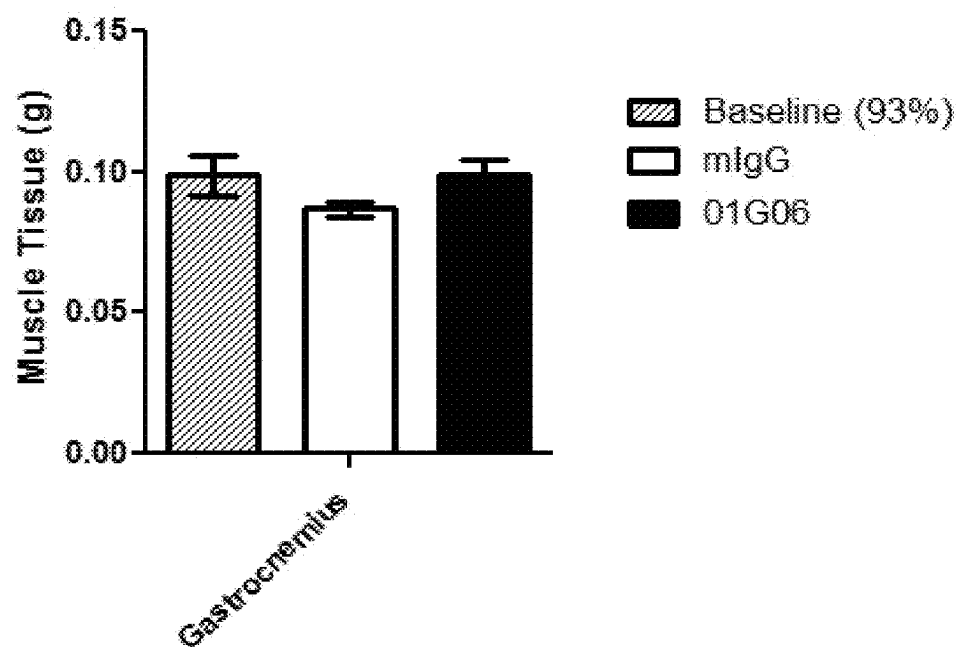


FIG. 16E

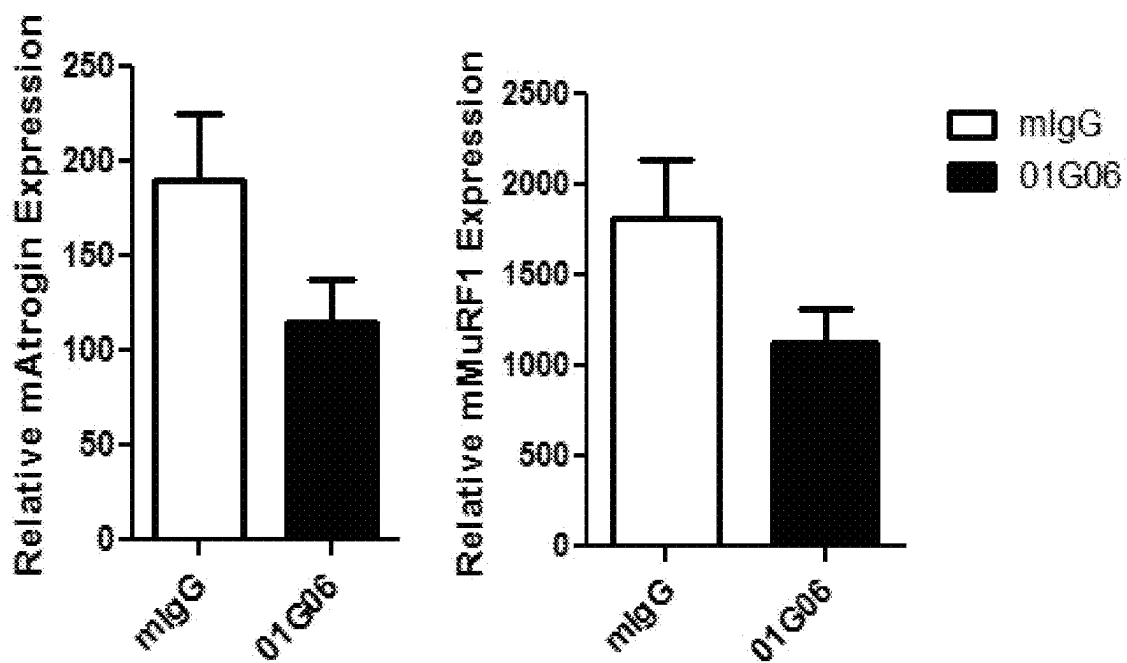


FIG. 17A

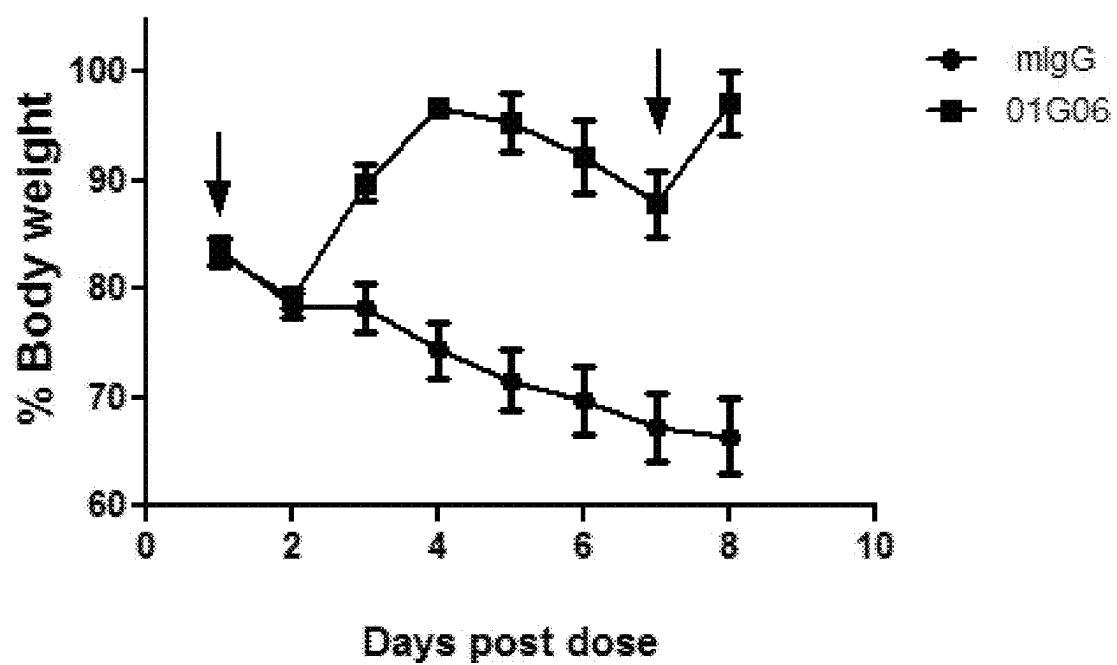


FIG. 17B

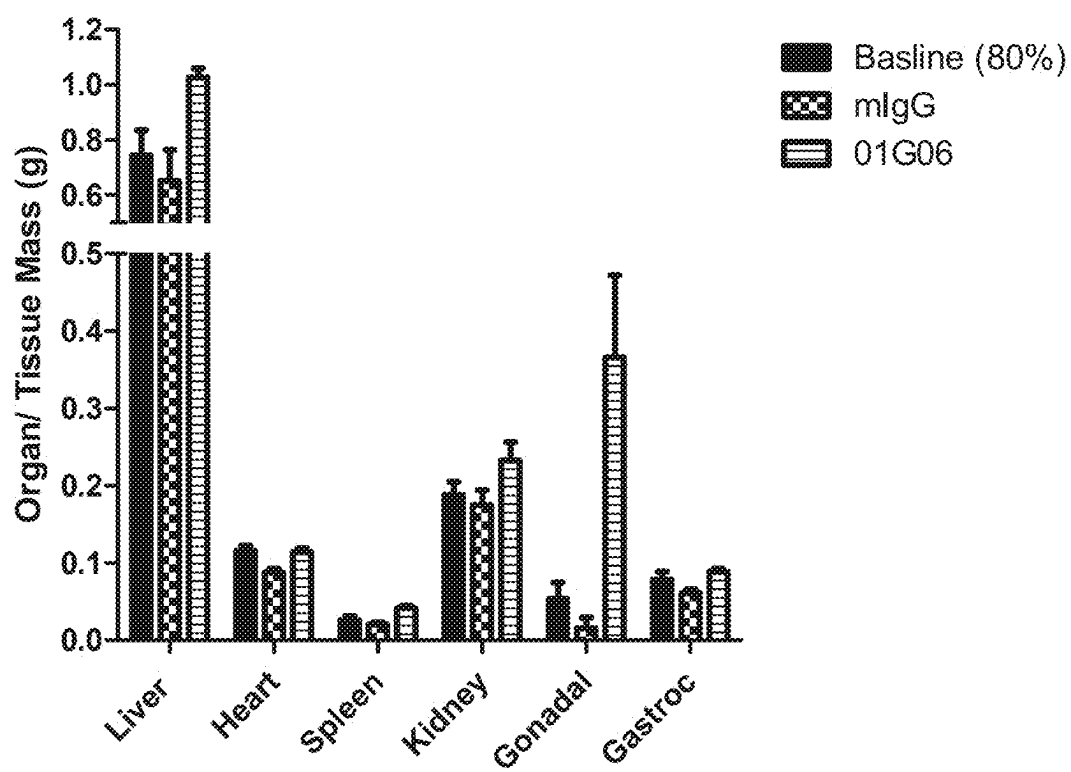
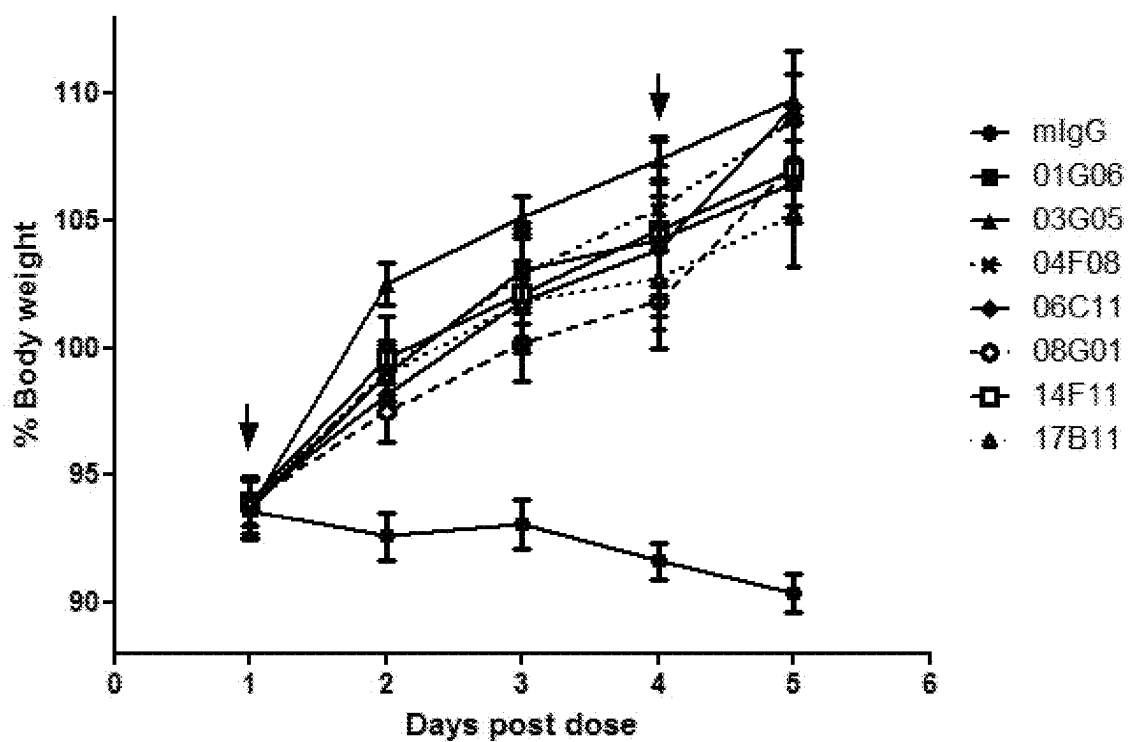


FIG. 18



Complete Humanized Heavy Chain Variable Region Amino Acid Alignments

Heavy Variable	CDRI							CDR2						
	1	2	3	4	5	6	7	1	2	3	4	5	6	7
	0	0	0	333	4	5	55	0	22	0	0	0	0	0
	AB													
Ch01G06 Chimeric	EVLLQSGPELVKPGASVKIPCKASGYTF	DYNMD--	WVKQSHGKSLEWIG	QINPNNGG	IFFNQKFKG	KATLT								
Hu01G06 IGHV1-18	QVQLVQSGAEVKKPGASVKVCKASGYTF	DYNMD--	WVRQAPGKKSLEWIG	QINPNNGG	IFFNQKFKG	RATLT								
Hu01G06 IGHV1-69	QVQLVQSGAEVKKPGSSVKVCKASGYTF	DYNMD--	WVRQAPGKKSLEWIG	QINPNNGG	IFFNQKFKG	RATLT								
Sh01G06 IGHV1-18 M69L	QVQLVQSGAEVKKPGASVKVCKASGYTF	DYNMD--	WVRQAPGQGLEWMC	QINPNNGG	IFFNQKFKG	RVTLT								
Sh01G06 IGHV1-18 M69L K64Q G44S	QVQLVQSGAEVKKPGASVKVCKASGYTF	DYNMD--	WVRQAPGQGLEWMC	QINPNNGG	IFFNQKFKG	RVTLT								
Sh01G06 IGHV1-18 M69L K64Q	QVQLVQSGAEVKKPGASVKVCKASGYTF	DYNMD--	WVRQAPGQGLEWMC	QINPNNGG	IFFNQKFKG	RVTLT								
Sh01G06 IGHV1-69 T30S I69L	QVQLVQSGAEVKKPGSSVKVCKASGYTF	DYNMD--	WVRQAPGQGLEWMC	QINPNNGG	IFFNQKFKG	RVTLT								
Sh01G06 IGHV1-69 T30S K64Q I69L	QVQLVQSGAEVKKPGSSVKVCKASGYTF	DYNMD--	WVRQAPGQGLEWMC	QINPNNGG	IFFNQKFKG	RVTLT								
Hu01G06 IGHV1-18 F1	QVQLVQSGAEVKKPGASVKVCKASGYTF	DYNMD--	WVRQAPGQSGLEWMC	QINPNNGG	IFFNQKFKG	RVTLT								
Hu01G06 IGHV1-18 F2	QVQLVQSGAEVKKPGASVKVCKASGYTF	DYNMD--	WVRQAPGQSGLEWMC	QINPNNGG	IFFNQKFKG	RVTLT								
Hu01G06 IGHV1-69 F1	QVQLVQSGAEVKKPGSSVKVCKASGYTF	DYNMD--	WVRQAPGQSGLEWMC	QINPNNGG	IFFNQKFKG	RVTLT								
Hu01G06 IGHV1-69 F2	QVQLVQSGAEVKKPGSSVKVCKASGYTF	DYNMD--	WVRQAPGQSGLEWMC	QINPNNGG	IFFNQKFKG	RVTLT								
Ch06C11 Chimeric	QVTLKESGPGILQPSQTLTLTCTFSGFSLN	TYGMGVSV	WIRQPSGKGLEWLAHIY	WDDDKRYNPSLKS	RLTIS									
HE LM 06C11 IGHV2-70	QVTLKESGPAIVKPTQTLTLTCTFSGFSLN	TYGMGVSV	WIRQPPGKALEWLAHIY	WDDDKRYNPSLKT	RLTIS									
Hu06C11 IGHV2-5	QVTLKESGPTLVKPTQTLTLTCTFSGFSLN	TYGMGVSV	WIRQPPGKGLEWLAHIY	WDDDKRYNPSLKS	RLTIT									
Ch14F11 Chimeric	QVTLKESGPGILQPSQTLTLTCTFSGFSLN	TYGMGVSV	WIRQPSGKGLEWLA	DIW-WDDDKYINPSLKS	RLTIS									
Sh14F11 IGHV2-5	QITLKESGPTLVKPTQTLTLTCTFSGFSLN	TYGMGVSV	WIRQPPGKALEWLA	DIW-WDDDKYINPSLKS	RLTIT									
Sh14F11 IGHV2-70	QVTLKESGPAIVKPTQTLTLTCTFSGFSLN	TYGMGVSV	WIRQPPGKALEWLA	DIW-WDDDKYINPSLKS	RLTIS									

FIG. 19

Complete Humanized Heavy Chain Variable Region Amino Acid Alignments

Heavy Variable	7	8	888	9	CDR3	1
	1	0	222	0	1111	1
					0000	1
			ABC		0000	0
					ABC	
Ch01G06 Chimeric	VDKSSNTAFMEVRS	LTSEDTAVYYCAREAI	TTVGAMDY	WCQGT	SVTVSS	(SEQ ID NO:40)
Hu01G06 IGHV1-18	VDTSTNTAYMELRSL	RSDDTAVYYCAREAI	TTVGAMDY	WCQGT	FLVTVSS	(SEQ ID NO:54)
Hu01G06 IGHV1-69	VDKSTNTAYMELSSL	RSEDTAVYYCAREAI	TTVGAMDY	WCQGT	FLVTVSS	(SEQ ID NO:56)
Sh01G06 IGHV1-18 M69L	TDTSTSTAYMELRSL	RSDDTAVYYCAREAI	TTVGAMDY	WCQGT	FLVTVSS	(SEQ ID NO:58)
Sh01G06 IGHV1-18 M69L K64Q	TDTSTSTAYMELRSL	RSDDTAVYYCAREAI	TTVGAMDY	WCQGT	FLVTVSS	(SEQ ID NO:60)
Sh01G06 IGHV1-18 M69L K64Q	TDTSTSTAYMELRSL	RSDDTAVYYCAREAI	TTVGAMDY	WCQGT	FLVTVSS	(SEQ ID NO:62)
Sh01G06 IGHV1-69 T30S I69L	ADKSTSTAYMELSSL	RSEDTAVYYCAREAI	TTVGAMDY	WCQGT	FLVTVSS	(SEQ ID NO:64)
Sh01G06 IGHV1-69 T30S K54Q I69L	ADKSTSTAYMELSSL	RSEDTAVYYCAREAI	TTVGAMDY	WCQGT	FLVTVSS	(SEQ ID NO:66)
Hu01G06 IGHV1-18 F1	TDTSTSTAYMELRSL	RSDDTAVYYCAREAI	TTVGAMDY	WCQGT	FLVTVSS	(SEQ ID NO:246)
Hu01G06 IGHV1-18 F2	TDTSTSTAYMELRSL	RSDDTAVYYCAREAI	TTVGAMDY	WCQGT	FLVTVSS	(SEQ ID NO:248)
Hu01G06 IGHV1-69 F1	ADKSTSTAYMELSSL	RSEDTAVYYCAREAI	TTVGAMDY	WCQGT	FLVTVSS	(SEQ ID NO:250)
Hu01G06 IGHV1-69 F2	ADKSTSTAYMELSSL	RSEDTAVYYCAREAI	TTVGAMDY	WCQGT	FLVTVSS	(SEQ ID NO:252)
Ch06C11 Chimeric	KDASNNRVFLKIT	SVDTADTAVYYCAQ	RGYDDY	--GY	WCQGT	FLVTISA
HE LM 06C11 IGHV2-70	KDTSKNQVVLIT	TNVDPPVDTAVYYCAQ	RGYDDY	--GY	WCQGT	FLVTISS
Hu06C11 IGHV2-5	KDTSKNQVVLIT	TNMDPVDATVYYCAQ	RGYDDY	--GY	WCQGT	FLVTVSS
Ch14F11 Chimeric	KDTSNEVFLKIAI	VDTADTAVYYCAREGHYSAM	--DY	WCQGT	SVTVSS	(SEQ ID NO:50)
Sh14F11 IGHV2-5	KDTSKNQVVLIT	TNMDPVDATVYYCAREGHYSAM	--DY	WCQGT	FLVTVSS	(SEQ ID NO:72)
Sh14F11 IGHV2-70	KDTSKNQVVLIT	TNMDPVDATVYYCAREGHYSAM	--DY	WCQGT	FLVTVSS	(SEQ ID NO:74)

FIG. 19 Continued

Humanized Heavy Chain CDR Amino Acid Alignments

Heavy Variable	CDR1	CDR2
Ch01G06 Chimeric	DYNMD-- (SEQ ID NO:1)	QINPNNGGIFFNQKFKG (SEQ ID NO:7)
Hu01G06 IGHV1-18	DYNMD-- (SEQ ID NO:1)	QINPNNGGIFFNQKFKG (SEQ ID NO:7)
Hu01G06 IGHV1-69	DYNMD-- (SEQ ID NO:1)	QINPNNGGIFFNQKFKG (SEQ ID NO:7)
Sh01G06 IGHV1-18 M69L	DYNMD-- (SEQ ID NO:1)	QINPNNGGIFFNQKFKG (SEQ ID NO:7)
Sh01G06 IGHV1-18 M69L K64Q G44S	DYNMD-- (SEQ ID NO:1)	QINPNNGGIFFNQKFKG (SEQ ID NO:13)
Sh01G06 IGHV1-18 M69L K64Q	DYNMD-- (SEQ ID NO:1)	QINPNNGGIFFNQKFKG (SEQ ID NO:13)
Sh01G06 IGHV1-69 T30S I69L	DYNMD-- (SEQ ID NO:1)	QINPNNGGIFFNQKFKG (SEQ ID NO:7)
Sh01G06 IGHV1-69 T30S K64Q I69L	DYNMD-- (SEQ ID NO:1)	QINPNNGGIFFNQKFKG (SEQ ID NO:13)
Hu01G06 IGHV1-18 F1	DYNMD-- (SEQ ID NO:1)	QINPNHLLIFFNQKFKG (SEQ ID NO:236)
Hu01G06 IGHV1-18 F2	DYNMD-- (SEQ ID NO:1)	QINPNNGLIFFNQKFKG (SEQ ID NO:237)
Hu01G06 IGHV1-69 F1	DYNMD-- (SEQ ID NO:1)	QINPNNGLIFFNQKFKG (SEQ ID NO:238)
Hu01G06 IGHV1-69 F2	DYNMD-- (SEQ ID NO:1)	QINPNHLLIFFNQKFKG (SEQ ID NO:239)
Ch06C11 Chimeric	TYGMSGVS (SEQ ID NO:4)	HIY-WDDDKRYNPSLKS (SEQ ID NO:9)
HE LM 06C11 IGHV2-70	TYGMSGVS (SEQ ID NO:4)	HIY-WDDDKRYNPSLKT (SEQ ID NO:14)
Hu06C11 IGHV2-5	TYGMSGVS (SEQ ID NO:4)	HIY-WDDDKRYNPSLKS (SEQ ID NO:9)
Ch14F11 Chimeric	TYGMSGVG (SEQ ID NO:5)	DIW-WDDDKRYNPSLKS (SEQ ID NO:11)
Sh14F11 IGHV2-5	TYGMSGVG (SEQ ID NO:5)	DIW-WDDDKRYNPSLKS (SEQ ID NO:11)
Sh14F11 IGHV2-70	TYGMSGVG (SEQ ID NO:5)	DIW-WDDDKRYNPSLKS (SEQ ID NO:11)

FIG. 20

Humanized Heavy Chain CDR Amino Acid Alignments

Heavy Variable	CDR3
Ch01G06 Chimeric	EAITTVGAMDY (SEQ ID NO:15)
Hu01G06 IGHV1-18	EAITTVGAMDY (SEQ ID NO:15)
Hu01G06 IGHV1-69	EAITTVGAMDY (SEQ ID NO:15)
Sh01G06 IGHV1-18 M69L	EAITTVGAMDY (SEQ ID NO:15)
Sh01G06 IGHV1-18 M69L K64Q G44S	EAITTVGAMDY (SEQ ID NO:15)
Sh01G06 IGHV1-18 M69L K64Q	EAITTVGAMDY (SEQ ID NO:15)
Sh01G06 IGHV1-69 T30S I69L	EAITTVGAMDY (SEQ ID NO:15)
Sh01G06 IGHV1-69 T30S K64Q I69L	EAITTVGAMDY (SEQ ID NO:15)
Hu01G06 IGHV1-18 F1	EAITTVGAMDY (SEQ ID NO:15)
Hu01G06 IGHV1-18 F2	EAITTVGAMDY (SEQ ID NO:15)
Hu01G06 IGHV1-69 F1	EAITTVGAMDY (SEQ ID NO:15)
Hu01G06 IGHV1-69 F2	EAITTVGAMDY (SEQ ID NO:15)
Ch06C11 Chimeric	RGYDDYW--GY (SEQ ID NO:18)
HE LM 06C11 IGHV2-70	RGYDDYW--GY (SEQ ID NO:18)
Hu06C11 IGHV2-5	RGYDDYW--GY (SEQ ID NO:18)
Ch14F11 Chimeric	RGHYSAM--DY (SEQ ID NO:19)
Sh14F11 IGHV2-5	RGHYSAM--DY (SEQ ID NO:19)
Sh14F11 IGHV2-70	RGHYSAM--DY (SEQ ID NO:19)

FIG. 20 Continued

Complete Humanized Light (Kappa) Chain Variable Region Amino Acid Alignments

Light (Kappa) Variable	CDR1							CDR2						
	1	2	3	4	5	6	7	1	2	3	4	5	6	7
Ch01G06 Chimeric	DIQMTQSPASLSASVGETVTIT	RTSEN	LHNYLA	WYQQKQKSPQLLVYDA	KTAD	GVPSR	FSGSGSGTQ	DIQMTQSPASLSASVGETVTIT	RTSEN	LHNYLA	WYQQKQKSPQLLVYDA	KTAD	GVPSR	FSGSGSGTQ
Hu01G06 IGKV1-39	DIQMTQSPSSLSASVGDRTVTIT	RTSEN	LHNYLA	WYQQKPKSPKLLVYDA	KTAD	GVPSR	FSGSGSGTD	DIQMTQSPSSLSASVGDRTVTIT	RTSEN	LHNYLA	WYQQKPKSPKLLVYDA	KTAD	GVPSR	FSGSGSGTD
Hu01G06 IGKV1-39 S43A V48I	DIQMTQSPSSLSASVGDRTVTIT	RTSEN	LHNYLA	WYQQKPKSPKLLVYDA	KTAD	GVPSR	FSGSGSGTD	DIQMTQSPSSLSASVGDRTVTIT	RTSEN	LHNYLA	WYQQKPKSPKLLVYDA	KTAD	GVPSR	FSGSGSGTD
Hu01G06 IGKV1-39 V48I	DIQMTQSPSSLSASVGDRTVTIT	RTSEN	LHNYLA	WYQQKPKSPKLLVYDA	KTAD	GVPSR	FSGSGSGTD	DIQMTQSPSSLSASVGDRTVTIT	RTSEN	LHNYLA	WYQQKPKSPKLLVYDA	KTAD	GVPSR	FSGSGSGTD
Hu01G06 IGKV1-39 F1	DIQMTQSPSSLSASVGDRTVTIT	RTSEN	LHNYLA	WYQQKPKSPKLLVYDA	KTAD	GVPSR	FSGSGSGTD	DIQMTQSPSSLSASVGDRTVTIT	RTSEN	LHNYLA	WYQQKPKSPKLLVYDA	KTAD	GVPSR	FSGSGSGTD
Hu01G06 IGKV1-39 F2	DIQMTQSPSSLSASVGDRTVTIT	RTSEN	LHNYLA	WYQQKPKSPKLLVYDA	KTAD	GVPSR	FSGSGSGTD	DIQMTQSPSSLSASVGDRTVTIT	RTSEN	LHNYLA	WYQQKPKSPKLLVYDA	KTAD	GVPSR	FSGSGSGTD
Ch06C11 Chimeric	DIVMTQSQKFMSTSVGDRTVTIT	CASQNV	GTNVAV	WFQQKPKSPKLLVYDA	KTAD	GVPSR	FSGSGSGTD	DIVMTQSQKFMSTSVGDRTVTIT	CASQNV	GTNVAV	WFQQKPKSPKLLVYDA	KTAD	GVPSR	FSGSGSGTD
Sh06C11 IGKV1-16	DIQMTQSPSSLSASVGDRTVTIT	CASQNV	GTNVAV	WFQQKPKSPKLLVYDA	KTAD	GVPSR	FSGSGSGTD	DIQMTQSPSSLSASVGDRTVTIT	CASQNV	GTNVAV	WFQQKPKSPKLLVYDA	KTAD	GVPSR	FSGSGSGTD
Ch14F11 Chimeric	DIVMTQSQKFMSTSVGDRTVTIT	CASQNV	GTNVAV	WFQQKPKSPKLLVYDA	KTAD	GVPSR	FSGSGSGTD	DIVMTQSQKFMSTSVGDRTVTIT	CASQNV	GTNVAV	WFQQKPKSPKLLVYDA	KTAD	GVPSR	FSGSGSGTD
Hu14F11 IGKV1-16	DIQMTQSPSSLSASVGDRTVTIT	CASQNV	GTNVAV	WFQQKPKSPKLLVYDA	KTAD	GVPSR	FSGSGSGTD	DIQMTQSPSSLSASVGDRTVTIT	CASQNV	GTNVAV	WFQQKPKSPKLLVYDA	KTAD	GVPSR	FSGSGSGTD

Light (Kappa) Variable

Light (Kappa) Variable	CDR3						
	7	8	9	1	2	3	4
Ch01G06 Chimeric	YSLKINS	LQPED	FGSYVC	QHEWSSPYT	FGG	G	T
Hu01G06 IGKV1-39	YTLTISS	LQPED	FATYYC	QHEWSSPYT	FGQ	G	T
Hu01G06 IGKV1-39 S43A V48I	YTLTISS	LQPED	FATYYC	QHEWSSPYT	FGQ	G	T
Hu01G06 IGKV1-39 V48I	YTLTISS	LQPED	FATYYC	QHEWSSPYT	FGQ	G	T
Hu01G06 IGKV1-39 F1	YTLTISS	LQPED	FATYYC	QHEWSSPYT	FGQ	G	T
Hu01G06 IGKV1-39 F2	YTLTISS	LQPED	FATYYC	QHEWSSPYT	FGQ	G	T
Ch06C11 Chimeric	FILTI	SNVQSE	DLAEYFC	QQYNNYPLT	FGAG	T	K
Sh06C11 IGKV1-16	FTLTISS	LQPED	FATYYC	QQYNNYPLT	FGQ	G	T
Ch14F11 Chimeric	FTLTI	SNVQSE	DLAEYFC	QQYNNYPLT	FGQ	G	T
Hu14F11 IGKV1-16	FTLTISS	LQPED	FATYYC	QQYNNYPLT	FGQ	G	T

FIG. 21

FIG. 23

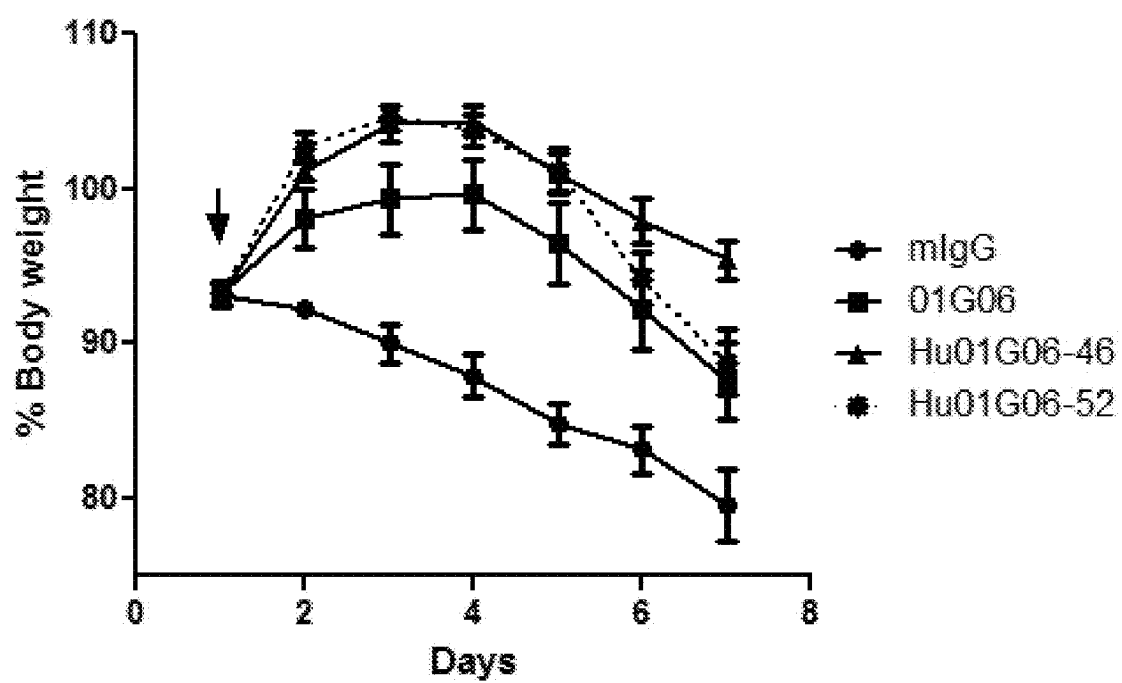


FIG. 24

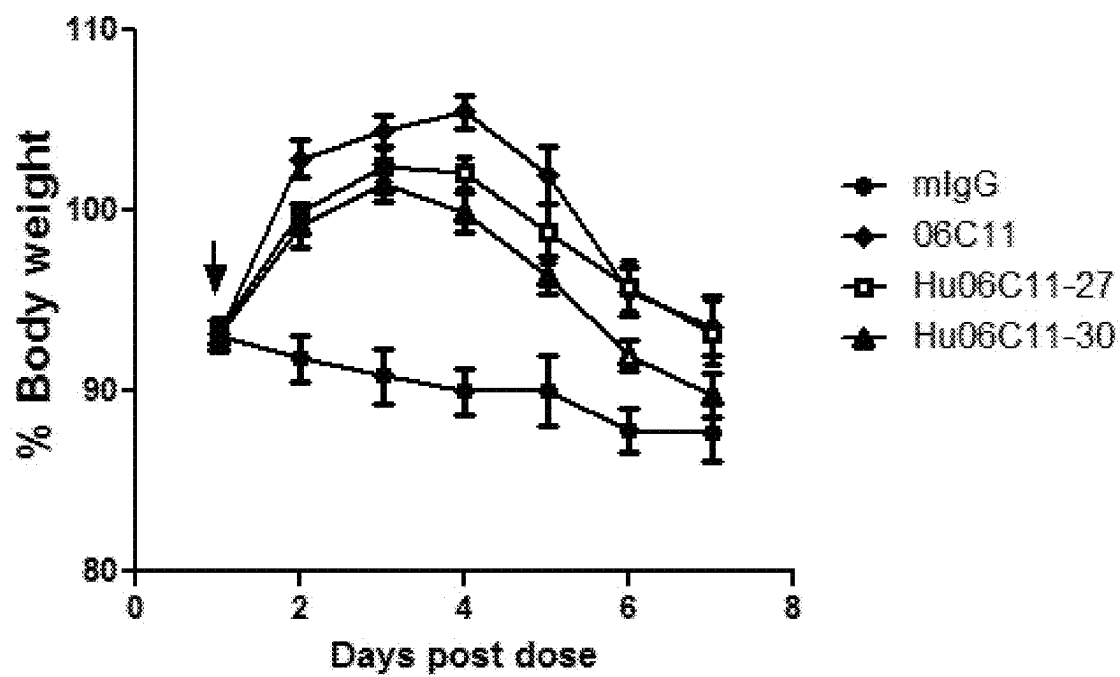


FIG. 25

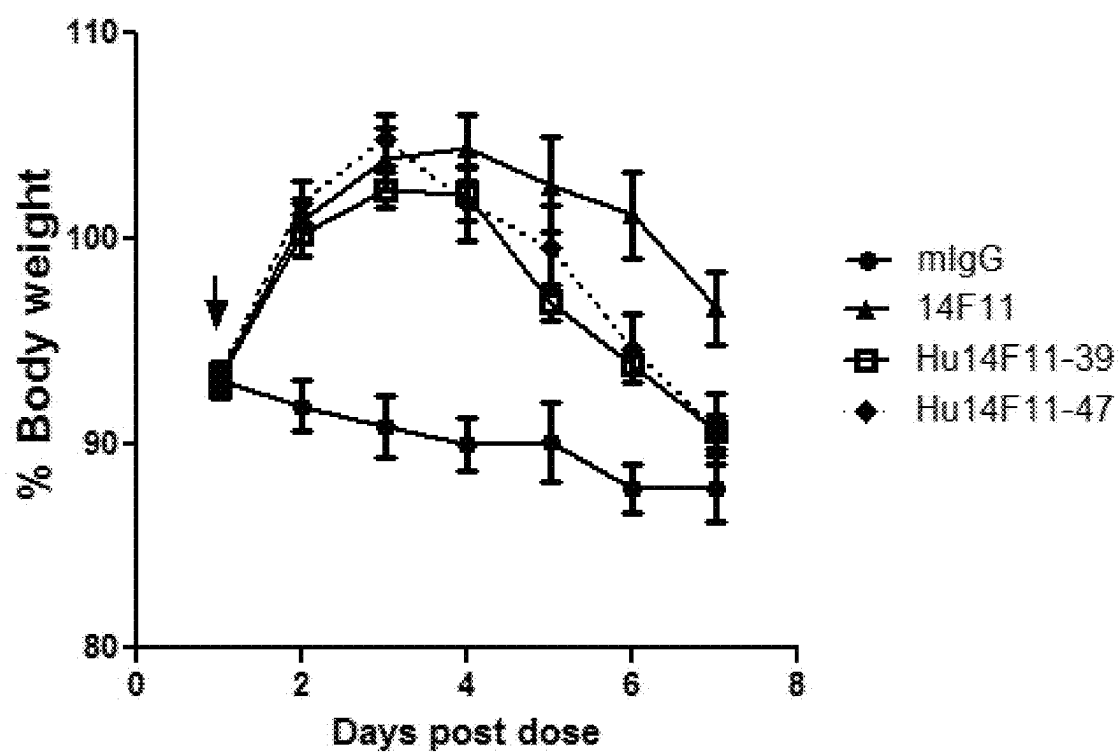


FIG. 26

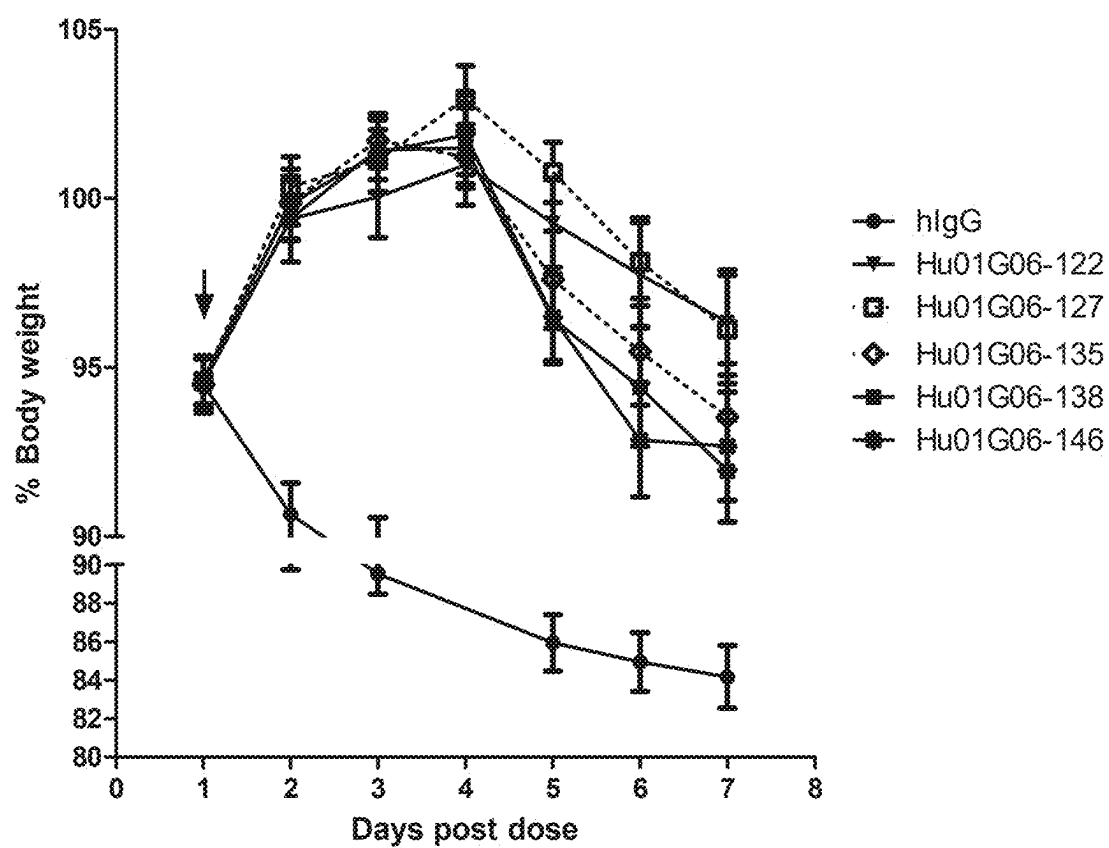


FIG. 27

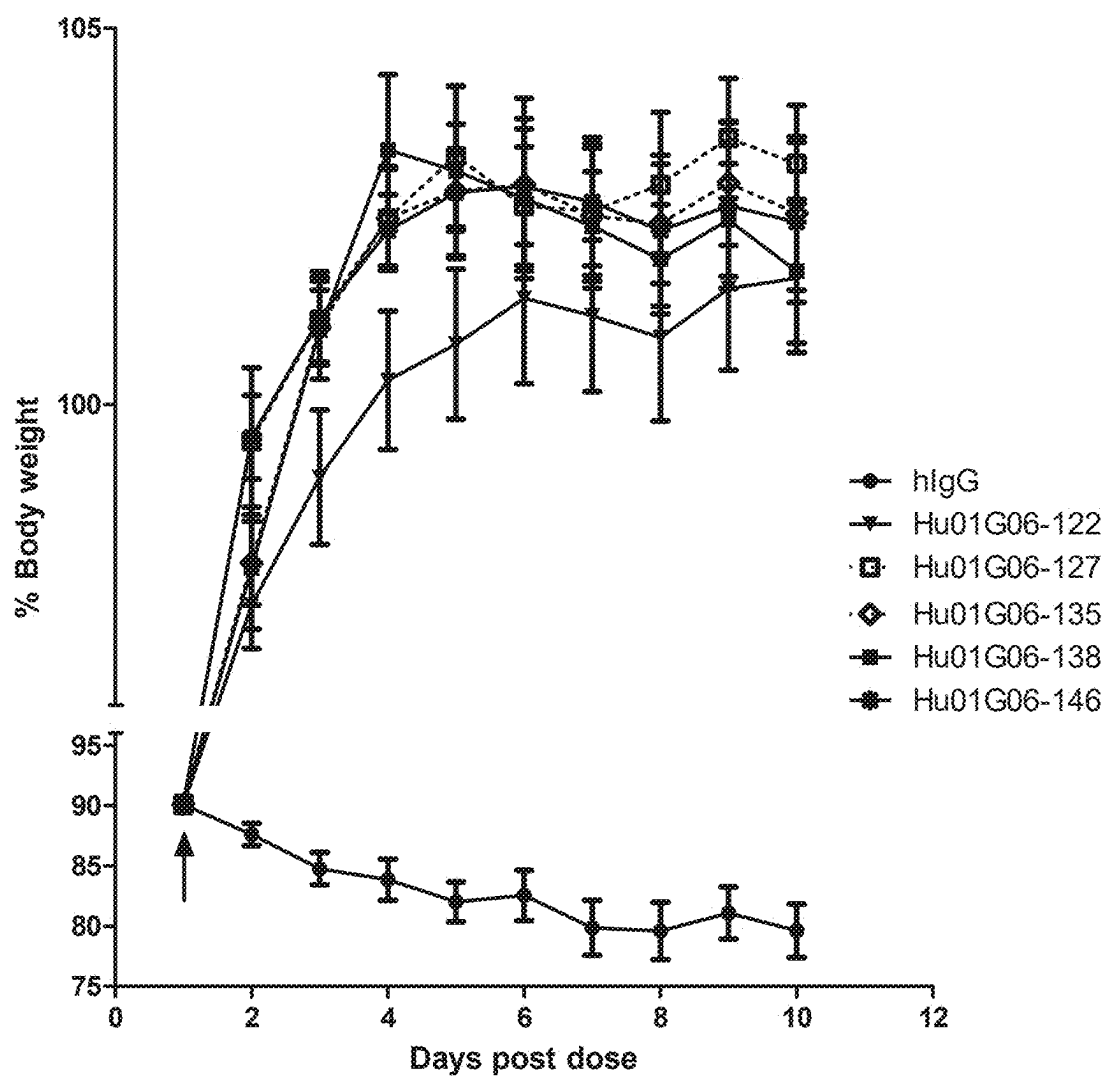


FIG. 28

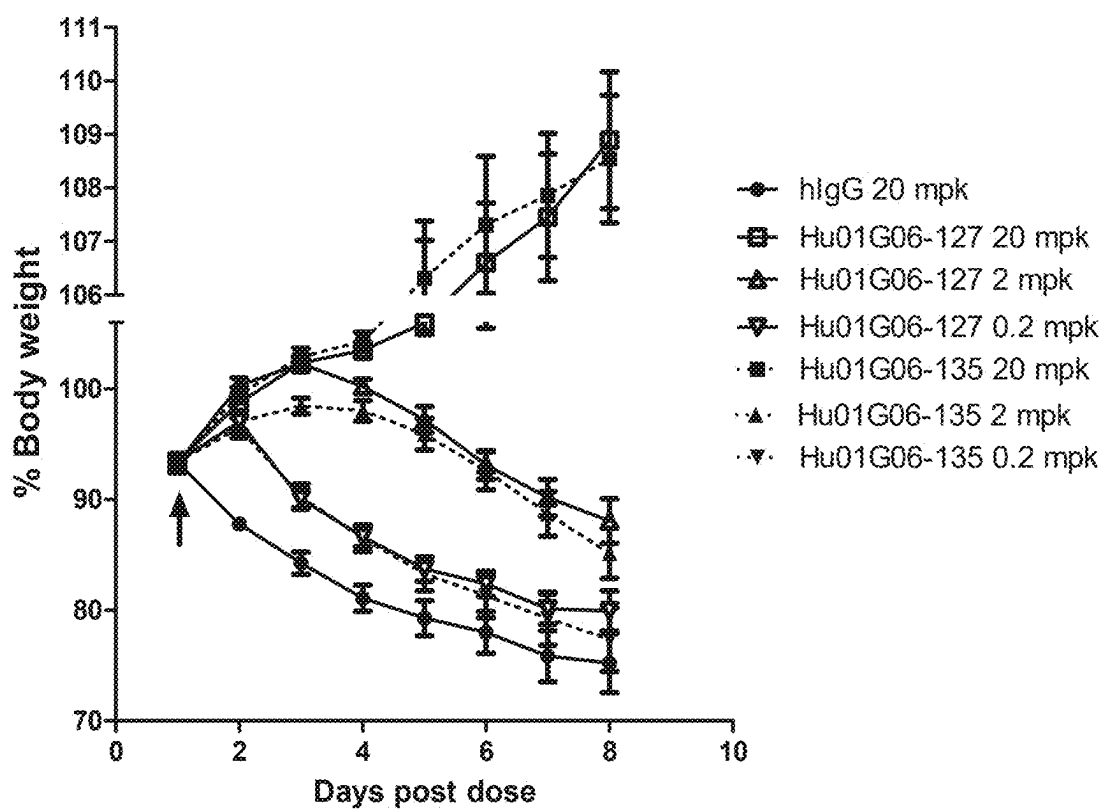


FIG. 29A

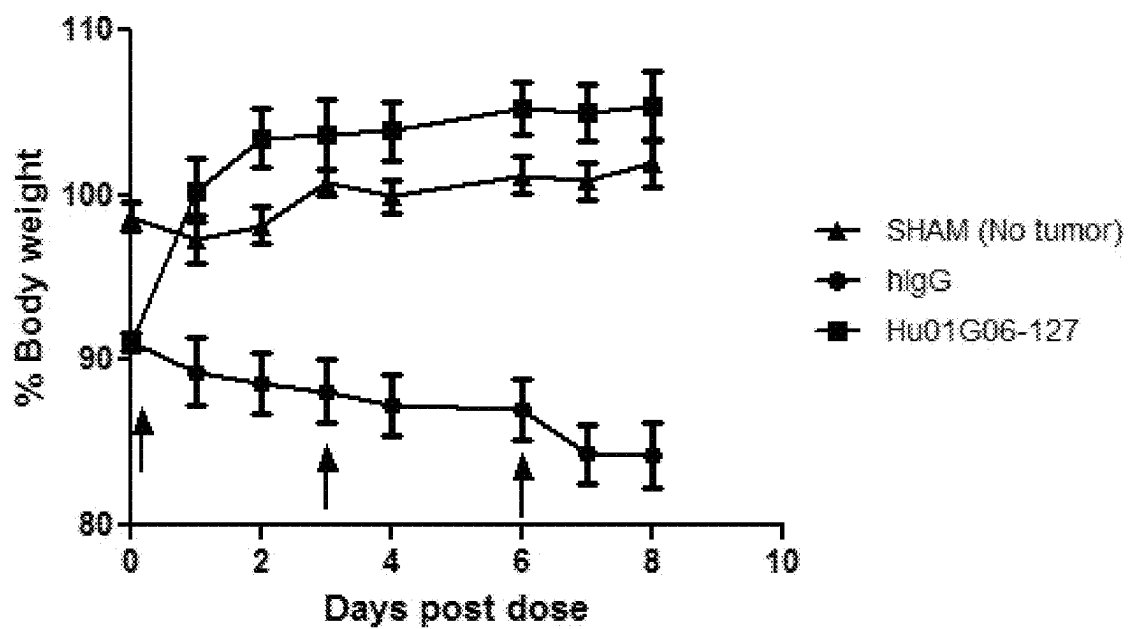


FIG. 29B

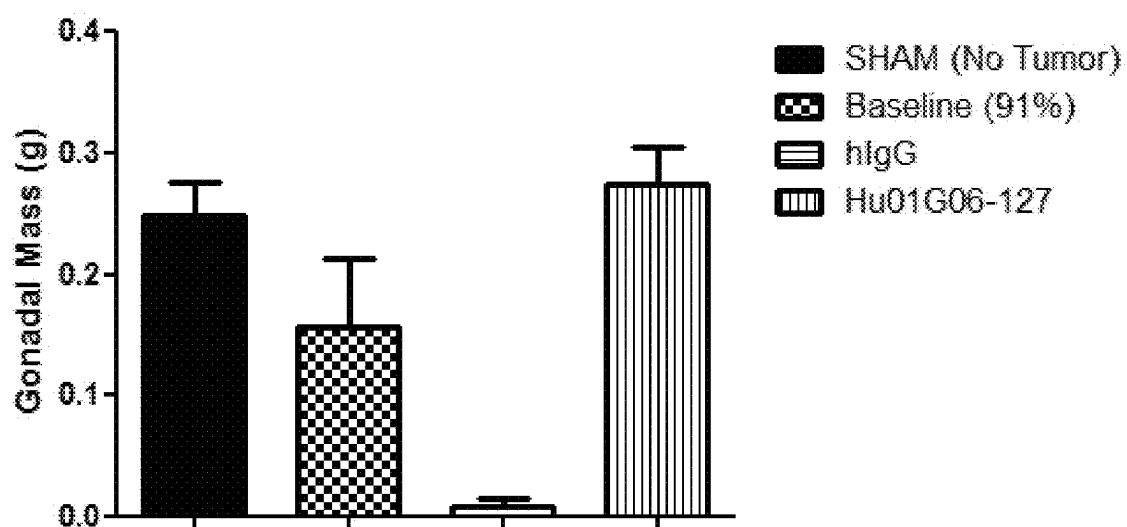
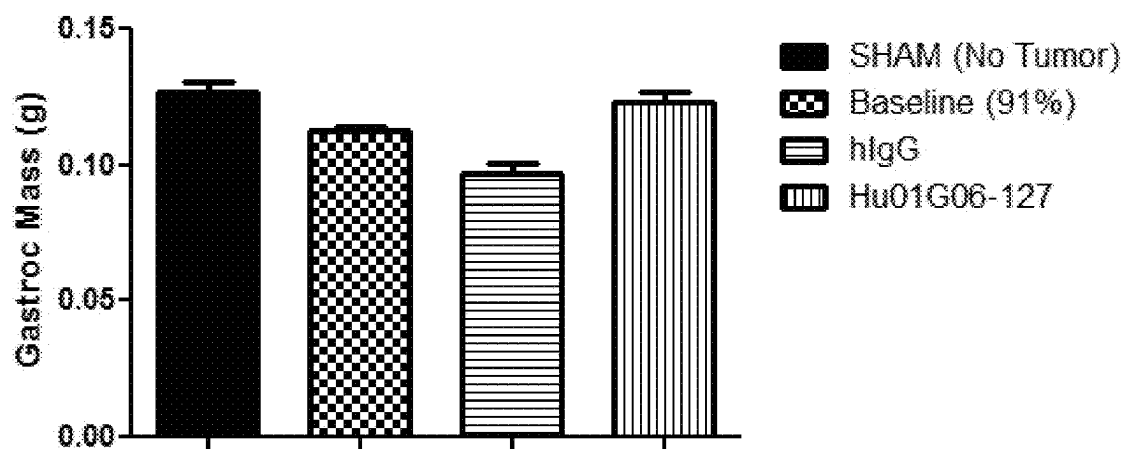


FIG. 29C



ANTI-GDF-15 ANTIBODIES

CROSS-REFERENCE TO RELATED APPLICATIONS

[0001] This application is a continuation of U.S. patent application Ser. No. 18/336,713, filed Jun. 16, 2023, which is a continuation of U.S. patent application Ser. No. 16/824,034, filed Mar. 19, 2020, issued as U.S. Pat. No. 11,725,047 on Aug. 15, 2023, which is a continuation of U.S. patent application Ser. No. 15/655,263, filed Jul. 20, 2017, issued as U.S. Pat. No. 10,597,444 on Mar. 24, 2020, which is a continuation of U.S. patent application Ser. No. 14/863,870, filed Sep. 24, 2015, issued as U.S. Pat. No. 9,725,505 on Aug. 8, 2017, which is a divisional of U.S. patent application Ser. No. 14/137,415, filed Dec. 20, 2013, issued as U.S. Pat. No. 9,175,076 on Nov. 3, 2015, which claims the benefit of and priority to U.S. Provisional Patent Application No. 61/827,325, filed May 24, 2013, and U.S. Provisional Patent Application No. 61/745,508, filed Dec. 21, 2012, the entire disclosures of each of which are incorporated by reference herein in their entireties.

SEQUENCE LISTING

[0002] The instant application contains a Sequence Listing which has been submitted electronically in ST.26 XML format and is hereby incorporated by reference in its entirety. Said ST.26 XML copy, created on Jun. 16, 2023, is named 406767_029USD1C3_SEQUENCE_LISTING_ST.26.XML and is 425,984 bytes in size.

FIELD OF THE INVENTION

[0003] The field of the invention is molecular biology, immunology, cachexia and cachexia-like disorders, and oncology. More particularly, the field is therapeutic antibodies.

BACKGROUND

[0004] Involuntary weight loss can be categorized into three primary etiologies that include, cachexia, sarcopenia and starvation. Cachexia is a debilitating metabolic syndrome associated with numerous diseases, including cancer, AIDS, chronic heart failure (also known as congestive heart failure), chronic obstructive pulmonary disease (COPD), chronic kidney disease, tuberculosis, sepsis and other forms of systemic inflammation. Cachexia varies in its manifestations, but generally involves involuntary loss of skeletal muscle mass and some form of underlying illness (Evans et al. (2008) CLIN. NUTR. 27:793-799). Cachexia is a wasting disorder involving involuntary weight loss and may be associated with systemic inflammation and/or an acute inflammatory response. Thomas (2007) CLIN. NUTRITION 26:389-399. Loss of fat mass as well as fat-free mass, such as muscle mass, often is a prominent clinical feature of cachexia. In many but not all cases, cachexia progresses through stages that have been designated precachexia, cachexia and refractory cachexia (Fearon et al. (2011) LANCET ONC. 12:489-495). Two different, but sometimes overlapping, processes appear to drive the development and progression of cachexia: (a) metabolic processes that act directly on muscle, reducing its mass and function; and (b) reduced food intake, which leads to loss of both fat and muscle (Tsai et al. (2012) J. CACHEXIA SARCOPENIA MUSCLE 3:239-243).

[0005] Although cachexia is a complex and incompletely understood syndrome, it is clear that GDF15 (also known as MIC-1, PLAB, PDF and NAG-1), a member of the TGF- β superfamily, is an important mediator of cachexia in various diseases (Tsai et al., supra). At least some tumors over-express and secrete GDF15, and elevated serum GDF15 levels have been associated with various cancers (Johnen et al. (2007) NAT. MED. 13:1333-1340; Bauskin et al. (2006) CANCER RES. 66:4983-4986). Monoclonal antibodies against GDF15 have been recognized as potential anti-cachexia therapeutic agents. See, e.g., U.S. Pat. No. 8,192,735.

[0006] Weight loss resulting from cachexia is associated with poor prognosis in various diseases (Evans et al., supra), and cachexia and its consequences are considered to be the direct cause of death in about 20% of cancer deaths (Tisdale (2002) NAT. REV. CANCER 2:862-871). Cachexia is infrequently reversed by nutritional intervention, and currently this syndrome is seldom treated with drug therapy (Evans et al., supra).

[0007] Sarcopenia is a clinical condition related to cachexia that is characterized by loss of skeletal muscle mass and muscle strength. The decrease in muscle mass can lead to functional impairment, with loss of strength, increased likelihood of falls, and loss of autonomy. Respiratory function may also be impaired with a reduced vital capacity. During metabolic stress, muscle protein is rapidly mobilized in order to provide the immune system, liver and gut with amino acids, particularly glutamine. Sarcopenia often is a disease of the elderly; however, its development may also be associated with muscle disuse and malnutrition, and may coincide with cachexia. Sarcopenia can be diagnosed based upon functional observations such as low muscle weight and low gait speed. See, e.g., Muscaritoli et al. (2010) CLIN. NUTRITION 29:154-159.

[0008] Starvation typically results in a loss of body fat and non-fat mass due to inadequate diet and/or nutritional uptake (Thomas (2007) supra). The effects of starvation often are reversed by improving diet and nutritional, for example, protein, uptake.

[0009] Naturally occurring antibodies are multimeric proteins that contain four polypeptide chains (FIG. 1). Two of the polypeptide chains are called heavy chains (H chains), and two of the polypeptide chains are called light chains (L chains). The immunoglobulin heavy and light chains are connected by an interchain disulfide bond. The immunoglobulin heavy chains are connected by interchain disulfide bonds. A light chain consists of one variable region (V_L in FIG. 1) and one constant region (C_L in FIG. 1). The heavy chain consists of one variable region (V_H in FIG. 1) and at least three constant regions (C_{H1} , C_{H2} and C_{H3} in FIG. 1). The variable regions determine the specificity of the antibody. Each variable region comprises three hypervariable regions also known as complementarity determining regions (CDRs) flanked by four relatively conserved framework regions (FRs). The three CDRs, referred to as CDR₁, CDR₂, and CDR₃, contribute to the antibody binding specificity. Naturally occurring antibodies have been used as starting material for engineered antibodies, such as chimeric antibodies and humanized antibodies.

[0010] There is a significant unmet need for effective therapeutic agents for treating cachexia and sarcopenia, including monoclonal antibodies targeting GDF15. Such

therapeutic agents have the potential to play an important role in the treatment of various cancers and other life-threatening diseases.

SUMMARY

[0011] The invention is based, in part, upon the discovery of a family of antibodies that specifically bind human GDF15 (hGDF15). The antibodies contain hGDF15 binding sites based on the CDRs of the antibodies. The antibodies can be used as therapeutic agents. When used as therapeutic agents, the antibodies are engineered, e.g., humanized, to reduce or eliminate an immune response when administered to a human patient.

[0012] The disclosed antibodies prevent or inhibit the activity of (i.e., neutralize) hGDF15. When administered to a mammal, the antibodies can inhibit the loss of muscle mass, for example, the loss of muscle mass associated with an underlying disease. The underlying disease may be selected from the group consisting of cancer, chronic heart failure, chronic kidney disease, COPD, AIDS, multiple sclerosis, rheumatoid arthritis, sepsis, and tuberculosis. In some embodiments, the loss of muscle mass may be accompanied by a loss of fat mass. The disclosed antibodies can also be used to inhibit involuntary weight loss in a mammal. In some embodiments, the disclosed antibodies may also be used to inhibit the loss of organ mass. Further, a method of treating cachexia and/or sarcopenia in a mammal comprising administering an effective amount of one of at least one of the disclosed antibodies to a mammal in need thereof is disclosed.

[0013] Also disclosed is a method for establishing a steady-state level of mature recombinant human GDF15 (rhGDF15) in plasma or serum in a mammal comprising administering a rhGDF15-immunoglobulin Fc (Fc-rhGDF15) fusion protein to the mammal. The Fc-rhGDF15 can be a mouse Fc mature recombinant human GDF15 (mFc-rhGDF15). In some embodiments, the mammal is a rodent, e.g., a mouse.

[0014] In another aspect, a method of treating obesity in a mammal, for example, a human, comprising administering a therapeutically effective amount of Fc-rhGDF15, e.g., a human Fc mature recombinant human GDF15 (hFc-rhGDF15), to the mammal in need thereof, is disclosed. Pharmaceutical compositions comprising an Fc-rhGDF15 fusion protein and a pharmaceutically acceptable carrier are also disclosed.

[0015] These and other aspects and advantages of the invention will become apparent upon consideration of the following figures, detailed description, and claims. As used herein, "including" means without limitation, and examples cited are non-limiting. As used herein, "antibody 01G06, 03G05, 04F08, 06C11, 08G01, 14F11, or 17B11" means antibody 01G06, 03G05, 04F08, 06C11, 08G01, 14F11, or 17B11, or humanized variants thereof.

DESCRIPTION OF THE DRAWINGS

[0016] The invention can be more completely understood with reference to the following drawings.

[0017] FIG. 1 (prior art) is a schematic representation of a typical naturally-occurring antibody.

[0018] FIG. 2 is a graph representing results from an experiment to measure hGDF15 serum levels in naïve mice

or mice bearing human xenograft tumors (Chago, RPMI7951, PC3, TOV21G, HT-1080, K-562, LS1034), as determined by ELISA.

[0019] FIG. 3 is a plot representing results from an experiment to determine the plasma pharmacokinetics (PK) of cleaved rhGDF15 administered by subcutaneous injection (1 µg/g) in naïve ICR-SCID mice, as determined by ELISA.

[0020] FIG. 4 is a graph summarizing results from an experiment to measure cachectic activity of cleaved rhGDF15 protein (■) and negative control (PBS; ●) to induce body weight loss in immune-incompetent mice, ICR-SCID. Arrows indicate subcutaneous doses of 1 µg/g of rhGDF15.

[0021] FIGS. 5A and 5B are graphs summarizing results from an experiment to measure cachectic activity of mFc-rhGDF15 (a mouse Fc fused to the amino terminus of a mature recombinant human GDF15; ■), rFc-rhGDF15 (a rabbit Fc fused to the amino terminus of a mature recombinant mouse GDF15; ▲), and negative control (PBS; ●) to induce body weight loss in immune-competent Balb/C mice (FIG. 5A) and immune-incompetent CB17-SCID mice (FIG. 5B). Arrows indicate subcutaneous doses of 1 µg/g of recombinant protein.

[0022] FIGS. 6A-6E are graphs summarizing results from an experiment to demonstrate cachectic activity of mFc-rhGDF15 (■) and negative control (PBS; ●) to induce body weight loss in immune-incompetent ICR-SCID mice (FIG. 6A; arrows indicate subcutaneous doses of 1 µg/g of mFc-rhGDF15); to induce loss of adipose tissue or gonadal fat mass (FIG. 6B); to induce loss of muscle mass of gastrocnemius muscle (FIG. 6C; Gastroc Mass); and to increase mRNA expression of muscle degradation molecular markers (mMuRF1 (FIG. 6D) and mAtrogin (FIG. 6E)).

[0023] FIG. 7 is a graph summarizing results from an experiment to measure cachectic activity of mFc-rhGDF15 (■) and negative control (PBS; ●) to induce body weight loss in immune-incompetent Balb/C nude mice. Arrows indicate subcutaneous doses of 1.33 µg/g of mFc-rhGDF15.

[0024] FIG. 8 is a graph summarizing results from an experiment to measure serum levels of mFc-rhGDF15 in mice dosed with the recombinant protein. The presence of mFc-rhGDF15 was determined by Western Blot. Two positive bands corresponding to mFc-rhGDF15 and rhGDF15 (according to the appropriate molecular size) were quantified by Licor. The percentage of released-rhGDF15 versus mFc-rhGDF15 was calculated.

[0025] FIG. 9A is a graph summarizing results from an experiment to measure cachectic activity of mFc-rhGDF15 (0.1 µg/g (■), 0.01 µg/g (Δ)) and negative control (mIgG 0.1p g/g (●)) to induce body weight loss in immune-incompetent ICR-SCID mice. Arrows indicate the intraperitoneal dose of the recombinant protein. FIG. 9B is a graph representing the total level of rhGDF15 in the plasma of mice dosed with mFc-rhGDF15 (0.1 µg/g (□), 0.01 µg/g (■)) five days post dose, as determined by ELISA.

[0026] FIG. 10 is a sequence alignment showing the amino acid sequence of the complete immunoglobulin heavy chain variable region of antibodies 01G06, 03G05, 04F08, 06C11, 08G01, 14F11, and 17B11. The amino acid sequences for each antibody are aligned against one another, and CDR₁, CDR₂, and CDR₃, are identified in boxes. The unboxed sequences represent framework (FR) sequences. Alignment positioning (gaps) is based on Kabat numbering,

rather than an alignment algorithm such as Clustal. Numbering above the sequences represents Kabat numbering.

[0027] FIG. 11 is a sequence alignment showing the CDR₁, CDR₂, and CDR₃ sequences for each of the immunoglobulin heavy chain variable region sequences in FIG. 10.

[0028] FIG. 12 is a sequence alignment showing the amino acid sequence of the complete immunoglobulin light chain variable region of antibodies 01G06, 03G05, 04F08, 06C11, 08G01, 14F11, and 17B11. The amino acid sequences for each antibody are aligned against one another, and CDR₁, CDR₂, and CDR₃, are identified in boxes. The unboxed sequences represent framework (FR) sequences. Alignment positioning (gaps) is based on Kabat numbering, rather than an alignment algorithm such as Clustal. Numbering above the sequences represents Kabat numbering.

[0029] FIG. 13 is a sequence alignment showing the CDR₁, CDR₂, and CDR₃ sequences for each of the immunoglobulin light chain variable region sequences in FIG. 12.

[0030] FIG. 14 is a graph summarizing results from an experiment to measure cachectic inhibitory activity of anti-GDF15 antibodies 01G06 (■), 03G05 (▲), 04F08 (▼), 06C11 (◇), 14F11 (◊), and 17B11 (□), and a murine IgG control (●; mIgG) dosed at 10 mg/kg in an mFc-rhGDF15 cachectic model in ICR-SCID mice. The arrow indicates intra-peritoneal injection of antibody.

[0031] FIG. 15 is a graph summarizing results from an experiment to measure cachectic inhibitory activity of anti-GDF15 antibodies 01G06 (▲), 03G05 (◇), 04F08 (▼), 06C11 (□), 08G01 (■), 14F11 (◆), and 17B11 (*), and a murine IgG control (●; mIgG), dosed at 10 mg/kg in an HT-1080 fibrosarcoma tumor xenograft model in ICR-SCID mice. The arrows indicate intra-peritoneal injection of antibody every three days.

[0032] FIGS. 16A-16E are graphs summarizing results from an experiment to demonstrate anti-cachectic activity of anti-GDF15 antibody 01G06 (■), dosed at 10 mg/kg, in immune-incompetent mice (ICR-SCID) bearing an HT-1080 fibrosarcoma tumor xenograft model. Treatment with antibody 01G06 reversed body weight loss (FIG. 16A); induced a significant increase in food consumption for up to three days post dose (FIG. 16B); induced a gain of gonadal fat mass (FIG. 16C); induced a gain of muscle mass of gastrocnemius muscle (FIG. 16D); and decreased mRNA expression of muscle degradation molecular markers (mMuRF1 and mAtrogin (FIG. 16E)) compared to negative control (murine IgG (●)). In FIG. 16A, the arrow indicates intra-peritoneal injection of antibody.

[0033] FIGS. 17A-17B are graphs summarizing results from an experiment to demonstrate anti-cachectic activity of anti-GDF15 antibody 01G06 (■), dosed at 2 mg/kg, in immune-incompetent mice (ICR-SCID) bearing an HT-1080 fibrosarcoma tumor xenograft model. Treatment with antibody 01G06 reversed body weight loss compared to murine IgG (●) (FIG. 17A); and induced a gain of organ mass (liver, heart, spleen, kidney) and induced a gain of tissue mass (gonadal and gastrocnemius) (FIG. 17B) compared to negative control (murine IgG) and baseline (day 1). The arrows in FIG. 17A indicate intra-peritoneal injection of antibody.

[0034] FIG. 18 is a graph summarizing results from an experiment to measure cachectic inhibitory activity of anti-GDF15 antibodies 01G06 (■), 03G05 (▲), 04F08 (X), 06C11 (◆), 08G01 (○), 14F11 (□), and 17B11 (Δ), and a

murine IgG control (●) dosed at 10 mg/kg in a K-562 leukemia tumor xenograft model in immune-incompetent (CB17SCRFMF) mice. The arrows indicate intra-peritoneal injection of antibody.

[0035] FIG. 19 is a sequence alignment showing the amino acid sequence of the complete immunoglobulin heavy chain variable region of chimeric 01G06 variable region denoted as Ch01G06 Chimeric; humanized 01G06 heavy chain variable regions denoted as Hu01G06 IGHV1-18, Hu01G06 IGHV1-69, Sh01G06 IGHV1-18 M69L, Sh01G06 IGHV1-18 M69L K64Q G44S, Sh01G06 IGHV1-18 M69L K64Q, Sh01G06 IGHV1-69 T30S I69L, Sh01G06 IGHV1-69 T30S K64Q I69L, Hu01G06 IGHV1-18 F1, Hu01G06 IGHV1-18 F2, Hu01G06 IGHV1-69 F1, and Hu01G06 IGHV1-69 F2; chimeric 06C11 denoted as Ch06C11 Chimeric; humanized 06C11 heavy chain variable regions denoted as HE LM 06C11 IGHV2-70, and Hu06C11 IGHV2-5; chimeric 14F11 denoted as Ch14F11 Chimeric; and humanized 14F11 heavy chain variable regions denoted as Sh14F11 IGHV2-5 and Sh14F11 IGHV2-70. The amino acid sequences for each antibody are aligned against one another, and CDR₁, CDR₂, and CDR₃, are identified in boxes. The unboxed sequences represent framework (FR) sequences. Alignment positioning (gaps) is based on Kabat numbering, rather than an alignment algorithm such as Clustal. Numbering above the sequences represents Kabat numbering.

[0036] FIG. 20 is a sequence alignment showing the CDR₁, CDR₂, and CDR₃ sequences for each of the immunoglobulin heavy chain variable region sequences in FIG. 19.

[0037] FIG. 21 is a sequence alignment showing the amino acid sequence of the complete immunoglobulin light chain variable region of chimeric 01G06 denoted as Ch01G06 Chimeric; humanized 01G06 light chain variable regions denoted as Hu01G06 IGKV1-39, Hu01G06 IGKV1-39 S43A V48I, Hu01G06 IGKV1-39 V48I, Hu01G06 IGKV1-39 F1, and Hu01G06 IGKV1-39 F2; chimeric 06C11 denoted as Ch06C11 Chimeric; humanized 06C11 light chain variable region denoted as Sh06C11 IGKV1-16; chimeric 14F11 denoted as Ch14F11 Chimeric; and humanized 14F11 light chain variable region denoted as Hu14F11 IGKV1-16. The amino acid sequences for each antibody are aligned against one another, and CDR₁, CDR₂, and CDR₃, are identified in boxes. The unboxed sequences represent framework (FR) sequences. Alignment positioning (gaps) is based on Kabat numbering, rather than an alignment algorithm such as Clustal. Numbering above the sequences represents Kabat numbering.

[0038] FIG. 22 is a sequence alignment showing the CDR₁, CDR₂, and CDR₃ sequences for each of the immunoglobulin light chain variable region sequences in FIG. 21.

[0039] FIG. 23 is a graph summarizing results from an experiment to measure cachectic inhibitory activity of anti-GDF15 antibodies 01G06 (■), Hu01G06-46 (▲), and Hu01G06-52 (*), and a murine IgG control (●) dosed at 2 mg/kg in an HT-1080 fibrosarcoma tumor xenograft model in ICR-SCID mice. The arrow indicates intra-peritoneal injection of antibody.

[0040] FIG. 24 is a graph summarizing results from an experiment to measure cachectic inhibitory activity of anti-GDF15 antibodies 06C11 (◆), Hu06C11-27 (□), and Hu06C11-30 (▲), and a murine IgG control (●) dosed at 2

mg/kg in an HT-1080 fibrosarcoma tumor xenograft model in ICR-SCID mice. The arrow indicates intra-peritoneal injection of antibody.

[0041] FIG. 25 is a graph summarizing results from an experiment to measure cachectic inhibitory activity of anti-GDF15 antibodies 14F11 (▲), Hu14F11-39 (□), and Hu14F11-47 (◆), and a murine IgG control (●) dosed at 2 mg/kg in an HT-1080 fibrosarcoma tumor xenograft model in ICR-SCID mice. The arrow indicates intra-peritoneal injection of antibody.

[0042] FIG. 26 is a graph summarizing results from an experiment to measure cachectic inhibitory activity of anti-GDF15 antibodies Hu01G06-122 (▼), Hu01G06-127 (□), Hu01G06-135 (◇), Hu01G06-138 (■), and Hu01G06-146 (*), and a human IgG control (●) dosed at 2 mg/kg in an HT-1080 fibrosarcoma tumor xenograft model in ICR-SCID mice. The arrow indicates intra-peritoneal injection of antibody.

[0043] FIG. 27 is a graph summarizing results from an experiment to measure cachectic inhibitory activity of anti-GDF15 antibodies Hu01G06-122 (▼), Hu01G06-127 (□), Hu01G06-135 (◇), Hu01G06-138 (■), and Hu01G06-146 (*), and a human IgG control (●) dosed at 2 mg/kg in an mFc-rhGDF15 cachectic model in ICR-SCID mice. The arrow indicates intra-peritoneal injection of antibody.

[0044] FIG. 28 is a graph summarizing results from an experiment to measure cachectic dose response inhibitory activity of anti-GDF15 antibodies Hu01G06-127 dosed at 20 mg/kg (□), 2 mg/kg (Δ), and 0.2 mg/kg (▽); Hu01G06-135 at 20 mg/kg (■), 2 mg/kg (▲), and 0.2 mg/kg (⋈), and a human IgG control at 20 mg/kg (●) in an HT-1080 fibrosarcoma tumor xenograft model in ICR-SCID mice. The arrow indicates intravenous injection of antibody.

[0045] FIGS. 29A-29C are graphs summarizing results from an experiment to demonstrate anti-cachectic activity of anti-GDF15 antibodies Hu01G06-127 (■), dosed at 10 mg/kg, in immune-incompetent mice (ICR-SCID) bearing an HT-1080 fibrosarcoma tumor xenograft model. Treatment with antibody Hu01G06-127 reversed body weight loss (FIG. 29A); induced a gain of gonadal fat mass (FIG. 29B); and induced a gain of muscle mass of gastrocnemius muscle (FIG. 29C) compared to negative control (hIgG (●); FIG. 29A) similar to levels found in non tumor bearing mice (SHAM (▲); FIG. 29A). The arrows in FIG. 29A indicate intra-peritoneal injection of antibody.

DETAILED DESCRIPTION

[0046] The anti-GDF15 antibodies disclosed herein are based on the antigen binding sites of certain monoclonal antibodies that have been selected on the basis of binding and neutralization of human GDF15 (hGDF15). The antibodies contain immunoglobulin variable region CDR sequences that define a binding site for hGDF15.

[0047] By virtue of the neutralizing activity of these antibodies, they are useful for treating cachexia and/or sarcopenia. For use as therapeutic agents, the antibodies can be engineered to minimize or eliminate an immune response when administered to a human patient. Various features and aspects of the invention are discussed in more detail below.

[0048] As used herein, “cachexia” means a metabolic syndrome associated with underlying disease and characterized by involuntary loss of muscle mass. Cachexia is often accompanied by involuntary weight loss, loss of fat mass, anorexia, inflammation, insulin resistance, fatigue, weak-

ness, significant loss of appetite, and/or increased muscle protein breakdown. Cachexia is distinct from starvation, age-related loss of muscle mass, malabsorption, and hyperthyroidism. Underlying diseases associated with cachexia include cancer, chronic heart failure, chronic kidney disease, COPD, AIDS, multiple sclerosis, rheumatoid arthritis, sepsis, and tuberculosis.

[0049] As used herein, “sarcopenia” is understood to be a condition characterized primarily by loss of skeletal muscle mass and muscle strength. Sarcopenia is frequently associated with aging. See, Ruegg and Glass (2011) ANNUAL REV. PHARMACOL. TOXICOL. 51:373-395. In one approach, sarcopenia can be identified in a subject if a value of the appendicular skeletal muscle mass of a subject divided by the height of the subject in meters is more than two standard deviations below the young normal mean. (Thomas (2007) supra; see also Baumgartner et al. (1999) MECH. AGEING DEV. 147:755-763).

[0050] As used herein, unless otherwise indicated, “antibody” means an intact antibody (e.g., an intact monoclonal antibody) or antigen-binding fragment of an antibody, including an intact antibody or antigen-binding fragment that has been modified or engineered, or that is a human antibody. Examples of antibodies that have been modified or engineered are chimeric antibodies, humanized antibodies, and multispecific antibodies (e.g., bispecific antibodies). Examples of antigen-binding fragments include Fab, Fab', F(ab')₂, Fv, single chain antibodies (e.g., scFv), minibodies and diabodies.

I. Antibodies that Bind GDF15

[0051] The antibodies disclosed herein comprise: (a) an immunoglobulin heavy chain variable region comprising the structure CDR_{H1}-CDR_{H2}-CDR_{H3} and (b) an immunoglobulin light chain variable region comprising the structure CDR_{L1}-CDR_{L2}-CDR_{L3}, wherein the heavy chain variable region and the light chain variable region together define a single binding site for binding hGDF15 protein.

[0052] In some embodiments, the antibody comprises: (a) an immunoglobulin heavy chain variable region comprising the structure CDR_{H1}-CDR_{H2}-CDR_{H3} and (b) an immunoglobulin light chain variable region, wherein the heavy chain variable region and the light chain variable region together define a single binding site for binding hGDF15. A CDR_{H1} comprises an amino acid sequence selected from the group consisting of SEQ ID NO:1 (01G06, 08G01, Ch01G06 Chimeric, Hu01G06 IGHV1-18, Hu01G06 IGHV1-69, Sh01G06 IGHV1-18 M69L, Sh01G06 IGHV1-18 M69L K64Q G44S, Sh01G06 IGHV1-18 M69L K64Q, Sh01G06 IGHV1-69 T30S I69L, Sh01G06 IGHV1-69 T30S K64Q I69L, Hu01G06 IGHV1-18 F1, Hu01G06 IGHV1-18 F2, Hu01G06 IGHV1-69 F1, Hu01G06 IGHV1-69 F2), SEQ ID NO:2 (03G05), SEQ ID NO:3 (04F08), SEQ ID NO:4 (06C11, Ch06C11 Chimeric, HE LM 06C11 IGHV2-70, Hu06C11 IGHV2-5), SEQ ID NO:5 (14F11, Ch14F11 Chimeric, Sh14F11 IGHV2-5, Sh14F11 IGHV2-70), and SEQ ID NO:6 (17B11); a CDR_{H2} comprises an amino acid sequence selected from the group consisting of SEQ ID NO:7 (01G06, Ch01G06 Chimeric, Hu01G06 IGHV1-18, Hu01G06 IGHV1-69, Sh01G06 IGHV1-18 M69L, Sh01G06 IGHV1-69 T30S I69L), SEQ ID NO:8 (03G05), SEQ ID NO:9 (04F08, 06C11, Ch06C11 Chimeric, Hu06C11 IGHV2-5), SEQ ID NO:10 (08G01), SEQ ID NO:11 (14F11, Ch14F11 Chimeric, Sh14F11 IGHV2-5, Sh14F11 IGHV2-70), SEQ ID NO:12 (17B11), SEQ ID

NO:13 (Sh01G06 IGHV1-18 M69L K64Q G44S, Sh01G06 IGHV1-18 M69L K64Q, Sh01G06 IGHV1-69 T30S K64Q I69L), SEQ ID NO:236 (Hu01G06 IGHV1-18 F1), SEQ ID NO:237 (Hu01G06 IGHV1-18 F2), SEQ ID NO:238 (Hu01G06 IGHV1-69 F1), SEQ ID NO:239 (Hu01G06 IGHV1-69 F2), and SEQ ID NO:14 (HE LM 06C11 IGHV2-70); and a CDR_{H3} comprises an amino acid sequence selected from the group consisting of SEQ ID NO:15 (01G06, 08G01, Ch01G06 Chimeric, Hu01G06 IGHV1-18, Hu01G06 IGHV1-69, Sh01G06 IGHV1-18 M69L, Sh01G06 IGHV1-18 M69L K64Q G44S, Sh01G06 IGHV1-18 M69L K64Q, Sh01G06 IGHV1-69 T30S I69L, Sh01G06 IGHV1-69 T30S K64Q I69L, Hu01G06 IGHV1-18 F1, Hu01G06 IGHV1-18 F2, Hu01G06 IGHV1-69 F1, Hu01G06 IGHV1-69 F2), SEQ ID NO:16 (03G05), SEQ ID NO:17 (04F08), SEQ ID NO:18 (06C11, Ch06C11 Chimeric, HE LM 06C11 IGHV2-70, Hu06C11 IGHV2-5), SEQ ID NO:19 (14F11, Ch14F11 Chimeric, Sh14F11 IGHV2-5, Sh14F11 IGHV2-70), and SEQ ID NO:20 (17B11). Throughout this specification, a particular SEQ ID NO. is followed in parentheses by the antibody that was the origin of that sequence. For example, “SEQ ID NO:2 (03G05)” means that SEQ ID NO:2 comes from antibody 03G05.

[0053] In some embodiments, the antibody comprises an immunoglobulin heavy chain variable region comprising a CDR_{H1} comprising the amino acid sequence of SEQ ID NO:1 (01G06, Ch01G06 Chimeric, Hu01G06 IGHV1-18, Hu01G06 IGHV1-69, Sh01G06 IGHV1-69 T30S I69L, Hu01G06 IGHV1-18 F1, Hu01G06 IGHV1-18 F2, Hu01G06 IGHV1-69 F1, Hu01G06 IGHV1-69 F2), a CDR_{H2} comprising the amino acid sequence of SEQ ID NO:7 (01G06, Ch01G06 Chimeric, Hu01G06 IGHV1-18, Hu01G06 IGHV1-69, Sh01G06 IGHV1-69 T30S I69L), and a CDR_{H3} comprising the amino acid sequence of SEQ ID NO:15 (01G06, Ch01G06 Chimeric, Hu01G06 IGHV1-18, Hu01G06 IGHV1-69, Sh01G06 IGHV1-69 T30S I69L, Hu01G06 IGHV1-18 F1, Hu01G06 IGHV1-18 F2, Hu01G06 IGHV1-69 F1, Hu01G06 IGHV1-69 F2).

[0054] In some embodiments, the antibody comprises an immunoglobulin heavy chain variable region comprising a CDR_{H1} comprising the amino acid sequence of SEQ ID NO:2 (03G05), a CDR_{H2} comprising the amino acid sequence of SEQ ID NO:8 (03G05), and a CDR_{H3} comprising the amino acid sequence of SEQ ID NO:16 (03G05).

[0055] In some embodiments, the antibody comprises an immunoglobulin heavy chain variable region comprising a CDR_{H1} comprising the amino acid sequence of SEQ ID NO:3 (04F08), a CDR_{H2} comprising the amino acid sequence of SEQ ID NO:9 (04F08), and a CDR_{H3} comprising the amino acid sequence of SEQ ID NO:17 (04F08).

[0056] In some embodiments, the antibody comprises an immunoglobulin heavy chain variable region comprising a CDR_{H1} comprising the amino acid sequence of SEQ ID NO:4 (06C11, Ch06C11 Chimeric, Hu06C11 IGHV2-5), a CDR_{H2} comprising the amino acid sequence of SEQ ID NO:9 (06C11, Ch06C11 Chimeric, Hu06C11 IGHV2-5), and a CDR_{H3} comprising the amino acid sequence of SEQ ID NO:18 (06C11, Ch06C11 Chimeric, Hu06C11 IGHV2-5).

[0057] In some embodiments, the antibody comprises an immunoglobulin heavy chain variable region comprising a CDR_{H1} comprising the amino acid sequence of SEQ ID NO:1 (08G01), a CDR_{H2} comprising the amino acid

sequence of SEQ ID NO:10 (08G01), and a CDR_{H3} comprising the amino acid sequence of SEQ ID NO:15 (08G01).

[0058] In some embodiments, the antibody comprises an immunoglobulin heavy chain variable region comprising a CDR_{H1} comprising the amino acid sequence of SEQ ID NO:5 (14F11, Ch14F11 Chimeric, Sh14F11 IGHV2-5, Sh14F11 IGHV2-70), a CDR_{H2} comprising the amino acid sequence of SEQ ID NO:11 (14F11, Ch14F11 Chimeric, Sh14F11 IGHV2-5, Sh14F11 IGHV2-70), and a CDR_{H3} comprising the amino acid sequence of SEQ ID NO:19 (14F11, Ch14F11 Chimeric, Sh14F11 IGHV2-5, Sh14F11 IGHV2-70).

[0059] In some embodiments, the antibody comprises an immunoglobulin heavy chain variable region comprising a CDR_{H1} comprising the amino acid sequence of SEQ ID NO:6 (17B11), a CDR_{H2} comprising the amino acid sequence of SEQ ID NO:12 (17B11), and a CDR_{H3} comprising the amino acid sequence of SEQ ID NO:20 (17B11).

[0060] In some embodiments, the antibody comprises an immunoglobulin heavy chain variable region comprising a CDR_{H1} comprising the amino acid sequence of SEQ ID NO:1 (Sh01G06 IGHV1-18 M69L K64Q G44S, Sh01G06 IGHV1-18 M69L K64Q, Sh01G06 IGHV1-69 T30S K64Q I69L, Hu01G06 IGHV1-18 F1, Hu01G06 IGHV1-18 F2, Hu01G06 IGHV1-69 F1, Hu01G06 IGHV1-69 F2), a CDR_{H2} comprising the amino acid sequence of SEQ ID NO:13 (Sh01G06 IGHV1-18 M69L K64Q G44S, Sh01G06 IGHV1-18 M69L K64Q, Sh01G06 IGHV1-69 T30S K64Q I69L), and a CDR_{H3} comprising the amino acid sequence of SEQ ID NO:15 (Sh01G06 IGHV1-18 M69L K64Q G44S, Sh01G06 IGHV1-18 M69L K64Q, Sh01G06 IGHV1-69 T30S K64Q I69L, Hu01G06 IGHV1-18 F1, Hu01G06 IGHV1-18 F2, Hu01G06 IGHV1-69 F1, Hu01G06 IGHV1-69 F2).

[0061] In some embodiments, the antibody comprises an immunoglobulin heavy chain variable region comprising a CDR_{H1} comprising the amino acid sequence of SEQ ID NO:1 (Sh01G06 IGHV1-18 M69L K64Q G44S, Sh01G06 IGHV1-18 M69L K64Q, Sh01G06 IGHV1-69 T30S K64Q I69L, Hu01G06 IGHV1-18 F1, Hu01G06 IGHV1-18 F2, Hu01G06 IGHV1-69 F1, Hu01G06 IGHV1-69 F2), a CDR_{H2} comprising the amino acid sequence of SEQ ID NO:236 (Hu01G06 IGHV1-18 F1), and a CDR_{H3} comprising the amino acid sequence of SEQ ID NO:15 (Sh01G06 IGHV1-18 M69L K64Q G44S, Sh01G06 IGHV1-18 M69L K64Q, Sh01G06 IGHV1-69 T30S K64Q I69L, Hu01G06 IGHV1-18 F1, Hu01G06 IGHV1-18 F2, Hu01G06 IGHV1-69 F1, Hu01G06 IGHV1-69 F2).

[0062] In some embodiments, the antibody comprises an immunoglobulin heavy chain variable region comprising a CDR_{H1} comprising the amino acid sequence of SEQ ID NO:1 (Sh01G06 IGHV1-18 M69L K64Q G44S, Sh01G06 IGHV1-18 M69L K64Q, Sh01G06 IGHV1-69 T30S K64Q I69L, Hu01G06 IGHV1-18 F1, Hu01G06 IGHV1-18 F2, Hu01G06 IGHV1-69 F1, Hu01G06 IGHV1-69 F2), a CDR_{H2} comprising the amino acid sequence of SEQ ID NO:237 (Hu01G06 IGHV1-18 F2), and a CDR_{H3} comprising the amino acid sequence of SEQ ID NO:15 (Sh01G06 IGHV1-18 M69L K64Q G44S, Sh01G06 IGHV1-18 M69L K64Q, Sh01G06 IGHV1-69 T30S K64Q I69L, Hu01G06 IGHV1-18 F1, Hu01G06 IGHV1-18 F2, Hu01G06 IGHV1-69 F1, Hu01G06 IGHV1-69 F2).

[0063] In some embodiments, the antibody comprises an immunoglobulin heavy chain variable region comprising a

CDR_{H1} comprising the amino acid sequence of SEQ ID NO:1 (Sh01G06 IGHV1-18 M69L K64Q G44S, Sh01G06 IGHV1-18 M69L K64Q, Sh01G06 IGHV1-69 T30S K64Q I69L, Hu01G06 IGHV1-18 F1, Hu01G06 IGHV1-18 F2, Hu01G06 IGHV1-69 F1, Hu01G06 IGHV1-69 F2), a CDR_{H2} comprising the amino acid sequence of SEQ ID NO:238 (Hu01G06 IGHV1-69 F1), and a CDR_{H3} comprising the amino acid sequence of SEQ ID NO:15 (Sh01G06 IGHV1-18 M69L K64Q G44S, Sh01G06 IGHV1-18 M69L K64Q, Sh01G06 IGHV1-69 T30S K64Q I69L, Hu01G06 IGHV1-18 F1, Hu01G06 IGHV1-18 F2, Hu01G06 IGHV1-69 F1, Hu01G06 IGHV1-69 F2).

[0064] In some embodiments, the antibody comprises an immunoglobulin heavy chain variable region comprising a CDR_{H1} comprising the amino acid sequence of SEQ ID NO:1 (Sh01G06 IGHV1-18 M69L K64Q G44S, Sh01G06 IGHV1-18 M69L K64Q, Sh01G06 IGHV1-69 T30S K64Q I69L, Hu01G06 IGHV1-18 F1, Hu01G06 IGHV1-18 F2, Hu01G06 IGHV1-69 F1, Hu01G06 IGHV1-69 F2), a CDR_{H2} comprising the amino acid sequence of SEQ ID NO:239 (Hu01G06 IGHV1-69 F2), and a CDR_{H3} comprising the amino acid sequence of SEQ ID NO:15 (Sh01G06 IGHV1-18 M69L K64Q G44S, Sh01G06 IGHV1-18 M69L K64Q, Sh01G06 IGHV1-69 T30S K64Q I69L, Hu01G06 IGHV1-18 F1, Hu01G06 IGHV1-18 F2, Hu01G06 IGHV1-69 F1, Hu01G06 IGHV1-69 F2).

[0065] In some embodiments, the antibody comprises an immunoglobulin heavy chain variable region comprising a CDR_{H1} comprising the amino acid sequence of SEQ ID NO:4 (HE LM 06C11 IGHV2-70), a CDR_{H2} comprising the amino acid sequence of SEQ ID NO:14 (HE LM 06C11 IGHV2-70), and a CDR_{H3} comprising the amino acid sequence of SEQ ID NO:18 (HE LM 06C11 IGHV2-70).

[0066] Preferably, the CDR_{H1}, CDR_{H2}, and CDR_{H3} sequences are interposed between fully human or humanized immunoglobulin FR sequences.

[0067] In some embodiments, the antibody comprises (a) an immunoglobulin light chain variable region comprising the structure CDR_{L1}-CDR_{L2}-CDR_{L3}, and (b) an immunoglobulin heavy chain variable region, wherein the immunoglobulin light chain variable region and the immunoglobulin heavy chain variable region together define a single binding site for binding hGDF15. A CDR_{L1} comprises an amino acid sequence selected from the group consisting of SEQ ID NO:21 (01G06, Ch01G06 Chimeric, Hu01G06 IGKV1-39, Hu01G06 IGKV1-39 S43A V48I, Hu01G06 IGKV1-39 V48I, Hu01G06 IGKV1-39 F1, Hu01G06 IGKV1-39 F2), SEQ ID NO:22 (03G05), SEQ ID NO:23 (04F08, 06C11, Ch06C11 Chimeric, Sh06C11 IGKV1-16, 14F11, Ch14F11 Chimeric, Hu14F11 IGKV1-16), SEQ ID NO:24 (08G01), and SEQ ID NO:25 (17B11); a CDR_{L2} comprises an amino acid sequence selected from the group consisting of SEQ ID NO:26 (01G06, Ch01G06 Chimeric, Hu01G06 IGKV1-39, Hu01G06 IGKV1-39 S43A V48I, Hu01G06 IGKV1-39 V48I, Hu01G06 IGKV1-39 F1, Hu01G06 IGKV1-39 F2), SEQ ID NO:27 (03G05), SEQ ID NO:28 (04F08, 06C11, Ch06C11 Chimeric, Sh06C11 IGKV1-16), SEQ ID NO:29 (08G01), SEQ ID NO:30 (14F11, Ch14F11 Chimeric, Hu14F11 IGKV1-16), and SEQ ID NO:31 (17B11); and a CDR_{L3} comprises an amino acid sequence selected from the group consisting of SEQ ID NO:32 (01G06, Ch01G06 Chimeric, Hu01G06 IGKV1-39, Hu01G06 IGKV1-39 S43A V48I, Hu01G06 IGKV1-39 V48I, 08G01, Hu01G06 IGKV1-39 F1), SEQ ID NO:244 (Hu01G06 IGKV1-39 F2),

SEQ ID NO:33 (03G05), SEQ ID NO:34 (04F08), SEQ ID NO:35 (06C11, Ch06C11 Chimeric, Sh06C11 IGKV1-16), SEQ ID NO:36 (14F11, Ch14F11 Chimeric, Hu14F11 IGKV1-16), and SEQ ID NO:37 (17B11).

[0068] In some embodiments, the antibody comprises an immunoglobulin light chain variable region comprising a CDR_{L1} comprising the amino acid sequence of SEQ ID NO:21 (01G06, Ch01G06 Chimeric, Hu01G06 IGKV1-39, Hu01G06 IGKV1-39 S43A V48I, Hu01G06 IGKV1-39 V48I, Hu01G06 IGKV1-39 F1, Hu01G06 IGKV1-39 F2), a CDR_{L2} comprising the amino acid sequence of SEQ ID NO:26 (01G06, Ch01G06 Chimeric, Hu01G06 IGKV1-39, Hu01G06 IGKV1-39 S43A V48I, Hu01G06 IGKV1-39 V48I, Hu01G06 IGKV1-39 F1, Hu01G06 IGKV1-39 F2), and a CDR_{L3} comprising the amino acid sequence of SEQ ID NO:32 (01G06, Ch01G06 Chimeric, Hu01G06 IGKV1-39, Hu01G06 IGKV1-39 S43A V48I, Hu01G06 IGKV1-39 V48I, Hu01G06 IGKV1-39 F1).

[0069] In some embodiments, the antibody comprises an immunoglobulin light chain variable region comprising a CDR_{L1} comprising the amino acid sequence of SEQ ID NO:21 (01G06, Ch01G06 Chimeric, Hu01G06 IGKV1-39, Hu01G06 IGKV1-39 S43A V48I, Hu01G06 IGKV1-39 V48I, Hu01G06 IGKV1-39 F1, Hu01G06 IGKV1-39 F2), a CDR_{L2} comprising the amino acid sequence of SEQ ID NO:26 (01G06, Ch01G06 Chimeric, Hu01G06 IGKV1-39, Hu01G06 IGKV1-39 S43A V48I, Hu01G06 IGKV1-39 V48I, Hu01G06 IGKV1-39 F1, Hu01G06 IGKV1-39 F2), and a CDR_{L3} comprising the amino acid sequence of SEQ ID NO:244 (Hu01G06 IGKV1-39 F2).

[0070] In some embodiments, the antibody comprises an immunoglobulin light chain variable region comprising a CDR_{L1} comprising the amino acid sequence of SEQ ID NO:22 (03G05), a CDR_{L2} comprising the amino acid sequence of SEQ ID NO:27 (03G05), and a CDR_{L3} comprising the amino acid sequence of SEQ ID NO:33 (03G05).

[0071] In some embodiments, the antibody comprises an immunoglobulin light chain variable region comprising a CDR_{L1} comprising the amino acid sequence of SEQ ID NO:23 (04F08), a CDR_{L2} comprising the amino acid sequence of SEQ ID NO:28 (04F08), and a CDR_{L3} comprising the amino acid sequence of SEQ ID NO:34 (04F08).

[0072] In some embodiments, the antibody comprises an immunoglobulin light chain variable region comprising a CDR_{L1} comprising the amino acid sequence of SEQ ID NO:23 (06C11, Ch06C11 Chimeric, Sh06C11 IGKV1-16), a CDR_{L2} comprising the amino acid sequence of SEQ ID NO:28 (06C11, Ch06C11 Chimeric, Sh06C11 IGKV1-16), and a CDR_{L3} comprising the amino acid sequence of SEQ ID NO:35 (06C11, Ch06C11 Chimeric, Sh06C11 IGKV1-16).

[0073] In some embodiments, the antibody comprises an immunoglobulin light chain variable region comprising a CDR_{L1} comprising the amino acid sequence of SEQ ID NO:24 (08G01), a CDR_{L2} comprising the amino acid sequence of SEQ ID NO:29 (08G01), and a CDR_{L3} comprising the amino acid sequence of SEQ ID NO:32 (08G01).

[0074] In some embodiments, the antibody comprises an immunoglobulin light chain variable region comprising a CDR_{L1} comprising the amino acid sequence of SEQ ID NO:23 (14F11, Ch14F11 Chimeric, Hu14F11 IGKV1-16), a CDR_{L2} comprising the amino acid sequence of SEQ ID NO:30 (14F11, Ch14F11 Chimeric, Hu14F11 IGKV1-16),

and a CDR_{L3} comprising the amino acid sequence of SEQ ID NO:36 (14F11, Ch14F11 Chimeric, Hu14F11 IGKV1-16).

[0075] In some embodiments, the antibody comprises an immunoglobulin light chain variable region comprising a CDR_{L1} comprising the amino acid sequence of SEQ ID NO:25 (17B11), a CDR_{L2} comprising the amino acid sequence of SEQ ID NO:31 (17B11), and a CDR_{L3} comprising the amino acid sequence of SEQ ID NO:37 (17B11).

[0076] Preferably, the CDR_{L1}, CDR_{L2}, and CDR_{L3} sequences are interposed between fully human or humanized immunoglobulin FR sequences.

[0077] In some embodiments, the antibody comprises: (a) an immunoglobulin heavy chain variable region comprising the structure CDR_{H1}-CDR_{H2}-CDR_{H3} and (b) an immunoglobulin light chain variable region comprising the structure CDR_{L1}-CDR_{L2}-CDR_{L3}, wherein the heavy chain variable region and the light chain variable region together define a single binding site for binding hGDF15. The CDR_{H1} is an amino acid sequence selected from the group consisting of SEQ ID NO:1 (01G06, 08G01, Ch01G06 Chimeric, Hu01G06 IGHV1-18, Hu01G06 IGHV1-69, Sh01G06 IGHV1-18 M69L, Sh01G06 IGHV1-18 M69L K64Q G44S, Sh01G06 IGHV1-18 M69L K64Q, Sh01G06 IGHV1-69 T30S I69L, Sh01G06 IGHV1-69 T30S K64Q I69L, Hu01G06 IGHV1-18 F1, Hu01G06 IGHV1-18 F2, Hu01G06 IGHV1-69 F1, Hu01G06 IGHV1-69 F2), SEQ ID NO:2 (03G05), SEQ ID NO:3 (04F08), SEQ ID NO:4 (06C11, Ch06C11 Chimeric, HE LM 06C11 IGHV2-70, Hu06C11 IGHV2-5), SEQ ID NO:5 (14F11, Ch14F11 Chimeric, Sh14F11 IGHV2-5, Sh14F11 IGHV2-70), and SEQ ID NO:6 (17B11); the CDR_{H2} is an amino acid sequence selected from the group consisting of SEQ ID NO:7 (01G06, Ch01G06 Chimeric, Hu01G06 IGHV1-18, Hu01G06 IGHV1-69, Sh01G06 IGHV1-18 M69L, Sh01G06 IGHV1-69 T30S I69L), SEQ ID NO:8 (03G05), SEQ ID NO:9 (04F08, 06C11, Ch06C11 Chimeric, Hu06C11 IGHV2-5), SEQ ID NO:10 (08G01), SEQ ID NO:11 (14F11, Ch14F11 Chimeric, Sh14F11 IGHV2-5, Sh14F11 IGHV2-70), SEQ ID NO:12 (17B11), SEQ ID NO:13 (Sh01G06 IGHV1-18 M69L K64Q G44S, Sh01G06 IGHV1-18 M69L K64Q, Sh01G06 IGHV1-69 T30S K64Q I69L), SEQ ID NO:236 (Hu01G06 IGHV1-18 F1), SEQ ID NO:237 (Hu01G06 IGHV1-18 F2), SEQ ID NO:238 (Hu01G06 IGHV1-69 F1), SEQ ID NO:239 (Hu01G06 IGHV1-69 F2), and SEQ ID NO:14 (HE LM 06C11 IGHV2-70); and the CDR_{H3} is an amino acid sequence selected from the group consisting of SEQ ID NO:15 (01G06, 08G01, Ch01G06 Chimeric, Hu01G06 IGHV1-18, Hu01G06 IGHV1-69, Sh01G06 IGHV1-18 M69L, Sh01G06 IGHV1-18 M69L K64Q G44S, Sh01G06 IGHV1-18 M69L K64Q, Sh01G06 IGHV1-69 T30S I69L, Sh01G06 IGHV1-69 T30S K64Q I69L, Hu01G06 IGHV1-18 F1, Hu01G06 IGHV1-18 F2, Hu01G06 IGHV1-69 F1, Hu01G06 IGHV1-69 F2), SEQ ID NO:16 (03G05), SEQ ID NO:17 (04F08), SEQ ID NO:18 (06C11, Ch06C11 Chimeric, HE LM 06C11 IGHV2-70, Hu06C11 IGHV2-5), SEQ ID NO:19 (14F11, Ch14F11 Chimeric, Sh14F11 IGHV2-5, Sh14F11 IGHV2-70), and SEQ ID NO:20 (17B11). The CDR_{L1} is an amino acid sequence selected from the group consisting of SEQ ID NO:21 (01G06, Ch01G06 Chimeric, Hu01G06 IGKV1-39, Hu01G06 IGKV1-39 S43A V48I, Hu01G06 IGKV1-39 V48I, Hu01G06 IGKV1-39 F1, Hu01G06 IGKV1-39 F2), SEQ ID NO:22 (03G05), SEQ ID NO:23 (04F08, 06C11,

Ch06C11 Chimeric, Sh06C11 IGKV1-16, 14F11, Ch14F11 Chimeric, Hu14F11 IGKV1-16), SEQ ID NO:24 (08G01), and SEQ ID NO:25 (17B11); the CDR_{L2} is an amino acid sequence selected from the group consisting of SEQ ID NO:26 (01G06, Ch01G06 Chimeric, Hu01G06 IGKV1-39, Hu01G06 IGKV1-39 S43A V48I, Hu01G06 IGKV1-39 V48I, Hu01G06 IGKV1-39 F1, Hu01G06 IGKV1-39 F2), SEQ ID NO:27 (03G05), SEQ ID NO:28 (04F08, 06C11, Ch06C11 Chimeric, Sh06C11 IGKV1-16), SEQ ID NO:29 (08G01), SEQ ID NO:30 (14F11, Ch14F11 Chimeric, Hu14F11 IGKV1-16), and SEQ ID NO:31 (17B11); and the CDR_{L3} is an amino acid sequence selected from the group consisting of SEQ ID NO:32 (01G06, Ch01G06 Chimeric, Hu01G06 IGKV1-39, Hu01G06 IGKV1-39 S43A V48I, Hu01G06 IGKV1-39 V48I, 08G01, Hu01G06 IGKV1-39 F1), SEQ ID NO:244 (Hu01G06 IGKV1-39 F2), SEQ ID NO:33 (03G05), SEQ ID NO:34 (04F08), SEQ ID NO:35 (06C11, Ch06C11 Chimeric, Sh06C11 IGKV1-16), SEQ ID NO:36 (14F11, Ch14F11 Chimeric, Hu14F11 IGKV1-16), and SEQ ID NO:37 (17B11).

[0078] In some embodiments, the antibody comprises an immunoglobulin heavy chain variable region comprising a CDR_{H1} comprising an amino acid sequence selected from the group consisting of SEQ ID NO:1 and SEQ ID NO:38 (Hu01G06 IGHV1-18 F1), a CDR_{H2} comprising an amino acid sequence selected from the group consisting of SEQ ID NO:236 and SEQ ID NO:240 (Hu01G06 IGHV1-18 F1), and a CDR_{H3} comprising the amino acid sequence of SEQ ID NO:15 (Hu01G06 IGHV1-18 F1); and an immunoglobulin light chain variable region comprising a CDR_{L1} comprising the amino acid sequence of SEQ ID NO:21 (Hu01G06 IGKV1-39 F1), a CDR_{L2} comprising the amino acid sequence of SEQ ID NO:26 (Hu01G06 IGKV1-39 F1), and a CDR_{L3} comprising the amino acid sequence of SEQ ID NO:32 (Hu01G06 IGKV1-39 F1).

[0079] In some embodiments, the antibody comprises an immunoglobulin heavy chain variable region comprising a CDR_{H1} comprising an amino acid sequence selected from the group consisting of SEQ ID NO:1 and SEQ ID NO:38 (Hu01G06 IGHV1-18 F2), a CDR_{H2} comprising an amino acid sequence selected from the group consisting of SEQ ID NO:237 and SEQ ID NO:241 (Hu01G06 IGHV1-18 F2), and a CDR_{H3} comprising the amino acid sequence of SEQ ID NO:15 (Hu01G06 IGHV1-18 F2); and an immunoglobulin light chain variable region comprising a CDR_{L1} comprising the amino acid sequence of SEQ ID NO:21 (Hu01G06 IGKV1-39 F2), a CDR_{L2} comprising the amino acid sequence of SEQ ID NO:26 (Hu01G06 IGKV1-39 F2), and a CDR_{L3} comprising the amino acid sequence of SEQ ID NO:244 (Hu01G06 IGKV1-39 F2).

[0080] In some embodiments, the antibody comprises an immunoglobulin heavy chain variable region comprising a CDR_{H1} comprising an amino acid sequence selected from the group consisting of SEQ ID NO:1 and SEQ ID NO:234 (Hu01G06 IGHV1-69 F1), a CDR_{H2} comprising an amino acid sequence selected from the group consisting of SEQ ID NO:238 and SEQ ID NO:241 (Hu01G06 IGHV1-69 F1), and a CDR_{H3} comprising the amino acid sequence of SEQ ID NO:15 (Hu01G06 IGHV1-69 F1); and an immunoglobulin light chain variable region comprising a CDR_{L1} comprising the amino acid sequence of SEQ ID NO:21 (Hu01G06 IGKV1-39 F1), a CDR_{L2} comprising the amino acid sequence of SEQ ID NO:26 (Hu01G06 IGKV1-39 F1),

and a CDR_{L3} comprising the amino acid sequence of SEQ ID NO:32 (Hu01G06 IGKV1-39 F1).

[0081] In some embodiments, the antibody comprises an immunoglobulin heavy chain variable region comprising a CDR_{H1} comprising an amino acid sequence selected from the group consisting of SEQ ID NO:1 and SEQ ID NO:234 (Hu01G06 IGHV1-69 F2), a CDR_{H2} comprising an amino acid sequence selected from the group consisting of SEQ ID NO:239 and SEQ ID NO:240 (Hu01G06 IGHV1-69 F2), and a CDR_{H3} comprising the amino acid sequence of SEQ ID NO:15 (Hu01G06 IGHV1-69 F2); and an immunoglobulin light chain variable region comprising a CDR_{L1} comprising the amino acid sequence of SEQ ID NO:21 (Hu01G06 IGKV1-39 F1), a CDR_{L2} comprising the amino acid sequence of SEQ ID NO:26 (Hu01G06 IGKV1-39 F1), and a CDR_{L3} comprising the amino acid sequence of SEQ ID NO:32 (Hu01G06 IGKV1-39 F1).

[0082] In some embodiments, the antibody comprises an immunoglobulin heavy chain variable region comprising a CDR_{H1} comprising an amino acid sequence selected from the group consisting of SEQ ID NO:1 and SEQ ID NO:234 (Hu01G06 IGHV1-69 F2), a CDR_{H2} comprising an amino acid sequence selected from the group consisting of SEQ ID NO:239 and SEQ ID NO:240 (Hu01G06 IGHV1-69 F2), and a CDR_{H3} comprising the amino acid sequence of SEQ ID NO:15 (Hu01G06 IGHV1-69 F2); and an immunoglobulin light chain variable region comprising a CDR_{L1} comprising the amino acid sequence of SEQ ID NO:21 (Hu01G06 IGKV1-39 F2), a CDR_{L2} comprising the amino acid sequence of SEQ ID NO:26 (Hu01G06 IGKV1-39 F2), and a CDR_{L3} comprising the amino acid sequence of SEQ ID NO:244 (Hu01G06 IGKV1-39 F2).

[0083] The antibodies disclosed herein comprise an immunoglobulin heavy chain variable region and an immunoglobulin light chain variable region. In some embodiments, the antibody comprises an immunoglobulin heavy chain variable region selected from the group consisting of SEQ ID NO:40 (01G06, Ch01G06 Chimeric), SEQ ID NO:42 (03G05), SEQ ID NO:44 (04F08), SEQ ID NO:46 (06C11, Ch06C11 Chimeric), SEQ ID NO:48 (08G01), SEQ ID NO:50 (14F11, Ch14F11 Chimeric), SEQ ID NO:52 (17B11), SEQ ID NO:54 (Hu01G06 IGHV1-18), SEQ ID NO:56 (Hu01G06 IGHV1-69), SEQ ID NO:58 (Sh01G06 IGHV1-18 M69L), SEQ ID NO:60 (Sh01G06 IGHV1-18 M69L K64Q G44S), SEQ ID NO:62 (Sh01G06 IGHV1-18 M69L K64Q), SEQ ID NO:64 (Sh01G06 IGHV1-69 T30S I69L), SEQ ID NO:66 (Sh01G06 IGHV1-69 T30S K64Q I69L), SEQ ID NO:246 (Hu01G06 IGHV1-18 F1), SEQ ID NO:248 (Hu01G06 IGHV1-18 F2), SEQ ID NO:250 (Hu01G06 IGHV1-69 F1), SEQ ID NO:252 (Hu01G06 IGHV1-69 F2), SEQ ID NO:68 (HE LM 06C11 IGHV2-70), SEQ ID NO:70 (Hu06C11 IGHV2-5), SEQ ID NO:72 (Sh14F11 IGHV2-5), and SEQ ID NO:74 (Sh14F11 IGHV2-70); and an immunoglobulin light chain variable region.

[0084] In other embodiments, the antibody comprises an immunoglobulin light chain variable region selected from the group consisting of SEQ ID NO:76 (01G06, Ch01G06 Chimeric), SEQ ID NO:78 (03G05), SEQ ID NO:80 (04F08), SEQ ID NO:82 (06C11, Ch06C11 Chimeric), SEQ ID NO:84 (08G01), SEQ ID NO:86 (14F11, Ch14F11 Chimeric), SEQ ID NO:88 (17B11), SEQ ID NO:90 (Hu01G06 IGKV1-39), SEQ ID NO:92 (Hu01G06 IGKV1-39 S43A V48I or Hu01G06 IGKV1-39 F1), SEQ ID NO:94

(Hu01G06 IGKV1-39 V48I), SEQ ID NO:96 (Sh06C11 IGKV1-16), SEQ ID NO:254 (Hu01G06 IGKV1-39 F2), and SEQ ID NO:98 (Hu14F11 IGKV1-16), and an immunoglobulin heavy chain variable region.

[0085] In some embodiments, the antibody comprises an immunoglobulin heavy chain variable region selected from the group consisting of SEQ ID NO:40 (01G06, Ch01G06 Chimeric), SEQ ID NO:42 (03G05), SEQ ID NO:44 (04F08), SEQ ID NO:46 (06C11, Ch06C11 Chimeric), SEQ ID NO:48 (08G01), SEQ ID NO:50 (14F11, Ch14F11 Chimeric), SEQ ID NO:52 (17B11), SEQ ID NO:54 (Hu01G06 IGHV1-18), SEQ ID NO:56 (Hu01G06 IGHV1-69), SEQ ID NO:58 (Sh01G06 IGHV1-18 M69L), SEQ ID NO:60 (Sh01G06 IGHV1-18 M69L K64Q G44S), SEQ ID NO:62 (Sh01G06 IGHV1-18 M69L K64Q), SEQ ID NO:64 (Sh01G06 IGHV1-69 T30S I69L), SEQ ID NO:66 (Sh01G06 IGHV1-69 T30S K64Q I69L), SEQ ID NO:246 (Hu01G06 IGHV1-18 F1), SEQ ID NO:248 (Hu01G06 IGHV1-18 F2), SEQ ID NO:250 (Hu01G06 IGHV1-69 F1), SEQ ID NO:252 (Hu01G06 IGHV1-69 F2), SEQ ID NO:68 (HE LM 06C11 IGHV2-70), SEQ ID NO:70 (Hu06C11 IGHV2-5), SEQ ID NO:72 (Sh14F11 IGHV2-5), and SEQ ID NO:74 (Sh14F11 IGHV2-70), and an immunoglobulin light chain variable region selected from the group consisting of SEQ ID NO:76 (01G06, Ch01G06 Chimeric), SEQ ID NO:78 (03G05), SEQ ID NO:80 (04F08), SEQ ID NO:82 (06C11, Ch06C11 Chimeric), SEQ ID NO:84 (08G01), SEQ ID NO:86 (14F11, Ch14F11 Chimeric), SEQ ID NO:88 (17B11), SEQ ID NO:90 (Hu01G06 IGKV1-39), SEQ ID NO:92 (Hu01G06 IGKV1-39 S43A V48I or Hu01G06 IGKV1-39 F1), SEQ ID NO:94 (Hu01G06 IGKV1-39 V48I), SEQ ID NO:96 (Sh06C11 IGKV1-16), SEQ ID NO:254 (Hu01G06 IGKV1-39 F2), and SEQ ID NO:98 (Hu14F11 IGKV1-16).

[0086] In some embodiments, the antibody comprises an immunoglobulin heavy chain variable region comprising the amino acid sequence of SEQ ID NO:40 (01G06, Ch01G06 Chimeric), and an immunoglobulin light chain variable region comprising the amino acid sequence of SEQ ID NO:76 (01G06, Ch01G06 Chimeric).

[0087] In some embodiments, the antibody comprises an immunoglobulin heavy chain variable region comprising the amino acid sequence of SEQ ID NO:42 (03G05), and an immunoglobulin light chain variable region comprising the amino acid sequence of SEQ ID NO:78 (03G05).

[0088] In some embodiments, the antibody comprises an immunoglobulin heavy chain variable region comprising the amino acid sequence of SEQ ID NO:44 (04F08), and an immunoglobulin light chain variable region comprising the amino acid sequence of SEQ ID NO:80 (04F08).

[0089] In some embodiments, the antibody comprises an immunoglobulin heavy chain variable region comprising the amino acid sequence of SEQ ID NO:46 (06C11), and an immunoglobulin light chain variable region comprising the amino acid sequence of SEQ ID NO:82 (06C11).

[0090] In some embodiments, the antibody comprises an immunoglobulin heavy chain variable region comprising the amino acid sequence of SEQ ID NO:48 (08G01), and an immunoglobulin light chain variable region comprising the amino acid sequence of SEQ ID NO:84 (08G01).

[0091] In some embodiments, the antibody comprises an immunoglobulin heavy chain variable region comprising the amino acid sequence of SEQ ID NO:50 (14F11), and an

amino acid sequence of SEQ ID NO:72 (Sh14F11 IGHV2-5), and an immunoglobulin light chain variable region comprising the amino acid sequence of SEQ ID NO:86 (Ch14F11 Chimeric).

[0135] In some embodiments, the antibody comprises an immunoglobulin heavy chain variable region comprising the amino acid sequence of SEQ ID NO:74 (Sh14F11 IGHV2-70), and an immunoglobulin light chain variable region comprising the amino acid sequence of SEQ ID NO:86 (Ch14F11 Chimeric).

[0136] In some embodiments, the antibody comprises an immunoglobulin heavy chain variable region comprising the amino acid sequence of SEQ ID NO:50 (Ch14F11 Chimeric), and an immunoglobulin light chain variable region comprising the amino acid sequence of SEQ ID NO:98 (Hu14F11 IGKV1-16).

[0137] In some embodiments, the antibody comprises an immunoglobulin heavy chain variable region comprising the amino acid sequence of SEQ ID NO:72 (Sh14F11 IGHV2-5), and an immunoglobulin light chain variable region comprising the amino acid sequence of SEQ ID NO:98 (Hu14F11 IGKV1-16).

[0138] In some embodiments, the antibody comprises an immunoglobulin heavy chain variable region comprising the amino acid sequence of SEQ ID NO:74 (Sh14F11 IGHV2-70), and an immunoglobulin light chain variable region comprising the amino acid sequence of SEQ ID NO:98 (Hu14F11 IGKV1-16).

[0139] In some embodiments, the antibody comprises an immunoglobulin heavy chain variable region comprising the amino acid sequence of SEQ ID NO:46 (Ch06C11 Chimeric), and an immunoglobulin light chain variable region comprising the amino acid sequence of SEQ ID NO:80 (04F08).

[0140] In some embodiments, the antibody comprises an immunoglobulin heavy chain variable region comprising the amino acid sequence of SEQ ID NO:50 (Ch14F11 Chimeric), and an immunoglobulin light chain variable region comprising the amino acid sequence of SEQ ID NO:80 (04F08).

[0141] In some embodiments, the antibody comprises an immunoglobulin heavy chain variable region comprising the amino acid sequence of SEQ ID NO:44 (04F08), and an immunoglobulin light chain variable region comprising the amino acid sequence of SEQ ID NO:82 (Ch06C11 Chimeric).

[0142] In some embodiments, the antibody comprises an immunoglobulin heavy chain variable region comprising the amino acid sequence of SEQ ID NO:50 (Ch14F11 Chimeric), and an immunoglobulin light chain variable region comprising the amino acid sequence of SEQ ID NO:82 (Ch06C11 Chimeric).

[0143] In some embodiments, the antibody comprises an immunoglobulin heavy chain variable region comprising the amino acid sequence of SEQ ID NO:44 (04F08), and an immunoglobulin light chain variable region comprising the amino acid sequence of SEQ ID NO:86 (Ch14F11 Chimeric).

[0144] In some embodiments, the antibody comprises an immunoglobulin heavy chain variable region comprising the amino acid sequence of SEQ ID NO:46 (Ch06C11 Chimeric), and an immunoglobulin light chain variable region comprising the amino acid sequence of SEQ ID NO:86 (Ch14F11 Chimeric).

[0145] In some embodiments, the antibody comprises an immunoglobulin heavy chain variable region comprising the amino acid sequence of SEQ ID NO:48 (08G01), and an immunoglobulin light chain variable region comprising the amino acid sequence of SEQ ID NO:76 (Ch01G06 Chimeric).

[0146] In some embodiments, the antibody comprises an immunoglobulin heavy chain variable region comprising the amino acid sequence of SEQ ID NO:40 (Ch01G06 Chimeric), and an immunoglobulin light chain variable region comprising the amino acid sequence of SEQ ID NO:84 (08G01).

[0147] In certain embodiments, the antibodies disclosed herein comprise an immunoglobulin heavy chain and an immunoglobulin light chain. In some embodiments, the antibody comprises an immunoglobulin heavy chain selected from the group consisting of SEQ ID NO:100 (01G06), SEQ ID NO:104 (03G05), SEQ ID NO:108 (04F08), SEQ ID NO:112 (06C11), SEQ ID NO:116 (08G01), SEQ ID NO:120 (14F11), SEQ ID NO:124 (17B11), SEQ ID NO:176 (Ch01G06 Chimeric), SEQ ID NO:178 (Hu01G06 IGHV1-18), SEQ ID NO:180 (Hu01G06 IGHV1-69), SEQ ID NO:182 (Sh01G06 IGHV1-18 M69L), SEQ ID NO:184 (Sh01G06 IGHV1-18 M69L K64Q G44S), SEQ ID NO:186 (Sh01G06 IGHV1-18 M69L K64Q), SEQ ID NO:188 (Sh01G06 IGHV1-69 T30S I69L), SEQ ID NO:190 (Sh01G06 IGHV1-69 T30S K64Q I69L), SEQ ID NO:256 (Hu01G06 IGHV1-18 F1), SEQ ID NO:258 (Hu01G06 IGHV1-18 F2), SEQ ID NO:260 (Hu01G06 IGHV1-69 F1), SEQ ID NO:262 (Hu01G06 IGHV1-69 F2), SEQ ID NO:192 (Ch06C11 Chimeric), SEQ ID NO:194 (HE LM 06C11 IGHV2-70), SEQ ID NO:196 (Hu06C11 IGHV2-5), SEQ ID NO:198 (Ch14F11 Chimeric), SEQ ID NO:200 (Sh14F11 IGHV2-5), and SEQ ID NO:202 (Sh14F11 IGHV2-70); and an immunoglobulin light chain.

[0148] In other embodiments, the antibody comprises an immunoglobulin light chain selected from the group consisting of SEQ ID NO:102 (01G06), SEQ ID NO:106 (03G05), SEQ ID NO:110 (04F08), SEQ ID NO:114 (06C11), SEQ ID NO:118 (08G01), SEQ ID NO:122 (14F11), SEQ ID NO:126 (17B11), SEQ ID NO:204 (Ch01G06 Chimeric), SEQ ID NO:206 (Hu01G06 IGKV1-39), SEQ ID NO:208 (Hu01G06 IGKV1-39 S43A V48I or Hu01G06 IGKV1-39 F1), SEQ ID NO:210 (Hu01G06 IGKV1-39 V48I), SEQ ID NO:264 (Hu01G06 IGKV1-39 F2), SEQ ID NO:212 (Ch06C11 Chimeric), SEQ ID NO:214 (Sh06C11 IGKV1-16), SEQ ID NO:216 (Ch14F11 Chimeric), and SEQ ID NO:218 (Hu14F11 IGKV1-16), and an immunoglobulin heavy chain.

[0149] In some embodiments, the antibody comprises (i) an immunoglobulin heavy chain selected from the group consisting of SEQ ID NO:100 (01G06), SEQ ID NO:104 (03G05), SEQ ID NO:108 (04F08), SEQ ID NO:112 (06C11), SEQ ID NO:116 (08G01), SEQ ID NO:120 (14F11), SEQ ID NO:124 (17B11), SEQ ID NO:176 (Ch01G06 Chimeric), SEQ ID NO:178 (Hu01G06 IGHV1-18), SEQ ID NO:180 (Hu01G06 IGHV1-69), SEQ ID NO:182 (Sh01G06 IGHV1-18 M69L), SEQ ID NO:184 (Sh01G06 IGHV1-18 M69L K64Q G44S), SEQ ID NO:186 (Sh01G06 IGHV1-18 M69L K64Q), SEQ ID NO:188 (Sh01G06 IGHV1-69 T30S I69L), SEQ ID NO:190 (Sh01G06 IGHV1-69 T30S K64Q I69L), SEQ ID NO:256 (Hu01G06 IGHV1-18 F1), SEQ ID NO:258 (Hu01G06

IGHV1-18 F2), SEQ ID NO:260 (Hu01G06 IGHV1-69 F1), SEQ ID NO:262 (Hu01G06 IGHV1-69 F2), SEQ ID NO:192 (Ch06C11 Chimeric), SEQ ID NO:194 (HE LM 06C11 IGHV2-70), SEQ ID NO:196 (Hu06C11 IGHV2-5), SEQ ID NO:198 (Ch14F11 Chimeric), SEQ ID NO:200 (Sh14F11 IGHV2-5), and SEQ ID NO:202 (Sh14F11 IGHV2-70), and (ii) an immunoglobulin light chain selected from the group consisting of SEQ ID NO:102 (01G06), SEQ ID NO:106 (03G05), SEQ ID NO:110 (04F08), SEQ ID NO:114 (06C11), SEQ ID NO:118 (08G01), SEQ ID NO:122 (14F11), SEQ ID NO:126 (17B11), SEQ ID NO:204 (Ch01G06 Chimeric), SEQ ID NO:206 (Hu01G06 IGKV1-39), SEQ ID NO:208 (Hu01G06 IGKV1-39 S43A V48I or Hu01G06 IGKV1-39 F1), SEQ ID NO:210 (Hu01G06 IGKV1-39 V48I), SEQ ID NO:264 (Hu01G06 IGKV1-39 F2), SEQ ID NO:212 (Ch06C11 Chimeric), SEQ ID NO:214 (Sh06C11 IGKV1-16), SEQ ID NO:216 (Ch14F11 Chimeric), and SEQ ID NO:218 (Hu14F11 IGKV1-16).

[0150] In some embodiments, the antibody comprises an immunoglobulin heavy chain comprising the amino acid sequence of SEQ ID NO:176 (Ch01G06 Chimeric), and an immunoglobulin light chain comprising the amino acid sequence of SEQ ID NO:204 (Ch01G06 Chimeric).

[0151] In some embodiments, the antibody comprises an immunoglobulin heavy chain comprising the amino acid sequence of SEQ ID NO:192 (Ch06C11 Chimeric), and an immunoglobulin light chain comprising the amino acid sequence of SEQ ID NO:212 (Ch06C11 Chimeric).

[0152] In some embodiments, the antibody comprises an immunoglobulin heavy chain comprising the amino acid sequence of SEQ ID NO:198 (Ch14F11 Chimeric), and an immunoglobulin light chain comprising the amino acid sequence of SEQ ID NO:216 (Ch14F11 Chimeric).

[0153] In some embodiments, the antibody comprises an immunoglobulin heavy chain comprising the amino acid sequence of SEQ ID NO:178 (Hu01G06 IGHV1-18), and an immunoglobulin light chain comprising the amino acid sequence of SEQ ID NO:206 (Hu01G06 IGKV1-39).

[0154] In some embodiments, the antibody comprises an immunoglobulin heavy chain comprising the amino acid sequence of SEQ ID NO:180 (Hu01G06 IGHV1-69), and an immunoglobulin light chain comprising the amino acid sequence of SEQ ID NO:206 (Hu01G06 IGKV1-39).

[0155] In some embodiments, the antibody comprises an immunoglobulin heavy chain comprising the amino acid sequence of SEQ ID NO:184 (Sh01G06 IGHV1-18 M69L K64Q G44S), and an immunoglobulin light chain comprising the amino acid sequence of SEQ ID NO:210 (Hu01G06 IGKV1-39 V48I).

[0156] In some embodiments, the antibody comprises an immunoglobulin heavy chain comprising the amino acid sequence of SEQ ID NO:188 (Sh01G06 IGHV1-69 T30S I69L), and an immunoglobulin light chain comprising the amino acid sequence of SEQ ID NO:210 (Hu01G06 IGKV1-39 V48I).

[0157] In some embodiments, the antibody comprises an immunoglobulin heavy chain comprising the amino acid sequence of SEQ ID NO:184 (Sh01G06 IGHV1-18 M69L K64Q G44S), and an immunoglobulin light chain comprising the amino acid sequence of SEQ ID NO:208 (Hu01G06 IGKV1-39 S43A V48I).

[0158] In some embodiments, the antibody comprises an immunoglobulin heavy chain comprising the amino acid

sequence of SEQ ID NO:188 (Sh01G06 IGHV1-69 T30S I69L), and an immunoglobulin light chain comprising the amino acid sequence of SEQ ID NO:208 (Hu01G06 IGKV1-39 S43A V48I).

[0159] In some embodiments, the antibody comprises an immunoglobulin heavy chain comprising the amino acid sequence of SEQ ID NO:256 (Hu01G06 IGHV1-18 F1), and an immunoglobulin light chain comprising the amino acid sequence of SEQ ID NO:208 (Hu01G06 IGKV1-39 F1).

[0160] In some embodiments, the antibody comprises an immunoglobulin heavy chain comprising the amino acid sequence of SEQ ID NO:258 (Hu01G06 IGHV1-18 F2), and an immunoglobulin light chain comprising the amino acid sequence of SEQ ID NO:264 (Hu01G06 IGKV1-39 F2).

[0161] In some embodiments, the antibody comprises an immunoglobulin heavy chain comprising the amino acid sequence of SEQ ID NO:260 (Hu01G06 IGHV1-69 F1), and an immunoglobulin light chain comprising the amino acid sequence of SEQ ID NO:208 (Hu01G06 IGKV1-39 F1).

[0162] In some embodiments, the antibody comprises an immunoglobulin heavy chain comprising the amino acid sequence of SEQ ID NO:262 (Hu01G06 IGHV1-69 F2), and an immunoglobulin light chain comprising the amino acid sequence of SEQ ID NO:208 (Hu01G06 IGKV1-39 F1).

[0163] In some embodiments, the antibody comprises an immunoglobulin heavy chain comprising the amino acid sequence of SEQ ID NO:262 (Hu01G06 IGHV1-69 F2), and an immunoglobulin light chain comprising the amino acid sequence of SEQ ID NO:264 (Hu01G06 IGKV1-39 F2).

[0164] In some embodiments, the antibody comprises an immunoglobulin heavy chain comprising the amino acid sequence of SEQ ID NO:194 (HE LM 06C11 IGHV2-70), and an immunoglobulin light chain comprising the amino acid sequence of SEQ ID NO:214 (Sh06C11 IGKV1-16).

[0165] In some embodiments, the antibody comprises an immunoglobulin heavy chain comprising the amino acid sequence of SEQ ID NO:196 (Hu06C11 IGHV2-5), and an immunoglobulin light chain comprising the amino acid sequence of SEQ ID NO:214 (Sh06C11 IGKV1-16).

[0166] In some embodiments, the antibody comprises an immunoglobulin heavy chain comprising the amino acid sequence of SEQ ID NO:200 (Sh14F11 IGHV2-5), and an immunoglobulin light chain comprising the amino acid sequence of SEQ ID NO:218 (Hu14F11 IGKV1-16).

[0167] In some embodiments, the antibody comprises an immunoglobulin heavy chain comprising the amino acid sequence of SEQ ID NO:202 (Sh14F11 IGHV2-70), and an immunoglobulin light chain comprising the amino acid sequence of SEQ ID NO:218 (Hu14F11 IGKV1-16).

[0168] In certain embodiments, an isolated antibody that binds hGDF15 comprises an immunoglobulin heavy chain variable region comprising an amino acid sequence that is at least 70%, 75%, 80%, 85%, 90%, 95%, 98%, or 99% identical to the entire variable region or the FR sequence of SEQ ID NO:40 (01G06, Ch01G06 Chimeric), SEQ ID NO:42 (03G05), SEQ ID NO:44 (04F08), SEQ ID NO:46 (06C11, Ch06C11 Chimeric), SEQ ID NO:48 (08G01), SEQ ID NO:50 (14F11, Ch14F11 Chimeric), SEQ ID NO:52 (17B11), SEQ ID NO:54 (Hu01G06 IGHV1-18), SEQ ID NO:56 (Hu01G06 IGHV1-69), SEQ ID NO:58 (Sh01G06 IGHV1-18 M69L), SEQ ID NO:60 (Sh01G06 IGHV1-18 M69L K64Q G44S), SEQ ID NO:62 (Sh01G06 IGHV1-18 M69L K64Q), SEQ ID NO:64 (Sh01G06 IGHV1-69 T30S I69L), SEQ ID NO:66 (Sh01G06 IGHV1-69 T30S K64Q

169L), SEQ ID NO:246 (Hu01G06 IGHV1-18 F1), SEQ ID NO:248 (Hu01G06 IGHV1-18 F2), SEQ ID NO:250 (Hu01G06 IGHV1-69 F1), SEQ ID NO:252 (Hu01G06 IGHV1-69 F2), SEQ ID NO:68 (HE LM 06C11 IGHV2-70), SEQ ID NO:70 (Hu06C11 IGHV2-5), SEQ ID NO:72 (Sh14F11 IGHV2-5), and SEQ ID NO:74 (Sh14F11 IGHV2-70).

[0169] In certain embodiments, an isolated antibody that binds hGDF15 comprises an immunoglobulin light chain variable region comprising an amino acid sequence that is at least 70%, 75%, 80%, 85%, 90%, 95%, 98%, or 99% identical to the entire variable region or the FR sequence of SEQ ID NO:76 (01G06, Ch01G06 Chimeric), SEQ ID NO:78 (03G05), SEQ ID NO:80 (04F08), SEQ ID NO:82 (06C11, Ch06C11 Chimeric), SEQ ID NO:84 (08G01), SEQ ID NO:86 (14F11, Ch14F11 Chimeric), SEQ ID NO:88 (17B11), SEQ ID NO:90 (Hu01G06 IGKV1-39), SEQ ID NO:92 (Hu01G06 IGKV1-39 S43A V48I or Hu01G06 IGKV1-39 F1), SEQ ID NO:94 (Hu01G06 IGKV1-39 V48I), SEQ ID NO:254 (Hu01G06 IGKV1-39 F2), SEQ ID NO:96 (Sh06C11 IGKV1-16), and SEQ ID NO:98 (Hu14F11 IGKV1-16).

[0170] Sequence identity may be determined in various ways that are within the skill of a person skilled in the art, e.g., using publicly available computer software such as BLAST, BLAST-2, ALIGN or Megalign (DNASTAR) software. BLAST (Basic Local Alignment Search Tool) analysis using the algorithm employed by the programs blastp, blastn, blastx, tblastn and tblastx (Karlin et al., (1990) *PROC. NATL. ACAD. SCI. USA* 87:2264-2268; Altschul, (1993) *J. MOL. EVOL.* 36:290-300; Altschul et al., (1997) *NUCLEIC ACIDS RES.* 25:3389-3402, incorporated by reference herein) are tailored for sequence similarity searching. For a discussion of basic issues in searching sequence databases see Altschul et al., (1994) *NATURE GENETICS* 6:119-129, which is fully incorporated by reference herein. Those skilled in the art can determine appropriate parameters for measuring alignment, including any algorithms needed to achieve maximal alignment over the full length of the sequences being compared. The search parameters for histogram, descriptions, alignments, expect (i.e., the statistical significance threshold for reporting matches against database sequences), cutoff, matrix and filter are at the default settings. The default scoring matrix used by blastp, blastx, tblastn, and tblastx is the BLOSUM62 matrix (Henikoff et al., (1992) *PROC. NATL. ACAD. SCI. USA* 89:10915-10919, fully incorporated by reference herein). Four blastn parameters may be adjusted as follows: Q=10 (gap creation penalty); R=10 (gap extension penalty); wink=1 (generates word hits at every wink.sup.th position along the query); and gapw=16 (sets the window width within which gapped alignments are generated). The equivalent Blastp parameter settings may be Q=9; R=2; wink=1; and gapw=32. Searches may also be conducted using the NCBI (National Center for Biotechnology Information) BLAST Advanced Option parameter (e.g.: -G, Cost to open gap [Integer]: default=5 for nucleotides/11 for proteins; -E, Cost to extend gap [Integer]: default=2 for nucleotides/1 for proteins; -q, Penalty for nucleotide mismatch [Integer]: default=-3; -r, reward for nucleotide match [Integer]: default=1; -e, expect value [Real]: default=10; -W, wordsize [Integer]: default=11 for nucleotides/28 for megablast/3 for proteins; -y, Dropoff (X) for blast extensions in bits: default=20 for blastn/7 for others; -X, X dropoff value for gapped alignment (in bits): default=15 for all programs,

not applicable to blastn; and -Z, final X dropoff value for gapped alignment (in bits): 50 for blastn, 25 for others). ClustalW for pairwise protein alignments may also be used (default parameters may include, e.g., Blosum62 matrix and Gap Opening Penalty=10 and Gap Extension Penalty=0.1). A Bestfit comparison between sequences, available in the GCG package version 10.0, uses DNA parameters GAP=50 (gap creation penalty) and LEN=3 (gap extension penalty). The equivalent settings in Bestfit protein comparisons are GAP=8 and LEN=2.

[0171] In each of the foregoing embodiments, it is contemplated herein that immunoglobulin heavy chain variable region sequences and/or light chain variable region sequences that together bind human GDF15 may contain amino acid alterations (e.g., at least 1, 2, 3, 4, 5, or 10 amino acid substitutions, deletions, or additions) in the framework regions of the heavy and/or light chain variable regions.

[0172] In some embodiments, the antibody binds hGDF15 with a K_D of about 300 pM, 250 pM, 200 pM, 190 pM, 180 pM, 170 pM, 160 pM, 150 pM, 140 pM, 130 pM, 120 pM, 110 pM, 100 pM, 90 pM, 80 pM, 70 pM, 60 pM, 50 pM, 40 pM, 30 pM, 20 pM, or 10 pM, or lower. Unless otherwise specified, K_D values are determined by surface plasmon resonance methods or biolayer interferometry under the conditions described in Examples 8, 14, and 15.

[0173] In some embodiments, a monoclonal antibody binds to the same epitope on hGDF15 (e.g., mature hGDF15 or cleaved rhGDF15) bound by one or more of the antibodies disclosed herein (e.g., antibodies 01G06, 03G05, 04F08, 06C11, 08G01, 14F11, or 17B11). In some embodiments, a monoclonal antibody competes for binding to hGDF15 with one or more of the antibodies disclosed herein (e.g., antibody 01G06, 03G05, 04F08, 06C11, 08G01, 14F11, or 17B11).

[0174] Competition assays for determining whether an antibody binds to the same epitope as, or competes for binding with, an anti-GDF15 antibody disclosed herein are known in the art. Exemplary competition assays include immunoassays (e.g., ELISA assays, RIA assays), surface plasmon resonance analysis (e.g., using a BIAcore™ instrument), biolayer interferometry and flow cytometry.

[0175] Typically, a competition assay involves the use of an antigen (e.g., a hGDF15 protein or fragment thereof) bound to a solid surface or expressed on a cell surface, a test anti-GDF15-binding antibody and a reference antibody (e.g., antibody 01G06, 03G05, 04F08, 06C11, 08G01, 14F11, or 17B11). The reference antibody is labeled and the test antibody is unlabeled. Competitive inhibition is measured by determining the amount of labeled reference antibody bound to the solid surface or cells in the presence of the test antibody. Usually the test antibody is present in excess (e.g., 1x, 5x, 10x, 20x or 100x). Antibodies identified by competition assay (i.e., competing antibodies) include antibodies binding to the same epitope, or similar (e.g., overlapping) epitopes, as the reference antibody, and antibodies binding to an adjacent epitope sufficiently proximal to the epitope bound by the reference antibody for steric hindrance to occur.

[0176] In an exemplary competition assay, a reference anti-GDF15 antibody (e.g., antibody 01G06, 03G05, 04F08, 06C11, 08G01, 14F11, or 17B11) is biotinylated using commercially available reagents. The biotinylated reference antibody is mixed with serial dilutions of the test antibody or unlabeled reference antibody (self-competition control)

resulting in a mixture of various molar ratios (e.g., 1×, 5×, 10×, 20× or 100×) of test antibody (or unlabeled reference antibody) to labeled reference antibody. The antibody mixture is added to a hGDF15 polypeptide coated-ELISA plate. The plate is then washed, and horseradish peroxidase (HRP)-streptavidin is added to the plate as the detection reagent. The amount of labeled reference antibody bound to the target antigen is detected following addition of a chromogenic substrate (e.g., TMB (3,3',5,5'-tetramethylbenzidine) or ABTS (2,2'-azino-di-(3-ethylbenzthiazoline-6-sulfonate)), which are known in the art. Optical density readings (OD units) are measured using a SpectraMax® M2 spectrometer (Molecular Devices). OD units corresponding to zero percent inhibition are determined from wells without any competing antibody. OD units corresponding to 100% inhibition, i.e., the assay background are determined from wells without any labeled reference antibody or test antibody. Percent inhibition of labeled reference antibody to GDF15 by the test antibody (or the unlabeled reference antibody) at each concentration is calculated as follows: % inhibition = $(1 - (\text{OD units} - 100\% \text{ inhibition}) / (0\% \text{ inhibition} - 100\% \text{ inhibition})) * 100$. Persons skilled in the art will appreciate that the competition assay can be performed using various detection systems known in the art.

[0177] A competition assay may be conducted in both directions to ensure that the presence of the label does not interfere or otherwise inhibit binding. For example, in the first direction the reference antibody is labeled and the test antibody is unlabeled, and in the second direction, the test antibody is labeled and the reference antibody is unlabeled.

[0178] A test antibody competes with the reference antibody for specific binding to the antigen if an excess of one antibody (e.g., 1×, 5×, 10×, 20× or 100×) inhibits binding of the other antibody, e.g., by at least 50%, 75%, 90%, 95% or 99%, as measured in a competitive binding assay.

[0179] Two antibodies bind to the same epitope if essentially all amino acid mutations in the antigen that reduce or eliminate binding of one antibody reduce or eliminate binding of the other. Two antibodies bind to overlapping epitopes if only a subset of the amino acid mutations that reduce or eliminate binding of one antibody reduce or eliminate binding of the other.

II. Production of Antibodies

[0180] Methods for producing antibodies, such as those disclosed herein, are known in the art. For example, DNA molecules encoding light chain variable regions and/or heavy chain variable regions can be chemically synthesized using the sequence information provided herein. Synthetic DNA molecules can be ligated to other appropriate nucleotide sequences, including, e.g., constant region coding sequences, and expression control sequences, to produce conventional gene expression constructs encoding the desired antibodies. Production of defined gene constructs is within routine skill in the art. Alternatively, the sequences provided herein can be cloned out of hybridomas by conventional hybridization techniques or polymerase chain reaction (PCR) techniques, using synthetic nucleic acid probes whose sequences are based on sequence information provided herein, or prior art sequence information regarding genes encoding the heavy and light chains of murine antibodies in hybridoma cells.

[0181] Nucleic acids encoding desired antibodies can be incorporated (ligated) into expression vectors, which can be

introduced into host cells through conventional transfection or transformation techniques. Exemplary host cells are *E. coli* cells, Chinese hamster ovary (CHO) cells, human embryonic kidney 293 (HEK 293) cells, HeLa cells, baby hamster kidney (BHK) cells, monkey kidney cells (COS), human hepatocellular carcinoma cells (e.g., Hep G2), and myeloma cells that do not otherwise produce IgG protein. Transformed host cells can be grown under conditions that permit the host cells to express the genes that encode the immunoglobulin light and/or heavy chain variable regions.

[0182] Specific expression and purification conditions will vary depending upon the expression system employed. For example, if a gene is to be expressed in *E. coli*, it is first cloned into an expression vector by positioning the engineered gene downstream from a suitable bacterial promoter, e.g., Trp or Tac, and a prokaryotic signal sequence. The expressed secreted protein accumulates in refractile or inclusion bodies, and can be harvested after disruption of the cells by French press or sonication. The refractile bodies then are solubilized, and the proteins refolded and cleaved by methods known in the art.

[0183] If the engineered gene is to be expressed in eukaryotic host cells, e.g., CHO cells, it is first inserted into an expression vector containing a suitable eukaryotic promoter, a secretion signal, a poly A sequence, and a stop codon. Optionally, the vector or gene construct may contain enhancers and introns. This expression vector optionally contains sequences encoding all or part of a constant region, enabling an entire, or a part of, a heavy or light chain to be expressed. The gene construct can be introduced into eukaryotic host cells using conventional techniques. The host cells express V_L or V_H fragments, V_L - V_H heterodimers, V_H - V_L or V_L - V_H single chain polypeptides, complete heavy or light immunoglobulin chains, or portions thereof, each of which may be attached to a moiety having another function (e.g., cytotoxicity). In some embodiments, a host cell is transfected with a single vector expressing a polypeptide expressing an entire, or part of, a heavy chain (e.g., a heavy chain variable region) or a light chain (e.g., a light chain variable region). In some embodiments, a host cell is transfected with a single vector encoding (a) a polypeptide comprising a heavy chain variable region and a polypeptide comprising a light chain variable region, or (b) an entire immunoglobulin heavy chain and an entire immunoglobulin light chain. In some embodiments, a host cell is co-transfected with more than one expression vector (e.g., one expression vector expressing a polypeptide comprising an entire, or part of, a heavy chain or heavy chain variable region, and another expression vector expressing a polypeptide comprising an entire, or part of, a light chain or light chain variable region).

[0184] A polypeptide comprising an immunoglobulin heavy chain variable region or light chain variable region can be produced by growing (culturing) a host cell transfected with an expression vector encoding such a variable region, under conditions that permit expression of the polypeptide. Following expression, the polypeptide can be harvested and purified or isolated using techniques known in the art, e.g., affinity tags such as glutathione-S-transferase (GST) or histidine tags.

[0185] A monoclonal antibody that binds hGDF15, or an antigen-binding fragment of the antibody, can be produced by growing (culturing) a host cell transfected with: (a) an expression vector that encodes a complete or partial immunoglobulin heavy chain, and a separate expression vector

that encodes a complete or partial immunoglobulin light chain; or (b) a single expression vector that encodes both chains (e.g., complete or partial heavy and light chains), under conditions that permit expression of both chains. The intact antibody (or antigen-binding fragment) can be harvested and purified or isolated using techniques known in the art, e.g., Protein A, Protein G, affinity tags such as glutathione-S-transferase (GST) or histidine tags. It is within ordinary skill in the art to express the heavy chain and the light chain from a single expression vector or from two separate expression vectors.

III. Antibody Modifications

[0186] Methods for reducing or eliminating the antigenicity of antibodies and antibody fragments are known in the art. When the antibodies are to be administered to a human, the antibodies preferably are “humanized” to reduce or eliminate antigenicity in humans. Preferably, each humanized antibody has the same or substantially the same affinity for the antigen as the non-humanized mouse antibody from which it was derived.

[0187] In one humanization approach, chimeric proteins are created in which mouse immunoglobulin constant regions are replaced with human immunoglobulin constant regions. See, e.g., Morrison et al., 1984, *PROC. NAT. ACAD. SCI.* 81:6851-6855; Neuberger et al., 1984, *NATURE* 312:604-608; U.S. Pat. No. 6,893,625 (Robinson); U.S. Pat. No. 5,500,362 (Robinson); and U.S. Pat. No. 4,816,567 (Cabilly).

[0188] In an approach known as CDR grafting, the CDRs of the light and heavy chain variable regions are grafted into frameworks from another species. For example, murine CDRs can be grafted into human FRs. In some embodiments, the CDRs of the light and heavy chain variable regions of an anti-GDF15 antibody are grafted into human FRs or consensus human FRs. To create consensus human FRs, FRs from several human heavy chain or light chain amino acid sequences are aligned to identify a consensus amino acid sequence. CDR grafting is described in U.S. Pat. No. 7,022,500 (Queen); U.S. Pat. No. 6,982,321 (Winter); U.S. Pat. No. 6,180,370 (Queen); U.S. Pat. No. 6,054,297 (Carter); U.S. Pat. No. 5,693,762 (Queen); U.S. Pat. No. 5,859,205 (Adair); U.S. Pat. No. 5,693,761 (Queen); U.S. Pat. No. 5,565,332 (Hoogenboom); U.S. Pat. No. 5,585,089 (Queen); U.S. Pat. No. 5,530,101 (Queen); Jones et al. (1986) *NATURE* 321: 522-525; Riechmann et al. (1988) *NATURE* 332: 323-327; Verhoeyen et al. (1988) *SCIENCE* 239: 1534-1536; and Winter (1998) *FEBS LETT* 430: 92-94.

[0189] In an approach called “SUPERHUMANIZATION™,” human CDR sequences are chosen from human germline genes, based on the structural similarity of the human CDRs to those of the mouse antibody to be humanized. See, e.g., U.S. Pat. No. 6,881,557 (Foote); and Tan et al., 2002, *J. IMMUNOL.* 169:1119-1125.

[0190] Other methods to reduce immunogenicity include “reshaping,” “hyperchimerization,” and “veneering/resurfacing.” See, e.g., Vaswami et al., 1998, *ANNALS OF ALLERGY, ASTHMA, & IMMUNOL.* 81:105; Roguska et al., 1996, *PROT. ENGINEER* 9:895-904; and U.S. Pat. No. 6,072,035 (Hardman). In the veneering/resurfacing approach, the surface accessible amino acid residues in the murine antibody are replaced by amino acid residues more frequently found at

the same positions in a human antibody. This type of antibody resurfacing is described, e.g., in U.S. Pat. No. 5,639,641 (Pedersen).

[0191] Another approach for converting a mouse antibody into a form suitable for medical use in humans is known as ACTIVMAB™ technology (Vaccinex, Inc., Rochester, NY), which involves a vaccinia virus-based vector to express antibodies in mammalian cells. High levels of combinatorial diversity of IgG heavy and light chains are said to be produced. See, e.g., U.S. Pat. No. 6,706,477 (Zauderer); U.S. Pat. No. 6,800,442 (Zauderer); and U.S. Pat. No. 6,872,518 (Zauderer).

[0192] Another approach for converting a mouse antibody into a form suitable for use in humans is technology practiced commercially by KaloBios Pharmaceuticals, Inc. (Palo Alto, CA). This technology involves the use of a proprietary human “acceptor” library to produce an “epitope focused” library for antibody selection.

[0193] Another approach for modifying a mouse antibody into a form suitable for medical use in humans is HUMAN ENGINEERING™ technology, which is practiced commercially by XOMA (US) LLC. See, e.g., PCT Publication No. WO 93/11794 and U.S. Pat. No. 5,766,886 (Studnicka); U.S. Pat. No. 5,770,196 (Studnicka); U.S. Pat. No. 5,821,123 (Studnicka); and U.S. Pat. No. 5,869,619 (Studnicka).

[0194] Any suitable approach, including any of the above approaches, can be used to reduce or eliminate human immunogenicity of an antibody.

[0195] In addition, it is possible to create fully human antibodies in mice. Fully human mAbs lacking any non-human sequences can be prepared from human immunoglobulin transgenic mice by techniques referenced in, e.g., Lonberg et al., *NATURE* 368:856-859, 1994; Fishwild et al., *NATURE BIOTECHNOLOGY* 14:845-851, 1996; and Mendez et al., *NATURE GENETICS* 15:146-156, 1997. Fully human mAbs can also be prepared and optimized from phage display libraries by techniques referenced in, e.g., Knappik et al., *J. MOL. BIOL.* 296:57-86, 2000; and Krebs et al., *J. Immunol. Meth.* 254:67-84 2001).

IV. Therapeutic Uses

[0196] The antibodies disclosed herein can be used to treat a variety of disorders, for example, cachexia and/or sarcopenia. In some embodiments, the antibodies disclosed herein (e.g., 01G06, 03G05, 04F08, 06C11, 08G01, 14F11, or 17B11) are used to inhibit the loss of muscle mass, for example, the loss of muscle mass associated with an underlying disease. Underlying diseases associated with cachexia include, but are not limited to, cancer, chronic heart failure, chronic kidney disease, COPD, AIDS, multiple sclerosis, rheumatoid arthritis, sepsis, and tuberculosis. In some embodiments, the disclosed antibodies inhibit loss of muscle mass by at least 40%, 50%, 60%, 70%, 80%, 90%, 95%, 98%, 99%, or 100%.

[0197] In some embodiments, a loss of muscle mass is accompanied by a loss of fat mass. The antibodies disclosed herein (e.g., 01G06, 03G05, 04F08, 06C11, 08G01, 14F11, or 17B11) may inhibit loss of fat mass by at least 40%, 50%, 60%, 70%, 80%, 90%, 95%, 98%, 99%, or 100%.

[0198] In other embodiments, the antibodies disclosed herein (e.g., 01G06, 03G05, 04F08, 06C11, 08G01, 14F11, or 17B11) are used to treat one or more features accompanying cachexia and/or sarcopenia, e.g., involuntary body

weight loss. In some embodiments, the antibodies revert involuntary body weight loss by at least 2%, 5%, 10%, 15%, 20%, 25%, 30% or 35%.

[0199] In another embodiment, the antibodies disclosed herein (e.g., 01G06, 03G05, 04F08, 06C11, 08G01, 14F11, or 17B11) are used to inhibit loss of organ mass, for example, loss of organ mass associated with an underlying disease. Underlying diseases associated with cachexia include, but are not limited to, cancer, chronic heart failure, chronic kidney disease, COPD, AIDS, multiple sclerosis, rheumatoid arthritis, sepsis, and tuberculosis. In some embodiments, the disclosed antibodies inhibit loss of organ mass by at least 40%, 50%, 60%, 70%, 80%, 90%, 95%, 98%, 99%, or 100%. In some embodiments, loss of organ mass is observed in heart, liver, kidney, and/or spleen. In some embodiments, the loss of organ mass is accompanied by a loss of muscle mass, a loss of fat mass and/or involuntary weight loss.

[0200] Antibody 01G06, 03G05, 04F08, 06C11, 08G01, 14F11, or 17B11 can be used in therapy. For example, antibody 01G06, 03G05, 04F08, 06C11, 08G01, 14F11, or 17B11 can be used to treat cachexia and/or sarcopenia. Use of antibody 01G06, 03G05, 04F08, 06C11, 08G01, 14F11, or 17B11 to treat cachexia and/or sarcopenia in a mammal comprises administering to the mammal a therapeutically effective amount of the antibody.

[0201] Sarcopenia, muscle wasting disorders and significant muscle weight loss may occur in the absence of cachexia, decreased appetite or body weight loss. In certain embodiments, therefore, one or more of the anti-GDF antibodies of the invention (for example, antibody 01G06, 03G05, 04F08, 06C11, 08G01, 14F11, or 17B11) can be used to treat a subject suffering from, or who has been diagnosed with, sarcopenia, a muscle wasting disorder and/or significant muscle weight loss, whether or not the subject has, or has been diagnosed with, cachexia or decreased appetite. Such a method comprises administering a therapeutically effective amount of one or more antibodies of the invention to the subject in need thereof.

[0202] The Fc-rhGDF15 fusion proteins disclosed herein can be used to treat obesity. In some embodiments, the hFc-rhGDF15 fusion proteins disclosed herein are used to inhibit weight gain or to reduce body weight by at least 5%, 10%, 15%, 20%, 25%, 30%, 35%, 40%, 45% or 50%. Use of an hFc-hGDF15 fusion protein to treat obesity in a mammal comprises administering to the mammal a therapeutically effective amount of the fusion protein.

[0203] As used herein, “treat,” “treating” and “treatment” mean the treatment of a disease in a mammal, e.g., in a human. This includes: (a) inhibiting the disease, i.e., arresting its development; and (b) relieving the disease, i.e., causing regression of the disease state.

[0204] Generally, a therapeutically effective amount of an active component (e.g., an antibody or a fusion protein) is in the range of 0.1 mg/kg to 100 mg/kg, e.g., 1 mg/kg to 100 mg/kg, e.g., 1 mg/kg to 10 mg/kg, e.g., 2.0 mg/kg to 10 mg/kg. The amount administered will depend on variables such as the type and extent of disease or indication to be treated, the overall health of the patient, the in vivo potency of the antibody or fusion protein, the pharmaceutical formulation, the serum half-life of the antibody or fusion protein, and the route of administration. The initial dosage can be increased beyond the upper level in order to rapidly achieve the desired blood-level or tissue level. Alternatively,

the initial dosage can be smaller than the optimum, and the dosage may be progressively increased during the course of treatment. Human dosage can be optimized, e.g., in a conventional Phase I dose escalation study designed to run from 0.5 mg/kg to 20 mg/kg. Dosing frequency can vary, depending on factors such as route of administration, dosage amount, serum half-life of the antibody or fusion protein, and the disease being treated. Exemplary dosing frequencies are once per day, once per week and once every two weeks. In some embodiments, dosing is once every two weeks. A preferred route of administration is parenteral, e.g., intravenous infusion. Formulation of monoclonal antibody-based drugs and fusion protein-based drugs are within ordinary skill in the art. In some embodiments, the antibody or fusion protein is lyophilized, and then reconstituted in buffered saline, at the time of administration. The effective amount of a second active agent, for example, an anti-cancer agent or the other agents discussed below, will also follow the principles discussed hereinabove and will be chosen so as to elicit the required therapeutic benefit in the patient.

[0205] For therapeutic use, an antibody preferably is combined with a pharmaceutically acceptable carrier. As used herein, “pharmaceutically acceptable carrier” means buffers, carriers, and excipients suitable for use in contact with the tissues of human beings and animals without excessive toxicity, irritation, allergic response, or other problem or complication, commensurate with a reasonable benefit/risk ratio. The carrier(s) should be “acceptable” in the sense of being compatible with the other ingredients of the formulations and not deleterious to the recipient. Pharmaceutically acceptable carriers include buffers, solvents, dispersion media, coatings, isotonic and absorption delaying agents, and the like, that are compatible with pharmaceutical administration. The use of such media and agents for pharmaceutically active substances is known in the art.

[0206] Pharmaceutical compositions containing antibodies or fusion proteins, such as those disclosed herein, can be presented in a dosage unit form and can be prepared by any suitable method. A pharmaceutical composition should be formulated to be compatible with its intended route of administration. Examples of routes of administration are intravenous (IV), intradermal, inhalation, transdermal, topical, transmucosal, and rectal administration. A preferred route of administration for monoclonal antibodies is IV infusion. Useful formulations can be prepared by methods known in the pharmaceutical art. For example, see *Remington's Pharmaceutical Sciences*, 18th ed. (Mack Publishing Company, 1990). Formulation components suitable for parenteral administration include a sterile diluent such as water for injection, saline solution, fixed oils, polyethylene glycols, glycerine, propylene glycol or other synthetic solvents; antibacterial agents such as benzyl alcohol or methyl parabens; antioxidants such as ascorbic acid or sodium bisulfite; chelating agents such as EDTA; buffers such as acetates, citrates or phosphates; and agents for the adjustment of tonicity such as sodium chloride or dextrose.

[0207] For intravenous administration, suitable carriers include physiological saline, bacteriostatic water, Cremophor EL™ (BASF, Parsippany, NJ) or phosphate buffered saline (PBS). The carrier should be stable under the conditions of manufacture and storage, and should be preserved against microorganisms. The carrier can be a solvent or dispersion medium containing, for example, water, ethanol,

polyol (for example, glycerol, propylene glycol, and liquid polyethylene glycol), and suitable mixtures thereof.

[0208] Pharmaceutical formulations preferably are sterile. Sterilization can be accomplished, for example, by filtration through sterile filtration membranes. Where the composition is lyophilized, filter sterilization can be conducted prior to or following lyophilization and reconstitution.

[0209] In addition to the GDF15 (i.e., MIC-1/PLAB/PDF/NAG-1) pathway, other cytokines implicated in cachexia include Activin A and IL-6. Increased activin levels have been associated with cancer-associated cachexia and gonadal tumors. See, e.g., Marino et al. (2013) *CYTOKINE & GROWTH FACTOR REV.* 24:477-484. Activin A is a member of the TGF-beta family, and is a ligand of the activin type 2 receptor, ActRIIB. See, e.g., Zhou et al. (2010) *CELL* 142: 531-543. Circulating levels of IL-6 have been shown to correlate with weight loss in cancer patients, as well as with reduced survival. See, e.g., Fearon et al. (2012) *CELL METABOLISM* 16:153-166.

[0210] Accordingly, in certain embodiments of the present invention, one or more inhibitors of Activin-A or the Activin-A receptor, ActRIIB, IL-6 or the IL-6 receptor (IL-6R), may be administered in combination with (for example, administered at the same time as, administered before, or administered after) an antibody of the present invention that inhibits GDF-15 activity. Exemplary inhibitors of Activin A or ActRIIB, include, for example, an anti-Activin-A antibody or an antigen binding fragment thereof, an anti-ActRIIB antibody or an antigen binding fragment thereof, a small molecule inhibitor of Activin-A, a small molecule inhibitor of ActRIIB, and a 'decoy' receptor of ActRIIB, such as a soluble ActRIIB receptor and a fusion of the soluble ActRIIB receptor with an Fc molecule (ActRIIB-Fc). See, for example, Zhou et al. (2010), *supra*. Suitable inhibitors of IL-6 or IL-6R, include an anti-IL-6 antibody or an antigen binding fragment thereof, an anti-IL-6R antibody or an antigen binding fragment thereof, a small molecule inhibitor of IL-6, a small molecule inhibitor of IL-6R, and a 'decoy' receptor of IL-6R, such as a soluble IL-6 receptor and a fusion of the soluble IL-6 receptor with an Fc molecule (IL6R-Fc). See, e.g., Enomoto et al. (2004) *BIOCHEM. AND BIOPHYS. RES. COMM.* 323:1096-1102; Argiles et al. (2011) *EUR. J. PHARMACOL.* 668:S81-S86; Tuca et al. (2013) *ONCOLOGY/HEMATOLOGY* 88:625-636. Suitable inhibitors of IL-6 or IL-6R may include, for example, Tocilizumab (Actemra®, Hoffmann-LaRoche), a humanized anti-IL-6R monoclonal antibody approved for treatment of rheumatoid arthritis, and Sarilumab/REGN88 (Regeneron), a humanized anti-IL6R antibody in clinical development for treatment of rheumatoid arthritis; and Selumetinib/AZD6244 (AstraZeneca), an allosteric inhibitor of MEK, which has been shown to inhibit IL-6 production. Prado et al. (2012) *BRITISH J. CANCER* 106:1583-1586.

[0211] TNF α and IL-1 are cytokines known to be involved in mediation of the proinflammatory response, which are also implicated in muscle depletion, anorexia and cachexia. Increased circulating levels of TNF α appear to inhibit myogenesis. TNF α , also known as "cachectin," stimulates interleukin-1 secretion and is implicated in the induction of cachexia. IL-1 is a potent trigger of the acute-phase inflammatory response, and it has been shown that infusion of IL-1 can lead to marked weight loss and appetite loss. IL-1 has been shown to contribute to the initiation of cancer cachexia in mice bearing a murine colon-26 adenocarcinoma (Strass-

mann et al. (1993) *J. IMMUNOL.* 150:2341). See also, Mathys and Billiau (1997) *NUTRITION* 13:763-770; Fong et al. (1989) *AM. J. PHYSIOL.—REGULATORY, INTEGRATIVE AND COMPARATIVE PHYSIOL.*, 256:R659-R665. Thus, TNF α inhibitors and IL-1 inhibitors that are used in the treatment of rheumatoid arthritis may also be useful in the treatment of cachexia.

[0212] Accordingly, in certain embodiments of the present invention, one or more inhibitors of TNF α or IL-1 may be administered in combination with (for example, administered at the same time as, administered before, or administered after) an antibody of the present invention that inhibits GDF-15 activity. Suitable inhibitors of TNF α or IL-1 include an anti-TNF α antibody or an antigen binding fragment thereof, an anti-IL-1 antibody or an antigen binding fragment thereof, a small molecule inhibitor of TNF α or IL-1, and a 'decoy' receptor of TNF α or IL-1, such as a soluble TNF α or IL-1 receptor and a fusion of the soluble form of TNF α or IL-1 with an Fc molecule. Suitable inhibitors of TNF α include for example, etanercept (Enbrel®, Pfizer/Amgen), infliximab (Remicade®, Janssen Biotech), adalimumab (Humira®, Abbvie), golimumab (Simpsoni®, Johnson and Johnson/Merck), and certolizumab pegol (Cimzia®, UCB). Suitable IL-1 inhibitors include, for example, Xilonix® antibody that targets IL-1 α (XBiotech), anakinra (Kinaret®, Amgen), canakinumab (Ilaris®, Novartis), and rilonacept (Arcalyst®, Regeneron). In certain embodiments, the TNF α inhibitor or IL-1 inhibitor, which is typically administered systemically for the treatment of rheumatoid arthritis may be administered locally and directly to the tumor site.

[0213] Myostatin, also known as GDF-8, is a member of the TGF- β family of peptides that is a negative regulator of muscle mass, as shown by increased muscle mass in myostatin deficient mammals. Myostatin is a ligand of the activin type 2 receptor, ActRIIB. Accordingly, in certain embodiments of the present invention, one or more inhibitors of myostatin or its receptor may be administered in combination with (for example, administered at the same time as, administered before, or administered after) an antibody of the invention that inhibits GDF-15 activity. Suitable inhibitors of myostatin or ActRIIB, include an anti-myostatin antibody or an antigen binding fragment thereof, an anti-ActRIIB antibody or an antigen binding fragment thereof, a small molecule inhibitor of myostatin, a small molecule inhibitor of ActRIIB, and a 'decoy' receptor of GDF-8, such as a soluble ActRIIB and a fusion of the soluble form of ActRIIB with an Fc molecule. See, e.g., Lokireddy et al. (2012) *BIOCHEM. J.* 446(1):23-26. Myostatin inhibitors that may be suitable for the present invention include REGN1033 (Regeneron); see Bauerlein et al. (2013) *J. CACHEXIA SARCOPENIA MUSCLE: Abstracts of the 7th Cachexia Conference, Kobe/Osaka, Japan, Dec. 9-11, 2013, Abstract 4-06; LY2495655 (Lilly), a humanized anti-myostatin antibody in clinical development by Eli Lilly; see also "A PHASE 2 STUDY OF LY2495655 IN PARTICIPANTS WITH PANCREATIC CANCER," available on the world wide web at clinicaltrials.gov/ct2/NCT01505530; NML identifier: NCT01505530; ACE-031 (Acceleron Pharma); and stamulumab (Pfizer).*

[0214] Agents such as Ghrelin or ghrelin mimetics, or other growth hormone secretagogues (GHS) which are able to activate the GHS receptor (GHS-R1a), also known as the ghrelin receptor, may be useful for increasing food intake and body weight in humans. See Guillory et al. (2013) in *VITAMINS AND HORMONES* vol. 92, chap. 3; and Steinman and

DeBoer (2013) VITAMINS AND HORMONES vol. 92, chap. 8. Suitable ghrelin mimetics include anamorelin (Helsinn, Lugano, CH); See Temel et al. (2013) J. CACHEXIA SARCOPENIA MUSCLE: Abstracts of the 7th Cachexia Conference, Kobe/Osaka, Japan, Dec. 9-11, 2013, Abstract 5-01. Other suitable GHS molecules can be identified, for example, using the growth hormone secretagogue receptor Ghrelin competition assay described in PCT Publication Nos. WO2011/117254 and WO2012/113103.

[0215] Agonists of the androgen receptor, including small molecules and other selective androgen receptor modulators (SARMs) may be useful in treating cachexia and/or sarcopenia. See, e.g., Mohler et al. (2009) J. MED. CHEM. 52:3597-3617; Nagata et al. (2011) BIOORGANIC AND MED. CHEM. LETTERS 21:1744-1747; and Chen et al. (2005) MOL. INTERV. 5:173-188. Ideally, SARMs should act as full agonists, like testosterone, in anabolic target tissues, such as muscle and bone, but should demonstrate only partial or pure androgen receptor antagonistic activities on prostate tissue. See, e.g., Bovee et al. (2010) J. STEROID BIOCHEM. & MOL. BIOL. 118:85-92. Suitable SARMs can be identified, for example, by use of the methods and assays described in Zhang et al. (2006) BIOORG. MED. CHEM. LETT. 16:5763-5766; and Zhang et al. (2007) BIOORG. MED. CHEM. LETT. 17:439-443. Suitable SARMs include, for example, GTx-024 (enobosarm, Ostarine®, GTx, Inc.), a SARM in phase II clinical development by GTx, Inc. See also, Dalton et al. (2011) J. CACHEXIA SARCOPENIA MUSCLE 2:153-161. Other suitable SARMs include 2-(2,2,2)-trifluoroethyl-benzimidazoles (Ng et al. (2007) BIOORG. MED. CHEM. LETT. 17:1784-1787) and JNJ-26146900 (Allan et al. (2007) J. STEROID BIOCHEM. & MOL. BIOL. 103:76-83).

[0216] β -adrenergic receptor blockers, or beta-blockers, have been studied for their effect on body weight in cachexic subjects, and have been associated with partial reversal of cachexia in patients with congestive heart failure. See, e.g., Hryniewicz et al. (2003) J. CARDIAC FAILURE 9:464-468. Beta-blocker MT-102 (PsiOxus Therapeutics, Ltd.) has been evaluated in a phase 2 clinical trial for subjects with cancer cachexia. See Coats et al. (2011) J. CACHEXIA SARCOPENIA MUSCLE 2:201-207.

[0217] Melanocortin receptor-knockout mice with a genetic defect in melanocortin signaling exhibit a phenotype opposite that of cachexia: increased appetite, increased lean body mass, and decreased metabolism. Thus, melanocortin antagonism has emerged as a potential treatment for cachexia associated with chronic disease (DeBoer and Marks (2006) TRENDS IN ENDOCRINOLOGY AND METABOLISM 17:199-204). Accordingly, in certain embodiments of the present invention, one or more inhibitors of a melanocortin peptide or a melanocortin receptor may be administered in combination (for example, administered at the same time as, administered before, or administered after) with an antibody of the invention that inhibits GDF-15 activity. Suitable inhibitors of melanocortins or melanocortin receptors include an anti-melanocortin peptide antibody or an antigen binding fragment thereof, an anti-melanocortin receptor antibody or an antigen binding fragment thereof, a small molecule inhibitor of a melanocortin peptide, a small molecule inhibitor of a melanocortin receptor, and a 'decoy' receptor of a melanocortin receptor, such as soluble melanocortin receptor and a fusion of a soluble melanocortin receptor with an Fc molecule. Suitable melanocortin receptor inhibitors include, for example, the melanocortin receptor

antagonist agouti-related peptide (AgRP(83-132)), which has been demonstrated to prevent cachexia-related symptoms in a mouse model of cancer-related cachexia (Joppa et al. (2007) PEPTIDES 28:636-642).

[0218] Anti-cancer agents, especially those that can cause cachexia and elevate GDF-15 levels, such as cisplatin, may be used in methods of the present invention in combination with (for example, administered at the same time as, administered before, or administered after) an anti-GDF-15 antibody of the invention. Many cancer patients are weakened by harsh courses of radio- and/or chemotherapy, which can limit the ability of the patient to tolerate such therapies, and hence restrict the dosage regimen. Certain cancer agents themselves, such as fluorouracil, Adriamycin, methotrexate and cisplatin, may contribute to cachexia, for example by inducing severe gastrointestinal complications. See, e.g., Inui (2002) CANCER J. FOR CLINICIANS 52:72-91. By the methods of the present invention, in which an anti-cancer agent is administered in combination with an anti-GDF-15 antibody of the invention, it is possible to decrease the incidence and/or severity of cachexia, and ultimately increase the maximum tolerated dose of such an anti-cancer agent. Accordingly, efficacy of treatment with anti-cancer agents that may cause cachexia can be improved by reducing the incidence of cachexia as a dose-limiting adverse effect, and by allowing administration of higher doses of a given anti-cancer agent.

[0219] Thus, the present invention includes pharmaceutical compositions comprising an anti-GDF-15 antibody of the present invention in combination with an agent selected from the group consisting of: an inhibitor of Activin-A, an inhibitor of ActRIIB, an inhibitor of IL-6 or an inhibitor of IL-6R, a ghrelin, a ghrelin mimetic or a GHS-R1a agonist, a SARM, a TNF α inhibitor, an IL-1 α inhibitor, a myostatin inhibitor, a beta-blocker, a melanocortin peptide inhibitor, a melanocortin receptor inhibitor, and an anti-cancer agent. The present invention also includes methods of treating, preventing or minimizing cachexia and/or sarcopenia in a mammal comprising administering to a mammal in need thereof a pharmaceutical composition or compositions comprising an effective amount of an anti-GDF-15 antibody of the invention in combination with an effective amount of an inhibitor of Activin-A, an inhibitor of ActRIIB, an inhibitor of IL-6 or an inhibitor of IL-6R, a ghrelin, a ghrelin mimetic or a GHS-R1a agonist, a SARM, a TNF α inhibitor, an IL-1 α inhibitor, a myostatin inhibitor, a beta-blocker, a melanocortin peptide inhibitor, or a melanocortin receptor inhibitor.

[0220] In another embodiment, the invention comprises a method of inhibiting loss of muscle mass associated with an underlying disease comprising administering to a mammal in need thereof a pharmaceutical composition or compositions comprising an effective amount of an anti-GDF-15 antibody of the invention in combination with an effective amount of an inhibitor of Activin-A, an inhibitor of ActRIIB, an inhibitor of IL-6 or an inhibitor of IL-6R, a ghrelin, a ghrelin mimetic or a GHS-R1a agonist, a SARM, a TNF α inhibitor, an IL-1 α inhibitor, a myostatin inhibitor, a beta-blocker, a melanocortin peptide inhibitor, or a melanocortin receptor inhibitor to prevent or reduce loss of muscle mass. The underlying disease may be selected from the group consisting of cancer, chronic heart failure, chronic kidney disease, COPD, AIDS, multiple sclerosis, rheumatoid arthritis,

tis, sepsis, and tuberculosis. Additionally, in certain embodiments, the loss of muscle mass is accompanied by a loss of fat mass.

[0221] In yet further embodiments, the present invention comprises a method of inhibiting or reducing involuntary weight loss in a mammal comprising administering to a mammal in need thereof a pharmaceutical composition or pharmaceutical compositions comprising an effective amount of an anti-GDF-15 antibody of the invention in combination with an effective amount of an inhibitor of Activin-A, an inhibitor of ActRIIB, an inhibitor of IL-6 or an inhibitor of IL-6R, a ghrelin, a ghrelin mimetic or a GHS-R1a agonist, a SARM, a TNF α inhibitor, a IL-1 α inhibitor, a myostatin inhibitor, a beta-blocker, a melanocortin peptide inhibitor, or a melanocortin receptor inhibitor.

[0222] Certain anti-cancer agents, such as cisplatin, have one or more undesirable adverse effects that involve causing or increasing one or more syndromes such as cachexia, sarcopenia, muscle wasting, bone wasting or involuntary body weight loss. Accordingly, in certain embodiments, the present invention comprises a method of treating cancer, while preventing, minimizing or reducing the occurrence, frequency or severity of cachexia, sarcopenia, or muscle wasting, bone wasting or involuntary loss of body weight in a mammal, comprising administering to a mammal in need thereof a pharmaceutical composition comprising an effective amount of an anti-GDF-15 antibody of the present invention in combination with one or more anti-cancer agents. In particular embodiments, the invention comprises a method of treating cancer, while preventing, minimizing or reducing the occurrence, frequency or severity of cachexia, sarcopenia or muscle wasting, bone wasting or involuntary loss of body weight in a mammal, comprising administering to a mammal in need thereof a pharmaceutical composition comprising an effective amount of an anti-GDF-15 antibody of the invention in combination with one or more anti-cancer agents known to cause or increase the occurrence, frequency or severity of cachexia, sarcopenia, or muscle wasting, bone wasting or involuntary loss of body weight in a mammal.

EXAMPLES

[0223] The following Examples are merely illustrative and are not intended to limit the scope or content of the invention in any way.

Example 1: Human GDF15 Serum Levels in Mouse Xenograft Tumor Models

[0224] In this example, the amount of hGDF15 in the serum of mice bearing various xenograft tumors was measured. Serum was collected from three mice for each of the following tumor xenograft models: Chago, RPMI7951, PC3, TOV21G, HT-1080, K-562, and LS1034. Serum was also collected from three naïve mice as a control. Human GDF15 serum levels were determined by ELISA (R&D Systems, Cat. No. DY957E). Mice bearing human xenograft tumors that induce cachexia had serum levels of hGDF15 above 2 ng/mL, while mice bearing human xenograft tumors that do not induce cachexia had serum levels of hGDF15 below 1 ng/mL (FIG. 2). Naïve mice had no detectable hGDF15 (control). These results indicate that a serum level of approximately 2 ng/mL GDF15 is a threshold for inducing cachexia in this mouse model. Similar levels of hGDF15 were also observed in plasma when determined by ELISA.

Example 2: Non-Tumor Bearing Mouse Model of Cachexia

[0225] An existing non-tumor bearing mouse model of cachexia is based on the injection of mature rhGDF15 into a mouse (Johnen et al. (1997) NAT. MED. 13:1333-1340). Mature rhGDF15 corresponds to amino acids 197 to 308 of hGDF15 protein. Mature rhGDF15 can be produced in the yeast *Pichia pastoris* as described in Fairlie et al. (2000) GENE 254:67-76). Cleaved-rhGDF15 corresponds to amino acids 197 to 308 of hGDF15 protein released from an Fc-rhGDF15 fusion protein. In FIGS. 3-4 described below, cleaved-rhGDF15 was produced by enzymatic digestion of mFc-rhGDF15 fusion protein with Factor Xa, and subsequent purification, prior to injection in mice.

[0226] To investigate the half-life of cleaved-rhGDF15, plasma was collected from a group of three mice after single dose of cleaved-rhGDF15 (1 μ g/g) at different time points (2, 5, 8, 11, and 23 hours). Human GDF15 plasma levels were determined by ELISA (R&D Systems, Cat. No. DY957E). As shown in FIG. 3, cleaved-rhGDF15 was rapidly cleared from the plasma following injection. Eleven hours post-injection, the amount of cleaved-rhGDF15 in the plasma was below 10 ng/mL, and, within 23-hours, cleaved-rhGDF15 was almost completely cleared from the plasma.

[0227] The rapid clearance of cleaved-rhGDF15 in non-tumor bearing mice was further investigated. Eight-week old female ICR-SCID mice were randomized into two groups of ten mice each. Mice were dosed subcutaneously into the flank every eight hours for three days (a total of nine doses) with one of the following treatments: PBS (control) or cleaved-rhGDF15 at 1 μ g/g. Body weight was measured daily. Statistical analyses were performed using a two-way ANOVA.

[0228] As shown in FIG. 4, cleaved-rhGDF15 induced body weight loss. After nine doses over a three day period, percent body weight dropped to 88% at day 4 ($p < 0.001$), but approximately 24 hours after the last dose the mice began to gain weight. On day 6, the last day of the experiment, percent body weight increased to 94.8 percent ($p < 0.001$). These results indicate that weight loss induced by cleaved-rhGDF15 is not sustained over long periods of time. The activity observed with cleaved-rhGDF15 described herein was similar to that observed with mature rhGDF15 in the existing mouse model (Johnen et al., supra).

[0229] The existing non-tumor bearing mouse model for cachexia relies on the injection of large amounts of mature rhGDF15 delivered in multiple doses per day to induce muscle loss and body weight loss (Johnen et al., supra). It appears that if mature rhGDF15 or cleaved-rhGDF15 is used, the mice do not sustain muscle weight loss or body weight loss for long periods of time without continuous dosing. This limits the usefulness of such models. Moreover, repeated dosing requires frequent handling of these mice which introduces stress that can compromise the reliability of body weight loss measurements. For example, as shown in FIG. 4, mice treated with multiple doses of PBS demonstrated a body weight drop due to the stress of repeated dosing and handling.

Example 3: GDF15 Fusion Proteins

[0230] In view of the large amounts of mature rhGDF15 (or cleaved-rhGDF15) and the labor intensity required to induce non-tumor bearing cachexia mouse models (as well

as the resulting limitations of these models), we investigated alternate forms of rhGDF15 to induce a cachectic phenotype in mice. This Example describes the construction and production of two fusion proteins consisting of GDF15 and an immunoglobulin Fc fragment, designated mFc-rhGDF15 (mouse IgG1 Fc fused to the amino terminus of mature human GDF15) and rFc-rmGDF15 (rabbit IgG1 Fc fused to the amino terminus of mature mouse GDF15). The GDF15 fusion proteins were designed using methods known in the art. The mFc-rhGDF15 DNA sequences were constructed from fragments using overlap extension PCR to include (in the following order): 5' HindIII restriction site, Kozak consensus sequence, amino terminal signal sequence, mouse IgG1 Fc, Factor Xa cleavage site, a polypeptide linker (GGGGS) (SEQ ID NO: 139), mature hGDF15, stop codon, and a 3' EcoRI restriction site. The rFc-rmGDF15 amino acid sequences were converted to codon-optimized DNA sequences and synthesized to include (in the following order): 5' HindIII restriction site, Kozak consensus sequence, amino terminal signal sequence, rabbit IgG1 Fc, a polypeptide linker (GGGG) (SEQ ID NO: 265), mature mouse GDF15, stop codon, and a 3' EcoRI restriction site.

[0231] The GDF15 fusion proteins were subcloned into the mammalian expression vector pEE14.4 (Lonza, Basel, Switzerland) via HindIII and EcoRI sites using In-Fusion™ PCR cloning (Clontech, Mountain View, CA). GDF15 fusion proteins were stably expressed in CHOK1SV cells using the GS System™ (Lonza Biologics) in order to produce large quantities of purified protein. Each expression vector was linearized and transfected into CHOK1SV cells. Stable clones were selected in the presence of methionine sulfoximine. Secreted proteins produced by CHOK1SV stably transfected cell lines were purified by Protein A and size exclusion chromatography.

[0232] The nucleic acid sequence and the encoded protein sequence defining the mouse IgG1 Fc-mature human GDF15 fusion protein (mFc-rhGDF15) are shown below. mFc-rhGDF15 contains mouse IgG1 Fc from amino acids 1-222, Factor Xa cleavage site from amino acids 223-228, an artificial linker sequence from amino acids 229-233, and mature hGDF15 from amino acids 234-345.

Nucleic Acid Sequence Encoding the Mouse IgG1 Fc—Mature Human GDF15 Fusion Protein (mFc-rhGDF15) (SEQ ID NO:219)

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1  ggggtgtaaac cctgcatctg cacggtgccg gaggtgtcct ccgcttttat ctteccctccc
61  aaaccaagg atgtgctgac aatcactttg actccaaaag tcacatgcgt agtcgtggac
121 atctcgaaag acgacccgga agtgacgttc tcgtggtttg ttgatgatgt agaagtgcgt
181 accgctcaaa cccagccgag ggaagaacag tttaacagca cgtttaggag tgtgtcggaa
241 ctgccccatta tgcaccagga ttggcttaat gggaaggagt tcaaatgtcg cgtgaatagt
301 gcggcggttc cagccctat tgaaaagact atttccaaaa cgaagggtcg gcccaaggct
361 ccccaagtat acacaatccc tccgccgaaa gaacaaatgg caaaagacaa agtgagtttg
421 acgtgcatga tcacggactt tttcccgag gatatacccg tcgaatggca atggaatggg
481 caacctgcgcg aaaactacaa gaatacacia ccattatgg ataccgatgg atcgtatttc
541 gtctactcaa agttgaacgt acagaagtca aattgggagg cagggaatac gttcacttgc
601 agtggttttg acgaaggcct ccataaccac catacggaaa agtcactgtc gactcccccg
661 ggaaaaatcg agggcagaat ggatggtgga ggagggtcgg cgcgcacagg ggaccactgt
721 ccgctcgggc cggggcggtg ctgccgtctg cacacgggtc cgcgctcgct ggaagacctg
781 ggctgggcgg attgggtgct gtcgccacgg gaggtgcaag tgaccatgtg catcggcgcg
841 tgcccagacc agttccgggc ggcaaacatg cagcgcaga tcaagacgag cctgcaccgc
901 ctgaagcccc acacgggtgc agcgccctgc tcggtgcccg ccagctacaa tcccatgggtg
961 ctcatcaaaa agaccgacac cggggtgtcg ctccagacct atgatgactt gttagccaaa
1021 gactgccact gcata

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Protein Sequence Defining the Mouse IgG1 Fc—Mature Human GDF15 Fusion Protein (mFc-rhGDF15) (SEQ ID NO:220)

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1  gckpictvp evssvfifpp kpkdvlitl tpkvtcvvd iskddpevqf swfvddvevh
61  taqtqpreeq fnstfrvse lpimhqdwln gkefkcrvns aafpapiekt isktkgrpka
121 pqvytipppk eqmakdkvsl tcmidffpe ditvewqwnq qpaenykntq pimtdgtsyf
181 vysklnvqks nweagntftc svlheglhnh htekslshsp gkiegrmdgg ggsarngdhc

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-continued

241 plpgpgrcrl htvrasledl gwadwvlspr evqvtmciga cpsqfraanm haqiktslhr
301 lkpdtvpapc cvpasynpmv liqktdtgvs lqtyddllak dchci

[0233] The nucleic acid sequence and the encoded protein sequence defining the rabbit IgG1 Fc-mature mouse GDF15 fusion protein (rFc-rmGDF15) are shown below. rFc-rmGDF15 contains rabbit IgG1 Fc from amino acids 1-223, an artificial linker sequence from amino acids 224-227, and mature mouse GDF15 from amino acids 228-342.

Nucleic Acid Sequence Encoding the Rabbit IgG1 Fc—Mature Mouse GDF15 Fusion Protein (rFc-rmGDF15) (SEQ ID NO:221)

1 tcgaaaccca cttgccctcc tccggagctg ttgggcggac cctccgtgtt tatctttccc
61 ccgaagccga aagataccct tatgatctca cggacgccgg aggtcacttg cgtagtagtg
121 gatgtgtcgg aggatgaccc cgaagtcag ttcacctggt atatcaataa cgagcaagtg
181 aggacagcga ggcacctact tagggagcag cagttcaact ccacaattcg ggtcgtcagc
241 actttgcccc tcgctcatga ggactggctc cgcggaaaag agttcaagtg taaggtgcat
301 aacaaggcat tgccagcgcc tattgaaaag acaatctcga aggcgcgagg gcagccgctc
361 gagcccaaa tgatacatga gggacccccg agggaagaat tgcgtcgcg ctcagtaagc
421 cttacgtgca tgattaacgg tttctaccct agcgacatca gcgtagagtg ggaaaagaat
481 ggaaaggcgg aggataacta caagacgact cccgcggtgc tggattcgga tgggtcgtac
541 tttctgtata gcaaattgtc agtcccgacc tcagaatggc agaggggtga cgtgttcacg
601 tgctccgtga tgcacgaagc acttcacaat cactacaccc agaaatcaat ctccggtccc
661 ccaggcaaa gtaggaggag gtccggtcac gccaccctc gcgattcgtg tccgtggggg
721 cctggtagat gctgtcatct cgagacagtc caggccacgc tggaggacct cgggtgggtca
781 gactgggtcc tgteccccag acaactgcag ctttcgatgt gcgtggggga atgtccgcac
841 ttgtacagat cggcgaatac ccacgctcag attaaggcac gactccatgg ttgcagcca
901 gataaagtc ccgcacctg ctgtgtcccc agctcatata ctctgtcgt actcatgcat
961 cggacagaca gcggcggtgc gcttcaaacg tatgacgacc tcgtagcgag aggatgtcat
1021 tgcgcc

Protein Sequence Defining the Rabbit IgG1 Fc—Mature Mouse GDF15 Fusion Protein (rFc-rmGDF15) (SEQ ID NO:222)

1 skptcpppel lggpsvfifp pkpkdtlmis rtpevtcvvv dvseddpevq ftwyinneqv
61 rtarpplreq qfnstirvvs tlpiahdwl rgkefkckvh nkalpapie kiskargqpl
121 epkvytmgpp reelssrsvs ltcmingfyp sdisvewekn gkaednyktt pavldsdgsy
181 flysklsvpt sewqrgdvft csvmhealhn hytqksisrs pgkggggsah ahprdsclp
241 pgrcchletv qatledlgws dwvlspqlq lsmcvgecph lyrsanthaq ikarlhglqp
301 dkvpapccvp ssytpvvlmh rtdsgvslqt yddlvargch ca

[02334] The following sequences represent exemplary protein sequences for human IgG1 Fc-mature human GDF15 fusion proteins (hFc-rhGDF15). hFc-rhGDF15 Xa consists of human IgG1 Fc from amino acids 1-227, Factor Xa cleavage site from amino acids 228-233, an artificial linker sequence from amino acids 234-238, and mature hGDF15 from amino acids 239-350. hFc-rhGDF15 consists of human IgG1 Fc from amino acids 1-227, an artificial linker sequence from amino acids 228-232, and mature hGDF15 from amino acids 233-344.

Protein Sequence Defining the Human IgG1 Fc—Mature Human GDF15 Fusion Protein with Xa Cleavage Site (hFc-hGDF15 Xa) (SEQ ID NO:223)

```

1  dkthtccppcp apellggpsv flfppkpkdt lmisrtpevt cvvvdvshed pevknfnyvd
61  gvevhnaktk preeqynsty rvsvltvlh qdwlngkeyk ckvsnkappa piektiskak
121  ggprepqvyt lppsreemtk nqvsltclvk gfypsdiave wesngqpenn ykttppvlds
181  dgsfflyskl tvdksrwqgg nvfscsvmhe alhnhytqks lslspgkieg rmdggggsar
241  ngdhcplpgp rccrlhtvra sledlgwadw vlsprevqvt mcigacpsqf raanmhaqik
301  tslhrlkpdv vpapccvpas ynpmvliqkt dtgvsllqtyd dllakdchci

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Protein Sequence Defining the Human IgG1 Fc—Mature Human GDF15 Fusion Protein with (hFc-hGDF15) (SEQ ID NO:224)

```

1  dkthtccppcp apellggpsv flfppkpkdt lmisrtpevt cvvvdvshed pevknfnyvd
61  gvevhnaktk preeqynsty rvsvltvlh qdwlngkeyk ckvsnkappa piektiskak
121  ggprepqvyt lppsreemtk nqvsltclvk gfypsdiave wesngqpenn ykttppvlds
181  dgsfflyskl tvdksrwqgg nvfscsvmhe alhnhytqks lslspgkggg gsarngdhcp
241  lpggrccrlh tvrasledlg wadwvlspre vqvtmcigac psqfraanmh aqiktshlrl
301  kpdvvpapcc vpasynpmvl iqktdtgvsll qtyddllakd chci

```

Example 4: Fc-rhGDF15 Induced Cachexia Model

[0235] This Example describes the generation of an Fc-GDF15-induced cachexia model in mice. Immune-competent (Balb/C) and immune-incompetent (CB17-Scid) mice were randomized into three groups of ten mice each. Each group received one of the following treatments: PBS (control), mFc-rhGDF15 (as described in Example 3), or rFc-rmGDF15 (as described in Example 3) at 1 µg/g. Eight-week old female mice were dosed subcutaneously into the flank for three days (Balb/C) or once (CB17-Scid). Body weight was measured daily.

[0236] As shown in FIG. 5A and FIG. 5B, administration of mFc-rhGDF15 or rFc-rmGDF15 induced body weight loss in immune-competent mice (FIG. 5A) and immune-incompetent mice (FIG. 5B). These results indicate that a steady-state level of active rhGDF15 was achieved, because regardless of dose (one vs. three doses), both mFc-rhGDF15 and rFc-rmGDF15 induced sustained weight loss over the measured time course (7 days).

[0237] The fusion proteins, mFc-rhGDF15 and rFc-rmGDF15, were further tested in additional immune-competent (C57BL6, Swiss Webster) and immune-incompetent (ICR-SCID) mouse strains. In each tested mouse strain, the administration of mFc-rhGDF15 or rFc-rmGDF15 induced

cachexia, as measured by body weight loss. Similar results were obtained regardless of whether mFc-rhGDF15 was dosed subcutaneously or intraperitoneally.

[0238] It was also investigated whether mFc-rhGDF15 induced weight loss regardless of the age of the mice treated with fusion protein. Swiss Webster (immune-competent) female mice of different ages (7, 13 and 25 weeks old) were divided into two groups of ten and treated with three doses per day of mFc-rhGDF15 or PBS (0.8 µg/g, 7 week old mice; 0.6 µg/g, 13 week old mice; or 0.4 µg/g, 25 week old mice). mFc-rhGDF15-induced weight loss was observed in all three mice age populations. In each age population, the

mice lost approximately 10% of their body weight following treatment with mFc-rhGDF15 measured at ten days post treatment.

[0239] In another experiment, mFc-rhGDF15 induction of cachexia was investigated by measuring the loss of body weight, the loss of muscle mass, the loss of fat mass, and the expression levels of two molecular markers indicative of muscle degradation (i.e., mMuRF1 and mAtrogin). MuRF1 and Atrogin are E3-ubiquitin ligases that are upregulated in multiple models of muscle atrophy and cachexia (Glass, D. (2010) CURR. OPIN. CLIN. NUTR. MET. CARE 13:225-229).

[0240] Eight-week old female ICR-SCID mice were randomly divided into ten groups of ten mice each. Five groups (ten mice each) were dosed subcutaneously in the flank with PBS (control) and five groups (ten mice each) were dosed subcutaneously in the flank with mFc-rhGDF15 at 1.6 µg/g on day one. Body weight was measured daily for up to 17 days. One control group and one treatment group were sacrificed at different time points (0, 1, 3, 7 and 16 days post dose). Gonadal fat and gastrocnemius muscles were removed surgically from each group of mice at the indicated sacrifice time, and weighed. Tissues were snap frozen in liquid nitrogen, and RNA was isolated from the gastrocnemius muscle samples. Levels of mMuRF1 and mAtrogin mRNA were measured by qRT-PCR in samples corresponding to groups collected after 1, 7, and 16 days post dose. Statistical analyses were performed using a two-way ANOVA.

[0241] As shown in FIG. 6A, mFc-rhGDF15 induced body weight loss in ICR-SCID mice. Percent body weight was 79.4 percent when measured after 16 days following one dose of mFc-rhGDF15 ($p<0.001$). mFc-rhGDF15 also induced loss of fat (adipose tissue), as observed by the loss of gonadal fat (FIG. 6B; $p<0.01$ at day 7 and $p<0.001$ at day 16) and loss of muscle, as observed by the loss of gastrocnemius muscle (FIG. 6C; $p<0.05$ at days 1 and 3, and $p<0.0001$ at days 7 and 16). Administration of mFc-rhGDF15 also elevated gene expression of two enzymes associated with muscle degradation and cachexia, mMuRF1 (FIG. 6D; $p<0.001$ at days 1, 7, and 16) and mAtrogin (FIG. 6E; $p<0.001$ at days 1 and 7, and $p<0.01$ at day 16).

[0242] These results indicated that mFc-rhGDF15 induces cachexia in mice.

Example 5: mFc-rhGDF15 Induces Cachexia with a Longer GDF15 Half-Life in Serum

[0243] In this Example, the serum hGDF15 levels were measured following administration of mFc-rhGDF15, to determine the half-life of rhGDF15 in this model. Eight-week old female Balb/C nude mice were randomly divided into two groups of twelve mice each. Mice were dosed subcutaneously in the flank every twelve hours for three days (a total of six doses) with one of the following treatments: PBS (control) or mFc-rhGDF15 at 1.33 $\mu\text{g/g}$. Body weight was measured daily. As shown in FIG. 7, mFc-rhGDF15 induced sustained body weight loss for at least one week after the final injection.

[0244] In this experiment, hGDF15 serum levels were measured 0.2, 5, and 8 days after the last dose of mFc-rhGDF15. Mice were sacrificed at the indicated time, and sera were collected. Human GDF15 serum levels were determined by ELISA (R&D Systems, Cat. No. DY957E). Table 1 provides the serum levels ($\mu\text{g/mL}$) for each mouse in the study.

TABLE 1

Days post	Mouse	Treatment		Serum GDF15
last dose	#	Agent	$\mu\text{g/g}$	($\mu\text{g/mL}$); ELISA
0.2	1	mFc-rhGDF15	1.33	10.02
0.2	2	mFc-rhGDF15	1.33	9.54
0.2	3	mFc-rhGDF15	1.33	9.36
5	4	mFc-rhGDF15	1.33	8.24
5	5	mFc-rhGDF15	1.33	8.01
5	6	mFc-rhGDF15	1.33	6.59
8	7	mFc-rhGDF15	1.33	5.60
8	8	mFc-rhGDF15	1.33	5.52
8	9	mFc-rhGDF15	1.33	5.57

[0245] The results in Table 1 reveal that strong, sustained levels of hGDF15 are present in the serum at least eight days after the last dose of mFc-rhGDF15.

[0246] Serum samples from day 0.2 and day 5 after the last dose were also analyzed by Western blot (reducing gel; blot with an antibody against hGDF15 (R&D Systems, Cat. No. AF957)) and quantified by Licor to determine the stability of mFc-rhGDF15 in the serum. Unexpectedly, two bands were observed. The upper band was approximately 40 kDa, and appeared to be mFc-rhGDF15. The lower band was approximately 15 kDa, and appeared to be cleaved mature rhGDF15. This indicated that mature rhGDF15 was released from mFc-rhGDF15 in the serum. Quantification of the two

bands showed that about 90% of the rhGDF15 present in the serum was in the form of mFc-rhGDF15, with about 10% of the total rhGDF15 in the serum being present as the cleaved mature form (FIG. 8). Quantification showed a slight decrease in mFc-rhGDF15 in the serum samples collected five days after the last dose, but, surprisingly, a constant level of mature rhGDF15 remained in the serum. The ratio of mature rhGDF15 to mFc-rhGDF15 slightly increased over time, as a result of a decrease in mFc-rhGDF15 in the serum. Similar results were observed when rFc-rmGDF15 was injected into mice.

[0247] The results presented in FIGS. 7-8 and Table 1 were unexpected. The expectation was that very little, if any, mature rhGDF15 would be cleaved (released) from the mFc-rhGDF15 by day 0.2, and that any cleaved rhGDF15 would be rapidly cleared from the serum, as had been previously observed. For example, in FIG. 4, a series of nine doses at 1 $\mu\text{g/g}$ per dose (for a total of 9 $\mu\text{g/g}$) of cleaved-rhGDF15 was required to induce significant body weight loss in mice. These mice gained weight, almost immediately when dosing stopped. In contrast, a single dose of mFc-rhGDF15 at 0.1 $\mu\text{g/g}$ was sufficient to induce significant body weight loss for at least eight days (FIG. 9A; ten ICR-SCID mice dosed intraperitoneally with 0.1 $\mu\text{g/g}$ on day 1). The data in Table 1 revealed that rhGDF15 serum levels were stable for at least eight days, when rhGDF15 was administered as an mFc-rhGDF15 fusion protein.

[0248] To determine the source of activity resulting in sustained body weight loss, we investigated whether the observed rhGDF15 activity was attributable to the mFc-rhGDF15 fusion protein, the released mature rhGDF15 form, or both. As shown in FIG. 9A, a low dose of mFc-rhGDF15 (0.1 $\mu\text{g/g}$) resulted in body weight loss continuing for at least eight days. A lower dose of mFc-rhGDF15 (0.01 $\mu\text{g/g}$) also induced body weight loss, but the effect was not sustained for longer than 3 days post dose.

[0249] In this experiment, plasma was collected from three mice each dosed with 0.1 $\mu\text{g/g}$ or 0.01 $\mu\text{g/g}$ at 5 days post dose. Total rhGDF15 was measured by ELISA as described above. Total rhGDF15 plasma levels in the mice dosed with 0.1 $\mu\text{g/g}$ were above 70 ng/mL, consistent with the observation that these mice had significant weight loss (FIG. 9B). Total rhGDF15 plasma levels in mice dosed with 0.01 $\mu\text{g/g}$ were approximately 3.3 ng/mL, but it was observed that these mice were gaining weight (FIG. 9A and FIG. 9B). As described in FIG. 2, the threshold for hGDF15 to induce cachexia in tumor bearing mice is approximately 2 ng/mL. Thus, if both forms of rhGDF15 were active (i.e., mFc-rhGDF15 and released mature rhGDF15), then these mice should be losing weight, not gaining weight (i.e., 3.3 ng/mL total rhGDF15 is above the threshold of approximately 2 ng/mL hGDF15).

[0250] To determine which form was the active form (i.e., either mFc-rhGDF15 or released mature rhGDF15), we considered the data from FIG. 8 which showed that approximately 90% of the total rhGDF15 in serum was in the mFc-rhGDF15 form, and the remaining 10% was the released mature form. Based on this extrapolation, approximately 3.0 ng/mL of rhGDF15 in the plasma was in the mFc-rhGDF15 form (i.e., 90% of 3.3 ng/mL). Once again, if mFc-rhGDF15 were active, these mice would be losing weight, not gaining weight because 3.3 ng/mL mFc-rhGDF15 is above the threshold of approximately 2 ng/mL hGDF15. The mice dosed with 0.1 $\mu\text{g/g}$ mFc-rhGDF15

served as an internal control, because these mice had sustained body weight loss indicating that at least one of the two forms must be active. A calculation of 10% of 70 ng/mL total rhGDF15 in these mice is 7 ng/mL released mature rhGDF15. This amount is consistent with inducing the observed body weight loss and the threshold observed in FIG. 2. Thus, the data indicate that the mFc-rhGDF15 is not an active form of the protein, and only the mature rhGDF15 is active. These results were unexpected, because: (a) there was no reason to predict that the Fc fusion protein (mFc-rhGDF15) would be inactive; and (b) there was no reason to predict that the Fc fusion protein would release mature rhGDF15 at the observed rate.

[0251] These results indicate that mFc-rhGDF15 sustains a cachectic phenotype by slowly releasing mature rhGDF15 into the serum. These results further indicate that a steady state level of mature rhGDF15 in the plasma or serum can be achieved in a non-tumor bearing mouse by administering mFc-rhGDF15 to the mouse. Therefore, administration of mFc-rhGDF15 to non-tumor bearing mice is particularly useful as a mouse model of cachexia with a robust and sustained loss of muscle mass, loss of fat mass, and body weight loss (see FIGS. 6A-C).

Example 6: Anti-GDF15 Antibodies

[0252] This Example describes the production of anti-GDF15 monoclonal antibodies. Immunizations, fusions, and primary screens were conducted using conventional methods following the Repetitive Immunization Multiple Sites (RIMMS) protocol. Five AJ mice and five Balb/c mice were immunized with 6xHis (SEQ ID NO: 266) tagged recombinant human GDF15 (His-rhGDF15) (R&D Systems, Inc., Minneapolis, MN). Two Balb/c mice with sera displaying the highest anti-GDF15 activity by Enzyme Linked Immunosorbent Assay (ELISA) were chosen for subsequent fusion. Spleens and lymph nodes from the appropriate mice were harvested. B-cells were harvested and fused with a myeloma line. Fusion products were serially diluted onto forty 96-well plates to near clonality. Two AJ mice with sera displaying the highest anti-GDF15 activity by ELISA were chosen for subsequent fusion. Spleens and lymph nodes from the appropriate mice were harvested. B-cells were harvested and fused with a myeloma line. Fusion products were serially diluted onto forty 96-well plates to near clonality.

[0253] Approximately 3,840 supernatants from the cell fusions were screened by ELISA for binding to rhGDF15. A total of 172 supernatants containing antibodies against GDF15 were further characterized in vitro. A panel of hybridomas was selected, subcloned and expanded. Antibodies were expressed and subsequently purified by affinity chromatography on Protein G resin, under standard conditions.

Example 7: Antibody Sequence Analysis

[0254] The light chain isotype and heavy chain isotype of each monoclonal antibody in Example 6 was determined using the IsoStrip™ Mouse Monoclonal Antibody Isotyping

Kit according to the kit vendor's instructions (Roche Applied Science, Indianapolis, IN). All antibodies were found to be kappa light chain, and IgG1 or IgG2b heavy chain.

[0255] The heavy and light chain variable regions of the mouse monoclonal antibodies were sequenced using 5' RACE (Rapid Amplification of cDNA Ends). Total RNA was extracted from each monoclonal hybridoma cell line using the RNeasy® Miniprep kit according to the kit vendor's instructions (Qiagen, Valencia, CA). Full-length first strand cDNA containing 5' ends was generated using the SMARTer™ RACE cDNA Amplification Kit (Clontech, Mountain View, CA) according to the kit vendor's instructions for 5' RACE.

[0256] The variable regions of the light (kappa) and heavy (IgG1 or IgG2b) chains were amplified by PCR using KOD Hot Start Polymerase (EMD Chemicals, Gibbstown, NJ) according to the kit vendor's instructions. For amplification of 5' cDNA ends in conjunction with the SMARTer™ RACE cDNA Amplification Kit, the Universal Primer Mix A primer (Clontech), a mix of: 5' CTAATACGACTCCTATAGGGCAAGCAGTGGTATCAACGCAGAGT 3' (SEQ ID NO:233) and 5' CTAATACGACTCACTATAGGGC 3' (SEQ ID NO:225), was used as a 5' primer. Heavy chain variable regions were amplified using the above 5' primers and a 3' IgG1 constant region specific primer, 5' TATGCAAGGCTTACAACCACA 3' (SEQ ID NO:226), or a 3' IgG2b constant region specific primer, 5' AGGACAGGGGTTGATTGTTGA 3' (SEQ ID NO:227). Kappa chain variable regions were first amplified with the above 5' primers and a 3' kappa constant region specific primer, 5' CTCATTCCTGTTGAAGCTCTTGACAAT 3' (SEQ ID NO:228). The light chains were subjected to a second, nested, round of PCR using the Nested Universal Primer A (Clontech) 5' primer, 5' AAGCAGTGGTATCAACGCAGAGT 3' (SEQ ID NO:229) and a nested 3' kappa constant region specific primer, 5' CGACTGAGGCACCTCCAGATGTT 3' (SEQ ID NO:230). Individual PCR products were either purified using the Qiaquick® PCR Purification kit or isolated by agarose gel electrophoresis and purified using the Qiaquick® Gel Purification kit according to the kit vendor's instructions (Qiagen). The PCR products were subsequently cloned into the pCR®4Blunt plasmid using the Zero Blunt® TOPO® PCR Cloning according to the kit vendor's instructions (Invitrogen) and transformed into DH5-α bacteria (Invitrogen) through standard molecular biology techniques. Plasmid DNA isolated from transformed bacterial clones was sequenced using M13 Forward (5' GTAAACGACGGCCAGT 3') (SEQ ID NO:231) and M13 Reverse primers (5' CAGGAAACAGCTATGACC 3') (SEQ ID NO:232) by Beckman Genomics (Danvers, MA), using standard dideoxy DNA sequencing methods to identify the sequence of the variable region sequences. The sequences were analyzed using Vector NTI software (Invitrogen) and the IMGT/V-Quest web server (imgt.cines.fr) to identify and confirm variable region sequences.

[0257] The nucleic acid sequences encoding and the protein sequences defining variable regions of the murine monoclonal antibodies are shown below (amino terminal signal peptide sequences are not shown). CDR sequences (Kabat definition) are indicated by bold font and underlining in the amino acid sequences.

Nucleic Acid Sequence Encoding the Heavy Chain Variable
Region of the 01G06 Antibody (SEQ ID NO:39)

1 gaggtcctgc tgcaacagtc tggacctgag ctggtgaagc ctggggcttc agtgaagata
61 ccctgcaagg cttctggata cacattcact gactacaaca tggactgggt gaagcagagc
121 catggaaaga gccttgagtg gattggacaa attaactcta acaatgggtg tattttcttc
181 aaccagaagt tcaagggcaa ggccacattg actgtagaca agtcctccaa tacagccttc
241 atggagggtc gcagcctgac atctgaggac actgcagtct attactgtgc aagagaggca
301 attactacgg taggcgctat ggactactgg ggtcaaggaa cctcagtcac cgtctcctca

Protein Sequence Defining the Heavy Chain Variable
Region of the 01G06 Antibody (SEQ ID NO:40)

1 evllqsgspe lvkpgasvki pckasgytft dynmdwvkqs hgkslewigg inpnnggiff
61 nqkfkkatl tvdkssntaf mevrsltsed tavyycarea ittvgamdyw gggtsvtvss

Nucleic Acid Sequence Encoding the Kappa Chain Variable
Region of the 01G06 Antibody (SEQ ID NO:75)

1 gacatccaga tgactcagtc tccagcctcc ctatctgcat ctgtgggaga aactgtcacc
61 atcacatgtc gaacaagtga gaatcttcac aattatttag catgggtatca gcagaaacag
121 ggaaaatctc ctcagctcct ggtctatgat gcaaaaacct tagcagatgg tgtgccatca
181 aggttcagtg gcagtggatc aggaacacaa tattctctca agatcaacag cctgcagcct
241 gaagattttg ggagttatta ctgtcaacat ttttgagta gtccttacac gttcggaggg
301 gggaccaagc tggaataaaa a

Protein Sequence Defining the Kappa Chain Variable
Region of the 01G06 Antibody (SEQ ID NO:76)

1 diqmtqspas lsasvgetvt itcrtsenlh nylawyqqkq gkspqllvyd aktladgvps
61 rfsgsgsgtg yslkinslqp edfgsyycqh fwsspytfpg gtleik

Nucleic Acid Sequence Encoding the Heavy Chain Variable
Region of the 03G05 Antibody (SEQ ID NO:41)

1 cagggtccaac tgcagcagcc tggggctgaa ctggtgaagc ctggggcttc agtgaagctg
61 tcctgcaagg cttctggcta caccttcacc agctactgga ttcactgggt gaaccagagg
121 cctggacaag gccttgagtg gattggagac attaactcta gcaacggcgc tagtaagtat
181 aatgagaagt tcaagaacaa ggccacaatg actgcagaca aatcctccaa cacagcctac
241 atgcaactca gcagcctgac atctgaggac tctgcggtct attactgtgc aagagaggtt
301 ctggatgggt ctatggacta ctgggggtcaa ggaacctcag tcaccgtctc ctca

Protein Sequence Defining the Heavy Chain Variable
Region of the 03G05 Antibody (SEQ ID NO:42)

1 qvqlqqpgae lvkpgasvkl sckasgytft sywihwvnqr pggglewigd inpsngrsky
61 nefkknkatm tadksntay mqlssltsed savyycarev ldgamdywgq gtsvtvss

Nucleic Acid Sequence Encoding the Kappa Chain Variable
Region of the 03G05 Antibody (SEQ ID NO:77)

```

1  gacattgtgt tgaccaatc tccagettct ttggtgtgt ctctaggga gagggccacc
61  atctcctgca gagccagcga aagtgttgat aattatggca ttagttttat gaactggttc
121 caacagaaac caggacagcc acccaaaactc ctcatctatg ctgcatccaa ccaaggetcc
181 ggggtccctg ccaggttttag tggcagtggg tctgggacag acttcagcct caacatccat
241 cctatggagg aggatgatac tgcaatgtat ttctgtcagc aaagtaagga ggttcctggg
301 acgttcggtg gaggtccaa gctggaaatc aaa

```

Protein Sequence Defining the Kappa Chain Variable
Region of the 03G05 Antibody (SEQ ID NO:78)

```

1  divltqspas lavslgqrat iscrasesvd nygisfmmwf qqkpgqppkl liyaasnqgs
61  gvparfsgsg sgtdfslnih pmeeddtamy fcqqskevpw tfgggsklei k

```

Nucleic Acid Sequence Encoding the Heavy Chain Variable
Region of the 04F08 Antibody (SEQ ID NO:43)

```

1  caggttactc tgaaagagtc tggccctggg atattgcagc cctcccagac cctcagtctg
61  acttgttctt tctctgggtt ttcactgagc acttatggta tgggtgtgac ctggattcgt
121 cagccttcag gaaaggtctt ggagtggctg gcacacattt actgggatga tgacaagcgc
181 tataacccat ccctgaagag ccggctcaca atctccaagg atacctccaa caaccaggta
241 ttctcaaga tcaccagtgt ggacactgca gatactgcca catactactg tgctcaaacg
301 gggatatagta acttgtttgc ttactggggc caagggactc tggtcactgt ctctgca

```

Protein Sequence Defining the Heavy Chain Variable
Region of the 04F08 Antibody (SEQ ID NO:44)

```

1  qvtlkesgpg ilqpsqtls l tcsfsgfsls tygmgtwir qpsgkglewl ahiywdddkr
61  ynpslksrlt iskdtstnnqv flkitsvdtat dtatyycagt gysnlfaywg qgtlvtvsa

```

Nucleic Acid Sequence Encoding the Kappa Chain Variable
Region of the 04F08 Antibody (SEQ ID NO:79)

```

1  gacattgtga tgaccagtc tcaaaaattc atgtccacat cagtaggaga cagggtcagc
61  gtcacctgca aggcagtcga gaatgtgggt actaatgtag cctggatatca acagaaatta
121 ggacaatctc ctaaaacact gatttactcg gcatcctacc ggtacagtgg agtccctgat
181 cgettcacag gcagtggatc tgggacagat ttcactctca ccatcagcaa tgtgcagtct
241 gaagacttgg cagagtatct ctgtcagcaa tataacagct atccgtacac gttcggaggg
301 gggaccaagc tggaaataaa a

```

Protein Sequence Defining the Kappa Chain Variable
Region of the 04F08 Antibody (SEQ ID NO:80)

```

1  divmtsqskf mstsvgdrvs vtckasqnv tnvawyqqkl gqspktliys asyrysgvpd
61  rftgsgsgtd ftltisnvqs edlaeyfcq q ynsyptfgg gtleik

```

Nucleic Acid Sequence Encoding the Heavy Chain Variable
Region of the 06C11 Antibody (SEQ ID NO:45)

1 caggttactc tgaaagagtc tggccctggg atattgcagc cctcccagac cctcagtctg
61 acttgttctt tctctgggtt ttcactgaac acttatggta tgggtgtgag ctggattcgt
121 cagccttcag gaaagggctc ggagtggctg gcacacattt actgggatga tgacaagcgc
181 tataacccat ccctgaagag ccggctcaca atctccaagg atgcctccaa caaccgggtc
241 ttctcaaga tcaccagtgt ggacactgca gatactgcca catactactg tgctcaaaga
301 ggttatgatg attactgggg ttactggggc caagggaactc tggtcactat ctctgca

Protein Sequence Defining the Heavy Chain Variable
Region of the 06C11 Antibody (SEQ ID NO:46)

1 qvtlkesgpg ilqpsqtlsl tcsfsgfsln tygmgswir qpsgkglewl ahiywdddkr
61 ynpslksrlt iskdasnrv flkitsvdtat dtatyycagr gyddywgywg qgtlvtisa

Nucleic Acid Sequence Encoding the Kappa Chain Variable
Region of the 06C11 Antibody (SEQ ID NO:81)

1 gacattgtga tgaccagtc tcaaaaattc atgtccacat cagtaggaga cagggtcagc
61 gtcacctgca aggccagtca gaatgtgggt actaatgtag cctggtttca acagaaacca
121 ggtcaatctc ctaaagcact gatttactcg gcattctacc ggtacagtgg agtccctgat
181 cgtttcacag gcagtggatc tgggacagat ttcattctca ccacagcaa tgtgcagtct
241 gaagacctgg cagagtattt ctgtcagcaa tataacaact atcctctcac gttcgggtct
301 gggaccaagc tggagctgaa a

Protein Sequence Defining the Kappa Chain Variable
Region of the 06C11 Antibody (SEQ ID NO:82)

1 divmtqsqkf mstsvgdrvs vtckasqnvg tnvawfqkp gqspkaliys asyrysgvpd
61 rftgsgsgtd filtisnvqs edlaeyfcqg ynnypltfga gtklelk

Nucleic Acid Sequence Encoding the Heavy Chain Variable
Region of the 08G01 Antibody (SEQ ID NO:47)

1 gaggtcctgc tgcaacagtc tggacctgag gtggtgaagc ctggggcttc agtgaagata
61 ccctgcaagg cttctggata cacattcact gactacaaca tggactgggt gaagcagagc
121 catggaaaga gccttgagtg gattggagag attaataccta acaatgggtg tactttctac
181 aaccagaagt tcaagggcaa ggccacattg actgtagaca agtcctocag cacagcctac
241 atggagctcc gcagcctgac atctgaggac actgcagtct attactgtgc aagagaggca
301 attactacgg taggcgctat ggactactgg ggtcaaggaa cctcagtcac cgtctoctca

Protein Sequence Defining the Heavy Chain Variable
Region of the 08G01 Antibody (SEQ ID NO:48)

1 evllqqsgpe vvkpgasvki pckasgytft dynmdwvkqs hgkslewise inpnnggtfy
61 nqkfkgkatl tvdkssstay melrsltsed tavyycaea ittvgamdyw ggtsvtvss

Nucleic Acid Sequence Encoding the Kappa Chain Variable
Region of the 08G01 Antibody (SEQ ID NO:83)

```

1  gacatccaga tgactcagtc tccagcctcc ctatctgcat ctgtgggaga aactgtcacc
61  atcacatgtc gagcaagtgg gaatattcac aattatttag catggtatca gcagaaacag
121 ggaaaatctc ctcagctcct ggtctataat gcaaaaacct tagcagatgg tgtgccatca
181 aggttcagtg gcagtggatc aggaacacaa tattctctca agatcaacag cctgcagcct
241 gaagattttg ggagttatta ctgtcaacat ttttgaggtt ctcttacac gttcggaggg
301 gggaccaagc tggaaataaa a

```

Protein Sequence Defining the Kappa Chain Variable
Region of the 08G01 Antibody (SEQ ID NO:84)

```

1  diqmtqspas lsasvgetvt itcrasgnih nylawyqqkq gkspqllvyn aktladgvps
61  rfsqsgsgtg yslkinslqp edfsgyyqgh fwsspytfpg gtleik

```

Nucleic Acid Sequence Encoding the Heavy Chain Variable
Region of the 14F11 Antibody (SEQ ID NO:49)

```

1  caggttactc tgaagagtc tggccctgga atattgcagc cctcccagac ctcagtcctg
61  acttggtctt tctctgggtt ttcactgagc acttatggta tgggtgtagg ctggattcgt
121 cagccttcag gaaagggctc agagtggctg gcagacattt ggtgggatga cgataagtac
181 tataacccat cctgaagag ccggctcaca atctccaagg atacctccag caatgaggta
241 ttctcaaga tcgccattgt ggacactgca gatactgcca ctactactg tgctcgaaga
301 ggtcactact ctgctatgga ctactggggc caaggaacct cagtcaccgt ctctca

```

Protein Sequence Defining the Heavy Chain Variable
Region of the 14F11 Antibody (SEQ ID NO:50)

```

1  qvtlkespgg ilqpsqtlll tcsfsgfsls tygmvgwir qpsgkglewl adiwdddky
61  ynpslksrlt iskdtsnev flkiaivda dtatyycarr ghysamywg qgtsvtvss

```

Nucleic Acid Sequence Encoding the Kappa Chain Variable
Region of the 14F11 Antibody (SEQ ID NO:85)

```

1  gacattgtaa tgaccagtc tcaaaaattc atgtccacat cagtaggaga cagggtcagc
61  gtcacctgca aggccagtca gaatgtgggt actaatgtag cctggatatca acagaaacca
121 gggcaatctc ctaaagcact gatttactcg ccacccatcc ggtacagtgg agtcacctgat
181 cgcttcacag gcagtggatc tgggacagat ttactctca ccacagcaa tgtgcagtct
241 gaagacttgg cagaatattt ctgtcagcaa tataacagct atctccacac gttcggaggg
301 gggaccaagc tggaaatgaa a

```

Protein Sequence Defining the Kappa Chain Variable
Region of the 14F11 Antibody (SEQ ID NO:86)

```

1  divmtqsgkf mstsvgdrvs vtckasqnvg tnvawyqqkp gqspkaliys psyrsgvpd
61  rftgsgsgtd ftltisnvqs edlaeyfcqg ynsyphtfg gtleik

```

Nucleic Acid Sequence Encoding the Heavy Chain Variable
Region of the 17B11 Antibody (SEQ ID NO:51)

```

1  caggttactc tgaaagagtc tggccctggg atattgcagc cctcccagac cctcagtctg
61  acttgttctt tctctgggtt ttcactgagc acttctggta tgggtgtgag ttggattcgt
121 cagccttcag gaaagggtct ggagtggctg gcacacaatg actgggatga tgacaagcgc
181 tataagtcac cctgaagag cgggtcaca atatccaagg atacctccag aaaccaggtg
241 ttctcaaga tcaccagtgt ggacactgca gatactgcca catactactg tgctcgaaga
301 gttgggggat tagagggtta ttttgattac tggggccaag gcaccactct cacagtctcc
361 tca

```

Protein Sequence Defining the Heavy Chain Variable
Region of the 17B11 Antibody (SEQ ID NO:52)

```

1  qvtlkespgg ilqpsqtlsl tcsfsgfsls tsgmgvswir qpsgkglewl ahndwdddkr
61  yksslksrlt iskdtsrnqv flkitsvdtat dtatyyccarr vgglegfydy wgqgttlts
121 s

```

Nucleic Acid Sequence Encoding the Kappa Chain Variable
Region of the 17B11 Antibody (SEQ ID NO:87)

```

1  gacattgtgc tgacacagtc tctctgttcc ttagctgtat ctctggggca gagggccacc
61  atctcatgca gggccagcca aagtgtcagt acatctaggt ttagttatat gcaactggtc
121 caacagaaac caggacaggc acccaaaactc ctcatcaagt atgcatccaa cctagaatct
181 ggggtccctg ccaggttcag tggcagtggt tctgggacag acttcaccct caacatccat
241 cctgtggagg gggaggatac tgcaacatat tactgtcagc acagttggga gattccgtac
301 acgttcggag gggggaccaa gctggaaata aaa

```

Protein Sequence Defining the Kappa Chain Variable
Region of the 17B11 Antibody (SEQ ID NO:88)

```

1  divltqspas layslgqrat iscrasqsvs tsrfsymhwf qqkpgqapk1 likyasnles
61  gvparfsgsg sgtddftlnih pvegedtaty ycghsweipy tfggggtklei k

```

[0258] The amino acid sequences defining the immunoglobulin heavy chain variable regions for the antibodies produced in Example 6 are aligned in FIG. 10. Amino terminal signal peptide sequences (for expression/secretion) are not shown. CDR₁, CDR₂, and CDR₃ (Kabat definition) are identified by boxes. FIG. 11 shows an alignment of the separate CDR₁, CDR₂, and CDR₃ sequences for each antibody.

[0259] The amino acid sequences defining the immunoglobulin light chain variable regions of the antibodies in Example 6 are aligned in FIG. 12. Amino terminal signal peptide sequences (for expression/secretion) are not shown. CDR₁, CDR₂ and CDR₃ are identified by boxes. FIG. 13 shows an alignment of the separate CDR₁, CDR₂, and CDR₃ sequences for each antibody.

[0260] Table 2 shows the SEQ ID NO. of each sequence discussed in this Example.

TABLE 2

SEQ. ID NO.	Nucleic Acid or Protein
39	01G06 Heavy Chain Variable Region-nucleic acid
40	01G06 Heavy Chain Variable Region-protein
75	01G06 Light (kappa) Chain Variable Region-nucleic acid
76	01G06 Light (kappa) Chain Variable Region-protein
1	01G06 Heavy Chain CDR ₁
7	01G06 Heavy Chain CDR ₂
15	01G06 Heavy Chain CDR ₃
21	01G06 Light (kappa) Chain CDR ₁
26	01G06 Light (kappa) Chain CDR ₂
32	01G06 Light (kappa) Chain CDR ₃
41	03G05 Heavy Chain Variable Region-nucleic acid
42	03G05 Heavy Chain Variable Region-protein
77	03G05 Light (kappa) Chain Variable Region-nucleic acid
78	03G05 Light (kappa) Chain Variable Region-protein
2	03G05 Heavy Chain CDR ₁
8	03G05 Heavy Chain CDR ₂
16	03G05 Heavy Chain CDR ₃

TABLE 2-continued

SEQ. ID NO.	Nucleic Acid or Protein
22	03G05 Light (kappa) Chain CDR ₁
27	03G05 Light (kappa) Chain CDR ₂
33	03G05 Light (kappa) Chain CDR ₃
43	04F08 Heavy Chain Variable Region-nucleic acid
44	04F08 Heavy Chain Variable Region-protein
79	04F08 Light (kappa) Chain Variable Region-nucleic acid
80	04F08 Light (kappa) Chain Variable Region-protein
3	04F08 Heavy Chain CDR ₁
9	04F08 Heavy Chain CDR ₂
17	04F08 Heavy Chain CDR ₃
23	04F08 Light (kappa) Chain CDR ₁
28	04F08 Light (kappa) Chain CDR ₂
34	04F08 Light (kappa) Chain CDR ₃
45	06C11 Heavy Chain Variable Region-nucleic acid
46	06C11 Heavy Chain Variable Region-protein
81	06C11 Light (kappa) Chain Variable Region-nucleic acid
82	06C11 Light (kappa) Chain Variable Region-protein
4	06C11 Heavy Chain CDR ₁
9	06C11 Heavy Chain CDR ₂
18	06C11 Heavy Chain CDR ₃
23	06C11 Light (kappa) Chain CDR ₁
28	06C11 Light (kappa) Chain CDR ₂
35	06C11 Light (kappa) Chain CDR ₃
47	08G01 Heavy Chain Variable Region-nucleic acid
48	08G01 Heavy Chain Variable Region-protein
83	08G01 Light (kappa) Chain Variable Region-nucleic acid
84	08G01 Light (kappa) Chain Variable Region-protein
1	08G01 Heavy Chain CDR ₁
10	08G01 Heavy Chain CDR ₂

TABLE 2-continued

SEQ. ID NO.	Nucleic Acid or Protein
15	08G01 Heavy Chain CDR ₃
24	08G01 Light (kappa) Chain CDR ₁
29	08G01 Light (kappa) Chain CDR ₂
32	08G01 Light (kappa) Chain CDR ₃
49	14F11 Heavy Chain Variable Region-nucleic acid
50	14F11 Heavy Chain Variable Region-protein
85	14F11 Light (kappa) Chain Variable Region-nucleic acid
86	14F11 Light (kappa) Chain Variable Region-protein
5	14F11 Heavy Chain CDR ₁
11	14F11 Heavy Chain CDR ₂
19	14F11 Heavy Chain CDR ₃
23	14F11 Light (kappa) Chain CDR ₁
30	14F11 Light (kappa) Chain CDR ₂
36	14F11 Light (kappa) Chain CDR ₃
51	17B11 Heavy Chain Variable Region-nucleic acid
52	17B11 Heavy Chain Variable Region-protein
87	17B11 Light (kappa) Chain Variable Region-nucleic acid
88	17B11 Light (kappa) Chain Variable Region-protein
6	17B11 Heavy Chain CDR ₁
12	17B11 Heavy Chain CDR ₂
20	17B11 Heavy Chain CDR ₃
25	17B11 Light (kappa) Chain CDR ₁
31	17B11 Light (kappa) Chain CDR ₂
37	17B11 Light (kappa) Chain CDR ₃

[0261] Mouse monoclonal antibody heavy chain CDR sequences (Kabat, Chothia, and IMGT definitions) are shown in Table 3.

TABLE 3

Kabat				
	CDR1	CDR2	CDR3	Variable Region SEQ ID NO
01G06	DYNMD (SEQ ID NO: 1)	QINPNNGGIFFNQKFKG (SEQ ID NO: 7)	EAITTVGAMDY (SEQ ID NO: 15)	40
03G05	SYWIH (SEQ ID NO: 2)	DINPSNGRSKYNEKFKN (SEQ ID NO: 8)	EVLDGAMDY (SEQ ID NO: 16)	42
04F08	TYGMGVT (SEQ ID NO: 3)	HIYWDDDKRYNPSLKS (SEQ ID NO: 9)	TGYSNLFAY (SEQ ID NO: 17)	44
06C11	TYGMGVS (SEQ ID NO: 4)	HIYWDDDKRYNPSLKS (SEQ ID NO: 9)	RGYDDYWG (SEQ ID NO: 18)	46
08G01	DYNMD (SEQ ID NO: 1)	EINPNNGGTFYNQKFKG (SEQ ID NO: 10)	EAITTVGAMDY (SEQ ID NO: 15)	48
14F11	TYGMGVG (SEQ ID NO: 5)	DIWDDDKRYNPSLKS (SEQ ID NO: 11)	RGHYSAMDY (SEQ ID NO: 19)	50
17B11	TSGMGVS (SEQ ID NO: 6)	HNDWDDDKRYKSSLKS (SEQ ID NO: 12)	RVGGLEGYFDY (SEQ ID NO: 20)	52
Chothia				
	CDR1	CDR2	CDR3	Variable Region SEQ ID NO
01G06	GYTFTDY (SEQ ID NO: 38)	NPNNGG (SEQ ID NO: 143)	EAITTVGAMDY (SEQ ID NO: 15)	40
03G05	GYTFTSY (SEQ ID NO: 128)	NPSNGR (SEQ ID NO: 144)	EVLDGAMDY (SEQ ID NO: 16)	42
04F08	GFSLSTYGM (SEQ ID NO: 130)	YWDDDD (SEQ ID NO: 145)	TGYSNLFAY (SEQ ID NO: 17)	44

TABLE 3-continued

06C11	GFSLNTYGM (SEQ ID NO: 132)	YWDDD (SEQ ID NO: 145)	RGYDDYWG (SEQ ID NO: 18)	46
08G01	GYTFTDY (SEQ ID NO: 38)	NPNNGG (SEQ ID NO: 143)	EAITTVGAMDY (SEQ ID NO: 15)	48
14F11	GFSLSTYGM (SEQ ID NO: 130)	WWDDD (SEQ ID NO: 146)	RGHYSAMDY (SEQ ID NO: 19)	50
17B11	GFSLSTSGM (SEQ ID NO: 134)	DWDDD (SEQ ID NO: 147)	RVGGLEGYFDY (SEQ ID NO: 20)	52
IMGT				
	CDR1	CDR2	CDR3	Variable Region SEQ ID NO
01G06	GYTFTDYN (SEQ ID NO: 136)	INPNNGGI (SEQ ID NO: 148)	AREAITTVGAMDY (SEQ ID NO: 154)	40
03G05	GYTFTSYW (SEQ ID NO: 138)	INPSNGRS (SEQ ID NO: 149)	AREVLDGAMDY (SEQ ID NO: 155)	42
04F08	GFSLSTYGMG (SEQ ID NO: 140)	IYWDDDK (SEQ ID NO: 150)	AQTGYSNLFAY (SEQ ID NO: 156)	44
06C11	GFSLNTYGMG (SEQ ID NO: 141)	IYWDDDK (SEQ ID NO: 150)	AQRGYDDYWG (SEQ ID NO: 157)	46
08G01	GYTFTDYN (SEQ ID NO: 136)	INPNNGGT (SEQ ID NO: 151)	AREAITTVGAMDY (SEQ ID NO: 154)	48
14F11	GFSLSTYGMG (SEQ ID NO: 140)	IWWDDDK (SEQ ID NO: 152)	ARRGHYSAMDY (SEQ ID NO: 158)	50
17B11	GFSLSTSGMG (SEQ ID NO: 142)	NDWDDDK (SEQ ID NO: 153)	ARRVGGLEGYFDY (SEQ ID NO: 159)	52

[0262] Mouse monoclonal antibody Kappa light chain CDR sequences (Kabat, Chothia, and IMGT definitions) are shown in Table 4.

TABLE 4

Kabat/Chothia				
	CDR1	CDR2	CDR3	Variable Region SEQ ID NO
01G06	RTSENLHNYLA (SEQ ID NO: 21)	DAKTLAD (SEQ ID NO: 26)	QHFWSPPYT (SEQ ID NO: 32)	76
03G05	RASEVDNYGISFMN (SEQ ID NO: 22)	AASNQGS (SEQ ID NO: 27)	QQSKEVPWT (SEQ ID NO: 33)	78
04F08	KASQNVGTNVA (SEQ ID NO: 23)	SASYRYS (SEQ ID NO: 28)	QQYNSYPYT (SEQ ID NO: 34)	80
06C11	KASQNVGTNVA (SEQ ID NO: 23)	SASYRYS (SEQ ID NO: 28)	QQYNNYPLT (SEQ ID NO: 35)	82
08G01	RASGNIHNYLA (SEQ ID NO: 24)	NAKTLAD (SEQ ID NO: 29)	QHFWSPPYT (SEQ ID NO: 32)	84
14F11	KASQNVGTNVA (SEQ ID NO: 23)	SPSYRYS (SEQ ID NO: 30)	QQYNSYPHT (SEQ ID NO: 36)	86
17B11	RASQSVSTSRFSYMH (SEQ ID NO: 25)	YASNLES (SEQ ID NO: 31)	QHSWEIPYT (SEQ ID NO: 37)	88

TABLE 4-continued

IMGT				
	CDR1	CDR2	CDR3	Variable Region SEQ ID NO
01G06	ENLHNY (SEQ ID NO: 160)	DAK	QHFWSPPYT (SEQ ID NO: 32)	76
03G05	ESVDNYGISF (SEQ ID NO: 161)	AAS	QQSKEVPWT (SEQ ID NO: 33)	78
04F08	QNVGTN (SEQ ID NO: 162)	SAS	QQYNSYPYT (SEQ ID NO: 34)	80
06C11	QNVGTN (SEQ ID NO: 162)	SAS	QQYNNYPLT (SEQ ID NO: 35)	82
08G01	GNIHNY (SEQ ID NO: 163)	NAK	QHFWSPPYT (SEQ ID NO: 32)	84
14F11	QNVGTN (SEQ ID NO: 162)	SPS	QQYNSYPHT (SEQ ID NO: 36)	86
17B11	QSVSTSRFSY (SEQ ID NO: 164)	YAS	QHSWEIPYT (SEQ ID NO: 37)	88

[0263] To create the complete heavy or kappa chain antibody sequences, each variable sequence above is combined with its respective constant region. For example, a complete heavy chain comprises a heavy variable sequence followed by the murine IgG1 or IgG2b heavy chain constant sequence, and a complete kappa chain comprises a kappa variable sequence followed by the murine kappa light chain constant sequence.

Nucleic Acid Sequence Encoding the Murine IgG1 Heavy Chain Constant Region (SEQ ID NO: 165)

```

1  gccaaaaacga cccccccatc tgtctatcca ctggcccttg gatctgctgc ccaaactaac
61  tccatggtga ccttgggatg cctgggtcaag ggctatttcc ctgagccagt gacagtgacc
121  tggaaactctg gatccctgtc cagcgggtgtg cacaccttcc cagctgtcct gcagtctgac
181  ctctacactc tgagcagctc agtgactgtc ccctccagca cctggcccag cgagaccgtc
241  acctgcaacg ttgccacccc ggccagcagc accaaggtgg acaagaaaat tgtgcccagg
301  gatttggtgt gtaagccttg catatgtaca gtcccagaag tatcatctgt cttcatcttc
361  cccccaaaagc ccaaggatgt gctcaccatt actctgactc ctaaggtcac gtgtgttggtg
421  gtagacatca gcaaggatga tcccagagtc cagttcagct ggttttaga tgatgtggag
481  gtgcacacag ctacagcgcg accccgggag gagcagttca acagcacttt ccgctcagtc
541  agtgaacttc ccatcatgca ccaggactgg ctcaatggca aggagttcaa atgcagggtc
601  aacagtgcag ctttcccttg ccccatcgag aaaaccatct ccaaaaccaa aggcagaccg
661  aaggctccac aggtgtacac cattccacct cccaaggagc agatggccaa ggataaagtc
721  agtctgacct gcatgataac agacttcttc cctgaagaca ttactgtgga gtggcagtg
781  aatgggcagc cagcggagaa ctacaagaac actcagccca tcatggacac agatggctct
841  tacttctgtc acagcaagct caatgtgcag aagagcaact gggaggcagg aaatactttc
901  acctgctctg tgttacatga gggcctgcac aaccaccata ctgagaagag cctctcccac
961  tctcctggta aa

```

Protein Sequence Defining the Murine IgG1 Heavy Chain
Constant Region (SEQ ID NO:166)

1 akttpsvvyp lapgsaaqtn smvtlgclvk gyfpepvtvt wnsqslssgv htfpavlqsd
61 lytlsssvtv psstwpsetv tcnvahpass tkvdtkivpr dcgckpcict vpevssvfif
121 ppkpkdvlti tltpkvtcvv vdiskddpev qfswfvddve vhtaqtqpre eqfnstfrsv
181 selpimhqdw lngkefkcrv nsaafpapie ktisktkgrp kapqvtyipp pkeqmakdkv
241 sltcmtdff peditvewqw ngqpaenykn tqpimtdggs yfvysklmv ksnweagntf
301 tcsvlheglh nhhtekslsh spgk

Nucleic Acid Sequence Encoding the Murine IgG2b Heavy
Chain Constant Region (SEQ ID NO:167)

1 gccaaaacaa ccccccatc agtctatcca ctggccctg ggtgtggaga tacaactggt
61 tcctccgtga ctctgggatg cctggcgaag ggctacttcc ctgagtcagt gactgtgact
121 tggaaactctg gatccctgtc cagcagtggt cacaccttcc cagctctcct gcagtcgtga
181 ctctacacta tgagcagctc agtgactgtc ccctccagca cctggccaag tcagaccgtc
241 acctgcagcg ttgctcacc agccagcagc accacgggtg acaaaaaact tgagcccagc
301 gggcccatct caacaatcaa cccctgtcct ccattgcaagg agtgtcaca atgccagct
361 cctaacctcg aggggtggacc atccgtcttc atcttcctc caaatatcaa ggatgtactc
421 atgatctccc tgacacccaa ggtcacgtgt gtgggtgggt atgtgagcga ggatgaccca
481 gacgtccaga tcagctggtt tgtgaacaac gtggaagtac acacagctca gacacaaacc
541 catagagagg attacaacag tactatccgg gtggtcagca cctcccat ccagcaccag
601 gactggatga gtggcaagga gttcaaatgc aaggtcaaca acaagacct cccatcacc
661 atcgagagaa ccatctcaaa aattaaagg ctagtcagag ctccacaagt atacatcttg
721 ccgccaccag cagagcagtt gtccaggaaa gatgtcagtc tcaattgcct ggctgtgggc
781 ttcaacctg gagacatcag tgtggagtgg accagcaatg ggcatacaga ggagaactac
841 aaggacaccg caccagtcct agactctgac ggttcttact tcatatatag caagctcaat
901 atgaaaacaa gcaagtggga gaaaacagat tccttctcat gcaacgtgag acacgaggg
961 ctgaaaaatt actacctgaa gaagaccatc tcccgtctc cgggtaaa

Protein Sequence Defining the Murine IgG2b Heavy Chain
Constant Region (SEQ ID NO:168)

1 akttpsvvyp lapgcgdtg ssvtlgclvk gyfpesvtvt wnsqslsssv htfpallqsg
61 lytmsssvtv psstwpstqv tcsvahpass ttvdkkleps gpistinpcp pkechkcpa
121 pnleggsvf ifppnikdvl misltpkvtc vvvdvsedd dvqiswfvnn vevhtaqtqt
181 hredynstir vstlpihq dwmsgkefk kvnnkdlps iertiskikg lvrappvyl
241 pppaeqlsrk dvsltclvvg fnpgdisvew tsnghteeny kdtapvlds gsyfiyskln
301 mktskwektd sfscnvrheg lknyylkkti srspgk

Nucleic Acid Sequence Encoding the Murine Kappa Light Chain Constant Region (SEQ ID NO:169)

```

1  cgggctgatg ctgcaccaac tgtatccatc ttcccacccat ccagtgagca gttaacatct
61  ggaggtgcct cagtcgtgtg cttcttgaac aacttctacc ccaaagacat caatgtcaag
121  tggagattg atggcagtga acgacaaaat ggcgtcctga acagttggac tgatcaggac
181  agcaaagaca gcacctacag catgagcagc accctcacgt tgaccaagga cgagtatgaa
241  cgacataaca gctatacctg tgaggccact cacaagacat caacttcacc cattgtcaag
301  agcttcaaca ggaatgagtg t

```

Protein Sequence Defining the Murine Kappa Light Chain Constant Region (SEQ ID NO:170)

```

1  radaaptvsi fppsseqqls ggasvvcln nfypkdivk wkidgserqn gvlswtdqd
61  skdstysmss tltltkdeye rhnsytceat hktstspivk sfnrnec

```

[0264] The following sequences represent the actual or contemplated full length heavy and light chain sequence (i.e., containing both the variable and constant regions sequences) for each antibody described in this Example. Signal sequences for proper secretion of the antibodies (e.g., signal sequences at the 5' end of the DNA sequences or the amino terminal end of the protein sequences) are not shown in the full length heavy and light chain sequences disclosed herein and are not included in the final secreted protein. Also not shown are stop codons for termination of translation required at the 3' end of the DNA sequences. It is within

ordinary skill in the art to select a signal sequence and/or a stop codon for expression of the disclosed full length immunoglobulin heavy chain and light chain sequences. It is also contemplated that the variable region sequences can be ligated to other constant region sequences to produce active full length immunoglobulin heavy and light chains.

Nucleic Acid Sequence Encoding the Full Length Heavy Chain Sequence (Heavy Chain Variable Region and IgG1 Constant Region) of 01G06 (SEQ ID NO:99)

```

1  gaggtcctgc tgcaacagtc tggacctgag ctggtgaagc ctggggcttc agtgaagata
61  ccctgcaagg cttctggata cacattcact gactacaaca tggactgggt gaagcagagc
121  catggaaga gccttgagtg gattggacaa attaatccta acaatgggtg tattttcttc
181  aaccagaagt tcaagggcaa ggccacattg actgtagaca agtcctccaa tacagccttc
241  atggagggtc gcagcctgac atctgaggac actgcagtct attactgtgc aagagaggca
301  attactacgg taggcgctat ggactactgg ggtcaaggaa cctcagtcac cgtctcctca
361  gccaaaacga caccctcatc tgtctatcca ctggccctcg gatctgtgac ccaaactaac
421  tccatgggtg cctggggatg cctgggtcaag ggctatttcc ctgagccagt gacagtgacc
481  tggaactctg gatccctgtc cagcgggtgt cacaccttcc cagctgtcct gcagctctgac
541  ctctacaact tgagcagctc agtgactgtc ccctccagca cctggcccag cgagaccgtc
601  acctgcaacg ttgccacccc ggccagcagc accaagggtg acaagaaaaa tgtgccaggg
661  gattgtgggt gtaagccttg catatgtaca gtcccagaag tatcatctgt cttcatcttc
721  cccccaagc ccaaggatgt gctcaccatt actctgactc ctaagggtcag gtgtgtgtgtg
781  gtagacatca gcaaggatga tcccagggtc cagttcagct ggttttaga tgatgtggag
841  gtgcacacag ctcagacgca accccgggag gagcagttca acagcacttt ccgctcagtc
901  agtgaacttc ccatcatgca ccaggactgg ctcaatggca aggagttaa atgcagggtc
961  aacagtgcag ctttccctgc ccccatcgag aaaaccatct ccaaaaccaa aggcagaccg
1021  aaggctccac aggtgtacac cattccacct cccaaggagc agatggccaa ggataaagtc
1081  agtctgacct gcatgataac agacttcttc cctgaagaca ttactgtgga gtggcagtg

```

-continued

1141 aatgggcagc cagcggagaa ctacaagaac actcagccca tcattggacac agatggctct
1201 tacttcgtct acagcaagct caatgtgcag aagagcaact gggaggcagg aaatactttc
1261 acctgctctg tggtacatga gggcctgcac aaccaccata ctgagaagag cctctccac
1321 tctcctggta aa

Protein Sequence Defining the Full Length Heavy Chain
Sequence (Heavy Chain Variable Region and IgG1 Constant
Region) of 01G06 (SEQ ID NO:100)

1 evllqsgpe lvkpgasvki pckasgytft dynmdwvkqs hgkslewigq inpnnggiff
61 nqkfkgtatl tvdkssntaf mevrsltsed tavyycarea ittvgamdyw gggtsvtvss
121 akttpsvvyp lapgsaaqtn smvtlglclvk gyfpepvtvt wnsqslssgv htfpavlqsd
181 lytlsssvtv psstwpsetv tcnvahpass tkvdtkivpr dcgckpcict vpevssvfif
241 ppkpkdvlti tltpkvtcvv vdiskddpev qfswfvddve vhtaqtqpre eqfnstfrsv
301 selpimhqdw lngkefkerv nsaaftpaple ktisktkgrp kapqvytipp pkeqmakdkv
361 sltcmtdff peditvewqw ngqpaenykn tqpimtdgsg yfvysklvq ksnweagntf
421 tcsvlheglh nhhtekslsh spgk

Nucleic Acid Sequence Encoding the Full Length Light
Chain Sequence (Kappa Chain Variable Region and
Constant Region) of 01G06 (SEQ ID NO:101)

1 gacatccaga tgactcagtc tccagcctcc ctatctgcat ctgtgggaga aactgtcacc
61 atcacatgtc gaacaagtga gaatcttcac aattatttag catgggtatca gcagaaacag
121 ggaaaatctc ctcagctcct ggtctatgat gcaaaaacct tagcagatgg tgtgccatca
181 aggttcagtg gcagtggatc aggaacacaa tattctctca agatcaacag cctgcagcct
241 gaagattttg ggagttatta ctgtcaacat ttttgagta gtccttacac gttcggaggg
301 gggaccaagc tggaaataaa acgggctgat gctgcaccaa ctgtatccat cttcccacca
361 tccagtgagc agttaacatc tggaggtgcc tcagtcgtgt gcttcttgaa caactctac
421 cccaaagaca tcaatgtcaa gtggaagatt gatggcagtg aacgacaaaa tggcgtcctg
481 aacagttgga ctgatcagga cagcaaagac agcacctaca gcatgagcag caccctcacg
541 ttgaccaagg acgagtatga acgacataac agctatacct gtgaggccac tcacaagaca
601 tcaacttcac ccattgtcaa gagcttcaac aggaatgagt gt

Protein Sequence Defining the Full Length Light Chain
Sequence (Kappa Chain Variable Region and Constant
Region) of 01G06 (SEQ ID NO:102)

1 diqmtqspas lsasvgetvt itcrtsenlh nylawyqqkq gkspqllvyd aktladgvps
61 rfsgsgsgtg yslkinslqp edfgsyycqh fwsspytfgg gtleikrad aaptvsifpp
121 sseqltsgga svvcflnnfy pkdinvkwi dgserqngvl nswtdqdsd stysmsstlt
181 ltkdeyerhn sytceathkt stspivksfn rnec

Nucleic Acid Sequence Encoding the Full Length Heavy
Chain Sequence (Heavy Chain Variable Region and IgG1
Constant Region) of 03G05 (SEQ ID NO:103)

```
1   caggtccaac tgcagcagcc tggggctgaa ctggtgaagc ctggggcttc agtgaagctg
61  tcttgcaagg cttctggcta caccttcacc agctactgga ttacttgggt gaaccagagg
121 cctggacaag gccttgagtg gattggagac attaataccta gcaacggccg tagtaagtat
181 aatgagaagt tcaagaacaa ggccacaatg actgcagaca aatcctccaa cacagcctac
241 atgcaactca gcagcctgac atctgaggac tctgcgtctt attactgtgc aagagaggtt
301 ctggatggtg ctatggacta ctgggggtcaa ggaacctcag tcaccgtctc ctcagccaaa
361 acgacacccc catctgtcta tccactggcc cctggatctg ctgcccaaac taactccatg
421 gtgaccctgg gatgcctggt caagggttat ttccctgagc cagtgcaggt gacctggaac
481 tctggatccc tgtccagcgg tgtgcacacc tcccagctg tctgcagtc tgacctctac
541 actctgagca gctcagtgac tgtccctccc agcacctggc ccagcgagac cgtcacctgc
601 aacgttgccc acccggccag cagcaccaag gtggacaaga aaattgtgcc cagggtattg
661 gggtgtaagc cttgcatatg tacagtccca gaagtatcat ctgtcttcat cttcccccca
721 aagcccaagg atgtgctcac cattactctg actcctaagg tcacgtgtgt tgtggtagac
781 atcagcaagg atgatccga ggtccagttc agctggtttg tagatgatgt ggaggtgcac
841 acagctcaga cgcaaccccg ggaggagcag ttcaacagca ctttccgctc agtcagtcaa
901 cttcccatac tgcaaccagga ctggctcaat ggcaaggagt tcaaatgcag ggtcaacagt
961 gcagctttcc ctgcccccat cgagaaaacc atctccaaaa ccaaaggcag accgaaggct
1021 ccacaggtgt acaccattcc acctcccaag gagcagatgg ccaaggataa agtcagtctg
1081 acctgcataa taacagactt cttccctgaa gacattactg tggagtggca gtggaatggg
1141 cagccagcgg agaactacaa gaacactcag cccatcatgg acacagatgg ctcttacttc
1201 gtctacagca agctcaatgt gcagaagagc aactgggagg caggaaatac tttcacctgc
1261 tctgtgttac atgagggcct gcacaaccac catactgaga agagcctctc ccaactctct
1321 ggtaaa
```

Protein Sequence Defining the Full Length Heavy Chain
Sequence (Heavy Chain Variable Region and IgG1 Constant
Region) of 03G05 (SEQ ID NO:104)

```
1   qvqlqpggae lvkpgasvkl sckasgytft sywihwnqr pgqglewlgd inpsngrsky
61  nekfkknatm tadvksntay mqlslstsed savyycarev ldgamdywgq gtsvtvssak
121 ttpsvvypla pgsaaqtnsm vtlgclvkgy fpepvtvtwn sgslssgvht fpavlsdly
181 tlsssvtvpv stwpsetvvc nvahpasstk vdkkivprdc gckpcictvp evssvfifpp
241 kpkdvlitil tpkvtcvvvd iskddpevqf swfvddvevh taqtqpreeq fnstfrsvse
301 lpimhqdwln gkefkcrvns aafpapiekt isktkgrpka pqvytipppk eqmakdkvsl
361 tcmtdffpe ditvewqwnq gpaenykntq pimtdgtsyf vysklvqks nweagntftc
421 svlhaglhnh htekslshsp gk
```

Nucleic Acid Sequence Encoding the Full Length Light Chain Sequence (Kappa Chain Variable Region and Constant Region) of 03G05 (SEQ ID NO:105)

```
1  gacattgtgt tgaccaate tccagcttct ttggtgtgt ctctaggga gagggccacc
61  atctcctgca gagccagcga aagtgttgat aattatggca ttagttttat gaactgggtc
121 caacagaaac caggacagcc acccaaaact ctcattatg ctgcatccaa ccaaggtccc
181 ggggtccctg ccaggtttag tggcagtggg tctgggacag acttcagcct caacatccat
241 cctatggagg aggatgatac tgcaatgtat ttctgtcagc aaagtaagga ggttccgtgg
301 acgttcggtg gaggtccaa gctggaaatc aaacgggctg atgctgcacc aactgtatcc
361 atcttcccac catccagtga gcagttaaca tctggaggtg cctcagtcgt gtgcttcttg
421 aacaacttct accccaaga catcaatgtc aagtgaaga ttgatggcag tgaacgacaa
481 aatggcgctc tgaacagttg gactgatcag gacagcaaag acagcaccta cagcatgagc
541 agcacctca cgttgaccaa ggacgagtat gaacgacata acagctatac ctgtgaggcc
601 actcacaaga catcaacttc acccattgtc aagagcttca acaggaatga gtgt
```

Protein Sequence Defining the Full Length Light Chain Sequence (Kappa Chain Variable Region and Constant Region) of 03G05 (SEQ ID NO:106)

```
1  divltqspas lavslgqrat iscrasesvd nygisfmnwf qqkpggppkl liyaasnqgs
61  gvparfsgsg sgtdfslnih pmeeddtamy fcqqskevpw tfgggsklei kradaaptvs
121 ifppsseqlt sggasvvcfl nnfypkdinv kwkidgserq ngvlnswtdq dskdstysms
181 stltltkdey erhnsytcea thktstspiv ksfnrnec
```

Nucleic Acid Sequence Encoding the Full Length Heavy Chain Sequence (Heavy Chain Variable Region and IgG1 Constant Region) of 04F08 (SEQ ID NO:107)

```
1  caggttactc tgaaagagtc tggccctggg atattgcagc cctcccagac cctcagtctg
61  acttgttctt tctctgggtt ttcactgagc acttatggta tgggtgtgac ctggattcgt
121 cagccttcag gaaaggtctt ggagtggctg gcacacattt actgggatga tgacaagcgc
181 tataacccat cctgaagag ccggctcaca atctccaagg atacctccaa caaccaggta
241 ttcctcaaga tcaccagtgt ggacactgca gatactgcca catactactg tgetcaaacg
301 gggtagatga acttgtttgc ttactggggc caagggaactc tggtcactgt ctctgcagcc
361 aaaacgacac ccccatctgt ctatccactg gccctggat ctgctgacca aactaactcc
421 atggtgaccc tgggatgcct ggtcaagggc tatttccttg agccagtga agtgacctgg
481 aactctggat cctgtccag cgggtgtgac accttcccag ctgtcctgca gtctgacctc
541 tacactctga gcagctcagt gactgtcccc tcagcacct ggcccagcga gaccgtcacc
601 tgcaacgttg cccaccggc cagcagcacc aaggtggaca agaaaattgt gcccagggat
661 tgtggttgta agccttgcat atgtacagtc ccagaagtat catctgtctt catcttcccc
721 ccaaagccca aggatgtgct caccattact ctgactccta aggtcacgtg tgttggtgta
781 gacatcagca aggatgatcc cgaggtccag ttcagctggt ttgtagatga tgtggaggtg
841 cacacagctc agacgcaacc ccgggaggag cagttcaaca gcactttccg ctcagtcagt
```

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```

901   gaacttecca tcatgcacca ggactggctc aatggcaagg agttcaaatg caggggtcaac
961   agtgcagctt tcctgcccc catcgagaaa accatctcca aaaccaaagg cagaccgaag
1021  gctccacagg tgtacaccat tccacctccc aaggagcaga tggccaagga taaagtcagt
1081  ctgacctgca tgataacaga cttcttccct gaagacatta ctgtggagtg gcagtggaat
1141  gggcagccag cggagaacta caagaacact cagcccatca tggacacaga tggtctctac
1201  ttcgtctaca gcaagctcaa tgtgcagaag agcaactggg aggcaggaaa tactttcacc
1261  tgctctgtgt tacatgaggg cctgcacaa caccatactg agaagagcct ctcccactct
1321  cctggtaaa

```

Protein Sequence Defining the Full Length Heavy Chain Sequence (Heavy Chain Variable Region and IgG1 Constant Region) of 04F08 (SEQ ID NO:108)

```

1   qvtlkesgpg ilqpsqtlsl tcsfsgfsls tygmgtwir qpsgkglewl ahiywdddkr
61  ynpslksrilt iskdtsnnqv flkitsvdtat datyycagt gysnlfaywg qgtlvtvsaa
121 kttpsvypl apgsaaqtns mvtlgclvkg yfpepvtvtw negslssgvh tfpavlqsd1
181 ytlsssvtvp sstwpsetvt cnvahpasst kvdkkivprd cgckpeictv pevssvfifp
241 pkpkdvltit ltpkvtcvvv diskddpevq fswfvddvev htaqtqpree qfnstfrsvs
301 elpimhqdlw ngkefkcrvn saafpapie ktsktkgrpk apqvtytippp kegmakdkvs
361 ltcmitdfff editvewqwn gqpaenyknt qpimdtgtsy fvysklvqk snweagntft
421 csvlheglhn hhtekslshs pgk

```

Nucleic Acid Sequence Encoding the Full Length Light Chain Sequence (Kappa Chain Variable Region and Constant Region) of 04F08 (SEQ ID NO:109)

```

1   gacattgtga tgaccagtc tcaaaaattc atgtccacat cagtaggaga cagggtcagc
61  gtcacctgca aggccagtca gaatgtgggt actaatgtag cctggatatca acagaaatta
121 ggacaatctc ctaaaacact gatttactcg gcatactacc ggtacagtgg agtcctgat
181 cgcttcacag gcagtggatc tgggacagat ttcactotca ccatcagcaa tgtgcagtct
241 gaagacttgg cagagtattt ctgtcagcaa tataacagct atccgtacac gttcggaggg
301 gggaccaagc tggaaataaa acgggctgat gctgcaccaa ctgtatccat cttcccacca
361 tccagtgagc agttaacatc tggaggtgcc tcagtcgtgt gcttcttgaa caacttctac
421 cccaaagaca tcaatgtcaa gtggaagatt gatggcagtg aacgacaaaa tggcgctcctg
481 aacagttgga ctgatcagga cagcaaagac agcacctaca gcatgagcag caccctcacg
541 ttgaccaagg acgagtatga acgacataac agctatacct gtgaggccac tcacaagaca
601 tcaacttcac ccattgtcaa gagcttcaac aggaatgagt gt

```

Protein Sequence Defining the Full Length Light Chain Sequence (Kappa Chain Variable Region and Constant Region) of 04F08 (SEQ ID NO:110)

```

1   divmtsqskf mstsvgdrvs vtckasqnv tnvawyqqkl gqspktliys asyrysgvpd
61  rftgsgsgtd ftltisnvqs edlaeyfcqq ynsypytfgg gtkleikrad aaptvsifpp

```

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121 sseqltsgga svvcflnnfy pkdinvkwki dgserqngvl nswtdqdskd stysmsstlt
181 ltkdeyerhn sytceathkt stspivksfn rnec

Nucleic Acid Sequence Encoding the Full Length Heavy
Chain Sequence (Heavy Chain Variable Region and IgG1
Constant Region) of 06C11 (SEQ ID NO:111)

1 caggttactc tgaagagtc tggccctggg atattgcagc cctcccagac cctcagtcgt
61 acttggtctt tctctgggtt ttcactgaac acttatggta tgggtgtgag ctggattcgt
121 cagccttcag gaaagggctt ggagtggtg gcacacattt actgggatga tgacaagcgc
181 tataacccat cctgaagag cgggtcaca atctccaagg atgcctocaa caaccgggtc
241 ttctcaaga tcaccagtgt ggacactgca gatactgcca catactactg tgctcaaaga
301 ggttatgatg attactgggg ttactggggc caagggactc tggtcactat ctctgcagcc
361 aaaacgacac ccccatctgt ctatccactg gccctggat ctgctgccc aactaactcc
421 atggtgaccc tgggatgcct ggtcaagggc tatttccttg agccagtgc agtgacctgg
481 aactctggat cctgtccag cgggtgtcac accttcccag ctgtcctgca gtctgacctc
541 tacactctga gcagctcagt gactgtcccc tccagcacct gggccagcga gaccgtcacc
601 tgcaacgttg cccaccggc cagcagcacc aaggtggaca agaaaattgt gcccagggat
661 tgtgggttga agccttgcat atgtacagtc ccagaagtat catctgtctt catcttcccc
721 ccaaagccca aggatgtgct caccattact ctgactccta aggtcacgtg tgttgtggta
781 gacatcagca aggatgatcc cgaggtccag ttcagctggt ttgtagatga tgtggaggtg
841 cacacagctc agacgcaacc cgggaggag cagttcaaca gcactttccg ctcagtcagt
901 gaacttccca tcatgacca ggactggctc aatggcaagg agttcaaatg cagggtcaac
961 agtgcagctt tccctgcccc catcgagaaa accatctcca aaaccaaagg cagaccgaag
1021 gctccacagg tgtacacat tccacctccc aaggagcaga tggccaagga taaagtcagt
1081 ctgacctgca tgataacaga cttcttcctt gaagacatta ctgtggagtg gcagtggaaat
1141 gggcagccag cggagaacta caagaacact cagcccatca tggacacaga tggctcttac
1201 ttctgtctaca gcaagctcaa tgtgcagaag agcaactggg aggcaggaaa tactttcacc
1261 tgctctgtgt tacatgaggg cctgcacaac caccatactg agaagagcct ctcccactct
1321 cctggtaaa

Protein Sequence Defining the Full Length Heavy Chain
Sequence (Heavy Chain Variable Region and IgG1 Constant
Region) of 06C11 (SEQ ID NO:112)

1 qvtlkesgpg ilqpsqtlsl tcsfsgfsln tygmgswwir qpsgkglewl ahiywdddkr
61 ynpslksrll iskdasnrv flkitsvda dtatycaqr gyddywgyw ggtlvtisaa
121 kttpsvypl apgsaaqtns mvtlgclvkg yfpepvtvtw nsgslssgvh tfpavlqsd1
181 ytlsssvtvp sstwpsetvt cnvahasst kvdkkivprd cgckpcictv pevssvfifp
241 pkpkdvltit ltpkvctcvv diskddpevq fswfvddvev htaqtqpree qfnstfrsvs
301 elpimhqdlw ngkefkcrvn saafpapie ktsktkgrp apqvtytippp keqmakdkvs
361 ltcmitdffp editvewqwn gqpaenyknt qpimdtgdy fvysklinvqk snweagntft
421 csvlheglhn hhtekslshs pgk

Nucleic Acid Sequence Encoding the Full Length Light Chain Sequence (Kappa Chain Variable Region and Constant Region) of 06C11 (SEQ ID NO:113)

```
1   gacattgtga tgaccagtc tcaaaaattc atgtccacat cagtaggaga cagggtcagc
61   gtcacctgca aggccagtc gaatgtgggt actaatgtag cctggtttca acagaaacca
121  ggtcaatctc ctaaagcact gatttactcg gcattctacc ggtacagtgg agtcctgat
181  cgcttcacag gcagtggatc tgggacagat ttcattctca ccatcagcaa tgtgcagtct
241  gaagacctgg cagagatatt ctgtcagcaa tataacaact atcctctcac gttcggtgct
301  gggaccaagc tggagctgaa acgggctgat gctgcaccaa ctgtatccat ctteccacca
361  tccagtgagc agttaacatc tggaggtgcc tcagtcgtgt gcttcttgaa caacttctac
421  cccaaagaca tcaatgtcaa gtggaagatt gatggcagtg aacgacaaaa tggcgtcctg
481  aacagttgga ctgatcagga cagcaaagac agcacctaca gcatgagcag caccctcagc
541  ttgaccaagg acgagtatga acgacataac agctatacct gtgaggccac tcacaagaca
601  tcaacttcac ccattgtcaa gagcttcaac aggaatgagt gt
```

Protein Sequence Defining the Full Length Light Chain Sequence (Kappa Chain Variable Region and Constant Region) of 06C11 (SEQ ID NO:114)

```
1   divmtsqkf mtsvsgdrvs vtckasqnvq tnvaufqqkp gqspkaliys asyrysgvpd
61   rftgsgsgtd filtisnvqs edlaeyfcqg ynnyppltfga gtklelkrad aaptvsifpp
121  sseqltsgga svvcflnnfy pkdinvkwki dgserqngvl nswtdqdskd stysmsstlt
181  ltkdeyerhn sytceathkt stspivksfn rnec
```

Nucleic Acid Sequence Encoding the Full Length Heavy Chain Sequence (Heavy Chain Variable Region and IgG2b Constant Region) of 08G01 (SEQ ID NO:115)

```
1   gaggtcctgc tgcaacagtc tggacctgag gtggtgaagc ctggggcttc agtgaagata
61   ccctgcaagg cttctggata cacattcact gactacaaca tggactgggt gaagcagagc
121  catggaaaga gccttgatgt gattggagag attaatacta acaatgggtg tactttctac
181  aaccagaagt tcaagggcaa ggccacattg actgtagaca agtcctccag cacagcctac
241  atggagctcc gcagcctgac atctgaggac actgcagtct attactgtgc aagagaggca
301  attactacgg taggcgctat ggactactgg ggtcaaggaa cctcagtcac cgtctcctca
361  gccaaaacaa ccccccatc agtctatcca ctggcccctg ggtgtggaga tacaactggt
421  tcctccgtga ctctgggatg cctggtcaag ggctacttcc ctgagtcagt gactgtgact
481  tggaactctg gatccctgtc cagcagtggt cacaccttcc cagctctcct gcagtctgga
541  ctctacacta tgagcagctc agtgactgtc ccctccagca cctggccaag tcagaccgtc
601  acctgcagcg ttgctcacc cagccagcagc accacgggtg aaaaaaact tgagcccagc
661  gggcccatct caacaatcaa cccctgtcct ccatgcaagg agtgtcacia atgcccagct
721  cctaacctcg aggggtggacc atccgtcttc atcttccctc caaatatcaa ggatgtactc
781  atgatctccc tgacacccaa ggtcacgtgt gtggtgggtg atgtgagcga ggatgaccca
841  gacgtccaga tcagctgggt tgtgaacaac gtggaagtac acacagctca gacacaaaacc
```

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```
901 catagagagg attacaacag tactatccgg gtggtcagca cctcccccatt ccagcaccag
961 gactggatga gtggcaagga gttcaaatgc aaggtcaaca acaaagacct cccatcaccc
1021 atcgagagaa ccatctcaaa aattaaagggt ctagtccagag ctccacaagt atacatcttg
1081 ccgccaccag cagagcagtt gtccaggaaa gatgtcagtc tcaattgcct ggctgtgggc
1141 ttcaaccctg gagacatcag tgtggagtgg accagcaatg ggcatacaga ggagaactac
1201 aaggacaccg caccagtcct agactctgac gggtcttact tcatatatag caagctcaat
1261 atgaaaaaaa gcaagtggga gaaaacagat tccttctcat gcaacgtgag acacgagggg
1321 ctgaaaaaatt actacctgaa gaagaccatc tcccgggtctc cgggtaaa
```

Protein Sequence Defining the Full Length Heavy Chain Sequence (Heavy Chain Variable Region and IgG2b Constant Region) of 08G01 (SEQ ID NO:116)

```
1 evllqqsgpe vvkpgasvki pckasgytft dynmdwvkqs hgkslewise inpnnggtfy
61 nqkfkgtatl tvdkssstay melrsltsed tavyyarea ittvgamdyw gqgtsvtvss
121 akttpsvvyp lapgcgdttg ssvtlgclvk gyfpevstvt wnsqslsssv htfpallqsg
181 lytmsssvtv psswpsqtv tcsvahpass ttvdkkleps gpistinpcp pckechkcpa
241 pnleggpsvf ifppnikdvl misltpkvtc vvdvseddp dvqiswfvnn vevhtaqtqt
301 hredynstir vvtlpihq dwmsgkefkc kvnnkdlpsp iertiskikg lvrpqvyil
361 pppaeqlsrk dvsltlclvvg fnpgdisvew tsnghteeny kdtapvldsd gsyfiyskln
421 mktskwektd sfscnvrheg lknyylkkti srspgk
```

Nucleic Acid Sequence Encoding the Full Length Light Chain Sequence (Kappa Chain Variable Region and Constant Region) of 08G01 (SEQ ID NO:117)

```
1 gacatccaga tgactcagtc tccagcctcc ctatctgcat ctgtgggaga aactgtcacc
61 atcacatgtc gagcaagtgg gaattattcac aattatttag catggtatca gcagaaacag
121 ggaaaatctc ctccagctcct ggtctataat gcaaaaacct tagcagatgg tgtgccatca
181 aggttcagtg gcagtggtac aggaacacaa tattctctca agatcaacag cctgcagcct
241 gaagattttg ggagttatta ctgtcaacat ttttgaggtt ctctctacac gttcggaggg
301 gggaccaagc tggaaataaa acgggctgat gctgcaccaa ctgtatccat cttcccacca
361 tccagtgcgc agttaacatc tggaggtgcc tcagtcgtgt gcttcttgaa caacttctac
421 cccaaagaca tcaatgtcaa gtggaagatt gatggcagtg aacgacaaaa tggcgtcctg
481 aacagttgga ctgatcagga cagcaaagac agcacctaca gcatgagcag caccctcacg
541 ttgaccaagg acgagtatga acgacataac agctatacct gtgaggccac tcacaagaca
601 tcaacttcac ccattgtcaa gagcttcaac aggaatgagt gt
```

Protein Sequence Defining the Full Length Light Chain Sequence (Kappa Chain Variable Region and Constant Region) of 08G01 (SEQ ID NO:118)

```
1 diqmtqspas lsasvgetvt itcrasgnih nylawyqqkq gkspqllvyn aktladgvps
61 rfsqsgsgtg yslkinslqp edfgyycqh fwsspytfgg gtleikrad aaptvsifpp
```

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121 sseqltsgga svvcflnnfy pkdinvkwki dgserqngvl nswtdqdskd stysmsstlt
181 ltkdeyerhn sytceathkt stspivksfn rnec

Nucleic Acid Sequence Encoding the Full Length Heavy
Chain Sequence (Heavy Chain Variable Region and IgG1
Constant Region) of 14F11 (SEQ ID NO:119)

1 caggttactc tgaagagtc tggccctgga atattgcagc cctcccagac cctcagtcgt
61 acttggtctt tctctgggtt ttcactgagc acttatggta tgggtgtagg ctggattcgt
121 cagccttcag gaaagggctc agagtggctg gcagacattt ggtgggatga cgataagtac
181 tataacccat cctgaagag cgggtcaca atctccaagg atacctocag caatgaggta
241 ttctcaaga tcgccattgt ggacactgca gatactgcca ctactactg tgctcgaaga
301 ggtcactact ctgctatgga ctactgggtt caaggaacct cagtcaccgt ctccctcagcc
361 aaaacgacac ccccatctgt ctatccactg gccctggat ctgctgccc aactaactcc
421 atggtgaccc tgggatgcct ggtcaagggc tatttccttg agccagtgc agtgacctgg
481 aactctggat cctgtccag cgggtgtcac accttcccag ctgtcctgca gtctgacctc
541 tacactctga gcagctcagt gactgtcccc tccagcacct ggcccagcga gaccgtcacc
601 tgcaacgttg cccaccggcg cagcagcacc aaggtggaca agaaaattgt gcccagggat
661 tgtggttgta agccttgcat atgtacagtc ccagaagtat catctgtctt catcttcccc
721 ccaaagccca aggatgtgct caccattact ctgactccta aggtcacgtg tgttggtgta
781 gacatcagca aggatgatcc cgaggtccag ttcagctggt ttgtagatga tgtggaggtg
841 cacacagctc agacgcaacc cgggaggag cagttcaaca gcactttccg ctcagtcagt
901 gaacttccca tcatgacca ggactggctc aatggcaagg agttcaaatg cagggtcaac
961 agtgcagctt tccctgcccc catcgagaaa accatctcca aaaccaaagg cagaccgaag
1021 gctccacagg tgtacacat tccacctccc aaggagcaga tggccaagga taaagtcagt
1081 ctgacctgca tgataacaga cttcttcctt gaagacatta ctgtggagtg gcagtggaat
1141 gggcagccag cggagaacta caagaacact cagcccatca tggacacaga tggctcttac
1201 ttctgtctaca gcaagctcaa tgtgcagaag agcaactggg aggcaggaaa tactttcacc
1261 tgctctgtgt tacatgaggg cctgcacaac caccatactg agaagagcct ctcccactct
1321 cctaataaaa

Protein Sequence Defining the Full Length Heavy Chain
Sequence (Heavy Chain Variable Region and IgG1 Constant
Region) of 14F11 (SEQ ID NO:120)

1 qvtlkesgpg ilqpsqtlsl tcsfsgfsls tygmvgwir qpsgkglewl adiwdddky
61 ynpslksrilt iskdtsnev flkiaivdta dtatyycarr ghysamdywg qgtsvtvssa
121 kttpsvypl apgsaaqtns mvtlgclvkg yfpepvtvtw nsgslssgvh tfpavlqsd1
181 ytlsssvtvp sstwpsetvt cnvahpasst kvdkkivprd cgckpcictv pevssvfifp
241 pkpkdvltit ltpkvctcvv diskddpevq fswfvddvev htaqtqpree qfnstfrsvs
301 elpimhqdlw ngkefkcrvn saafpapie ktsktkgrp apqvtytippp keqmakdkvs
361 ltcmitdffp editvewqwn gqpaenyknt qpimdtgdy fvysklvqk snweagntft
421 csvlheglhn hhtekslshs pgk

Nucleic Acid Sequence Encoding the Full Length Light Chain Sequence (Kappa Chain Variable Region and Constant Region) of 14F11 (SEQ ID NO:121)

```
1   gacattgtaa tgaccagtc tcaaaaattc atgtccacat cagtaggaga cagggtcagc
61  gtcacctgca aggccagtc gaattgtggg actaatgtag cctggtatca acagaaacca
121 gggcaatctc ctaaagcact gatttactcg ccatactacc ggtacagtgg agtcctgat
181 cgcttcacag gcagtggatc tgggacagat ttcactctca ccatacagaa tgtgcagtct
241 gaagacttgg cagaatattt ctgtcagcaa tataacagct atcctcacac gttcggaggg
301 gggaccaagc tggaaatgaa acgggctgat gctgcaccaa ctgtatccat ctteccacca
361 tccagtgagc agttaacatc tggaggtgcc tcagtcgtgt gcttcttgaa caacttctac
421 cccaaagaca tcaatgtcaa gtggaagatt gatggcagtg aacgacaaaa tggcgctcctg
481 aacagttgga ctgatcagga cagcaaagac agcacctaca gcatgagcag caccctcacg
541 ttgaccaagg acgagtatga acgacataac agctatacct gtgaggccac tcacaagaca
601 tcaacttcac ccattgtcaa gagcttcaac aggaatgagt gt
```

Protein Sequence Defining the Full Length Light Chain Sequence (Kappa Chain Variable Region and Constant Region) of 14F11 (SEQ ID NO:122)

```
1   divmtsqkf mtsvsgdrvs vtckasqnvq tnvaWyqqkp gqspkaliys psyrysgvpd
61  rftgsgsgtd ftltisnvqs edlaeyfcqg ynsyphtfgg gtklemkrad aaptvsifpp
121 sseqltsgga svvcflnnfy pkdinvkwi dgserqngvl nswtdqdskd stysmsstlt
181 ltkdeyerhn sytceathkt stspivksfn rnec
```

Nucleic Acid Sequence Encoding the Full Length Heavy Chain Sequence (Heavy Chain Variable Region and IgG1 Constant Region) of 17B11 (SEQ ID NO:123)

```
1   caggttactc tgaagagtc tggccctggg atattgcagc cctcccagac cctcagtctg
61  acttgttctt tctctggggt ttcactgagc acttctggta tgggtgtgag ttggattcgt
121 cagccttcag gaaaggtctt ggagtggctg gcacacaatg actgggatga tgacaagcgc
181 tataagtcat ccttgaagag ccggctcaca atatccaagg atacctccag aaaccaggta
241 ttcctcaaga tcaccagtgt ggacactgca gatactgcca catactactg tgctcgaaga
301 gttgggggat tagagggcta ttttgattac tggggccaag gcaccactct cacagtctcc
361 tcagccaaaa cgacaccccc atctgtctat ccactggccc ctggatctgc tgccaaaact
421 aactccatgg tgaccctggg atgcctggtc aagggtatt tccctgagcc agtgacagtg
481 acctggaact ctggatccct gtccagcggg gtgcacacct tcccagctgt cctgcagtct
541 gacctctaca ctctgagcag ctccagtact gtccctcca gcacctggcc cagcgagacc
601 gtcacctgca acgttgccca ccgggccagc agcaccaagg tggacaagaa aattgtgccc
661 agggattgtg gttgtaagcc ttgcatatgt acagteccag aagtatcctc tgtcttcac
721 tcccccccaa agcccaagga tgtgctcacc attactctga ctccaaaggc cagtggtgtt
781 gtggtagaca tcagcaagga tgatcccag gtccagtcca gctggtttgt agatgatgtg
841 gagggtgcaca cagctcagac gcaaccccgg gaggagcagt tcaacagcac tttccgctca
```

-continued

```

901   gtcagtgaac ttcccatcat gcaccaggac tggctcaatg gcaaggagtt caaatgcagg
961   gtcaacagtg cagctttccc tgcccccato gagaaaacca tctccaaaac caaaggcaga
1021  ccgaaggctc cacaggtgta caccattcca cctcccaagg agcagatggc caaggataaa
1081  gtcagtctga cctgcatgat aacagacttc ttccctgaag acattactgt ggagtggcag
1141  tggaatgggc agccagcgga gaactacaag aacactcagc ccatcatgga cacagatggc
1201  tcttacttcg tctacagcaa gctcaatgtg cagaagagca actgggagggc aggaaatact
1261  ttcacctgct ctgtgttaca tgagggcctg cacaaccacc atactgagaa gagcctctcc
1321  cactctctcg gtaaa

```

Protein Sequence Defining the Full Length Heavy Chain Sequence (Heavy Chain Variable Region and IgG1 Constant Region) of 17B11 (SEQ ID NO:124)

```

1   qvtlkesgpg ilqpsqtlsl tcsfsgfsls tsgmgvswir qpsgkglewl ahndwddkr
61  yksslksrll iskdtsrnqv flkitsvdta dtatyycarr vgglegyfdy wgqgttltsv
121  sakttppsvy plapgsaaqt nsmvtlgclv kgyfpepvtv twnsqslssg vhtfpavlqs
181  dlytlsssvt vpstwpset vtcnvahpas stkvdkiivp rdcgckpcic tpevssvfi
241  fppkpkdvlv itltpkvtcv vdiskddpe vqfswfvddv evhtaqtqpr eeqfnstfrs
301  vselpimhqd wlngkefkcr vnsaafpapi ektisktkgr pkapqvtyip ppkeqmakdk
361  vsltcmitdf fpeditvewq wngqpaenyk ntqpimtdtg syfvysklmv qksnweagnt
421  ftcsvlhagl hnhhteksll hspgk

```

Nucleic Acid Sequence Encoding the Full Length Light Chain Sequence (Kappa Chain Variable Region and Constant Region) of 17B11 (SEQ ID NO:125)

```

1   gacattgtgc tgacacagtc tectgettec ttagctgtat ctctggggca gagggccacc
61  atctcatgca gggccagcca aagtgtcagt acatctaggt ttagttatat gcaactgttc
121  caacagaaac caggacaggc acccaaaact ctcataagat atgcatccaa cctagaatct
181  ggggtccctg ccaggttcag tggcagtggtg tctgggacag acttcaccct caacatccat
241  cctgtggagg gggaggatac tgcaacatat tactgtcagc acagttggga gattccgtac
301  acgttcggag gggggaccaa gctggaaata aaacgggctg atgctgcacc aactgtatcc
361  atcttcccac catccagtga gcagttaaca tctggaggtg cctcagtcgt gtgcttcttg
421  aacaacttct accccaaaga catcaatgtc aagtgaaga ttgatggcag tgaacgaaa
481  aatggcgtcc tgaacagttg gactgatcag gacagcaaag acagcaccta cagcatgagc
541  agcacctca cgttgaccaa ggacgagtat gaacgacata acagctatac ctgtgaggcc
601  actcacaaga catcaacttc acccattgtc aagagcttca acaggaatga gtgt

```

Protein Sequence Defining the Full Length Light Chain Sequence (Kappa Chain Variable Region and Constant Region) of 17B11 (SEQ ID NO:126)

```

1   divltqspas lavslgqrat iscrasqsvs tsrfsymhfw qqkpggapkl likyasnlcs
61  gvparfsgsg sgtdftlnih pvegedtaty yqhsweipy tfgggtklei kradaaptvs

```

-continued
121 ifppsseqlt sggasvvcf1 nnfypkdlv kwkidgserq ngvlswtdq dskdstysms
181 stltltkdey erhnsytcea thktstspiv ksfnrnec

[0265] Table 5 shows the correspondence between the full-length sequences of the antibodies discussed in this Example with those presented in the Sequence Listing.

TABLE 5	
SEQ ID NO.	Nucleic Acid or Protein
99	01G06_Heavy Variable + IgG1 Constant-nucleic acid
100	01G06_Heavy Variable + IgG1 Constant-protein
101	01G06_Kappa Variable + Constant-nucleic acid
102	01G06_Kappa Variable + Constant-protein
103	03G05_Heavy Variable + IgG1 Constant-nucleic acid
104	03G05_Heavy Variable + IgG1 Constant-protein
105	03G05_Kappa Variable + Constant-nucleic acid
106	03G05_Kappa Variable + Constant-protein
107	04F08_Heavy Variable + IgG1 Constant-nucleic acid
108	04F08_Heavy Variable + IgG1 Constant-protein
109	04F08_Kappa Variable + Constant-nucleic acid
110	04F08_Kappa Variable + Constant-protein
111	06C11_Heavy Variable + IgG1 Constant-nucleic acid
112	06C11_Heavy Variable + IgG1 Constant-protein
113	06C11_Kappa Variable + Constant-nucleic acid
114	06C11_Kappa Variable + Constant-protein
115	08G01_Heavy Variable + IgG2b Constant-nucleic acid
116	08G01_Heavy Variable + IgG2b Constant-protein
117	08G01_Kappa Variable + Constant-nucleic acid
118	08G01_Kappa Variable + Constant-protein
119	14F11_Heavy Variable + IgG1 Constant-nucleic acid
120	14F11_Heavy Variable + IgG1 Constant-protein
121	14F11_Kappa Variable + Constant-nucleic acid
122	14F11_Kappa Variable + Constant-protein
123	17B11_Heavy Variable + IgG1 Constant-nucleic acid
124	17B11_Heavy Variable + IgG1 Constant-protein
125	17B11_Kappa Variable + Constant-nucleic acid
126	17B11_Kappa Variable + Constant-protein

Example 8: Binding Affinities

[0266] The binding affinities and kinetics of binding of antibodies to 6xHis tagged (SEQ ID NO: 266) recombinant human GDF15 (His-rhGDF15 (R&D Systems, Inc.)), untagged recombinant human GDF15 (rhGDF15 (Pepro-tech, Rocky Hill, NJ), and recombinant human GDF15 produced as either mouse Fc fused to human GDF15 (mFc-rhGDF15) or a version in which the Fc was enzymatically removed (cleaved-rhGDF15) were measured by surface plasmon resonance, using a Biacore® T100 instrument (GE Healthcare, Piscataway, NJ).

[0267] Rabbit anti-mouse IgGs (GE Healthcare) were immobilized on carboxymethylated dextran CM4 sensor chips (GE Healthcare) by amine coupling, according to a standard protocol. Analyses were performed at 37° C. using PBS containing 0.05% surfactant P20 as running buffer. The antibodies were captured in individual flow cells at a flow rate of 10 µL/minute. Injection time was varied for each antibody to yield an Rmax between 30 and 60 RU. 250 µg/mL mouse Fc (Jackson ImmunoResearch, West Grove, PA) was injected at 30 µL/minute for 120 seconds to block non-specific binding of capture antibodies to mouse Fc portion of the recombinant GDF15 protein when needed. Buffer, mFc-rhGDF15, cleaved-rhGDF15, His-rhGDF15, or rhGDF15 diluted in running buffer was injected sequentially over a reference surface (no antibody captured) and the active surface (antibody to be tested) for 240 seconds at 60

µL/minute. The dissociation phase was monitored for up to 1500 seconds. The surface was then regenerated with two 60-second injections of 10 mM Glycine-HCl, pH 1.7, at a flow rate of 30 µL/minute. The GDF15 concentration range tested was 30 nM to 0.625 nM.

[0268] Kinetic parameters were determined using the kinetic function of the BLAevaluation software (GE Healthcare) with double reference subtraction. Kinetic parameters for each antibody, k_a (association rate constant), k_d (dissociation rate constant), and K_D (equilibrium dissociation constant) were determined. Kinetic values of the monoclonal antibodies on mFc-rhGDF15, cleaved rhGDF15, His-rhGDF15, or rhGDF15 are summarized in Tables 6, 7, 8, and 9, respectively.

TABLE 6				
Antibody Binding to mFc-rhGDF15				
Antibody	k_a (1/Ms)	k_d (1/s)	K_D (M)	n
01G06	5.6E+06	7.0E-04	2.1E-10	7
03G05	1.0E+07	6.4E-04	6.9E-11	3
04F08	3.6E+06	6.4E-04	1.9E-10	3
06C11	4.5E+06	6.8E-04	1.7E-10	5
08G01	6.0E+06	1.1E-03	1.9E-10	4
14F11	1.7E+06	3.3E-04	2.2E-10	4
17B11	3.7E+06	5.1E-04	1.4E-10	3

[0269] The data in Table 6 demonstrate that antibodies bind mFc-rhGDF15 with a K_D of about 250 pM or less, 200 pM or less, 150 pM or less, 100 pM or less, 75 pM or less, or 50 pM or less.

[0270] Kinetic values of the monoclonal antibodies on cleaved-rhGDF15 are summarized in Table 7.

TABLE 7				
Antibody Binding to Cleaved-rhGDF15				
Antibody	k_a (1/Ms)	k_d (1/s)	K_D (M)	n
01G06	7.5E+06	8.6E-04	1.1E-10	1
06C11	1.2E+07	2.0E-03	1.7E-10	2
14F11	5.7E+06	6.0E-04	1.1E-10	1

[0271] The data in Table 7 demonstrate that antibodies 01G06, 06C11 and 14F11 bind cleaved-rhGDF15 with a K_D of about 200 pM or less, 150 pM or less, or 100 pM or less.

[0272] Kinetic values of the monoclonal antibodies on His-rhGDF15 are summarized in Table 8.

TABLE 8				
Antibody Binding to His-rhGDF15				
Antibody	k_a (1/Ms)	k_d (1/s)	K_D (M)	n
01G06	1.4E+07	1.1E-03	8.1E-11	2
06C11	2.9E+07	1.5E-03	5.1E-11	2
14F11	4.4E+06	4.2E-04	9.6E-11	1

[0273] The data in Table 8 demonstrate that antibodies 01G06, 06C11 and 14F11 bind His-rhGDF15 with a K_D of about 150 pM or less, 100 pM or less, 75 pM or less, or 50 pM or less.

[0274] Kinetic values of the monoclonal antibodies on rhGDF15 are summarized in Table 9.

TABLE 9				
Antibody Binding to rhGDF15				
Antibody	k_a (1/Ms)	k_d (1/s)	K_D (M)	n
01G06	2.1E+07	1.9E-03	9.3E-11	1
06C11	2.2E+07	4.6E-03	2.1E-10	1
14F11	3.1E+07	2.2E-03	7.1E-11	1

[0275] The data in Table 9 demonstrate that antibodies 01G06, 06C11 and 14F11 bind rhGDF15 with a K_D of about 250 pM or less, 200 pM or less, 150 pM or less, 100 pM or less, 75 pM or less, or 50 pM or less.

Example 9: Reversal of Cachexia in an mFc-rhGDF15-Induced Model

[0276] This Example demonstrates the reversal of cachexia (as indicated by body weight loss) by antibody 01G06, 03G05, 04F08, 06C11, 14F11, or 17B11 in an mFc-rhGDF15-induced cachexia model. mFc-rhGDF15 (2 μ g/g) was administered subcutaneously into the flank of 8-week old female ICR-SCID mice. Body weight was measured daily. When body weight reached 93%, the mice were randomized into seven groups of ten mice each. Each group received one of the following treatments: murine IgG control, 01G06, 03G05, 04F08, 06C11, 14F11, or 17B11 at 10 mg/kg. Treatment was administered once by intra-peritoneal injection. Treatment with antibody 01G06, 03G05, 04F08, 06C11, 14F11, or 17B11 resulted in body weight increase relative to initial weight or about 100% ($p < 0.001$) (FIG. 14 and Table 10).

TABLE 10				
Gr.	Treatment		% Body	ANOVA Analysis
	Agent	mg/kg	Weight	(compared to mIgG)
1	mIgG	10	77.1	NA
2	01G06	10	94.1	$p < 0.001$
3	03G05	10	95.1	$p < 0.001$
4	04F08	10	95.8	$p < 0.001$
5	06C11	10	93.8	$p < 0.001$
7	14F11	10	95.4	$p < 0.001$
8	17B11	10	92.8	$p < 0.001$

[0277] The data in FIG. 14 and Table 10 indicate that the disclosed anti-GDF15 antibodies can reverse cachexia in an mFc-rhGDF15-induced mouse model (i.e., a non-tumor bearing mouse model).

Example 10: Reversal of Cachexia in an HT-1080 Xenograft Tumor Model

[0278] This Example demonstrates the reversal of cachexia (as indicated by body weight loss) by antibody 01G06, 03G05, 04F08, 06C11, 08G01, 14F11 or 17B11 in an HT-1080 fibrosarcoma xenograft model. HT-1080 cells were grown in culture at 37° C. in an atmosphere containing 5% CO₂, using Eagle's Minimum Essential Medium

(ATCC, Catalog No. 30-2003) containing 10% FBS. Cells were inoculated subcutaneously into the flank of 8-week old female ICR SCID mice with 5×10^6 cells per mouse in 50% matrigel. Body weight was measured daily. When body weight reached 93%, the mice were randomized into eight groups of ten mice each. Each group received one of the following treatments: murine IgG control, 01G06, 03G05, 04F08, 06C11, 08G01, 14F11 or 17B11 at 10 mg/kg. Treatment was administered every three days by intra-peritoneal injection. Treatment with antibody 01G06, 03G05, 04F08, 06C11, 08G01, 14F11 or 17B11 resulted in body weight increase relative to initial weight or about 100% ($p < 0.001$) (FIG. 15 and Table 11).

TABLE 11				
GR.	TREATMENT		% BODY	ANOVA ANALYSIS
	AGENT	MG/KG	WEIGHT	(COMPARED TO MIGG)
1	MIGG	10	81.4	NA
2	01G06	10	103.3	$P < 0.001$
3	03G05	10	106.1	$p < 0.001$
4	04F08	10	104.3	$p < 0.001$
5	06C11	10	106.6	$p < 0.001$
6	08G01	10	105.3	$p < 0.001$
7	14F11	10	99.6	$p < 0.001$
8	17B11	10	103.7	$p < 0.001$

[0279] The data in FIG. 15 and Table 11 indicate that the disclosed anti-GDF15 antibodies can reverse cachexia in an HT-1080 fibrosarcoma xenograft model.

[0280] Additional studies were conducted with antibody 01G06 to demonstrate the reversal of cachexia in this mouse model. HT-1080 cells were grown and inoculated subcutaneously into the flank of 8-week old female ICR SCID mice as described above. When body weight reached 93%, the mice were randomized into two groups of ten mice each. Each group received one of the following treatments: murine IgG control or 01G06 at 10 mg/kg. Treatment was administered once by intra-peritoneal injection. As shown in FIG. 16A, treatment with antibody 01G06 resulted in body weight increase to initial weight or 100% ($p < 0.001$) (FIG. 16A).

[0281] Food consumption was determined by weighing the food supply given to the mice daily (FIG. 16B). A significant increase in food consumption was observed in the 01G06 treated group for the first three days post treatment. After that time, no significant change was observed compared to the control group (mIgG).

[0282] Water consumption was determined by weighing the water supply given to the mice daily. No significant change in water consumption was observed between groups.

[0283] In this experiment, a group of ten mice were sacrificed at the time of the dose (baseline or 93% body weight, without treatment) and at the end of study (seven days post dose, either mIgG or 01G06). Gonadal fat and the gastrocnemius muscles were removed surgically and weighed as described above in Example 4 and tissues were snap frozen in liquid nitrogen. RNA was isolated from the gastrocnemius muscle samples to determine the levels of mMuRF1 and mAtrogin mRNA by RT-PCR, as described in Example 4.

[0284] As shown in FIG. 16C, a significant reduction in gonadal fat mass was observed seven days post dose with

mIgG, but not in the group treated with antibody 01G06. In addition, mice treated with mIgG displayed significant gastrocnemius muscle loss compared to the baseline group, while the group of mice treated with antibody 01G06 did not (FIG. 16D). Further, the levels of muscular degradation markers, mMuRF1 and mAtrogin, were significantly higher in the mIgG group compared to the 01G06 group (FIG. 16E).

[0285] These results indicate that the disclosed anti-GDF15 antibodies can reverse cachexia measured by the loss of muscle mass, the loss of fat and involuntary weight loss in an HT-1080 xenograft tumor model.

Example 11: Reversal of Cachexia in an HT-1080 Xenograft Tumor Model

[0286] This Example demonstrates the reversal of cachexia (as indicated by body weight loss) by antibody 01G06 in an HT-1080 fibrosarcoma xenograft model. HT-1080 cells were grown in culture at 37° C. in an atmosphere containing 5% CO₂, using Eagle's Minimum Essential Medium (ATCC, Catalog No. 30-2003) containing 10% FBS. Cells were inoculated subcutaneously into the flank of 8-week old female ICR SCID mice with 5×10⁶ cells per mouse in 50% matrigel. Body weight was measured daily. When body weight reached 80%, the mice were randomized into two groups of five mice each. Each group received one of the following treatments: murine IgG control, 01G06 dosed at 2 mg/kg on day 1 and day 7. Treatment was administered by intra-peritoneal injection. Treatment with antibody 01G06 resulted in body weight increase relative to initial weight or about 100% (p<0.001) (FIG. 17A and Table 12).

TABLE 12

Gr.	Treatment		% Body Weight	ANOVA Analysis (compared to mIgG)
	Agent	mg/kg		
1	mIgG	2	66.4	NA
2	01G06	2	97.16	p < 0.001

[0287] The data in FIGS. 17A-B and Table 12 indicate that the disclosed anti-GDF15 antibodies can reverse cachexia in an HT-1080 fibrosarcoma xenograft model.

[0288] In this experiment, a group of five mice were sacrificed at the time of dosing (baseline or 80% body weight loss, without treatment) and at the end of study (seven days post dose, either mIgG or 01G06). Liver, heart, spleen, kidney, gonadal fat and the gastrocnemius muscles were removed surgically and weighed. As shown in FIG. 17B, a significant loss in liver, heart, spleen, kidney, gonadal fat and gastrocnemius muscle mass was observed seven days post dose with mIgG, but not in the group treated with antibody 01G06. In addition, mice treated with antibody 01G06 displayed significant liver and gonadal muscle gain compared to the baseline group (FIG. 17B).

[0289] These results indicate that the disclosed anti-GDF15 antibodies can reverse cachexia measured by the loss of key organ mass, loss of muscle mass, loss of fat and involuntary weight loss in an HT-1080 xenograft tumor model.

Example 12: Reversal of Cachexia in a K-562 Xenograft Tumor Model

[0290] This Example demonstrates the reversal of cachexia (as indicated by body weight loss) by antibody 01G06, 03G05, 04F08, 06C11, 08G01, 14F11 or 17B11 in a K-562 leukemia xenograft model. K-562 cells were grown in culture at 37° C. in an atmosphere containing 5% CO₂, using Iscove's Modified Dulbecco's Medium (ATCC Catalog No. 30-2005) containing 10% FBS. Cells were inoculated subcutaneously into the flank of 8-week old female CB17SCRFMF mice with 2.5×10⁶ cells per mouse in 50% matrigel. Body weight was measured daily. When body weight reached 93%, the mice were randomly distributed into eight groups of ten mice each. Each group received one of the following treatments: murine IgG control, 01G06, 03G05, 04F08, 06C11, 08G01, 14F11 or 17B11 at 10 mg/kg. Treatment was administered every three days by intra-peritoneal injection. Treatment with antibody 01G06, 03G05, 04F08, 06C11, 08G01, 14F11 or 17B11 resulted in body weight increase relative to initial weight or about 100% (p<0.001) (FIG. 18 and Table 13).

TABLE 13

Gr.	Treatment		% Body Weight	ANOVA Analysis (compared to mIgG)
	Agent	mg/kg		
1	mIgG	10	90.4	NA
2	01G06	10	106.5	p < 0.001
3	03G05	10	109.8	p < 0.001
4	04F08	10	108.9	p < 0.001
5	06C11	10	109.5	p < 0.001
6	08G01	10	107.2	p < 0.001
7	14F11	10	107.0	p < 0.001
8	17B11	10	105.3	p < 0.001

[0291] The data in FIG. 18 and Table 13 indicate that the disclosed anti-GDF15 antibodies can reverse cachexia in a K-562 xenograft tumor model.

Example 13: Additional Xenograft Tumor Models

[0292] Antibody 01G06 was tested in additional tumor xenograft models including the TOV-21G ovarian xenograft model and the LS1034 colon xenograft model. In each model, antibody 01G06 reversed body weight loss compared to a PBS control (p<0.001 for the TOV-21G model and p<0.01 for the LS1034 model).

Example 14: Humanization of Anti-GDF15 Antibodies

[0293] This Example describes the humanization and chimerization of three murine antibodies, designated 01G06, 06C11, and 14F11, and the characterization of the resulting humanized antibodies. The humanized anti-GDF15 antibodies were designed, affinity matured by targeted CDR mutagenesis, and optimized using methods known in the art. The amino acid sequences were converted to codon-optimized DNA sequences and synthesized to include (in the following order): 5' HindIII restriction site, Kozak consensus sequence, amino terminal signal sequence, humanized variable region, human IgG1 or Kappa constant region, stop codon, and a 3' EcoRI restriction site.

[0294] Chimeric (murine variable region and human constant region) 01G06, 06C11, and 14F11 heavy (human IgG1) and light (human Kappa) chains were also constructed. To generate chimeric antibodies, the murine variable regions were fused to the human constant region, and codon-optimized DNA sequences were synthesized, including (in the following order): 5' HindIII restriction site, Kozak consensus sequence, amino terminal signal sequence, mouse variable region, human IgG1 or Kappa constant region, stop codon, and 3' EcoRI restriction site.

[0295] The humanized and chimeric heavy chains were subcloned into pEE6.4 (Lonza, Basel, Switzerland) via HindIII and EcoRI sites using In-Fusion™ PCR cloning (Clontech, Mountain View, CA). The humanized and chi-

meric Kappa light chains were subcloned into pEE14.4 (Lonza) via HindIII and EcoRI sites using In-Fusion™ PCR cloning.

[0296] Humanized antibody chains or chimeric antibody chains were transiently transfected into 293T cells to produce antibody. Antibody was either purified or used in cell culture media supernatant for subsequent in vitro analysis. Binding of the chimeric and humanized antibodies to human GDF15 was measured as described below. The results are summarized in Tables 24-27.

[0297] Each of the possible combinations of the chimeric or humanized K1G06 immunoglobulin heavy chain and immunoglobulin light chain variable regions is set forth below in Table 14.

TABLE 14

Antibody Name	Light Chain Variable Region	Heavy Chain Variable Region
Hu01G06-1	Ch01G06 Chimeric Kappa (SEQ ID NO: 76)	Ch01G06 Chimeric Heavy (SEQ ID NO: 40)
Hu01G06-14	Ch01G06 Chimeric Kappa (SEQ ID NO: 76)	Hu01G06 IGHV1-18 Heavy (SEQ ID NO: 54)
Hu01G06-15	Ch01G06 Chimeric Kappa (SEQ ID NO: 76)	Hu01G06 IGHV1-69 Heavy (SEQ ID NO: 56)
Hu01G06-147	Ch01G06 Chimeric Kappa (SEQ ID NO: 76)	Sh01G06 IGHV1-18 M69L Heavy (SEQ ID NO: 58)
Hu01G06-148	Ch01G06 Chimeric Kappa (SEQ ID NO: 76)	Sh01G06 IGHV1-18 M69L K64Q G44S Heavy (SEQ ID NO: 60)
Hu01G06-149	Ch01G06 Chimeric Kappa (SEQ ID NO: 76)	Sh01G06 IGHV1-18 M69L K64Q Heavy (SEQ ID NO: 62)
Hu01G06-150	Ch01G06 Chimeric Kappa (SEQ ID NO: 76)	Sh01G06 IGHV1-69 T30S I69L Heavy (SEQ ID NO: 64)
Hu01G06-151	Ch01G06 Chimeric Kappa (SEQ ID NO: 76)	Sh01G06 IGHV1-69 T30S K64Q I69L Heavy (SEQ ID NO: 66)
Hu01G06-4	Hu01G06 IGKV1-39 Kappa (SEQ ID NO: 90)	Ch01G06 Chimeric Heavy (SEQ ID NO: 40)
Hu01G06-46	Hu01G06 IGKV1-39 Kappa (SEQ ID NO: 90)	Hu01G06 IGHV1-18 Heavy (SEQ ID NO: 54)
Hu01G06-52	Hu01G06 IGKV1-39 Kappa (SEQ ID NO: 90)	Hu01G06 IGHV1-69 Heavy (SEQ ID NO: 56)
Hu01G06-100	Hu01G06 IGKV1-39 Kappa (SEQ ID NO: 90)	Sh01G06 IGHV1-18 M69L Heavy (SEQ ID NO: 58)
Hu01G06-102	Hu01G06 IGKV1-39 Kappa (SEQ ID NO: 90)	Sh01G06 IGHV1-18 M69L K64Q G44S Heavy (SEQ ID NO: 60)
Hu01G06-101	Hu01G06 IGKV1-39 Kappa (SEQ ID NO: 90)	Sh01G06 IGHV1-18 M69L K64Q Heavy (SEQ ID NO: 62)
Hu01G06-103	Hu01G06 IGKV1-39 Kappa (SEQ ID NO: 90)	Sh01G06 IGHV1-69 T30S I69L Heavy (SEQ ID NO: 64)
Hu01G06-104	Hu01G06 IGKV1-39 Kappa (SEQ ID NO: 90)	Sh01G06 IGHV1-69 T30S K64Q I69L Heavy (SEQ ID NO: 66)
Hu01G06-152	Hu01G06 IGKV1-39 S43A V48I Kappa (SEQ ID NO: 92)	Ch01G06 Chimeric Heavy (SEQ ID NO: 40)
Hu01G06-71	Hu01G06 IGKV1-39 S43A V48I Kappa (SEQ ID NO: 92)	Hu01G06 IGHV1-18 Heavy (SEQ ID NO: 54)
Hu01G06-77	Hu01G06 IGKV1-39 S43A V48I Kappa (SEQ ID NO: 92)	Hu01G06 IGHV1-69 Heavy (SEQ ID NO: 56)
Hu01G06-110	Hu01G06 IGKV1-39 S43A V48I Kappa (SEQ ID NO: 92)	Sh01G06 IGHV1-18 M69L Heavy (SEQ ID NO: 58)
Hu01G06-112	Hu01G06 IGKV1-39 S43A V48I Kappa (SEQ ID NO: 92)	Sh01G06 IGHV1-18 M69L K64Q G44S Heavy (SEQ ID NO: 60)
Hu01G06-111	Hu01G06 IGKV1-39 S43A V48I Kappa (SEQ ID NO: 92)	Sh01G06 IGHV1-18 M69L K64Q Heavy (SEQ ID NO: 62)
Hu01G06-113	Hu01G06 IGKV1-39 S43A V48I Kappa (SEQ ID NO: 92)	Sh01G06 IGHV1-69 T30S I69L Heavy (SEQ ID NO: 64)
Hu01G06-114	Hu01G06 IGKV1-39 S43A V48I Kappa (SEQ ID NO: 92)	Sh01G06 IGHV1-69 T30S K64Q I69L Heavy (SEQ ID NO: 66)
Hu01G06-122	Hu01G06 IGKV1-39 S43A V48I Kappa (SEQ ID NO: 92)	Hu01G06 IGHV1-18 F1 Heavy (SEQ ID NO: 246)
Hu01G06-119	Hu01G06 IGKV1-39 S43A V48I Kappa (SEQ ID NO: 92)	Hu01G06 IGHV1-18 F2 Heavy (SEQ ID NO: 248)
Hu01G06-135	Hu01G06 IGKV1-39 S43A V48I Kappa (SEQ ID NO: 92)	Hu01G06 IGHV1-69 F1 Heavy (SEQ ID NO: 250)
Hu01G06-138	Hu01G06 IGKV1-39 S43A V48I Kappa (SEQ ID NO: 92)	Hu01G06 IGHV1-69 F2 Heavy (SEQ ID NO: 252)

TABLE 14-continued

Antibody Name	Light Chain Variable Region	Heavy Chain Variable Region
Hu01G06-153	Hu01G06 IGKV1-39 V48I Kappa (SEQ ID NO: 94)	Ch01G06 Chimeric Heavy (SEQ ID NO: 40)
Hu01G06-69	Hu01G06 IGKV1-39 V48I Kappa (SEQ ID NO: 94)	Hu01G06 IGHV1-18 Heavy (SEQ ID NO: 54)
Hu01G06-75	Hu01G06 IGKV1-39 V48I Kappa (SEQ ID NO: 94)	Hu01G06 IGHV1-69 Heavy (SEQ ID NO: 56)
HU01G06-105	Hu01G06 IGKV1-39 V48I Kappa (SEQ ID NO: 94)	Sh01G06 IGHV1-18 M69L Heavy (SEQ ID NO: 58)
Hu01G06-107	Hu01G06 IGKV1-39 V48I Kappa (SEQ ID NO: 94)	Sh01G06 IGHV1-18 M69L K64Q G44S Heavy (SEQ ID NO: 60)
Hu01G06-106	Hu01G06 IGKV1-39 V48I Kappa (SEQ ID NO: 94)	Sh01G06 IGHV1-18 M69L K64Q Heavy (SEQ ID NO: 62)
Hu01G06-108	Hu01G06 IGKV1-39 V48I Kappa (SEQ ID NO: 94)	Sh01G06 IGHV1-69 T30S I69L Heavy (SEQ ID NO: 64)
Hu01G06-109	Hu01G06 IGKV1-39 V48I Kappa (SEQ ID NO: 94)	Sh01G06 IGHV1-69 T30S K64Q I69L Heavy (SEQ ID NO: 66)
Hu01G06-154	Hu01G06 IGKV1-39 F2 Kappa (SEQ ID NO: 254)	Ch01G06 Chimeric Heavy (SEQ ID NO: 40)
Hu01G06-155	Hu01G06 IGKV1-39 F2 Kappa (SEQ ID NO: 254)	Hu01G06 IGHV1-18 Heavy (SEQ ID NO: 54)
Hu01G06-156	Hu01G06 IGKV1-39 F2 Kappa (SEQ ID NO: 254)	Hu01G06 IGHV1-69 Heavy (SEQ ID NO: 56)
Hu01G06-157	Hu01G06 IGKV1-39 F2 Kappa (SEQ ID NO: 254)	Sh01G06 IGHV1-18 M69L Heavy (SEQ ID NO: 58)
Hu01G06-158	Hu01G06 IGKV1-39 F2 Kappa (SEQ ID NO: 254)	Sh01G06 IGHV1-18 M69L K64Q G44S Heavy (SEQ ID NO: 60)
Hu01G06-159	Hu01G06 IGKV1-39 F2 Kappa (SEQ ID NO: 254)	Sh01G06 IGHV1-18 M69L K64Q Heavy (SEQ ID NO: 62)
Hu01G06-160	Hu01G06 IGKV1-39 F2 Kappa (SEQ ID NO: 254)	Sh01G06 IGHV1-69 T30S I69L Heavy (SEQ ID NO: 64)
Hu01G06-161	Hu01G06 IGKV1-39 F2 Kappa (SEQ ID NO: 254)	Su01G06 IGHV1-69 T30S K64Q I69L Heavy (SEQ ID NO: 66)
Hu01G06-130	Hu01G06 IGKV1-39 F2 Kappa (SEQ ID NO: 254)	Hu01G06 IGHV1-18 F1 Heavy (SEQ ID NO: 246)
Hu01G06-127	Hu01G06 IGKV1-39 F2 Kappa (SEQ ID NO: 254)	Hu01G06 IGHV1-18 F2 Heavy (SEQ ID NO: 248)
Hu01G06-143	Hu01G06 IGKV1-39 F2 Kappa (SEQ ID NO: 254)	Hu01G06 IGHV1-69 F1 Heavy (SEQ ID NO: 250)
Hu01G06-146	Hu01G06 IGKV1-39 F2 Kappa (SEQ ID NO: 254)	Hu01G06 IGHV1-69 F2 Heavy (SEQ ID NO: 252)

[0298] Each of the possible combinations of the chimeric or humanized 06C11 immunoglobulin heavy chain and immunoglobulin light chain variable regions is set forth below in Table 15.

TABLE 15

Antibody Name	Light Chain Variable Region	Heavy Chain Variable Region
Hu06C11-1	Ch06C11 Chimeric Kappa (SEQ ID NO: 82)	Ch06C11 Chimeric Heavy (SEQ ID NO: 46)
Hu06C11-7	Ch06C11 Chimeric Kappa (SEQ ID NO: 82)	HE LM 06C11 IGHV2-70 Heavy (SEQ ID NO: 68)
Hu06C11-10	Ch06C11 Chimeric Kappa (SEQ ID NO: 82)	Hu06C11 IGHV2-5 Heavy (SEQ ID NO: 70)
Hu06C11-12	Sh06C11 IGKV1-16 Kappa (SEQ ID NO: 96)	Ch06C11 Chimeric Heavy (SEQ ID NO: 46)
Hu06C11-27	Sh06C11 IGKV1-16 Kappa (SEQ ID NO: 96)	HE LM 06C11 IGHV2-70 Heavy (SEQ ID NO: 68)
Hu06C11-30	Sh06C11 IGKV1-16 Kappa (SEQ ID NO: 96)	Hu06C11 IGHV2-5 Heavy (SEQ ID NO: 70)

[0299] Each of the possible combinations of the chimeric or humanized 14F11 immunoglobulin heavy chain and immunoglobulin light chain variable regions is set forth below in Table 16.

TABLE 16

Antibody Name	Light Chain Variable Region	Heavy Chain Variable Region
Hu14F11-1	Ch14F11 Chimeric Kappa (SEQ ID NO: 86)	Ch14F11 Chimeric Heavy (SEQ ID NO: 50)
Hu14F11-14	Ch14F11 Chimeric Kappa (SEQ ID NO: 86)	Sh14F11 IGHV2-5 Heavy (SEQ ID NO: 72)
Hu14F11-15	Ch14F11 Chimeric Kappa (SEQ ID NO: 86)	Sh14F11 IGHV2-70 Heavy (SEQ ID NO: 74)
Hu14F11-11	Hu14F11 IGKV1-16 Kappa (SEQ ID NO: 98)	Ch14F11 Chimeric Heavy (SEQ ID NO: 50)
Hu14F11-39	Hu14F11 IGKV1-16 Kappa (SEQ ID NO: 98)	Sh14F11 IGHV2-5 Heavy (SEQ ID NO: 72)
Hu14F11-47	Hu14F11 IGKV1-16 Kappa (SEQ ID NO: 98)	Sh14F11 IGHV2-70 Heavy (SEQ ID NO: 74)

[0300] Each of the possible combinations of the chimeric 04F08, 06C11, and 14F11 immunoglobulin heavy chain and immunoglobulin light chain variable regions is set forth below in Table 17.

TABLE 17

Light Chain Variable Region	Heavy Chain Variable Region
04F08 Chimeric Kappa (SEQ ID NO: 80)	Ch06C11 Chimeric Heavy (SEQ ID NO: 46)
04F08 Chimeric Kappa (SEQ ID NO: 80)	Ch14F11 Chimeric Heavy (SEQ ID NO: 50)
Ch06C11 Chimeric Kappa (SEQ ID NO: 82)	04F08 Chimeric Heavy (SEQ ID NO: 44)
Ch06C11 Chimeric Kappa (SEQ ID NO: 82)	Ch14F11 Chimeric Heavy (SEQ ID NO: 50)
Ch14F11 Chimeric Kappa (SEQ ID NO: 86)	04F08 Chimeric Heavy (SEQ ID NO: 44)
Ch14F11 Chimeric Kappa (SEQ ID NO: 86)	Ch06C11 Chimeric Heavy (SEQ ID NO: 46)
Light Chain Variable Region (SEQ ID NO: 86)	Heavy Chain Variable Region (SEQ ID NO: 46)

[0301] Each of the possible combinations of the chimeric 01G06 and chimeric 08G01 immunoglobulin heavy chain and immunoglobulin light chain variable regions is set forth below in Table 18.

TABLE 18

Light Chain Variable Region	Heavy Chain Variable Region
Ch01G06 Chimeric Kappa (SEQ ID NO: 76)	08G01 Chimeric Heavy (SEQ ID NO: 48)
08G01 Chimeric Kappa (SEQ ID NO: 84)	Ch01G06 Chimeric Heavy (SEQ ID NO: 40)

[0302] The nucleic acid sequences and the encoded protein sequences defining variable regions of the chimeric and humanized 01G06, 06C11, and 14F11 antibodies are summarized below (amino terminal signal peptide sequences are not shown). CDR sequences (Kabat definition) are shown in bold and are underlined in the amino acid sequences.

Nucleic Acid Sequence Encoding the Ch01G06 Chimeric Heavy Chain Variable Region (SEQ ID NO:127)

```

1  gaagtgttgt tgcagcagtc agggccggag ttggtaaaaa cgggagcgtc ggtgaaaatc
61  ccgtgcaaag cgctggggta tacgtttacg gactataaca tggattgggt gaaacagtcg
121 catgggaaat cgcttgaatg gattggtcag atcaatccga ataattggag aatcttcttt
181 aatcagaagt ttaaaggaaa agcgacgctt acagtcgata agtcgctgaa caccggcgtt
241 atggaagtac ggtcgcttac gtcggaagat acggcgggtc attactgtgc gagggaggcg
301 attacgacgg tgggagcgat ggactattgg ggacaaggga cgctcggtcac ggtatcgctc

```

Protein Sequence Defining the Ch01G06 Chimeric Heavy Chain Variable Region (SEQ ID NO:40)

```

1  evllqsgspe lvkpgasvki pckasgytft dynmdwvkqs hgkslewig inpnnggiff
61 nqkfkqkatl tvdkssntaf mevrsltsed tavyycare ittvgamdyw gggtstvtss

```

Nucleic Acid Sequence Encoding the Hu01G06 IGHV1-18 Heavy Chain Variable Region (SEQ ID NO:53)

```

1  caagtgcaac ttgtgcagtc ggggtcggaa gtcaaaaagc cgggagcgtc ggtgaaagta
61  tcgtgtaaag cgctgggata tacgtttacg gactataaca tggactgggt acgacaggca
121 ccggggaaat cgttggaatg gatcgacagc attaatccga acaatggggg aattttcttt
181 aatcagaaat tcaaaggacg ggcgacgttg acggtcgata catcgacgaa tacggcgtat
241 atggaattga ggtcgcttcg ctcggacgat acggcgggtc attactgcgc cagggaggcg
301 atcacgacgg taggggagat ggattattgg ggacagggga cgcttgtgac ggtatcgctc

```

Protein Sequence Defining the Hu01G06 IGHV1-18 Heavy Chain Variable Region (SEQ ID NO:54)

```

1  qvqlvqsgae vkkpgasvkv sckasgytft dynmdwvrqa pgkslewig inpnnggiff
61 nqkfkgrat1 tvdtstntay melrslrsdd tavyycare ittvgamdyw gggtltvtss

```

Nucleic Acid Sequence Encoding the Hu01G06 IGHV1-69
Heavy Chain Variable Region (SEQ ID NO:55)

```
1   caagtccagc ttgtccagtc gggagcggaa gtgaagaaac cggggtcgtc ggtcaaagta
61  tcgtgtaaag cgtcggggata tacgtttacg gactataaca tggattgggt acgacaggct
121 ccgggaaaat cattggaatg gattggacag attaatacca ataatggggg tatcttcttt
181 aatcaaaagt ttaaaggag ggcgacgttg acggtggaca aatcgacaaa tacggcgtat
241 atggaattgt cgtcgcttcg gtcggaggac acggcgggtg attactgcgc gagggaggcg
301 atcacgacgg tcggggcgat ggattattgg ggacagggaa cgcttgtagc ggtatcgtag
```

Protein Sequence Defining the Hu01G06 IGHV1-69 Heavy
Chain Variable Region (SEQ ID NO:56)

```
1   qvqlvqsgae vkkgpssvkv sckasgytft dynmdwvrqa pgkslewig inpnnggiff
61  nqkfkgratrl tvdkstntay melsslrsed tavyycarea ittvgamdyw gggtlvtvss
```

Nucleic Acid Sequence Encoding the Sh01G06 IGHV1-18
M69L Heavy Chain Variable Region (SEQ ID NO:57)

```
1   caggtccagc ttgtgcaatc gggagcggaa gtgaagaaac cgggagcgtc ggtaaaagtc
61  tcgtgcaaag cgtcggggta tacgtttacg gactataaca tggactgggt gcgccaagcg
121 cctggacagg gtcttgaatg gatggggcag attaatacca ataatggagg gatcttcttt
181 aatcagaaat tcaaaggaag ggtaacgctg acgacagaca cgtcaacatc gacggcctat
241 atggaattgc ggctgttcg atcagatgat acggcgggtc actattgtgc gagggaggcg
301 attacgacgg tgggagcgat ggattattgg ggacagggga cgttggtaac ggtatcgtag
```

Protein Sequence Defining the Sh01G06 IGHV1-18 M69L
Heavy Chain Variable Region (SEQ ID NO:58)

```
1   qvqlvqsgae vkkgpaskvk sckasgytft dynmdwvrqa pgqglewmgg inpnnggiff
61  nqkfkgrrvtl ttdtststay melrslrsdd tavyycarea ittvgamdyw gggtlvtvss
```

Nucleic Acid Sequence Encoding the Sh01G06 IGHV1-18
M69L K64Q G44S Heavy Chain Variable Region (SEQ ID
NO:59)

```
1   caggtccagc ttgtgcaatc gggagcggaa gtgaagaaac cgggagcgtc ggtaaaagtc
61  tcgtgcaaag cgtcggggta tacgtttacg gactataaca tggactgggt gcgccaagcg
121 cctggacaga gccttgaatg gatggggcag attaatacca ataatggagg gatcttcttt
181 aatcagaaat tccaggaag ggtaacgctg acgacagaca cgtcaacatc gacggcctat
241 atggaattgc ggctgttcg atcagatgat acggcgggtc actattgtgc gagggaggcg
301 attacgacgg tgggagcgat ggattattgg ggacagggga cgttggtaac ggtatcgtag
```

Protein Sequence Defining the Sh01G06 IGHV1-18 M69L
K64Q G44S Heavy Chain Variable Region (SEQ ID NO:60)

```
1   qvqlvqsgae vkkgpaskvk sckasgytft dynmdwvrqa pgqglewmgg inpnnggiff
61  nqkfkgrrvtl ttdtststay melrslrsdd tavyycarea ittvgamdyw gggtlvtvss
```

Nucleic Acid Sequence Encoding the Sh01G06 IGHV1-18
M69L K64Q Heavy Chain Variable Region (SEQ ID
NO:61)

```
1  caggtccagc ttgtgcaatc gggagcggaa gtgaagaaac cgggagcgtc ggtaaaagtc
61  tcgtgcaaag cgtcggggta tacgtttacg gactataaca tggactgggt gcgccaagcg
121  cctggacagg gtcttgaatg gatggggcag attaatecga ataatggagg gatcttcttt
181  aatcagaaat tccaggggaag ggtaacgctg acgacagaca cgtcaacatc gacggcctat
241  atggaattgc ggtcgttgcg atcagatgat acggcggctc actattgtgc gagggaggcg
301  attacgacgg tgggagcgat ggattattgg ggacagggga cgttggtaac ggtatcgtcg
```

Protein Sequence Defining the Sh01G06 IGHV1-18 M69L
K64Q Heavy Chain Variable Region (SEQ ID NO:62)

```
1  qvqlvqsgae vkkgpvasvk sckasgytft dynmdwvrqa pgqglewmgg inpnnggiff
61  nqkfqgrvtl tdtststay melrslrdd tavvycarea ittvgamdyw gqgtlvtvss
```

Nucleic Acid Sequence Encoding the Sh01G06 IGHV1-69
T30S I69L Heavy Chain Variable Region (SEQ ID NO:63)

```
1  caagtacagc ttgtacagtc gggagcggaa gtcaagaaac cgggatcgtc ggtcaaagtg
61  tcgtgtaaag cgtcggggata tacgtttagc gactataaca tggattgggt gcgacaagcg
121  cctgggcagg gacttgaatg gatgggtcag atcaatccga ataatggggg aatctttttc
181  aatcagaagt ttaaagggag ggtaacgctg acggcggata aaagcacgtc aacggcgtat
241  atggagtgtg cgtcgttgcg gtcggaggac acggcggctc attactgcgc gaggggaagcg
301  attacgacgg tgggagcgat ggattattgg gggcagggaa cgcttgtaac ggtgtcatcg
```

Protein Sequence Defining the Sh01G06 IGHV1-69 T30S
I69L Heavy Chain Variable Region (SEQ ID NO:64)

```
1  qvqlvqsgae vkkgpssvk sckasgytfs dynmdwvrqa pgqglewmgg inpnnggiff
61  nqkfkgrvtl tadkststay melsslrsed tavvycarea ittvgamdyw gqgtlvtvss
```

Nucleic Acid Sequence Encoding the Sh01G06 IGHV1-69
T30S K64Q I69L Heavy Chain Variable Region (SEQ ID
NO:65)

```
1  caagtacagc ttgtacagtc gggagcggaa gtcaagaaac cgggatcgtc ggtcaaagtg
61  tcgtgtaaag cgtcggggata tacgtttagc gactataaca tggattgggt gcgacaagcg
121  cctgggcagg gacttgaatg gatgggtcag atcaatccga ataatggggg aatctttttc
181  aatcagaagt ttcaggggag ggtaacgctg acggcggata aaagcacgtc aacggcgtat
241  atggagtgtg cgtcgttgcg gtcggaggac acggcggctc attactgcgc gaggggaagcg
301  attacgacgg tgggagcgat ggattattgg gggcagggaa cgcttgtaac ggtgtcatcg
```

Protein Sequence Defining the Sh01G06 IGHV1-69 T30S
K64Q I69L Heavy Chain Variable Region (SEQ ID NO:66)

1 qvqlvqsgae vkkgpgssvk sckasgytfs dynmdwvrqa pgqglewmqg inpnnngiff
61 nqkfqgrvtl tadkststay melsslrse davyycarea ittvgamdyw gggtlvtvss

Nucleic Acid Sequence Encoding the Hu01G06 IGHV1-18
F1 Heavy Chain Variable Region (SEQ ID NO:245)

1 cagggtccagc ttgtgcaatc gggagcggaa gtgaagaaac cgggagcgtc ggtaaaagtc
61 tcgtgcaaag cgtcggggta tacgtttacg gactataaca tggactgggt gcgccaagcg
121 cctggacaga gccttgaatg gatggggcag attaatacgt acaatcacct gatcttcttt
181 aatcagaaat tccaggaag ggtaacgctg acgacagaca cgtcaacatc gacggcctat
241 atggaattgc ggtcgttgcg atcagatgat acggcggtct actattgtgc gagggaggcg
301 attacgacgg tgggagcgat ggattattgg ggacagggga cgttggtaac ggtatcgtcg

Protein Sequence Defining the Hu01G06 IGHV1-18 F1
Heavy Chain Variable Region (SEQ ID NO:246)

1 qvqlvqsgae vkkgpgasvk sckasgytft dynmdwvrqa pgqglewmqg inpynhliff
61 nqkfqgrvtl ttdtststay melrslrse davyycarea ittvgamdyw gggtlvtvss

Nucleic Acid Sequence Encoding the Hu01G06 IGHV1-18
F2 Heavy Chain Variable Region (SEQ ID NO:247)

1 cagggtccagc ttgtgcaatc gggagcggaa gtgaagaaac cgggagcgtc ggtaaaagtc
61 tcgtgcaaag cgtcggggta tacgtttacg gactataaca tggactgggt gcgccaagcg
121 cctggacaga gccttgaatg gatggggcag attaatacga ataatggact gatcttcttt
181 aatcagaaat tccaggaag ggtaacgctg acgacagaca cgtcaacatc gacggcctat
241 atggaattgc ggtcgttgcg atcagatgat acggcggtct actattgtgc gagggaggcg
301 attacgacgg tgggagcgat ggattattgg ggacagggga cgttggtaac ggtatcgtcg

Protein Sequence Defining the Hu01G06 IGHV1-18 F2
Heavy Chain Variable Region (SEQ ID NO:248)

1 qvqlvqsgae vkkgpgasvk sckasgytft dynmdwvrqa pgqglewmqg inpnnliff
61 nqkfqgrvtl ttdtststay melrslrse davyycarea ittvgamdyw gggtlvtvss

Nucleic Acid Sequence Encoding the Hu01G06 IGHV1-69
F1 Heavy Chain Variable Region (SEQ ID NO:249)

1 caagtacagc ttgtacagtc gggagcggaa gtcaagaaac cgggacgtc ggtcaaagtg
61 tcgtgtaaa cgtcgggata tacgtttacg gactataaca tggattgggt gcgacaagcg
121 cctgggcagg gacttgaatg gatgggtcag atcaatccga ataatgggt gatctttttc
181 aatcagaagt ttaaggagg ggtaacgctg acggcggata aaagcacgtc aacggcgtat
241 atggagttgt cgtcgttgcg gtcggaggac acggcggtct attactgcgc gagggaagcg
301 attacgacgg tgggagcgat ggattattgg gggcaggga cgttggtaac ggtgtcatcg

Protein Sequence Defining the Hu01G06 IGHV1-69 F1
Heavy Chain Variable Region (SEQ ID NO:250)

1 qvqlvqsgae vkkgpgssvkv sckasgytfs dynmdwvrqa pgqglewmqg inpnnqliff
61 nqkfkgrrvtl tadkststay melsslrse tavyycarea ittvgamdyw gggtlvtvss

Nucleic Acid Sequence Encoding the Hu01G06 IGHV1-69
F2 Heavy Chain Variable Region (SEQ ID NO:251)

1 caagtacagc ttgtacagtc gggagcggaa gtcaagaaac cgggatcgtc ggtcaaagtg
61 tcgtgtaaag cgctcgggata tacgttttagc gactataaca tggattgggt gcgacaagcg
121 cctgggcagg gacttgaatg gatgggtcag atcaatccgt acaatcacct gatctttttc
181 aatcagaagt ttaaagggag ggtaacgctg acggcggata aaagcacgtc aacggcgtat
241 atggagttgt cgctcgttgcg gtcggaggac acggcggctc attactgcgc gagggaagcg
301 attacgacgg tgggagcgat ggattattgg gggcagggaa cgcttgtaac ggtgtcatcg

Protein Sequence Defining the Hu01G06 IGHV1-69 F2
Heavy Chain Variable Region (SEQ ID NO:252)

1 qvqlvqsgae vkkgpgssvkv sckasgytfs dynmdwvrqa pgqglewmqg inpynhliff
61 nqkfkgrrvtl tadkststay melsslrse tavyycarea ittvgamdyw gggtlvtvss

Nucleic Acid Sequence Encoding the Ch06C11 Chimeric
Heavy Chain Variable Region (SEQ ID NO:129)

1 caggtgacac tcaaagaatc aggaccgga atccttcagc ccagccagac cttgtcgtcg
61 acttgttcgt tctccggttt cagcctgaat acttatggga tgggtgtgtc atggatcagg
121 caaccgtccg ggaaggatt ggagtggctc gcgcacatct actgggacga tgacaaaagc
181 tacaatcctt cgctgaagag ccgattgacg atttccaagg atgcctcgaa caaccgggta
241 tttcttaaga tcacgtcggc cgatacggca gacacggcga cctattactg cgcccaaaga
301 gggtacgatg actattgggg atattggggc caggggacac tcgtcacaat ttcagct

Protein Sequence Defining the Ch06C11 Chimeric Heavy
Chain Variable Region (SEQ ID NO:46)

1 qvtlkesgpg ilqpsqtllsl tcsfsgfsln tygmgswir qpsgglewl ahiywddkr
61 ynpslksrlt iskdasnrv flkitsvda dtatycaqr gyddywywg qgtlvtisa

Nucleic Acid Sequence Encoding the HE LM 06C11
IGHV2-70 Heavy Chain Variable Region (SEQ ID NO:67)

1 caggtgactt tgaagaatc cggccccga ttggtaaagc caaccagac acttacgctc
61 acatgtacat tttccggatt cagcttgaac acttacggga tgggagtgtc gtggattcgg
121 caacctccg ggaaggctct ggagtggctg gcgcacatct actgggatga tgacaaaagg
181 tataaccct cacttaaac gagactgacg atctcgaagg acacaagcaa gaatcaggtc
241 gtctcacga ttacgaatgt agaccgggtg gatactgccg tetattactg cgcgcaacgc
301 gggtatgatg actactgggg atattggggc cagggcacc tcgtgaccat ctgcgtca

Protein Sequence Defining the HE LM 06C11 IGHV2-70
Heavy Chain Variable Region (SEQ ID NO:68)

1 qvtlkesgpa lvkptqtltl tctfsgfsln tygmgvswir qppgkalewl ahiywdddkr
61 ynpslksrlt iskdt sknqv vltitnvpv dtavyycar gyddywgywg qgtltvtiss

Nucleic Acid Sequence Encoding the Hu06C11 IGHV2-5
Heavy Chain Variable Region (SEQ ID NO:69)

1 caagtaacgc tcaaggagtc cggacccacc ttggtgaagc cgacgcagac cttgactctt
61 acgtgcactt tctcgggggtt ttcactgaat acgtacggga tgggtgtctc atggatcagg
121 caacctccgg gaaaggatt ggaatggctg gcgcacatct actgggatga cgataagaga
181 tataacccaa gcctcaagtc gcggctcacc attacaaaag atacatcgaa aaatcaggtc
241 gtacttacta tcacgaacat ggaccccggtg gacacagcaa catattactg tgcccagcgc
301 ggctatgacg attattgggg ttactgggga cagggaaacac tggtcacggt gtccagc

Protein Sequence Defining the Hu06C11 IGHV2-5 Heavy
Chain Variable Region (SEQ ID NO:70)

1 qvtlkesgpt lvkptqtltl tctfsgfsln tygmgvswir qppgkglewl ahiywdddkr
61 ynpslksrlt itkdt sknqv vltitnmdpv dtatyycar gyddywgywg qgtltvtvss

Nucleic Acid Sequence Encoding the Ch14F11 Chimeric
Heavy Chain Variable Region (SEQ ID NO:131)

1 caggtcacgc tgaagagtc aggtcccgga atccttcaac ctcgcagac attgtcactc
61 acatgttctt tctccgggtt ctcgctctcg acttatggca tgggtgtagg atggattcgg
121 cagcccagcg ggaaggggct tgagtgggtg gcggatatct ggtgggacga cgacaaatac
181 tacaatccga gcctgaagtc ccgcctcacc atttcgaaag atacgtcacc aaacgaagtc
241 tttttgaaga tcgccatcgt ggacacggcg gatacagcga cgtattactg cgccagaagg
301 ggacactaca gcgcaatgga ttattgggga caggggacct cggtgactgt gtcgtcc

Protein Sequence Defining the Ch14F11 Chimeric Heavy
Chain Variable Region (SEQ ID NO:50)

1 qvtlkesgpg ilqpsqtlsl tcsfsgfsls tygmvgvwir qpsgkglewl adiwdddky
61 ynpslksrlt iskdtssnev flkiaivdta dtatyycar ghysamdywg qgtsvtvss

Nucleic Acid Sequence Encoding the Sh14F11 IGHV2-5
Heavy Chain Variable Region (SEQ ID NO:71)

1 cagatcactt tgaagaaaag cggaccgacc ttggtcaagc ccacacaaac cctcacgctc
61 acgtgtacat ttccgggggtt ctcgctttca acttacggga tgggagtagg gtggattcgc
121 cagccgctcg gtaaagcgtt ggagtggctt gcagacatct ggtgggacga cgataagtac
181 tataatccct cgctcaagtc cagactgacc atcacgaaag atacgagcaa gaaccaggtc
241 gtgctgacaa tgactaacat ggacccagtg gatacggcta catattactg cgccaggcgg
301 ggtcactact cagcgatgga ttattggggc cagggaaacac tggtaacggt gtcgtcc

Protein Sequence Defining the Sh14F11 IGHV2-5 Heavy
Chain Variable Region (SEQ ID NO:72)

1 qitlkesgpt lvkptqtltl tctfsgfsls tygmvgwir qppgkalewl adiwwdddky
61 ynpslksrlt itkdtsknqv vltmtndpv dtatyycarr ghysamdywg qgtlvtvss

Nucleic Acid Sequence Encoding the Sh14F11 IGHV2-70
Heavy Chain Variable Region (SEQ ID NO:73)

1 caagtgactc tcaaggagtc cggacccgcc ctggtcaaac caacgcagac actgacgctc
61 acatgcacct tcagcggatt ttcgttgta acgtacggca tgggtgtggg gtggattcgc
121 cagcctccgg gaaagccct tgaatggttg gcggacatct ggtgggatga tgacaagtac
181 tataatccct cacttaagtc acggttgacg atctcgaaag acaccagcaa gaaccaggta
241 gtgctgacaa tgactaacat ggaccggtc gatacagcgg tctactattg tgctagaagg
301 ggacactact ccgcaatgga ttattggggt caggggacgc tcgtaaccgt gtcgtcg

Protein Sequence Defining the Sh14F11 IGHV2-70 Heavy
Chain Variable Region (SEQ ID NO:74)

1 qvtlkesgpa lvkptqtltl tctfsgfsls tygmvgwir qppgkalewl adiwwdddky
61 ynpslksrlt iskdtsknqv vltmtndpv dtavyycarr ghysamdywg qgtlvtvss

Nucleic Acid Sequence Encoding the Ch01G06 Chimeric
Kappa Chain Variable Region (SEQ ID NO:133)

1 gacatccaaa tgaccagtc acccgcgagc ctttcggcgt cggtcggaga aacggtcacg
61 atcagctgcc ggacatcaga gaatctccat aactacctcg cgtggtatca acagaagcag
121 gggaagtgcg cccagttgct tgtatacgat gcgaaaacgt tggcggatgg ggtgccgtcc
181 agattctcgg gatcgggctc ggggaocgag tactcgctca agatcaattc gctgcagcgg
241 gaggactttg ggtcgtaacta ttgtcagcat ttttggtcat caccgtatac atttgagggt
301 ggaacgaaac ttgagattaa g

Protein Sequence Defining the Ch01G06 Chimeric Kappa
Chain Variable Region (SEQ ID NO:76)

1 diqmtqspas lsasvgetvt itcrtsenlh nylawyqqkq gkspqllvyd aktladgvps
61 rfsqsgsgtgq yslkinslqp edfgyycqh fwsspytfpg gtleik

Nucleic Acid Sequence Encoding the Hu01G06 IGKV1-39
Kappa Chain Variable Region (SEQ ID NO:89)

1 gacatccaaa tgaccagtc gccgtcgctg ctttcagcgt cggtagggga tcgggtcaca
61 attacgtgcc gaacgtcaga gaatttgcac aactacctcg cgtggtatca gcagaagccc
121 gggaagtcac cgaaactcct tgtctacgat gcgaaaacgc tggcggatgg agtgccgtcg
181 agattctcgg gaagcggatc cggtaaggac tatacgttta cgatctcacc gctccagccc
241 gaggactttg cgacgtacta ttgtcagcat ttttggtcgt cgccctacac atttgggcag
301 gggaaccaagt tggaaatcaa g

Protein Sequence Defining the Hu01G06 IGKV1-39 Kappa Chain Variable Region (SEQ ID NO:90)

```

1  diqmtqspss lsasvgdrvt itcrtsenlh nylawyqqkp gkspkllvyd aktladgvps
61  rfsqsgsgtd ytlitisslqp edfatyyqgh fwsspytfgq gtkleik

```

Nucleic Acid Sequence Encoding the Hu01G06 IGKV1-39 S43A V48I Kappa Chain Variable Region (Also Referred to Herein as Hu01G06 IGKV1-39 F1 Kappa Chain Variable Region; SEQ ID NO:91)

```

1  gacatccaaa tgaccagtc gccgtcgctg ctttcagcgt cggtagggga tcgggtcaca
61  attacgtgcc gaacgtcaga gaatttgcac aactacctcg cgtggtatca gcagaagccc
121  gggaaggccc cgaaactcct tatctacgat gcgaaaacgc tggcggatgg agtgccgctcg
181  agattctcgg gaagcggatc cggtagcgac tatacgttta cgatctcacc gctccagccc
241  gaggactttg cgacgtacta ttgtcagcat ttttggtcgt cgcctacac atttgggcag
301  gggaccaagt tggaaatcaa g

```

Protein Sequence Defining the Hu01G06 IGKV1-39 S43A V48I Kappa Chain Variable Region (Also Referred to Herein as Hu01G06 IGKV1-39 F1 Kappa Chain Variable Region; SEQ ID NO:92)

```

1  diqmtqspss lsasvgdrvt itcrtsenlh nylawyqqkp gkapklliyd aktladgvps
61  rfsqsgsgtd ytlitisslqp edfatyyqgh fwsspytfgq gtkleik

```

Nucleic Acid Sequence Encoding the Hu01G06 IGKV1-39 V48I Kappa Chain Variable Region (SEQ ID NO:93)

```

1  gacatccaaa tgaccagtc gccgtcgctg ctttcagcgt cggtagggga tcgggtcaca
61  attacgtgcc gaacgtcaga gaatttgcac aactacctcg cgtggtatca gcagaagccc
121  gggaagtcac cgaaactcct tatctacgat gcgaaaacgc tggcggatgg agtgccgctcg
181  agattctcgg gaagcggatc cggtagcgac tatacgttta cgatctcacc gctccagccc
241  gaggactttg cgacgtacta ttgtcagcat ttttggtcgt cgcctacac atttgggcag
301  gggaccaagt tggaaatcaa g

```

Nucleic Acid Sequence Encoding the Hu01G06 IGKV1-39 V48I Kappa Chain Variable Region (SEQ ID NO:94)

```

1  diqmtqspss lsasvgdrvt itcrtsenlh nylawyqqkp gkspklliyd aktladgvps
61  rfsqsgsgtd ytlitisslqp edfatyyqgh fwsspytfgq gtkleik

```

Nucleic Acid Sequence Encoding the Full Length Hu01G06 IGKV1-39 F1 Kappa Chain Variable Region (Also Referred to Herein as Hu01G06 IGKV1-39 S43A V48I Kappa Chain Variable Region; SEQ ID NO:91)

```

1  gacatccaaa tgaccagtc gccgtcgctg ctttcagcgt cggtagggga tcgggtcaca
61  attacgtgcc gaacgtcaga gaatttgcac aactacctcg cgtggtatca gcagaagccc
121  gggaaggccc cgaaactcct tatctacgat gcgaaaacgc tggcggatgg agtgccgctcg

```

-continued

```
181 agattctcgg gaagcggatc cggtaacggac tatacgctta cgatctcacc gctccagccc
241 gaggactttg cgacgtacta ttgtcagcat ttttggtcgt cgcctacac atttgggcag
301 gggaccaagt tggaaatcaa g
```

Protein Sequence Defining the Hu01G06 IGKV1-39 F1
Kappa Chain Variable Region (Also Referred to Herein as
Hu01G06 IGKV1-39 S43A V48I Kappa Chain Variable
Region; SEQ ID NO:92)

```
1 diqmtqspss lsasvgrvt itcrtsenlh nylawyqqkp gkapklliyl aktladgvps
61 rfsqsgsgtd ytlitisslqp edfatyyqgh fwsspytfgq gtleik
```

Nucleic Acid Sequence Encoding the Hu01G06 IGKV1-39
F2 Kappa Chain Variable Region (SEQ ID NO:253)

```
1 gacatccaaa tgaccagtc gccgtcgtcg ctttcagcgt cggtagggga tggggtcaca
61 attacgtgcc gaacgtcaga gaatttgcac aactacctcg cgtggtatca gcagaagccc
121 gggaagtca cgaactcct tatctacgat gcgaaaacgc tggcggatgg agtgccgtcg
181 agattctcgg gaagcggatc cggtaacggac tatacgctta cgatctcacc gctccagccc
241 gaggactttg cgacgtacta ttgtcagcat ttttggtcgg acccctacac atttgggcag
301 gggaccaagt tggaaatcaa g
```

Protein Sequence Defining the Hu01G06 IGKV1-39 F2
Kappa Chain Variable Region (SEQ ID NO:254)

```
1 diqmtqspss lsasvgrvt itcrtsenlh nylawyqqkp gkspklliyl aktladgvps
61 rfsqsgsgtd ytlitisslqp edfatyyqgh fwsspytfgq gtleik
```

Nucleic Acid Sequence Encoding the Ch06C11 Chimeric
Kappa Chain Variable Region (SEQ ID NO:135)

```
1 gatatcgtea tgaccagtc ccagaagttc atgtcaactt cagtgggaga cagagtgtcc
61 gtcacatgta aagcctcgca aaatgtggga accaacgtag cgtggttcca gcagaaacct
121 ggccaatcac cgaaggcact gatctactcg gccagctata ggtactcggg agtaccagat
181 cggtttacgg ggtcggggag cgggaacggac ttatctctca ctatttccaa tgtccagtcg
241 gaggaccttg cggaatactt ctgccagcag tataacaact atcccctcac gtttggtgct
301 ggtacaaaaat tggagttgaa g
```

Protein Sequence Defining the Ch06C11 Chimeric Kappa
Chain Variable Region (SEQ ID NO:82)

```
1 divmtqsqkf mtsvgrvrs vtckasqnv tnvawfqkp gqspkaliys asyrysgvpd
61 rftgsgsgtd fildisnvqs edlaeyfcqg ynnypltfga gtleik
```

Nucleic Acid Sequence Encoding the Sh06C11 IGKV1-16
Kappa Chain Variable Region (SEQ ID NO:95)

```
1  gacatccaaa tgaccaatc gccctcctcc ctctccgcat cagtagggga ccgcgtcaca
61  attacttgca aagcgtcgca gaacgtcgga acgaatgtgg cgtggtttca gcagaagccc
121 ggaaaagctc cgaagagctt gatctactcg gcctcatata ggtattcggg tgtgccgagc
181 cggtttagcg ggtcggggtc aggtactgat ttcacgtca caatttcac gttgcagcca
241 gaagatttcg ccacatatta ctgtcagcag tacaacaatt accctctgac gttcggccag
301 ggaaccaaac ttgagatcaa g
```

Protein Sequence Defining the Sh06C11 IGKV1-16 Kappa
Chain Variable Region (SEQ ID NO:96)

```
1  diqmtqspss lsasvgrvt itckasqnvg tnvawfqqkp gkapksliys asyrysgvps
61  rfsqsgsgtd ftltisslqp edfatyycqq ynnypltfqq gtleik
```

Nucleic Acid Sequence Encoding the Ch14F11 Chimeric
Kappa Chain Variable Region (SEQ ID NO:137)

```
1  gacatcgtga tgacacagtc acagaaattc atgtccacat ccgtcgggtga tagagtatcc
61  gtcacgtgta aggcctcgca aaacgtagga actaatgtgg cgtgggtatca acagaagcca
121 ggacagtcac ccaaagcact catctacagc ccctcatatc ggtacagcgg ggtgccggac
181 aggttcacgg gatcggggag cgggaccgat tttacactga ccatttcgaa tgtccagtcg
241 gaggaacctg cggaataact ctgccagcag tataactcgt accctcacac gtttgagggt
301 ggcactaagt tggagatgaa a
```

Protein Sequence Defining the Ch14F11 Chimeric Kappa
Chain Variable Region (SEQ ID NO:86)

```
1  divmtqsqkf mtsvgrvrs vtckasqnvg tnvawyqqkp gqspkaliys psyrysgvpd
61  rftqsgsgtd ftltisnvqs edlaeyfcqq ynsyphtfgg gtleik
```

Nucleic Acid Sequence Encoding the Hu14F11 IGKV1-16
Kappa Chain Variable Region (SEQ ID NO:97)

```
1  gatatccaga tgacacagtc accctcgtcg ctctcagctt ccgtaggcga cagggtcact
61  attacgtgta aagcatcaca gaacgtcgga acgaatgtgg cgtggtttca gcagaagccc
121 gggaagagcc ccaaagcgct tatctactcc ccgtcgtatc ggtattccgg tgtccaagc
181 agatttttcg ggtcagggtc gggaactgac tttaccctga ccatctcgtc cctccaacgg
241 gaagatttcg ccacgtactt ctgccagcag tacaacagct atcctcacac attcggacaa
301 gggaacaaag ttgagattaa a
```

Protein Sequence Defining the Hu14F11 IGKV1-16 Kappa
Chain Variable Region (SEQ ID NO:98)

```
1  diqmtqspss lsasvgrvt itckasqnvg tnvawfqqkp gkspkaliys psyrysgvps
61  rfsqsgsgtd ftltisslqp edfatyfcqq ynsyphtfgg gtleik
```

[0303] The amino acid sequences defining the immunoglobulin heavy chain variable regions for the antibodies produced in Example 13 are aligned in FIG. 19. Amino terminal signal peptide sequences (for proper expression/secretion) are not shown. CDR₁, CDR₂, and CDR₃ (Kabat definition) are identified by boxes. FIG. 20 show an alignment of the separate CDR₁, CDR₂, and CDR₃ sequences for each of the variable region sequences shown in FIG. 19.

[0304] The amino acid sequences defining the immunoglobulin light chain variable regions for the antibodies in Example 13 are aligned in FIG. 21. Amino terminal signal peptide sequences (for proper expression/secretion) are not shown. CDR₁, CDR₂ and CDR₃ are identified by boxes. FIG. 22 shows an alignment of the separate CDR₁, CDR₂, and CDR₃ sequences for each of the variable region sequences shown in FIG. 21.

[0305] Table 19 is a concordance chart showing the SEQ ID NO. of each sequence discussed in this Example.

TABLE 19

SEQ. ID NO.	Nucleic Acid or Protein
127	Ch01G06 Chimeric Heavy Chain Variable Region - nucleic acid
40	Ch01G06 Chimeric Heavy Chain Variable Region - protein
1	Ch01G06 Chimeric Heavy Chain CDR ₁
7	Ch01G06 Chimeric Heavy Chain CDR ₂
15	Ch01G06 Chimeric Heavy Chain CDR ₃
53	Hu01G06 IGHV1-18 Heavy Chain Variable Region - nucleic acid
54	Hu01G06 IGHV1-18 Heavy Chain Variable Region - protein
1	Hu01G06 IGHV1-18 Heavy Chain CDR ₁
7	Hu01G06 IGHV1-18 Heavy Chain CDR ₂
15	Hu01G06 IGHV1-18 Heavy Chain CDR ₃
55	Hu01G06 IGHV1-69 Heavy Chain Variable Region - nucleic acid
56	Hu01G06 IGHV1-69 Heavy Chain Variable Region - protein
1	Hu01G06 IGHV1-69 Heavy Chain CDR ₁
7	Hu01G06 IGHV1-69 Heavy Chain CDR ₂
15	Hu01G06 IGHV1-69 Heavy Chain CDR ₃
57	Sh01G06 IGHV1-18 M69L Heavy Chain Variable Region - nucleic acid
58	Sh01G06 IGHV1-18 M69L Heavy Chain Variable Region - protein
1	Sh01G06 IGHV1-18 M69L Heavy Chain CDR ₁
7	Sh01G06 IGHV1-18 M69L Heavy Chain CDR ₂
15	Sh01G06 IGHV1-18 M69L Heavy Chain CDR ₃
59	Sh01G06 IGHV1-18 M69L K64Q G44S Heavy Chain Variable Region - nucleic acid
60	Sh01G06 IGHV1-18 M69L K64Q G44S Heavy Chain Variable Region - protein
1	Sh01G06 IGHV1-18 M69L K64Q G44S Heavy Chain CDR ₁
13	Sh01G06 IGHV1-18 M69L K64Q G44S Heavy Chain CDR ₂
15	Sh01G06 IGHV1-18 M69L K64Q G44S Heavy Chain CDR ₃
61	Sh01G06 IGHV1-18 M69L K64Q Heavy Chain Variable Region - nucleic acid
62	Sh01G06 IGHV1-18 M69L K64Q Heavy Chain Variable Region - protein
1	Sh01G06 IGHV1-18 M69L K64Q Heavy Chain CDR ₁
13	Sh01G06 IGHV1-18 M69L K64Q Heavy Chain CDR ₂
15	Sh01G06 IGHV1-18 M69L K64Q Heavy Chain CDR ₃
63	Sh01G06 IGHV1-69 T30S I69L Heavy Chain Variable Region - nucleic acid
64	Sh01G06 IGHV1-69 T30S I69L Heavy Chain Variable Region - protein
1	Sh01G06 IGHV1-69 T30S I69L Heavy Chain CDR ₁
7	Sh01G06 IGHV1-69 T30S I69L Heavy Chain CDR ₂
15	Sh01G06 IGHV1-69 T30S I69L Heavy Chain CDR ₃
65	Sh01G06 IGHV1-69 T30S K64Q I69L Heavy Chain Variable Region - nucleic acid
66	Sh01G06 IGHV1-69 T30S K64Q I69L Heavy Chain Variable Region - protein
1	Sh01G06 IGHV1-69 T30S K64Q I69L Heavy Chain CDR ₁
13	Sh01G06 IGHV1-69 T30S K64Q I69L Heavy Chain CDR ₂
15	Sh01G06 IGHV1-69 T30S K64Q I69L Heavy Chain CDR ₃
245	Hu01G06 IGHV1-18 F1 Heavy Chain Variable Region - nucleic acid
246	Hu01G06 IGHV1-18 F1 Heavy Chain Variable Region - protein
1	Hu01G06 IGHV1-18 F1 Heavy Chain CDR ₁
236	Hu01G06 IGHV1-18 F1 Heavy Chain CDR ₂
15	Hu01G06 IGHV1-18 F1 Heavy Chain CDR ₃
247	Hu01G06 IGHV1-18 F2 Heavy Chain Variable Region - nucleic acid
248	Hu01G06 IGHV1-18 F2 Heavy Chain Variable Region - protein
1	Hu01G06 IGHV1-18 F2 Heavy Chain CDR ₁
237	Hu01G06 IGHV1-18 F2 Heavy Chain CDR ₂
15	Hu01G06 IGHV1-18 F2 Heavy Chain CDR ₃
259	Hu01G06 IGHV1-69 F1 Heavy Chain Variable Region - nucleic acid
250	Hu01G06 IGHV1-69 F1 Heavy Chain Variable Region - protein
1	Hu01G06 IGHV1-69 F1 Heavy Chain CDR ₁
238	Hu01G06 IGHV1-69 F1 Heavy Chain CDR ₂
15	Hu01G06 IGHV1-69 F1 Heavy Chain CDR ₃
251	Hu01G06 IGHV1-69 F2 Heavy Chain Variable Region - nucleic acid
252	Hu01G06 IGHV1-69 F2 Heavy Chain Variable Region - protein
1	Hu01G06 IGHV1-69 F2 Heavy Chain CDR ₁
239	Hu01G06 IGHV1-69 F2 Heavy Chain CDR ₂
15	Hu01G06 IGHV1-69 F2 Heavy Chain CDR ₃
129	Ch06C11 Chimeric Heavy Chain Variable Region - nucleic acid
46	Ch06C11 Chimeric Heavy Chain Variable Region - protein

TABLE 19-continued

SEQ. ID NO.	Nucleic Acid or Protein
4	Ch06C11 Chimeric Heavy Chain CDR ₁
9	Ch06C11 Chimeric Heavy Chain CDR ₂
18	Ch06C11 Chimeric Heavy Chain CDR ₃
67	HE LM 06C11 IGHV2-70 Heavy Chain Variable Region - nucleic acid
68	HE LM 06C11 IGHV2-70 Heavy Chain Variable Region - protein
4	HE LM 06C11 IGHV2-70 Heavy Chain CDR ₁
14	HE LM 06C11 IGHV2-70 Heavy Chain CDR ₂
18	HE LM 06C11 IGHV2-70 Heavy Chain CDR ₃
69	Hu06C11 IGHV2-5 Heavy Chain Variable Region - nucleic acid
70	Hu06C11 IGHV2-5 Heavy Chain Variable Region - protein
4	Hu06C11 IGHV2-5 Heavy Chain CDR ₁
9	Hu06C11 IGHV2-5 Heavy Chain CDR ₂
18	Hu06C11 IGHV2-5 Heavy Chain CDR ₃
131	Ch14F11 Chimeric Heavy Chain Variable Region - nucleic acid
50	Ch14F11 Chimeric Heavy Chain Variable Region - protein
5	Ch14F11 Chimeric Heavy Chain CDR ₁
11	Ch14F11 Chimeric Heavy Chain CDR ₂
19	Ch14F11 Chimeric Heavy Chain CDR ₃
71	Sh14F11 IGHV2-5 Heavy Chain Variable Region - nucleic acid
72	Sh14F11 IGHV2-5 Heavy Chain Variable Region - protein
5	Sh14F11 IGHV2-5 Heavy Chain CDR ₁
11	Sh14F11 IGHV2-5 Heavy Chain CDR ₂
19	Sh14F11 IGHV2-5 Heavy Chain CDR ₃
73	Sh14F11 IGHV2-70 Heavy Chain Variable Region - nucleic acid
74	Sh14F11 IGHV2-70 Heavy Chain Variable Region - protein
5	Sh14F11 IGHV2-70 Heavy Chain CDR ₁
11	Sh14F11 IGHV2-70 Heavy Chain CDR ₂
19	Sh14F11 IGHV2-70 Heavy Chain CDR ₃
133	Ch01G06 Chimeric Light (kappa) Chain Variable Region - nucleic acid
76	Ch01G06 Chimeric Light (kappa) Chain Variable Region - protein
21	Ch01G06 Chimeric Light (kappa) Chain CDR ₁
26	Ch01G06 Chimeric Light (kappa) Chain CDR ₂
32	Ch01G06 Chimeric Light (kappa) Chain CDR ₃
89	Hu01G06 IGKV1-39 Light (kappa) Chain Variable Region - nucleic acid
90	Hu01G06 IGKV1-39 Light (kappa) Chain Variable Region - protein
21	Hu01G06 IGKV1-39 Light (kappa) Chain CDR ₁
26	Hu01G06 IGKV1-39 Light (kappa) Chain CDR ₂
32	Hu01G06 IGKV1-39 Light (kappa) Chain CDR ₃
91	Hu01G06 IGKV1-39 S43A V48I Light (kappa) Chain Variable Region - nucleic acid
92	Hu01G06 IGKV1-39 S43A V48I Light (kappa) Chain Variable Region - protein
21	Hu01G06 IGKV1-39 S43A V48I Light (kappa) Chain CDR ₁
26	Hu01G06 IGKV1-39 S43A V48I Light (kappa) Chain CDR ₂
32	Hu01G06 IGKV1-39 S43A V48I Light (kappa) Chain CDR ₃
93	Hu01G06 IGKV1-39 V48I Light (kappa) Chain Variable Region - nucleic acid
94	Hu01G06 IGKV1-39 V48I Light (kappa) Chain Variable Region - protein
21	Hu01G06 IGKV1-39 V48I Light (kappa) Chain CDR ₁
26	Hu01G06 IGKV1-39 V48I Light (kappa) Chain CDR ₂
32	Hu01G06 IGKV1-39 V48I Light (kappa) Chain CDR ₃
91	Hu01G06 IGKV1-39 F1 Light (kappa) Chain Variable Region - nucleic acid
92	Hu01G06 IGKV1-39 F1 Light (kappa) Chain Variable Region - protein
21	Hu01G06 IGKV1-39 F1 Light (kappa) Chain CDR ₁
26	Hu01G06 IGKV1-39 F1 Light (kappa) Chain CDR ₂
32	Hu01G06 IGKV1-39 F1 Light (kappa) Chain CDR ₃
253	Hu01G06 IGKV1-39 F2 Light (kappa) Chain Variable Region - nucleic acid
254	Hu01G06 IGKV1-39 F2 Light (kappa) Chain Variable Region - protein
21	Hu01G06 IGKV1-39 F2 Light (kappa) Chain CDR ₁
26	Hu01G06 IGKV1-39 F2 Light (kappa) Chain CDR ₂
244	Hu01G06 IGKV1-39 F2 Light (kappa) Chain CDR ₃
135	Ch06C11 Chimeric Light (kappa) Chain Variable Region - nucleic acid
82	Ch06C11 Chimeric Light (kappa) Chain Variable Region - protein
23	Ch06C11 Chimeric Light (kappa) Chain CDR ₁
28	Ch06C11 Chimeric Light (kappa) Chain CDR ₂
35	Ch06C11 Chimeric Light (kappa) Chain CDR ₃
95	Sh06C11 IGKV1-16 Light (kappa) Chain Variable Region - nucleic acid
96	Sh06C11 IGKV1-16 Light (kappa) Chain Variable Region - protein
23	Sh06C11 IGKV1-16 Light (kappa) Chain CDR ₁
28	Sh06C11 IGKV1-16 Light (kappa) Chain CDR ₂
35	Sh06C11 IGKV1-16 Light (kappa) Chain CDR ₃
137	Ch14F11 Chimeric Light (kappa) Chain Variable Region - nucleic acid
86	Ch14F11 Chimeric Light (kappa) Chain Variable Region - protein
23	Ch14F11 Chimeric Light (kappa) Chain CDR ₁
30	Ch14F11 Chimeric Light (kappa) Chain CDR ₂
36	Ch14F11 Chimeric Light (kappa) Chain CDR ₃
97	Hu14F11 IGKV1-16 Light (kappa) Chain Variable Region - nucleic acid
98	Hu14F11 IGKV1-16 Light (kappa) Chain Variable Region - protein

TABLE 19-continued

SEQ. ID NO.	Nucleic Acid or Protein
23	Hu14F11 IGKV1-16 Light (kappa) Chain CDR ₁
30	Hu14F11 IGKV1-16 Light (kappa) Chain CDR ₂
36	Hu14F11 IGKV1-16 Light (kappa) Chain CDR ₃

[0306] Humanized monoclonal antibody heavy chain CDR sequences (Kabat, Chothia, and IMGT definitions) are shown in Table 20.

TABLE 20

	CDR1	CDR2	CDR3	Variable Region SEQ ID NO:
	Kabat			
Ch01G06 Chimeric	DYNMD (SEQ ID NO: 1)	QINPNNGGIFFNQKFKG (SEQ ID NO: 7)	EAITTVGAMDY (SEQ ID NO: 15)	40
Hu01G06 IGHV1-18	DYNMD (SEQ ID NO: 1)	QINPNNGGIFFNQKFKG (SEQ ID NO: 7)	EAITTVGAMDY (SEQ ID NO: 15)	54
Hu01G06 IGHV1-69	DYNMD (SEQ ID NO: 1)	QINPNNGGIFFNQKFKG (SEQ ID NO: 7)	EAITTVGAMDY (SEQ ID NO: 15)	56
Sh01G06 IGHV1-18 M69L	DYNMD (SEQ ID NO: 1)	QINPNNGGIFFNQKFKG (SEQ ID NO: 7)	EAITTVGAMDY (SEQ ID NO: 15)	58
Sh01G06 IGHV1-18 M69L K64Q G44S	DYNMD (SEQ ID NO: 1)	QINPNNGGIFFNQKFKG (SEQ ID NO: 13)	EAITTVGAMDY (SEQ ID NO: 15)	60
Sh01G06 IGHV1-18 M69L K64Q	DYNMD (SEQ ID NO: 1)	QINPNNGGIFFNQKFKG (SEQ ID NO: 13)	EAITTVGAMDY (SEQ ID NO: 15)	62
Sh01G06 IGHV1-69 T30S I69L	DYNMD (SEQ ID NO: 1)	QINPNNGGIFFNQKFKG (SEQ ID NO: 7)	EAITTVGAMDY (SEQ ID NO: 15)	64
Sh01G06 IGHV1-69 T30S K64Q I69L	DYNMD (SEQ ID NO: 1)	QINPNNGGIFFNQKFKG (SEQ ID NO: 13)	EAITTVGAMDY (SEQ ID NO: 15)	66
Hu01G06 IGHV1-18F1	DYNMD (SEQ ID NO: 1)	QINPNHLIFFNQKFKG (SEQ ID NO: 236)	EAITTVGAMDY (SEQ ID NO: 15)	246
Hu01G06 IGHV1-18F2	DYNMD (SEQ ID NO: 1)	QINPNGLIFFNQKFKG (SEQ ID NO: 237)	EAITTVGAMDY (SEQ ID NO: 15)	248
Hu01G06 IGHV1-69F1	DYNMD (SEQ ID NO: 1)	QINPNGLIFFNQKFKG (SEQ ID NO: 238)	EAITTVGAMDY (SEQ ID NO: 15)	250
Hu01G06 IGHV1-69F2	DYNMD (SEQ ID NO: 1)	QINPNHLIFFNQKFKG (SEQ ID NO: 239)	EAITTVGAMDY (SEQ ID NO: 15)	252
Ch06C11 Chimeric	TYGMGVS (SEQ ID NO: 4)	HIYWDDDKRYNPSLKS (SEQ ID NO: 9)	RGYDDYWG (SEQ ID NO: 18)	46
HE LM 06C11 IGHV2-70	TYGMGVS (SEQ ID NO: 4)	HIYWDDDKRYNPSLKT (SEQ ID NO: 14)	RGYDDYWG (SEQ ID NO: 18)	68
Hu06C11 IGHV2-5	TYGMGVS (SEQ ID NO: 4)	HIYWDDDKRYNPSLKS (SEQ ID NO: 9)	RGYDDYWG (SEQ ID NO: 18)	70
Ch14F11 Chimeric	TYGMGVG (SEQ ID NO: 5)	DIWDDDKYYNPSLKS (SEQ ID NO: 11)	RGHYSAMDY (SEQ ID NO: 19)	50
Sh14F11 IGHV2-5	TYGMGVG (SEQ ID NO: 5)	DIWDDDKYYNPSLKS (SEQ ID NO: 11)	RGHYSAMDY (SEQ ID NO: 19)	72
Sh14F11 IGHV2-70	TYGMGVG (SEQ ID NO: 5)	DIWDDDKYYNPSLKS (SEQ ID NO: 11)	RGHYSAMDY (SEQ ID NO: 19)	74

TABLE 20-continued

	CDR1	CDR2	CDR3	Variable Region SEQ ID NO:
Chothia				
Ch01G06 Chimeric	GYTFTDY (SEQ ID NO: 38)	NPNNGG (SEQ ID NO: 143)	EAITTVGAMDY (SEQ ID NO: 15)	40
Hu0106 IGHV1- 18	GYTFTDY (SEQ ID NO: 38)	NPNNGG (SEQ ID NO: 143)	EAITTVGAMDY (SEQ ID NO: 15)	54
Hu01G06 IGHV1- 69	GYTFTDY (SEQ ID NO: 38)	NPNNGG (SEQ ID NO: 143)	EAITTVGAMDY (SEQ ID NO: 15)	56
Sh01G06 IGHV1- 18 M69L	GYTFTDY (SEQ ID NO: 38)	NPNNGG (SEQ ID NO: 143)	EAITTVGAMDY (SEQ ID NO: 15)	58
Sh01G06 IGHV1- 18 M69L K64Q G44S	GYTFTDY (SEQ ID NO: 38)	NPNNGG (SEQ ID NO: 143)	EAITTVGAMDY (SEQ ID NO: 15)	60
Sh01G06 IGHV1- 18 M69L K64Q	GYTFTDY (SEQ ID NO: 38)	NPNNGG (SEQ ID NO: 143)	EAITTVGAMDY (SEQ ID NO: 15)	62
Sh01G06 IGHV1- 69 T30S I69L	GYTFSDY (SEQ ID NO: 234)	NPNNGG (SEQ ID NO: 143)	EAITTVGAMDY (SEQ ID NO: 15)	64
Sh01G06 IGHV1- 69 T30S K64Q I69L	GYTFSDY (SEQ ID NO: 234)	NPNNGG (SEQ ID NO: 143)	EAITTVGAMDY (SEQ ID NO: 15)	66
Hu01G06 IGHV1- 18 F1	GYTFTDY (SEQ ID NO: 38)	NPYNHL (SEQ ID NO: 240)	EAITTVGAMDY (SEQ ID NO: 15)	246
Hu01G06 IGHV1- 18 F2	GYTFTDY (SEQ ID NO: 38)	NPNNGL (SEQ ID NO: 241)	EAITTVGAMDY (SEQ ID NO: 15)	248
Hu01G06 IGHV1- 69 F1	GYTFSDY (SEQ ID NO: 234)	NPNNGL (SEQ ID NO: 241)	EAITTVGAMDY (SEQ ID NO: 15)	250
Hu01G06 IGHV1- 69 F2	GYTFSDY (SEQ ID NO: 234)	NPYNHL (SEQ ID NO: 240)	EAITTVGAMDY (SEQ ID NO: 15)	252
Ch06C11 Chimeric	GFSLNTYGM (SEQ ID NO: 132)	YWDDD (SEQ ID NO: 145)	RGYDDYWGY (SEQ ID NO: 18)	46
HE LM 06C11 IGHV2-70	GFSLNTYGM (SEQ ID NO: 132)	YWDDD (SEQ ID NO: 145)	RGYDDYWGY (SEQ ID NO: 18)	68
Hu06C11 IGHV2- 5	GFSLNTYGM (SEQ ID NO: 132)	YWDDD (SEQ ID NO: 145)	RGYDDYWGY (SEQ ID NO: 18)	70
Ch14F11 Chimeric	GFSLSTYGM (SEQ ID NO: 130)	WWDDD (SEQ ID NO: 146)	RGHYSAMDY (SEQ ID NO: 19)	50
Sh14F11 IGHV2-5	GFSLSTYGM (SEQ ID NO: 130)	WWDDD (SEQ ID NO: 146)	RGHYSAMDY (SEQ ID NO: 19)	72
Sh14F11 IGHV2- 70	GFSLSTYGM (SEQ ID NO: 130)	WWDDD (SEQ ID NO: 146)	RGHYSAMDY (SEQ ID NO: 19)	74
IMGT				
Ch01G06 Chimeric	GYTFTDYN (SEQ ID NO: 136)	INPNNGGI (SEQ ID NO: 148)	AREAITTVGAMDY (SEQ ID NO: 154)	40
Hu01G06 IGHV1-18	GYTFTDYN (SEQ ID NO: 136)	INPNNGGI (SEQ ID NO: 148)	AREAITTVGAMDY (SEQ ID NO: 154)	54
Hu01G06 IGHV1-69	GYTFTDYN (SEQ ID NO: 136)	INPNNGGI (SEQ ID NO: 148)	AREAITTVGAMDY (SEQ ID NO: 154)	56
Sh01G06 IGHV1-18 M69L	GYTFTDYN (SEQ ID NO: 136)	INPNNGGI (SEQ ID NO: 148)	AREAITTVGAMDY (SEQ ID NO: 154)	58

TABLE 20-continued

	CDR1	CDR2	CDR3	Variable Region SEQ ID NO:
Sh01G06 IGHV1-18 M69L K64Q G44S	GYTFTDYN (SEQ ID NO: 136)	INPNNGGI (SEQ ID NO: 148)	AREAITTVGAMDY (SEQ ID NO: 154)	60
Sh01G06 IGHV1-18 M69L K64Q	GYTFTDYN (SEQ ID NO: 136)	INPNNGGI (SEQ ID NO: 148)	AREAITTVGAMDY (SEQ ID NO: 154)	62
Sh01G06 IGHV1-69 T30S I69L	GYTFSDYN (SEQ ID NO: 235)	INPNNGGI (SEQ ID NO: 148)	AREAITTVGAMDY (SEQ ID NO: 154)	64
Sh01G06 IGHV1-69 T30S K64Q I69L	GYTFSDYN (SEQ ID NO: 235)	INPNNGGI (SEQ ID NO: 148)	AREAITTVGAMDY (SEQ ID NO: 154)	66
Hu01G06 IGHV1-18 F1	GYTFTDYN (SEQ ID NO: 136)	INPNHLLI (SEQ ID NO: 242)	AREAITTVGAMDY (SEQ ID NO: 154)	246
Hu01G06 IGHV1-18 F2	GYTFTDYN (SEQ ID NO: 136)	INPNNGLI (SEQ ID NO: 243)	AREAITTVGAMDY (SEQ ID NO: 154)	248
Hu01G06 IGHV1-69 F1	GYTFSDYN (SEQ ID NO: 235)	INPNNGLI (SEQ ID NO: 243)	AREAITTVGAMDY (SEQ ID NO: 154)	250
Hu01G06 IGHV1-69 F2	GYTFSDYN (SEQ ID NO: 235)	INPNHLLI (SEQ ID NO: 242)	AREAITTVGAMDY (SEQ ID NO: 154)	252
Ch06C11 Chimeric	GFSLNTYGMG (SEQ ID NO: 141)	IYWDDDK (SEQ ID NO: 150)	AQRGYDDYWG (SEQ ID NO: 157)	46
HE LM 06C11 IGHV2-70	GFSLNTYGMG (SEQ ID NO: 141)	IYWDDDK (SEQ ID NO: 150)	AQRGYDDYWG (SEQ ID NO: 157)	68
Hu06C11 IGHV2-5	GFSLNTYGMG (SEQ ID NO: 141)	IYWDDDK (SEQ ID NO: 150)	AQRGYDDYWG (SEQ ID NO: 157)	70
Ch14F11 Chimeric	GFSLNTYGMG (SEQ ID NO: 140)	IWWDDDK (SEQ ID NO: 152)	ARRGHYSAMDY (SEQ ID NO: 158)	50
Sh14F11 IGHV2- 5	GFSLNTYGMG (SEQ ID NO: 140)	IWWDDDK (SEQ ID NO: 152)	ARRGHYSAMDY (SEQ ID NO: 158)	72
Sh14F11 IGHV2- 70	GFSLNTYGMG (SEQ ID NO: 140)	IWWDDDK (SEQ ID NO: 152)	ARRGHYSAMDY (SEQ ID NO: 158)	74

[0307] Humanized monoclonal antibody Kappa light chain CDR sequences (Kabat, Chothia, and IMGT definitions) are shown in Table 21.

TABLE 21

	CDR1	CDR2	CDR3	Variable Region SEQ ID NO:
Kabat/Chothia				
Ch01G06 Chimeric	RTSENHNYLA (SEQ ID NO: 21)	DAKTLAD (SEQ ID NO: 26)	QHFWSPPYT (SEQ ID NO: 32)	76
Hu01G06 IGKV1-39	RTSENHNYLA (SEQ ID NO: 21)	DAKTLAD (SEQ ID NO: 26)	QHFWSPPYT (SEQ ID NO: 32)	90
Hu01G06 IGKV1-39 S43A V48I (also known as Hu0106 IGKV1-39 F1)	RTSENHNYLA (SEQ ID NO: 21)	DAKTLAD (SEQ ID NO: 26)	QHFWSPPYT (SEQ ID NO: 32)	92
Hu01G06 IGKV1-39 V48I	RTSENHNYLA (SEQ ID NO: 21)	DAKTLAD (SEQ ID NO: 26)	QHFWSPPYT (SEQ ID NO: 32)	94

TABLE 21-continued

	CDR1	CDR2	CDR3	Variable Region SEQ ID NO:
Hu01G06 IGKV1-39 F1 (also known as Hu01G06 IGKV1-39 S43A V481)	RTSENLHNYLA (SEQ ID NO: 21)	DAKTLAD (SEQ ID NO: 26)	QHFWSSPYT (SEQ ID NO: 32)	92
Hu01G06 IGKV1-39 F2	RTSENLHNYLA (SEQ ID NO: 21)	DAKTLAD (SEQ ID NO: 26)	QHFWSDPYT (SEQ ID NO: 244)	254
Ch06C11 Chimeric	KASQNVGTNVA (SEQ ID NO: 23)	SASYRYS (SEQ ID NO: 28)	QQYNNYPLT (SEQ ID NO: 35)	82
Sh06C11 IGKV1-16	KASQNVGTNVA (SEQ ID NO: 23)	SASYRYS (SEQ ID NO: 28)	QQYNNYPLT (SEQ ID NO: 35)	96
Ch14F11 Chimeric	KASQNVGTNVA (SEQ ID NO: 23)	SPSYRYS (SEQ ID NO: 30)	QQYNSYPHT (SEQ ID NO: 36)	86
Hu14F11 IGKV1-16	KASQNVGTNVA (SEQ ID NO: 23)	SPSYRYS (SEQ ID NO: 30)	QQYNSYPHT (SEQ ID NO: 36)	98
IMGT				
Ch01G06 Chimeric	ENLHNY (SEQ ID NO: 160)	DAK	QHFWSSPYT (SEQ ID NO: 32)	76
Hu01G06 IGKV1-39	ENLHNY (SEQ ID NO: 160)	DAK	QHFWSSPYT (SEQ ID NO: 32)	90
Hu01G06 IGKV1-39 S43A V48I (also known as Hu01G06 IGKV1-39 F1)	ENLHNY (SEQ ID NO: 160)	DAK	QHFWSSPYT (SEQ ID NO: 32)	92
Hu01G06 IGKV1-39 V48I	ENLHNY (SEQ ID NO: 160)	DAK DAK	QHFWSSPYT (SEQ ID NO: 32)	94
Hu01G06 IGKV1-39 F1 (also known as Hu01G06 IGKV1-39 S43A V48I)	ENLHNY (SEQ ID NO: 160)		QHFWSSPYT (SEQ ID NO: 32)	92
Hu01G06 IGKV1-39 F2	ENLHNY (SEQ ID NO: 160)	DAK	QHFWSDPYT (SEQ ID NO: 244)	254
Ch06C11 Chimeric	QNVGTN (SEQ ID NO: 162)	SAS	QQYNNYPLT (SEQ ID NO: 35)	82
Sh06C11 IGKV1-16	QNVGTN (SEQ ID NO: 162)	SAS	QQYNNYPLT (SEQ ID NO: 35)	96
Ch14F11 Chimeric	QNVGTN (SEQ ID NO: 162)	SPS	QQYNSYPHT (SEQ ID NO: 36)	86
Hu14F11 IGKV1-16	QNVGTN (SEQ ID NO: 162)	SPS	QQYNSYPHT (SEQ ID NO: 36)	98

[0308] To create the complete chimeric and humanized heavy or kappa chain antibody sequences, each variable sequence above is combined with its respective human constant region. For example, a complete heavy chain comprises a heavy variable sequence followed by a human

IgG1 heavy chain constant sequence. A complete kappa chain comprises a kappa variable sequence followed by the human kappa light chain constant sequence.

Nucleic Acid Sequence Encoding the Human IgG1 Heavy Chain Constant Region (SEQ ID NO:171)

```

1  gcctcaacaa aaggaccaag tgtgttccca ctgcccccta gcagcaagag tacatccggg
61  ggcaactgcag cactcggtcg cctcgtcaag gattattttc cagagccagt aacctgagc
121  tggaacagtg gaggactcac ttctggtgtc catacttttc ctgetgtcct gcaaagctct

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-continued

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181  ggccgtgact cactcagctc cgtcgtgacc gtgccatctt catctctggg cactcagacc
241  tacatctgta atgtaaacca caagcctagc aatactaagg tcgataagcg ggtggaaccc
301  aagagctgcg acaagactca cacttgctcc ccatgccctg cccctgaact tctgggcggt
361  cccagcgtct ttttgttccc accaaagcct aaagatactc tgatgataag tagaacaccc
421  gaggtgacat gtgttggtgt agacgtttcc cagcaggacc cagaggttaa gttcaactgg
481  tacgttgatg gagtcgaagt acataatgct aagaccaagc ctagagagga gcagtataat
541  agtacatacc gtgtagttag ttttctcaca gtgctgcacc aagactggct caacggcaaa
601  gaatacaaat gcaaagtgtc caacaaagca ctcccagccc ctatcgagaa gactattagt
661  aaggcaaagg ggagcctcgc tgaaccacag gtgtacactc tgccaccag tagagaggaa
721  atgacaaaaga accaagtctc attgacctgc ctggtgaaag gcttctaccc cagcgacatc
781  gccgttgagt gggagagtaa cggtcagcct gagaacaatt acaagacaac cccccagtg
841  ctggatagtg acgggtcttt ctttctgtac agtaagctga ctgtggacaa gtcccgctgg
901  cagcagggta acgtcttcag ctgttcctg atgcacgagg cattgcacaa ccactacacc
961  cagaagtcac tgagcctgag cccagggaag

```

Protein Sequence Defining the Human IgG1 Heavy Chain
Constant Region (SEQ ID NO:172)

```

1  astkgpsvfp lapsskstsg gtaalgclvk dyfpepvtvs wnsqaltsgv htfpavlqss
61  glyslssvvt vpssslgtqt yicnvnhkps ntkvdkrvep kscdkthtct pcpapellgg
121  psvflfppkp kdtlmisrtp evtcvvvdvs hedpevkfnw yvdgvevhna ktkpreeqyn
181  styrvsvvlt vlhqdwlngk eykckvsnka lpapiektis kakggprepq vytlppsree
241  mtknqvsltc lvkgfypsdi avewesngqp ennykttppv ldsdgsffly skltvdksrw
301  qqgnvfscsv mhealhnhyt qkslsispqk

```

[0309] Nucleic Acid Sequence Encoding the Human
Kappa Light Chain Constant Region (SEQ ID NO:173)

```

1  cgcacagttg ctgccccag cgtgttcatt ttcccaccta gcgatgagca gctgaaaagc
61  ggtactgcct ctgtcgtatg ctgtctcaac aacttttacc cagctgaggc taaggtgcag
121  tggaaagtgg ataatgcact tcaatctgga aacagtcaag agtccgtgac agaacaggac
181  agcaaagact caacttattc actctcttcc accctgactc tgtccaaggc agactatgaa
241  aaacacaagg tatacgctg cgaggttaca caccagggtt tgtctagtcc tgtcaccaag
301  tccttcaata ggggcgaatg t

```

Protein Sequence Defining the Human Kappa Light Chain
Constant Region (SEQ ID NO:174)

```

1  rtvaapsvfi fppsdeqlks gtasvvc1ln nfypreakvq wkvdnalqsg nsqesvteqd
61  skdstyslss tltlskadye khkvacevt hqglsspvtk sfnrgec

```

[0310] The following sequences represent the actual or contemplated full length heavy and light chain sequence (i.e., containing both the variable and constant regions sequences) for each antibody described in this Example. Signal sequences for proper secretion of the antibodies (e.g.,

signal sequences at the 5' end of the DNA sequences or the amino terminal end of the protein sequences) are not shown in the full length heavy and light chain sequences disclosed herein and are not included in the final secreted protein. Also not shown are stop codons for termination of translation

required at the 3' end of the DNA sequences. It is within ordinary skill in the art to select a signal sequence and/or a stop codon for expression of the disclosed full length immunoglobulin heavy chain and light chain sequences. It is also contemplated that the variable region sequences can be

ligated to other constant region sequences to produce active full length immunoglobulin heavy and light chains.

Nucleic Acid Sequence Encoding the Full Length Ch01G06 Chimeric Heavy Chain (Mouse Heavy Chain Variable Region and Human IgG1 Constant Region) (SEQ ID NO:175)

```

1 gaagtgttgt tgcagcagtc agggccggag ttggtaaaac cgggagcgtc ggtgaaaatc
61 ccgtgcaaaag cgtcggggta tacgtttacg gactataaca tggattgggt gaaacagtcg
121 catgggaaat cgcttgaatg gattggtcag atcaatccga ataaggagg aatcttcttt
181 aatcagaagt ttaaaggaaa agcgacgctt acagtcgata agtcgtcgaa cacggcgctt
241 atggaagtac ggtcgcttac gtcggaagat acggcgggtct attactgtgc gagggaggcg
301 attacgacgg tgggagcgat ggactattgg ggacaaggga cgtcggctac ggtatcgctg
361 gcctcaacaa aaggaccaag tgtgttccca ctgcgcccta cgacgaagag tacatccggg
421 ggcaactgcag cactcggctg cctcgtcaag gattattttc cagagccagt aaccgtgagc
481 tggaacagtg gagcactcac ttctggtgtc catacttttc ctgtgtctct gcaaagctct
541 ggctgtact cactcagctc cgtcgtgacc gtgccatctt catctctggg cactcagacc
601 tacatctgta atgtaaacca caagcctagc aatactaagg tcgataagcg ggtggaaccc
661 aagagctgcg acaagactca cacttgctcc ccatgccctg cccctgaact tctgggcggg
721 ccagcgtct ttttgttccc accaaagcct aaagatactc tgatgataag tagaacccc
781 gaggtgacat gtgttggtgt agacgtttcc cagcaggacc cagaggttaa gttcaactgg
841 tacgttgatg gagtcgaagt acataatgct aagaccaagc ctagagagga gcagtataat
901 agtacatacc gtgtagttag tgttctcaca gtgctgcacc aagactggct caacggcaaa
961 gaatacaaat gcaaagtgtc caacaaagca ctcccagccc ctatcgagaa gactattagt
1021 aaggcaaaag ggcagcctcg tgaaccacag gtgtacacto tgccacccag tagagaggaa
1081 atgacaaaag accaagtctc attgacctgc ctgggtgaaa gcttctaccc cagcgacatc
1141 gccgttgagt gggagagtaa cggtcagcct gagaacaatt acaagacaac cccccagtg
1201 ctggatagtg acgggtcttt ctttctgtac agtaagctga ctgtggacaa gtcccgtgg
1261 cagcagggta acgtcttcag ctgttcgtg atgcacgagg cattgcacaa ccactacacc
1321 cagaagtcac tgagcctgag cccagggaag

```

Protein Sequence Defining the Full Length Ch01G06 Chimeric Heavy Chain (Mouse Heavy Chain Variable Region and Human IgG1 Constant Region) (SEQ ID NO:176)

```

1 evllqsggpe lvkpgasvki pckasgytft dynmdwvkqs hgkslewigq inpnnggiff
61 nqkfkkgkatl tvdkssntaf mevrsltsed tavyycarea ittvgamdyw gggtsvtvss
121 astkgpsvfp lapsskstsg gtaalgclvk dyfpepvtvs wnsгалtsgv htfpavqlss
181 glyslssvvt vpssslgtqt yicnvnhkps ntkvdkrvep kscdkthtcp pcpapellgg
241 psvflfppkp kdtlmisrtp evtcvvvdvs hedpevkfnw yvdgvevhna ktkpreeqyn
301 styrvsvlt vlhqdwlngk eyckvsnka lpapiektis kakgqppeqp vytlppsree
361 mtknqvsltc lvkgfypsdi avewesngqp ennykttppv ldsdgsffly skltvdksrw
421 qggnvfscsv mhealhnhyt qkslslepgk

```

Nucleic Acid Sequence Encoding the Full Length Hu01G06
IGHV1-18 Heavy Chain (Humanized Heavy Chain Variable
Region and Human IgG1 Constant Region) (SEQ ID
NO:177)

```
1 caagtgaac ttgtgcagtc ggggtgcgga gtcaaaaagc cgggagcgtc ggtgaaagta
61 tcgtgtaaag cgtcgggata tacgtttacg gactataaca tggactgggt acgacaggca
121 ccggggaaat cgttggaaat gatcggacag attaatccga acaatggggg aattttcttt
181 aatcagaaat tcaaaggacg ggcgacgttg acggtcgata catcgacgaa tacggcgatat
241 atggaattga ggtcgcttcg ctccgacgat acggcggtct attactgcgc cagggaggcg
301 atcacgacgg taggggagat ggattattgg ggacagggga cgttctgac ggtatcgctg
361 gcctcaacaa aaggaccaag tgtgttccca ctgcgcccta gcagcaagag tacatccggg
421 ggccactgcag cactcggctg cctcgtcaag gattattttc cagagccagt aacctgagc
481 tggaacagtg gagcactcac ttctgggtgc catacttttc ctgctgtcct gcaaagctct
541 ggctgtact cactcagctc cgtcgtgacc gtgccatctt catctctggg cactcagacc
601 tacatctgta atgtaaacca caagcctagc aatactaagg tcgataagcg ggtggaaccc
661 aagagctgcg acaagactca cacttgctcc ccattgccctg cccctgaact tctgggcggt
721 cccagcgtct ttttgttccc accaaagcct aaagatactc tgatgataag tagaacacc
781 gaggtgacat gtgtgtttgt agacgtttcc cagcaggacc cagaggttaa gttcaactgg
841 tacgttgatg gagtcgaagt acataatgct aagaccaagc ctagagagga gcagtataat
901 agtacatacc gtgtagtcag tgttctcaca gtgctgcacc aagactggct caacggcaaa
961 gaatacaaat gcaaagtgtc caacaaagca ctcccagccc ctatcgagaa gactattagt
1021 aaggcaaaag ggcagcctcg tgaaccacag gtgtacactc tgccaccag tagagaggaa
1081 atgacaaaga accaagtctc attgacctgc ctgggtgaaag gcttctaccc cagcgacatc
1141 gccgttgagt gggagagtaa cggtcagcct gagaacaatt acaagacaac cccccagtg
1201 ctggatagtg acgggtcttt ctttctgtac agtaagctga ctgtggacaa gtcccgtgg
1261 cagcagggta acgtcttcag ctgttcctg atgcacgagg cattgcacaa ccactacacc
1321 cagaagtcac tgagcctgag cccagggaag
```

Protein Sequence Defining the Full Length Hu01G06
IGHV1-18 Heavy Chain (Humanized Heavy Chain Variable
Region and Human IgG1 Constant Region) (SEQ ID
NO:178)

```
1 qvqlvqsgae vkkpgasvkv sckasgytft dynmdwvrqa pgkslewigq inpnnggiff
61 nqkfkgratl tvdtstntay melrslrsdd tavyycarea ittvgamdyw gqgtlvtvss
121 astkgpsvfp lapsskstsg gtaalgclvk dyfpepvtvs wnsгалtsgv htfpavlqss
181 glyslssvvt vpssslgtqt yicnvnhkps ntkvdkrvep kscdkthtcp pcpapellgg
241 psvflfppkp kdtlmisrtp evtcvvvds hedpevkfnw yvdgvevhna ktkpreeqyn
301 styrvsvlt vlhqdwlngk eykckvsnka lpapiektis kakgqprepq vytlppsree
361 mtknqvsltc lvkgfypsdi avewesngqp ennykttppv ldsdgsffly skltvdksrw
421 qqgnvfscsv mhealhnhyt qkslsispk
```

Nucleic Acid Sequence Encoding the Full Length Hu01G06
IGHV1-69 Heavy Chain (Humanized Heavy Chain Variable
Region and Human IgG1 Constant Region) (SEQ ID
NO:179)

```
1 caagtccagc ttgtccagtc gggagcggaa gtgaagaaac cggggtcgtc ggtcaaagta
61 tcgtgtaaag cgtcgggata tacgtttacg gactataaca tggattgggt acgacaggct
121 ccgggaaaat cattggaatg gattggacag attaatccga ataatggggg tatcttcttt
181 aatcaaaagt ttaaaggagg ggcgacgttg acggtggaca aatcgacaaa tacggcgatat
241 atggaattgt cgtcgcttcg gtcggaggac acggcgggtg attactgcgc gagggaggcg
301 atcacgacgg tcggggcgat ggattattgg ggacagggaa cgcttgtagc ggtatcgtcg
361 gcctcaacaa aaggaccaag tgtgttccca ctgcgcccta gcagcaagag tacatccggg
421 ggccactgcag cactcggctg cctcgtcaag gattattttc cagagccagt aaccgtgagc
481 tggaacagtg gagcactcac ttctgggtgc catacttttc ctgctgtcct gcaaagctct
541 ggctgtact cactcagctc cgtcgtgacc gtgccatctt catctctggg cactcagacc
601 tacatctgta atgtaaacca caagcctagc aatactaagg tcgataagcg ggtggaaccc
661 aagagctgcg acaagactca cacttgctcc ccattgccctg cccctgaact totgggcggt
721 cccagcgtct ttttgttccc accaaagcct aaagatactc tgatgataag tagaacacc
781 gaggtgacat gtgtgtttgt agacgtttcc cagcaggacc cagaggttaa gttcaactgg
841 tacgttgatg gagtcgaagt acataatgct aagaccaagc ctagagagga gcagtataat
901 agtacatacc gtgtagtcag tgttctcaca gtgctgcacc aagactggct caacggcaaa
961 gaatacaaat gcaaagtgtc caacaaagca ctcccagccc ctatcgagaa gactattagt
1021 aaggcaaaagg ggcagcctcg tgaaccacag gtgtacactc tgccaccag tagagaggaa
1081 atgacaaaga accaagtctc attgacctgc ctggtgaaag gcttctaccc cagcgacatc
1141 gccgttgagt gggagagtaa cggtcagcct gagaacaatt acaagacaac cccccagtg
1201 ctggatagtg acgggtcttt ctttctgtac agtaagctga ctgtggacaa gtcccgtgg
1261 cagcagggta acgtcttcag ctgttcctg atgcacgagg cattgcacaa ccactacacc
1321 cagaagtcac tgagcctgag cccagggaag
```

Protein Sequence Defining the Full Length Hu01G06
IGHV1-69 Heavy Chain (Humanized Heavy Chain Variable
Region and Human IgG1 Constant Region) (SEQ ID
NO:180)

```
1 qvqlvqsgae vkkpgssvkv sckasgytft dynmdwvrqa pgkslewigq inpnnggiff
61 nqkfkgratl tvdkstntay melsslr sed tavyycarea ittvgamdyw gqgtlvtvss
121 astkgpsvfp lapsskstsg gtaalglvk dyfpepvtvs wnsгалtsgv htfpavlqss
181 glyslssvvt vpssslgtqt yicnvnhkps ntkvdkrvep kscdkthtcp pcpapellgg
241 psvflfpkp kdtlmisrtp evtcvvvds hedpevkfnw yvdgvevhna ktkpreeqyn
301 styrvsvlt vlhqdwlngk eykckvsnka lpapiektis kakgqprepq vytlppsree
361 mtknqvsltc lvkgfypsdi avewesngqp ennykttppv ldsdgsffly skltvdksrw
421 qqgnvfscsv mhealhnhyt qkslsispk
```

Nucleic Acid Sequence Encoding the Full Length Sh01G06
IGHV1-18 M69L Heavy Chain (Humanized Heavy Chain
Variable Region and Human IgG1 Constant Region) (SEQ
ID NO:181)

```
1 caggtccagc ttgtgcaatc gggagcggaa gtgaagaaac cgggagcgtc ggtaaaagtc
61 tcgtgcaaag cgtcggggta tacgtttacg gactataaca tggactgggt gcgccaagcg
121 cctggacagg gtcttgaatg gatggggcag attaatccga ataatggagg gatcttcttt
181 aatcagaaat tcaaaggaag ggtaacgctg acgacagaca cgtcaacatc gacggcctat
241 atggaattgc ggtcgttgcg atcagatgat acggcgggtc actattgtgc gagggaggcg
301 attacgacgg tgggagcgat ggattattgg ggacagggga cgttggtaac ggtatcgtcg
361 gcctcaacaa aaggaccaag tgtgttccca ctgcgcccta gcagcaagag tacatccggg
421 ggcaactgcag cactcggctg cctcgtcaag gattattttc cagagccagt aaccgtgagc
481 tggaaacagt gagcactcac ttctgggtgc catacttttc ctgctgtcct gcaaagctct
541 ggctgtact cactcagctc cgtcgtgacc gtgccatctt catctctggg cactcagacc
601 tacatctgta atgtaaacca caagcctagc aatactaagg tcgataagcg ggtggaaccc
661 aagagctgcg acaagactca cacttgctcc ccattgccctg cccctgaact tctgggcggg
721 cccagcgtct ttttgttccc accaaagcct aaagatactc tgatgataag tagaacaccc
781 gaggtgacat gtgtgtttgt agacgtttcc cagcaggacc cagaggttaa gttcaactgg
841 tacgttgatg gagtcgaagt acataatgct aagaccaagc ctagagagga gcagtataat
901 agtacatacc gtgtagtcag tgttctcaca gtgctgcacc aagactggct caacggcaaa
961 gaatacaaat gcaaagtgtc caacaaagca ctcccagccc ctatcgagaa gactattagt
1021 aaggcaaaagg ggcagcctcg tgaaccacag gtgtacactc tgccaccag tagagaggaa
1081 atgacaaaga accaagtctc attgacctgc ctggtgaaag gcttctaccc cagcgacatc
1141 gccgttgagt gggagagtaa cggtcagcct gagaacaatt acaagacaac cccccagtg
1201 ctggatagtg acgggtcttt ctttctgtac agtaagctga ctgtggacaa gtcccgtgg
1261 cagcagggta acgtcttcag ctgttcctg atgcacgagg cattgcacaa ccactacacc
1321 cagaagtcac tgagcctgag cccagggaag
```

Protein Sequence Defining the Full Length Sh01G06
IGHV1-18 M69L Heavy Chain (Humanized Heavy Chain
Variable Region and Human IgG1 Constant Region) (SEQ
ID NO:182)

```
1 qvqlvqsgae vkkpgasvkv sckasgytft dynmdwvrqa pgqglewmqg inpnnggiff
61 nqkfkgrvtl ttdtststay melrslrdd tavyyarea ittvgamdyw gqgtlvtvss
121 astkgspsvfp lapsskstsg gtaalgclvk dyfpepvtvs wnsгалtsgv htfpavlqss
181 glyslssvvt vpssslgtqt yicnvnhkps ntkvdkrvep kscdkthtct pcpapellgg
241 psvflfpkp kdtlmisrtp evtcvvvds hedpevkfnw yvdgvevhna ktkpreeqyn
301 styrvsvlt vlhqdwlngk eykckvsnka lpapiektis kakgqprepq vytlppsree
361 mtknqvsltc lvkgfypsdi avewesngqp ennykttppv ldsdgsffly skltvdksrw
421 qqgnvfscsv mhealhnhyt qkslspsgk
```

Nucleic Acid Sequence Encoding the Full Length Sh01G06
IGHV1-18 M69L K640 G44S Heavy Chain (Humanized
Heavy Chain Variable Region and Human IgG1 Constant
Region) (SEQ ID NO:183)

```
1 caggtccagc ttgtgcaatc gggagcggaa gtgaagaaac cgggagcgtc ggtaaaagtc
61 tcgtgcaaag cgtcggggta tacgtttacg gactataaca tggactgggt gcgccaagcg
121 cctggacaga gccttgaatg gatggggcag attaatccga ataatggagg gatcttcttt
181 aatcagaaat tccaggaag ggtaacgctg acgacagaca cgtcaacatc gacggcctat
241 atggaattgc ggtcgttgcg atcagatgat acggcggctc actattgtgc gagggaggcg
301 attacgacgg tgggagcgat ggattattgg ggacagggga cgttggtaac ggtatcgtcg
361 gcctcaacaa aaggaccaag tgtgttccca ctgcgcccta gcagcaagag tacatccggg
421 ggcaactgcag cactcggctg cctcgtcaag gattattttc cagagccagt aaccgtgagc
481 tggaaacagt gagcactcac ttctgggtgc catacttttc ctgctgtcct gcaaagctct
541 ggctgtact cactcagctc cgtcgtgacc gtgccatctt catctctggg cactcagacc
601 tacatctgta atgtaaacca caagcctagc aatactaagg tcgataagcg ggtggaaccc
661 aagagctgcg acaagactca cacttgctcc ccattgccctg cccctgaact tctgggcggg
721 cccagcgtct ttttgttccc accaaagcct aaagatactc tgatgataag tagaacaccc
781 gaggtgacat gtgtgtttgt agacgtttcc cagcaggacc cagagggttaa gttcaactgg
841 tacgttgatg gagtcgaagt acataatgct aagaccaagc ctagagagga gcagtataat
901 agtacatacc gtgtagtcag tgttctcaca gtgctgcacc aagactggct caacggcaaa
961 gaatacaaat gcaaagtgtc caacaaagca ctcccagccc ctatcgagaa gactattagt
1021 aaggcaaaag ggcagcctcg tgaaccacag gtgtacactc tgccaccag tagagaggaa
1081 atgacaaaga accaagtctc attgacctgc ctgggtgaaag gcttctaccc cagcgacatc
1141 gccgttgagt gggagagtaa cggtcagcct gagaacaatt acaagacaac cccccagtg
1201 ctggatagtg acgggtcttt ctttctgtac agtaagctga ctgtggacaa gtcccgtgg
1261 cagcagggta acgtcttcag ctgttcctg atgcacgagg cattgcacaa ccactacacc
1321 cagaagtcac tgagcctgag cccagggaag
```

Protein Sequence Defining the Full Length Sh01G06
IGHV1-18 M69L K64Q G44S Heavy Chain (Humanized
Heavy Chain Variable Region and Human IgG1 Constant
Region) (SEQ ID NO:184)

```
1 qvqlvqsgae vkkpgasvkv sckasgytft dynmdwvrqa pgqslwmgq inpnnggiff
61 nqkfqgrvtl ttdtststay melrslrdd tavyyarea ittvgamdyw gqgtlvtvss
121 astkgspsvfp lapsskstsg gtaalgclvk dyfpepvtvs wnsгалtsgv htfpavlqss
181 glyslssvvt vpssslgtqt yicnvnhkps ntkvdkrvep kscdkthtcp pcpapellgg
241 psvflfpkp kdtlmisrtp evtcvvvds hedpevkfnw yvdgvevhna ktkpreeqyn
301 styrvsvlt vlhqdwlngk eykckvsnka lpapiektis kakgqprepq vytlppsree
361 mtknqvsltc lvkgfypsdi avewesngqp ennykttppv ldsdgsffly skltvdksrw
421 qqgnvfscsv mhealhnhyt qkslsispk
```


Nucleic Acid Sequence Encoding the Full Length Sh01G06
IGHV1-18 M69L K64Q Heavy Chain (Humanized Heavy
Chain Variable Region and Human IgG1 Constant Region)
(SEQ ID NO:185)

```
1 caggtccagc ttgtgcaatc gggagcggaa gtgaagaaac cgggagcgtc ggtaaaagtc
61 tcgtgcaaag cgtcggggta tacgtttacg gactataaca tggactgggt gcgccaagcg
121 cctggacagg gtcttgaatg gatggggcag attaatccga ataatggagg gatcttcttt
181 aatcagaaat tccagggaag ggtaacgctg acgacagaca cgtcaacatc gacggcctat
241 atggaattgc ggtcgttgcg atcagatgat acggcggctc actattgtgc gagggaggcg
301 attacgacgg tgggagcgat ggattattgg ggacagggga cgttggtaac ggtatcgtcg
361 gcctcaacaa aaggaccaag tgtgttccca ctgcgcccta gcagcaagag tacatccggg
421 ggcaactgcag cactcggctg cctcgtcaag gattattttc cagagccagt aacctgagc
481 tggaaacagt gagcactcac ttctgggtgc catacttttc ctgctgtcct gcaaagctct
541 ggctgtact cactcagctc cgtcgtgacc gtgccatctt catctctggg cactcagacc
601 tacatctgta atgtaaacca caagcctagc aatactaagg tcgataagcg ggtggaaccc
661 aagagctgcg acaagactca cacttgctcc ccattgccctg cccctgaact tctgggcggg
721 cccagcgtct ttttgttccc accaaagcct aaagatactc tgatgataag tagaacaccc
781 gaggtgacat gtgtgtttgt agacgtttcc cagcaggacc cagaggttaa gttcaactgg
841 tacgttgatg gagtcgaagt acataatgct aagaccaagc ctagagagga gcagtataat
901 agtacatacc gtgtagtcag tgttctcaca gtgctgcacc aagactggct caacggcaaa
961 gaatacaaat gcaaagtgtc caacaaagca ctcccagccc ctatcgagaa gactattagt
1021 aaggcaaaagg ggcagcctcg tgaaccacag gtgtacactc tgccaccag tagagaggaa
1081 atgacaaaga accaagtctc attgacctgc ctgggtgaaag gcttctaccc cagcgacatc
1141 gccgttgagt gggagagtaa cggtcagcct gagaacaatt acaagacaac cccccagtg
1201 ctggatagtg acgggtcttt ctttctgtac agtaagctga ctgtggacaa gtcccgtgg
1261 cagcagggta acgtcttcag ctgttcctg atgcacgagg cattgcacaa ccactacac
1321 cagaagtcac tgagcctgag cccagggaag
```

Protein Sequence Defining the Full Length Sh01G06
IGHV1-18 M69L K64Q Heavy Chain (Humanized Heavy
Chain Variable Region and Human IgG1 Constant Region)
(SEQ ID NO: 186)

```
1 qvqlvqsgae vkkpgasvkv sckasgytft dynmdwvrqa pgqglewmgq inpnnggiff
61 nqkfqgrvtl ttdtststay melrslrdd tavyyarea ittvgamdyw gqgtivtvss
121 astkgspsvfp lapsskstsg gtaalgclvk dyfpepvtvs wnsгалtsgv htfpavlqss
181 glyslssvvt vpssslgtqt yicnvnhkps ntkvdkrvep kscdkthtct pcpapellgg
241 psvflfpkp kdtlmisrtp evtcvvvdvs hedpevkfnw yvdgvevhna ktkpreeqyn
301 styrvsvlt vlhqdwlngk eykckvsnka lpapiektis kakgqprepq vytlppsree
361 mtknqvsltc lvkgfypsdi avewesngqp ennykttppv ldsdgsffly skltvdksrw
421 qqgnvfscsv mhealnhnyt qkslspsgk
```

Nucleic Acid Sequence Encoding the Full Length Sh01G06
IGHV1-69 T30S I69L Heavy Chain (Humanized Heavy
Chain Variable Region and Human IgG1 Constant Region)
(SEQ ID NO:187)

```
1 caagtacagc ttgtacagtc gggagcggaa gtcaagaaac cgggatcgtc ggtcaaagtg
61 tcgtgtaaag cgtcgggata tacgttttagc gactataaca tggattgggt gcgacaagcg
121 cctgggcagg gacttgaatg gatgggtcag atcaatccga ataatggggg aatctttttc
181 aatcagaagt ttaaaggagg ggtaacgctg acggcggata aaagcacgtc aacggcgat
241 atggagttgt cgtcgttgcg gtcggaggac acggcggctc attactgcgc gagggaagcg
301 attacgacgg tgggagcgat ggattattgg gggcagggaa cgcttgtaac ggtgtcatcg
361 gcctcaacaa aaggaccaag tgtgttccca ctgcgcccta gcagcaagag tacatccggg
421 ggccactgcag cactcggctg cctcgtcaag gattattttc cagagccagt aaccgtgagc
481 tggaacagtg gagcactcac ttctgggtgc catacttttc ctgctgtcct gcaaagctct
541 ggccctgtact cactcagctc cgtcgtgacc gtgccatctt catctctggg cactcagacc
601 tacatctgta atgtaaacca caagcctagc aatactaagg tcgataagcg ggtggaaccc
661 aagagctgcg acaagactca cacttgctcc ccattgccctg cccctgaact tctgggcggg
721 cccagcgtct ttttgttccc accaaagcct aaagatactc tgatgataag tagaacaccc
781 gaggtgacat gtgtgtttgt agacgtttcc cagcaggacc cagaggttaa gttcaactgg
841 tacgttgatg gagtcgaagt acataatgct aagaccaagc ctagagagga gcagtataat
901 agtacatacc gtgtagtcag tgttctcaca gtgctgcacc aagactggct caacggcaaa
961 gaatacaaat gcaaagtgtc caacaaagca ctcccagccc ctatcgagaa gactattagt
1021 aaggcaaaagg ggcagcctcg tgaaccacag gtgtacactc tgccaccag tagagaggaa
1081 atgacaaaga accaagtctc attgacctgc ctggtgaaag gcttctaccc cagcgacatc
1141 gccgttgagt gggagagtaa cggtcagcct gagaacaatt acaagacaac cccccagtg
1201 ctggatagtg acgggtcttt ctttctgtac agtaagctga ctgtggacaa gtcccgtgg
1261 cagcagggta acgtcttcag ctgttcctg atgcacgagg cattgcacaa ccactacac
1321 cagaagtcac tgagcctgag cccagggaag
```

Protein Sequence Defining the Full Length Sh01G06
IGHV1-69 T30S I69L Heavy Chain (Humanized Heavy
Chain Variable Region and Human IgG1 Constant Region)
(SEQ ID NO:188)

```
1 qvqlvqsgae vkkpgsssvkv sckasgytfs dynmdwvrqa pgqglewmgq inpnnggiff
61 nqkfkgrvtl tadkststay melsslrsed tavyycarea ittvgamdyw gqgtivtvss
121 astkgspsvfp lapsskstsg gtaalgclvk dyfpepvtvs wnsгалtsgv htfpavlqss
181 glyslssvvt vpssslgtqt yicnvnhkps ntkvdkrvep kscdkthtct pcpapellgg
241 psvflfpkp kdtlmisrtp evtcvvvdvs hedpevkfnw yvdgvevhna ktkpreeqyn
301 styrvsvlt vlhqdwlngk eykckvsnka lpapiektis kakgqprepq vytlppsree
361 mtknqvsltc lvkgfypsdi avewesngqp ennykttppv ldsdgsffly skltvdksrw
421 qqgnvfscsv mhealhnhyt qkslsispk
```

Nucleic Acid Sequence Encoding the Full Length Sh01G06
IGHV1-69 T30S K64Q I69L Heavy Chain (Humanized
Heavy Chain Variable Region and Human IgG1 Constant
Region) (SEQ ID NO:189)

```
1 caagtacagc ttgtacagtc gggagcggaa gtcaagaaac cgggatcgtc ggtcaaagtg
61 tcgtgtaaag cgtcgggata tacgttttagc gactataaca tggattgggt gcgacaagcg
121 cctgggcagg gacttgaatg gatgggtcag atcaatccga ataatggggg aatctttttc
181 aatcagaagt ttcaggggag ggtaacgctg acggcggata aaagcacgtc aacggcgat
241 atggagttgt cgtcgttgcg gtcggaggac acggcggctc attactgcgc gagggaagcg
301 attacgacgg tgggagcgat ggattattgg gggcagggaa cgcttgtaac ggtgtcatcg
361 gcctcaacaa aaggaccaag tgtgttccca ctgcgcccta gcagcaagag tacatccggg
421 ggcaactgcag cactcggctg cctcgtcaag gattattttc cagagccagt aaccgtgagc
481 tggaacagtg gagcactcac ttctgggtgc catacttttc ctgctgtcct gcaaagctct
541 ggctgtact cactcagctc cgtcgtgacc gtgccatctt catctctggg cactcagacc
601 tacatctgta atgtaaacca caagcctagc aatactaagg tcgataagcg ggtggaaccc
661 aagagctgcg acaagactca cacttgctcc ccattgccctg cccctgaact tctgggcggg
721 cccagcgtct ttttgttccc accaaagcct aaagatactc tgatgataag tagaacaccc
781 gaggtgacat gtgtgtttgt agacgtttcc cagcaggacc cagaggttaa gttcaactgg
841 tacgttgatg gagtcgaagt acataatgct aagaccaagc ctagagagga gcagtataat
901 agtacatacc gtgtagtcag tgttctcaca gtgctgcacc aagactggct caacggcaaa
961 gaatacaaat gcaaagtgtc caacaaagca ctcccagccc ctatcgagaa gactattagt
1021 aaggcaaaag ggcagcctcg tgaaccacag gtgtacactc tgccaccag tagagaggaa
1081 atgacaaaga accaagtctc attgacctgc ctgggtgaaag gcttctaccc cagcgacatc
1141 gccgttgagt gggagagtaa cggtcagcct gagaacaatt acaagacaac cccccagtg
1201 ctggatagtg acgggtcttt ctttctgtac agtaagctga ctgtggacaa gtcccgtgg
1261 cagcagggta acgtcttcag ctgttcctg atgcacgagg cattgcacaa ccactacacc
1321 cagaagtcac tgagcctgag cccagggaag
```

Protein Sequence Defining the Full Length Sh01G06
IGHV1-69 T30S K64Q I69L Heavy Chain (Humanized
Heavy Chain Variable Region and Human IgG1 Constant
Region) (SEQ ID NO:190)

```
1 qvqlvqsgae vkkpgssvkv sckasgytfs dynmdwvrqa pgqglewmqg inpnnggiff
61 nqkfqgrvtl tadkststay melsslrse d tavyycarea ittvgamdyw gqgtivtvss
121 astkgspsvfp lapsskstsg gtaalgclvk dyfpepvtvs wnsгалtsgv htfpavlqss
181 glyslssvvt vpssslgtqt yicnvnhkps ntkvdkrvep kscdkthtct pcpapellgg
241 psvflfpkp kdtlmisrtp evtcvvvdvs hedpevkfnw yvdgvevhna ktkpreeqyn
301 styrvsvlt vlhqdwlngk eykckvsnka lpapiektis kakgqprep q vytlppsree
361 mtknqvsltc lvkgfypsdi avewesngqp ennykttppv ldsdgsffly skltvdksrw
421 qqgnvfscsv mhealhnhyt qkslspspgk
```

Nucleic Acid Sequence Encoding the Full Length Hu01G06
IGHV1-18 F1 Heavy Chain (Humanized Heavy Chain
Variable Region and Human IgG1 Constant Region) (SEQ
ID NO:255)

```
1 caggtccagc ttgtgcaatc gggagcggaa gtgaagaaac cgggagcgtc ggtaaaagtc
61 tcgtgcaaag cgtcggggta tacgtttacg gactataaca tggactgggt gcgccaagcg
121 cctggacaga gccttgaatg gatggggcag attaatccgt acaatcacct gatcttcttt
181 aatcagaaat tccaggaag ggtaacgctg acgacagaca cgtcaacatc gacggcctat
241 atggaattgc ggtcgttgcg atcagatgat acggcgggtc actattgtgc gagggaggcg
301 attacgacgg tgggagcgat ggattattgg ggacagggga cgttggtaac ggtatcgtcg
361 gcctcaacaa aaggaccaag tgtgttccca ctgcgcccta gcagcaagag tacatccggg
421 ggcaactgcag cactcggctg cctcgtcaag gattattttc cagagccagt aaccgtgagc
481 tggaaacagt gagcactcac ttctgggtgc catacttttc ctgctgtcct gcaaagctct
541 ggctgtgact cactcagctc cgtcgtgacc gtgccatctt catctctggg cactcagacc
601 tacatctgta atgtaaacca caagcctagc aatactaagg tcgataagcg ggtggaaccc
661 aagagctgcg acaagactca cacttgctcc ccattgccctg cccctgaact tctgggcggg
721 cccagcgtct ttttgttccc accaaagcct aaagatactc tgatgataag tagaacaccc
781 gaggtgacat gtgtgtttgt agacgtttcc cagcaggacc cagagggtta gttcaactgg
841 tacgttgatg gagtcgaagt acataatgct aagaccaagc ctagagagga gcagtataat
901 agtacatacc gtgtagtcag tgttctcaca gtgctgcacc aagactggct caacggcaaa
961 gaatacaaat gcaaagtgtc caacaaagca ctcccagccc ctatcgagaa gactattagt
1021 aaggcaaaag ggcagcctcg tgaaccacag gtgtacactc tgccaccag tagagaggaa
1081 atgacaaaga accaagtctc attgacctgc ctggtgaaag gcttctaccc cagcgacatc
1141 gccgttgagt gggagagtaa cggtcagcct gagaacaatt acaagacaac cccccagtg
1201 ctggatagtg acgggtcttt ctttctgtac agtaagctga ctgtggacaa gtcccgtgg
1261 cagcagggta acgtcttcag ctgttcctg atgcacgagg cattgcacaa ccactacacc
1321 cagaagtcac tgagcctgag cccagggaag
```

Protein Sequence Defining the Full Length Hu01G06
IGHV1-18 F1 Heavy Chain (Humanized Heavy Chain
Variable Region and Human IgG1 Constant Region) (SEQ
ID NO:256)

```
1 qvqlvqsgae vkkpgasvkv sckasgytft dynmdwvrqa pgqslwmgq inpynhliff
61 nqkfqgrvtl ttdtststay melrslrdd tavyycarea ittvgamdyw gqgtivtvss
121 astkgspsvfp lapsskstsg gtaalgclvk dyfpepvtvs wnsгалtsgv htfpavlqss
181 glyslssvvt vpssslgtqt yicnvnhkps ntkvdkrvep kscdkthtct pcpapellgg
241 psvflfpkp kdtlmisrtp evtcvvvdvs hedpevkfnw yvdgvevhna ktkpreeqyn
301 styrvsvlt vlhqdwlngk eykckvsnka lpapiektis kakgqprepq vytlppsree
361 mtknqvsltc lvkgfypsdi avewesngqp ennykttppv ldsdgsffly skltvdksrw
421 qqgnvfscsv mhealhnhyt qkslsispk
```

Nucleic Acid Sequence Encoding the Full Length Hu01G06
IGHV1-18 F2 Heavy Chain (Humanized Heavy Chain
Variable Region and Human IgG1 Constant Region) (SEQ
ID NO:257)

```
1 caggtccagc ttgtgcaatc gggagcggaa gtgaagaaac cgggagcgtc ggtaaaagtc
61 tcgtgcaaag cgctggggta tacgtttacg gactataaca tggactgggt gcgccaagcg
121 cctggacaga gccttgaatg gatggggcag attaatccga ataatggact gatcttcttt
181 aatcagaaat tccaggaag ggtaacgctg acgacagaca cgtcaacatc gacggcctat
241 atggaattgc ggtcgttgcg atcagatgat acggcgggtc actattgtgc gagggaggcg
301 attacgacgg tgggagcgat ggattattgg ggacagggga cgttggtaac ggtatcgtcg
361 gcctcaacaa aaggaccaag tgtgttccca ctgcgcccta gcagcaagag tacatccggg
421 ggcaactgcag cactcggctg cctcgtcaag gattattttc cagagccagt aaccgtgagc
481 tggaaacagt gagcactcac ttctgggtgc catacttttc ctgctgtcct gcaaagctct
541 ggctgtgact cactcagctc cgctcgtgacc gtgccatctt catctctggg cactcagacc
601 tacatctgta atgtaaacca caagcctagc aatactaagg tcgataagcg ggtggaaccc
661 aagagctgcg acaagactca cacttgctcc ccattgccctg cccctgaact tctgggcggg
721 cccagcgtct ttttgttccc accaaagcct aaagatactc tgatgataag tagaacaccc
781 gaggtgacat gtgtgtttgt agacgtttcc cagcaggacc cagaggttaa gttcaactgg
841 tacgttgatg gagtcgaagt acataatgct aagaccaagc ctagagagga gcagtataat
901 agtacatacc gtgtagtcag tgttctcaca gtgctgcacc aagactggct caacggcaaa
961 gaatacaaat gcaaagtgtc caacaaagca ctcccagccc ctatcgagaa gactattagt
1021 aaggcaaaag ggcagcctcg tgaaccacag gtgtacactc tgccaccag tagagaggaa
1081 atgacaaaga accaagtctc attgacctgc ctgggtgaaag gcttctaccc cagcgacatc
1141 gccgttgagt gggagagtaa cggtcagcct gagaacaatt acaagacaac cccccagtg
1201 ctggatagtg acgggtcttt ctttctgtac agtaagctga ctgtggacaa gtcccgtgg
1261 cagcagggta acgtcttcag ctgttcctg atgcacgagg cattgcacaa ccactacac
1321 cagaagtcac tgagcctgag cccagggaag
```

Protein Sequence Defining the Full Length Hu01G06
IGHV1-18 F2 Heavy Chain (Humanized Heavy Chain
Variable Region and Human IgG1 Constant Region) (SEQ
ID NO:258)

```
1 qvqlvqsgae vkkpgasvkv sckasgytft dynmdwvrqa pgqslwmgq inpnngliff
61 nqkfqgrvtl ttdtststay melrslrdd tavyyarea ittvgamdyw gggativtvs
121 astkgspsvfp lapsskstsg gtaalgclvk dyfpepvtvs wnsгалtsgv htfpavlqss
181 glyslssvvt vpssslgtqt yicnvnhkps ntkvdkrvep kscdkthtct pcpapellgg
241 psvflfpkp kdtlmisrtp evtcvvvds hedpevkfnw yvdgvevhna ktkpreeqyn
301 styrvsvlt vlhqdwlngk eykckvsnka lpapiektis kakgqprepq vytlppsree
361 mtknqvsltc lvkgfypsdi avewesngqp ennykttppv ldsdgsffly skltvdksrw
421 qqgnvfscsv mhealhnhyt qkslsispk
```

Nucleic Acid Sequence Encoding the Full Length Hu01G06
IGHV1-69 F1 Heavy Chain (Humanized Heavy Chain
Variable Region and Human IgG1 Constant Region) (SEQ
ID NO:259)

```
1 caagtacagc ttgtacagtc gggagcggaa gtcaagaaac cgggatcgtc ggtcaaagtg
61 tcgtgtaaag cgtcgggata tacgttttagc gactataaca tggattgggt gcgacaagcg
121 cctgggcagg gacttgaatg gatgggtcag atcaatccga ataatgggct gatctttttc
181 aatcagaagt ttaaaggagg ggtaacgctg acggcggata aaagcacgtc aacggcgat
241 atggagttgt cgtcgttgcg gtcggaggac acggcggctc attactgcgc gagggaagcg
301 attacgacgg tgggagcgat ggattattgg gggcagggaa cgcttgtaac ggtgtcatcg
361 gcctcaacaa aaggaccaag tgtgttccca ctgcgcccta gcagcaagag tacatccggg
421 ggccactgcag cactcggctg cctcgtcaag gattattttc cagagccagt aaccgtgagc
481 tggaacagtg gagcactcac ttctgggtgc catacttttc ctgctgtcct gcaaagctct
541 ggccctgtact cactcagctc cgtcgtgacc gtgccatctt catctctggg cactcagacc
601 tacatctgta atgtaaacca caagcctagc aatactaagg tcgataagcg ggtggaaccc
661 aagagctgcg acaagactca cacttgctcc ccattgccctg cccctgaact tctgggcggg
721 cccagcgtct ttttgttccc accaaagcct aaagatactc tgatgataag tagaacaccc
781 gaggtgacat gtgtgtttgt agacgtttcc cagcaggacc cagaggttaa gttcaactgg
841 tacgtttgatg gagtcgaagt acataatgct aagaccaagc ctagagagga gcagtataat
901 agtacatacc gtgtagtcag tgttctcaca gtgctgcacc aagactggct caacggcaaa
961 gaatacaaat gcaaagtgtc caacaaagca ctcccagccc ctatcgagaa gactattagt
1021 aaggcaaaagg ggcagcctcg tgaaccacag gtgtacactc tgccaccag tagagaggaa
1081 atgacaaaga accaagtctc attgacctgc ctggtgaaag gcttctaccc cagcgacatc
1141 gccgttgagt gggagagtaa cggtcagcct gagaacaatt acaagacaac cccccagtg
1201 ctggatagtg acgggtcttt ctttctgtac agtaagctga ctgtggacaa gtcccgtgg
1261 cagcagggta acgtcttcag ctgttcctg atgcacgagg cattgcacaa ccactacacc
1321 cagaagtcac tgagcctgag cccagggaag
```

Protein Sequence Defining the Full Length Hu01G06
IGHV1-69 F1 Heavy Chain (Humanized Heavy Chain
Variable Region and Human IgG1 Constant Region) (SEQ
ID NO:260)

```
1 qvqlvqsgae vkkpgsssvkv sckasgytfs dynmdwvrqa pgqglewmqg inpnngliff
61 nqkfkgrvtl tadkststay melsslrsed tavyycarea ittvgamdyw gqgtivtvss
121 astkgspsvfp lapsskstsg gtaalgclvk dyfpepvtvs wnsгалtsqv htfpavlqss
181 glyslssvvt vpssslgtqt yicnvnhkps ntkvdkrvep kscdkthtct pcpapellgg
241 psvflfpkp kdtlmisrtp evtcvvvdvs hedpevkfnw yvdgvevhna ktkpreeqyn
301 styrvsvlt vlhqdwlngk eykckvsnka lpapiektis kakgqprepq vytlppsree
361 mtknqvsltc lvkgfypsdi avewesngqp ennykttppv ldsdgsffly skltvdksrw
421 qqgnvfscsv mhealhnhyt qkslsispk
```

Nucleic Acid Sequence Encoding the Full Length Hu01G06
IGHV1-69 F2 Heavy Chain (Humanized Heavy Chain
Variable Region and Human IgG1 Constant Region) (SEQ
ID NO:261)

```
1 caagtacagc ttgtacagtc gggagcggaa gtcaagaaac cgggatcgtc ggtcaaagtg
61 tcgtgtaaag cgtcgggata tacgttttagc gactataaca tggattgggt gcgacaagcg
121 cctgggcagg gacttgaatg gatgggtcag atcaatccgt acaatcacct gatctttttc
181 aatcagaagt ttaaaggagg ggtaacgctg acggcggata aaagcacgtc aacggcgat
241 atggagttgt cgtcgttgcg gtcggaggac acggcggctc attactgcgc gagggagcgc
301 attacgacgg tgggagcgat ggattattgg gggcagggaa cgcttgtaac ggtgtcatcg
361 gcctcaacaa aaggaccaag tgtgttccca ctgcgcccta gcagcaagag tacatccggg
421 ggccactgcag cactcggctg cctcgtcaag gattattttc cagagccagt aaccgtgagc
481 tggaacagtg gagcactcac ttctgggtgc catacttttc ctgctgtcct gcaaagctct
541 ggccctgtact cactcagctc cgtcgtgacc gtgccatctt catctctggg cactcagacc
601 tacatctgta atgtaaacca caagcctagc aatactaagg tcgataagcg ggtggaaccc
661 aagagctgcg acaagactca cacttgctcc ccattgccctg cccctgaact tctgggcggg
721 cccagcgtct ttttgttccc accaaagcct aaagatactc tgatgataag tagaacaccc
781 gaggtgacat gtgtgtttgt agacgtttcc cagcaggacc cagaggttaa gttcaactgg
841 tacgtttgatg gagtcgaagt acataatgct aagaccaagc ctagagagga gcagtataat
901 agtacatacc gtgtagtcag tgttctcaca gtgctgcacc aagactggct caacggcaaa
961 gaatacaaat gcaaagtgtc caacaaagca ctcccagccc ctatcgagaa gactattagt
1021 aaggcaaaagg ggcagcctcg tgaaccacag gtgtacactc tgccaccag tagagaggaa
1081 atgacaaaga accaagtctc attgacctgc ctggtgaaag gcttctaccc cagcgacatc
1141 gccgttgagt gggagagtaa cggtcagcct gagaacaatt acaagacaac cccccagtg
1201 ctggatagtg acgggtcttt ctttctgtac agtaagctga ctgtggacaa gtcccgtgg
1261 cagcagggta acgtcttcag ctgttcctg atgcacgagg cattgcacaa ccactacacc
1321 cagaagtcac tgagcctgag cccagggaag
```

Protein Sequence Defining the Full Length Hu01G06
IGHV1-69 F2 Heavy Chain (Humanized Heavy Chain
Variable Region and Human IgG1 Constant Region) (SEQ
ID NO:262)

```
1 qvqlvqsgae vkkpgsssvkv sckasgytfs dynmdwvrqa pgqglewmqg inpynhliff
61 nqkfkgrvtl tadkststay melsslrseed tavyycarea ittvgamdyw gqgtivtvss
121 astkgspsvfp lapsskstsg gtaalgclvk dyfpepvtvs wnsгалtsgv htfpavlqss
181 glyslssvvt vpssslgtqt yicnvnhkps ntkvdkrvep kscdkthtct pcpapellgg
241 psvflfpkp kdtlmisrtp evtcvvvdvs hedpevkfnw yvdgvevhna ktkpreeqyn
301 styrvsvlt vlhqdwlngk eykckvsnka lpapiektis kakgqprepq vytlppsree
361 mtknqvsltc lvkgfypsdi avewesngqp ennykttppv ldsdgsffly skltvdksrw
421 qqgnvfscsv mhealhnhyt qkslsispk
```

Nucleic Acid Sequence Encoding the Full Length Ch06C11
Chimeric Heavy Chain (Mouse Heavy Chain Variable
Region and Human IgG1 Constant Region) (SEQ ID
NO:191)

```
1 caggtgacac tcaaagaatc aggaccgga atccttcagc ccagccagac cttgtcgctg
61 acttgttcgt tctccggttt cagcctgaat acttatggga tgggtgtgtc atggatcagg
121 caaccgtccg ggaaaggatt ggagtggctc gcgcacatct actgggacga tgacaaacgc
181 tacaatcctt cgctgaagag ccgattgacg atttccaagg atgcctcgaa caaccgggta
241 tttcttaaga tcacgtcggg cgatacggca gacacggcga cctattactg cgcccaaaga
301 gggtagcatg actattgggg atattggggc caggggacac tcgtcacaat ttcagctgcc
361 tcaacaaaag gaccaagtgt gttcccactc gcccttagca gcaagagtac atccgggggc
421 actgcagcac tcggctgcct cgtcaaggat tattttccag agccagtaac cgtgagctgg
481 aacagtggag cactcacttc tgggtgccat acttttctct ctgtcctgca aagctctggc
541 ctgtactcac tcagctccgt cgtgaccgtg ccactctcat ctctgggcac tcagacctac
601 atctgtaatg taaaccacaa gcctagcaat actaaggctc ataagcgggt ggaacccaag
661 agctgcgaca agactcacac ttgtcccca tgccctgccc ctgaacttct gggcggtccc
721 agcgtctttt tgttcccacc aaagcctaaa gatactctga tgataagtag aacccccgag
781 gtgacatgtg ttgtttaga cgtttccac gaggaccag aggttaagtt caactggtac
841 gttgatggag tcgaagtaca taatgctaag accaagccta gagaggagca gtataatag
901 acataccgtg tagtcagtgt tctcacagt ctgcaccaag actggctcaa cggcaaagaa
961 taaaaatgca aagtgtccaa caaagcactc ccagccccta tcgagaagac tattagtaag
1021 gcaaaggggc agcctcgtga accacaggtg tacactctgc caccagtag agaggaaatg
1081 acaagaacc aagtctcatt gacctgcctg gtgaaaggct tctacccag cgacatcgcc
1141 gttgagtggt agagtaacgg tcagcctgag aacaattaca agacaacccc ccagtgctg
1201 gatagtgcg ggtctttctt tctgtacagt aagctgactg tggacaagtc ccgctggcag
1261 cagggtaacg tcttcagctg ttccgtgatg cagaggcat tgcacaacca ctacaccag
1321 aagtcactga gctgagccc agggaag
```

Protein Sequence Defining the Full Length Ch06C11
Chimeric Heavy Chain (Mouse Heavy Chain Variable
Region and Human IgG1 Constant Region) (SEQ ID
NO:192)

```
1 qvtlkesgpg ilqpsqtlsl tcsfsgfsln tygmgswir qpsgkglew1 ahiywdddkr
61 ynpelkerlt iskdaennry flkitsvda dtatycaqr gyddygywg qgtivtisaa
121 stkgpsvfpl apssktsagg taalglvkd yfpepvtvsw nsaltsgvh tfpavlqssg
181 lyslssvvtv pssslgtqty icnvnhkpsn tkvdkrvepk scdkthtcpp cpapellggp
241 svflfpkpkp dtlmisrtpe vtcvvvdvsh edpevkfnwy vdgvevhnak tkpreeqyns
301 tyrvvsvltv lhqdwlngke ykckvsnkai papiektisk akggprepqv ytlppsreem
361 tknqvsltcl vkgyfysdia vewesngqpe nnykttppvl dsdgsfflys kltvdksrwq
421 qgnvfscsvm healhnyhtq kslslspgk
```


Nucleic Acid Sequence Encoding the Full Length HE LM
06C11 IGHV2-70 Heavy Chain (Humanized Heavy Chain
Variable Region and Human IgG1 Constant Region) (SEQ
ID NO:193)

```
1  caggtgactt tgaagaatc cggccccga ttggtaaagc caaccagac acttacgctc
61  acatgtacat ttcccgatt cagcttgaaac acttacggga tgggagtgtc gtggattcgg
121 caacctccgg ggaaggtctt ggagtggctg gcgcacatct actgggatga tgacaaaagg
181 tataaccctt cacttaaaac gagactgacg atctcgaagg acacaagcaa gaatcaggtc
241 gtcttcacga ttacgaatgt agaccgggtg gatactgccg tctattactg cgcgcaacgc
301 ggggtatgat actactgggg atattggggg cagggcacc cctgtgacct ctcgtcagcc
361 tcaacaaaag gaccaagtgt gtccccactc gcccttagca gcaagagtac atccgggggc
421 actgcagcac tcggctgcct cgtcaaggat tattttccag agccagtaac cgtgagctgg
481 aacagtggag cactcacttc tgggtgccat acttttctct ctgtcctgca aagctctggc
541 ctgtactcac tcagctccgt cgtgaccgtg ccatcttcat ctctgggcac tcagacctac
601 atctgtaatg taaaccacaa gcctagcaat actaaggctc ataagcgggt ggaacccaag
661 agctgcgaca agactcacac ttgtcccca tgccctgcc ctgaacttct gggcgggtccc
721 agcgtctttt tgttcccacc aaagcctaaa gatactctga tgataagtag aacacccgag
781 gtgacatgtg ttgtttaga cgtttccac gaggaccag aggttaagtt caactggtac
841 gttgatggag tcgaagtaca taatgctaag accaagccta gagaggagca gtataatagt
901 acataccgtg tagtcagtgt tctcacagtg ctgcaccaag actggctcaa cggcaaagaa
961 taaaaatgca aagtgtccaa caaagcactc ccagccccta tcgagaagac tattagtaag
1021 gcaaaggggc agcctcgtga accacaggtg tacactctgc caccagtag agaggaaatg
1081 acaagaacc aagtctcatt gacctgcctg gtgaaaggct tctacccag cgacatcgcc
1141 gttgagtggg agagtaacgg tcagcctgag aacaattaca agacaacccc cccagtgtctg
1201 gatagtgcag ggtctttctt tctgtacagt aagctgactg tggacaagtc ccgctggcag
1261 cagggtaacg tcttcagctg ttccgtgatg cagaggcat tgcacaacca ctacaccag
1321 aagtcactga gcctgagccc agggaag
```

Protein Sequence Defining the Full Length HE LM 06C11
IGHV2-70 Heavy Chain (Humanized Heavy Chain Variable
Region and Human IgG1 Constant Region) (SEQ ID
NO:194)

```
1  qvtlkesgpa lvkptqtltl tctfsgfsln tygmgsuir qpgkalewl ahlywdddkr
61  ynpslktrlt iskdtsknqv vltitnvdpv dtavyycagr gyddywgywg qgtivtissa
121 stkgpsvfpl apssktsagg taalgcclvk yfpepvtvsw nsgaltsgvh tfpavlqssg
181 lyslssvvtv pssslgtqty icnvnhkpsn tkvdkrvepk scdkthtcpp cpapellggp
241 svflfppkpk dtlmisrtpv vtcvvdvsh edpevkfnwy vdgvevhnak tkpreeqyns
301 tyrvvsvltv lhqdwlngke ykckvsnkai papiektisk akqgprepqv ytlppsreem
361 tknqvsltcl vkgyfysdia vewesnggpe nnykttppvl dsdgsfflys kltvdksrwg
421 qgnvfscsvm healhnhytq kslslspgk
```

Nucleic Acid Sequence Encoding the Full Length Hu06C11
IGHV2-5 Heavy Chain (Humanized Heavy Chain Variable
Region and Human IgG1 Constant Region) (SEQ ID
NO:195)

```
1 caagtaacgc tcaaggagtc cggacccacc ttggtgaagc cgacgcagac cttgactctt
61 acgtgcactt tctcggggtt ttcactgaat acgtacggga tgggtgtctc atggatcagg
121 caacctccgg ggaaaggatt ggaatggctg gcgcacatct actgggatga cgataagaga
181 tataacccaa gcctcaagtc gcggctcacc attacaaaag atacatcgaa aaatcaggtc
241 gtacttacta tcacgaacat ggaccccgctg gacacagcaa catattactg tgcccagcgc
301 ggctatgacg attattgggg ttactgggga cagggaacac tggtcacggg gtccagcgcc
361 tcaacaaaag gaccaagtgt gttcccactc gccctagca gcaagagtac atccgggggc
421 actgcagcac tcggctgcct cgtcaaggat tattttccag agccagtaac cgtgagctgg
481 aacagtggag cactcacttc tgggtgccat acttttctcg ctgtcctgca aagctctggc
541 ctgtactcac tcagctccgt cgtgaccgtg ccattctcat ctctgggcac tcagacctac
601 atctgtaatg taaaccacaa gcctagcaat actaaggctg ataagcgggt ggaacccaag
661 agctgcgaca agactcacac ttgtcccca tgccctgcc ctgaacttct gggcgggtccc
721 agcgtctttt tgttccacc aaagcctaaa gatactctga tgataagtag aacacccgag
781 gtgacatgtg ttgtttaga cgtttccac gaggaccag aggttaagtt caactggtac
841 gttgatggag tcgaagtaca taatgctaag accaagccta gagaggagca gtataatagt
901 acataccgtg tagtcagtgt tctcacagt ctgcaccaag actggctcaa cggcaaagaa
961 taaaaatgca aagtgtccaa caaagcactc ccagccccta tcgagaagac tattagtaag
1021 gcaaaggggc agcctcgtga accacaggtg tacactctgc caccagtag agaggaaatg
1081 acaaagaacc aagtctcatt gacctgcctg gtgaaaggct tctacccag cgacatcgcc
1141 gttgagtggg agagtaacgg tcagcctgag aacaattaca agacaacccc ccagtgctg
1201 gatagtgacg ggtctttctt tctgtacagt aagctgactg tggacaagtc ccgctggcag
1261 cagggtaacg tcttcagctg ttcggtgatg cagaggcat tgcacaacca ctacaccag
1321 aagtcaactga gcctagagccc agggaag
```

Protein Sequence Defining the Full Length Hu06C11
IGHV2-5 Heavy Chain (Humanized Heavy Chain Variable
Region and Human IgG1 Constant Region) (SEQ ID
NO:196)

```
1 qvtlkesgpt lvkptqtltl tctfsgfsln tygmgvswir qppgkglewl ahlywdddkr
61 ynpslksrlt itkdtsknqv vltitnmdpv dtatyycagr gyddywgywg qgtlvtvssa
121 stkgpsvfpl apssktsagg taalgcclvk yfpepvtvsw nsgaltsgvh tfpavlqssg
181 lyslssvvtv pssslgtqty icnvnhkpsn tkvdkrvepk scdkthtcpp cpapellggp
241 svflfppkpk dtlmisrtp vtcvvdvsh edpevkfnwy vdgvevhnak tkpreeqyns
301 tyrvvsvltv lhqdwlngke ykckvsnkai papiektisk akqgprepqv ytlppsreem
361 tknqvsltcl vkgyfysdia vewesnggpe nnykttppvl dsdgsfflys kltvdksrwg
421 qgnvfscsvm healhnhytq kslslspgk
```

Nucleic Acid Sequence Encoding the Full Length Ch14F11
Chimeric Heavy Chain (Mouse Heavy Chain Variable
Region and Human IgG1 Constant Region) (SEQ ID
NO:197)

```
1  cagggtcacgc tgaagagtc aggtcccga atccttcaac ctctgcagac attgtcactc
61  acatgttctc tctccgggtt ctctgtctcg acttatggca tgggtgtagg atggattcgg
121 cagcccagcg ggaaggggct tgagtgggtg gcgatatct ggtgggacga cgacaaatac
181 tacaatccga gctgaagtc ccgctcacc atttcgaaag atacgtcatc aaacgaagtc
241 tttttgaaga tcgccatcgt ggacacggcg gatacagcga cgtattactg cgccagaagg
301 ggacactaca gcgaatgga ttattgggga caggggacct cgtgactgtg gtcgtccgcc
361 tcaacaaaag gaccaagtgt gttcccactc gccctagca gcaagagtac atccgggggc
421 actgcagcac tcggctgcct cgtcaaggat tattttccag agccagtaac cgtgagctgg
481 aacagtggag cactcacttc tgggtgccat acttttctcg ctgtcctgca aagctctggc
541 ctgtactcac tcagctccgt cgtgaccgtg ccactctcat ctctgggcac tcagacctac
601 atctgtaatg taaaccacaa gcctagcaat actaaggctg ataagcgggt ggaacccaag
661 agctgcgaca agactcacac ttgtcccca tgccctgcc ctgaacttct gggcgggtcc
721 agcgtctttt tgttccacc aaagcctaaa gatactctga tgataagtag aacacccgag
781 gtgacatgtg ttgtttaga cgtttccac gaggaccag aggttaagtt caactggtac
841 gttgatggag tcgaagtaca taatgctaag accaagccta gagaggagca gtataatagt
901 acataccgtg tagtcagtgt tctcacagtg ctgcaccaag actggctcaa cggcaaagaa
961 tacaatgca aagtgtccaa caaagcactc ccagccccta tcgagaagac tattagtaag
1021 gcaaaggggc agcctcgtga accacaggtg tactctctgc caccagtag agaggaaatg
1081 acaagaacc aagtctcatt gacctgcctg gtgaaaggct tctacccag cgacatcgcc
1141 gttgagtggg agagtaacgg tcagcctgag aacaattaca agacaacccc ccagtgctg
1201 gatagtgcg ggtctttctt tctgtacagt aagctgactg tggacaagtc ccgctggcag
1261 cagggtaacg tcttcagctg ttcgtgatg cagaggcat tgcacaacca ctacaccag
1321 aagtcactga gcctgagccc agggaag
```

Protein Sequence Defining the Full Length Ch14F11
Chimeric Heavy Chain (Mouse Heavy Chain Variable
Region and Human IgG1 Constant Region) (SEQ ID
NO:198)

```
1  qvtlkespgp ilqpsqtlsl tcsfsgfsls tygmvgwir qpsgkglewl adiwdddky
61  ynpslksrlt iskdtsnev flkiaivdta dtatyycarr ghysamdywg qgtsvtvssa
121 stkgpsvfpl apssktsagg taalgcclvk yfpepvtvsw nsgaltsgvh tfpavlqssg
181 lyslssvvtv pssslgtqty icnvnhkpsn tkvdkrvepk scdkthtcpp cpapellgpp
241 svflfppkpk dtlmisrtp vtcvvdvsh edpevkfnwy vdgvevhnak tkpreeqyns
301 tyrvvsvltv lhqdwlngke ykckvsnkai papiektisk akqgprepqv ytlppsreem
361 tknqvsltlc vkgyfysdia vewesnggpe nnykttppvl dsdgsfflys kltvdksrwg
421 qgnvfscsvm healhnhytq kslslspgk
```

Nucleic Acid Sequence Encoding the Full Length Sh14F11
IGHV2-5 Heavy Chain (Humanized Heavy Chain Variable
Region and Human IgG1 Constant Region) (SEQ ID
NO:199)

```
1  cagatcactt tgaagaaaag cggaccgacc ttggtcaagc ccacacaaac cctcacgctc
61  acgtgtacat ttctgggggt ctcgctttca acttacggga tgggagtagg gtggattcgc
121 cagccgcctg gtaaagcggt ggagtggctt gcagacatct ggtgggacga cgataagtac
181 tataatccct cgctcaagtc cagactgacc atcacgaaag atacgagcaa gaaccaggtc
241 gtgtgacaa tgactaacat ggaccagtg gatacggcta catattactg cgccaggcgg
301 ggctactact cagcgatgga ttattggggc cagggaacac tggtaacggt gtcgtccgcc
361 tcaacaaaag gaccaagtgt gttcccactc gccctagca gcaagagtac atccgggggc
421 actgcagcac tcggctgcct cgtcaaggat tattttccag agccagtaac cgtgagctgg
481 aacagtggag cactcacttc tgggtgccat acttttctcg ctgtcctgca aagctctggc
541 ctgtactcac tcagctccgt cgtgaccgtg ccatcttcat ctctgggcac tcagacctac
601 atctgtaatg taaaccacaa gcctagcaat actaaggctc ataagcgggt ggaacccaag
661 agctgcgaca agactcacac ttgtcccca tgccctgcc ctgaacttct gggcgggtccc
721 agcgtctttt tgttccacc aaagcctaaa gatactctga tgataagtag aacacccgag
781 gtgacatgtg ttgtttaga cgtttccac gaggaccag aggttaagtt caactggtac
841 gttgatggag tcgaagtaca taatgctaag accaagccta gagaggagca gtataatagt
901 acataccgtg tagtcagtgt tctcacagtg ctgcaccaag actggctcaa cggcaaagaa
961 taaaaatgca aagtgtccaa caaagcactc ccagccccta tcgagaagac tattagtaag
1021 gcaaaggggc agcctcgtga accacaggtg tacactctgc caccagtag agaggaaatg
1081 acaaagaacc aagtctcatt gacctgcctg gtgaaaggct tctacccag cgacatcgcc
1141 gttgagtggg agagtaacgg tcagcctgag aacaattaca agacaacccc cccagtgtctg
1201 gatagtgcag ggtctttctt tctgtacagt aagctgactg tggacaagtc ccgctggcag
1261 cagggtaacg tcttcagctg ttcgtgatg cagaggcat tgcacaacca ctacaccag
1321 aagtcaactga gcctgagccc agggaag
```

Protein Sequence Defining the Full Length Sh14F11
IGHV2-5 Heavy Chain (Humanized Heavy Chain Variable
Region and Human IgG1 Constant Region) (SEQ ID
NO:200)

```
1  qitlkesgpt lvkptqtltl tctfsgfsls tygmvgwir qpgkalewl adiwdddky
61  ynpslksrlt itkdtsknqv vltmtndpv dtatyycarr ghysamdyw qgtlvtvssa
121 stkgpsvfpl apsskctsgg taalgcclvk yfpepvtvsw nsgaltsgvh tfpavlqssg
181 lyslssvvtv pssslgtqty icnvnhkpsn tkvdkrvepk scdkthtcpp cpapellggy
241 svflfppkpk dtlmisrtpe vtcvvvdvsh edpevkfnwy vdgvevhnak tkpreeqyns
301 tyrvvsvltv lhqdwlngke ykckvsnkai papiektisk akqgprepqv ytlppsreem
361 tknqvsltcl vkgyfysdia vewesnggpe nnykttppvl dsdgsfflys kltvdksrwg
421 qgnvfscsvm healhnhytq kslslspgk
```

Nucleic Acid Sequence Encoding the Full Length Sh14F11
IGHV2-70 Heavy Chain (Humanized Heavy Chain Variable
Region and Human IgG1 Constant Region) (SEQ ID
NO:201)

```
1  caagtgactc tcaaggagtc cggacccgcc ctggtcaaac caacgcagac actgacgctc
61  acatgcacct tcagcggatt ttcgttgtea acgtacggca tgggtgtggg gtggattcgc
121 cagcctccgg ggaagccct tgaatggtg gcggacatct ggtgggatga tgacaagtac
181 tataatccct cacttaagtc acggttgacg atctcgaaag acaccagcaa gaaccaggta
241 gtgctgacaa tgactaacat ggaccggtc gatacagcgg tctactattg tgctagaagg
301 ggacactact ccgaatgga ttattggggc caggggacgc tcgtaaccgt gtcgtcggcc
361 tcaacaaaag gaccaagtgt gttcccactc gccccctagc gcaagagtac atccgggggc
421 actgcagcac tcggctgcct cgtcaaggat tattttccag agccagtaac cgtgagctgg
481 aacagtggag cactcacttc tgggtgccat acttttctcg ctgtcctgca aagctctggc
541 ctgtactcac tcagctccgt cgtgaccgtg ccatcttcat ctctgggcac tcagacctac
601 atctgtaatg taaaccacaa gcctagcaat actaaggctc ataagcgggt ggaacccaag
661 agctgcgaca agactcacac ttgtccccc tgccctgccc ctgaacttct gggcgggtccc
721 agcgtctttt tgttcccacc aaagcctaaa gatactctga tgataagtag aacacccgag
781 gtgacatgtg ttgtttaga cgtttccac gaggaccag aggttaagtt caactggtac
841 gttgatggag tcgaagtaca taatgctaag accaagccta gagaggagca gtataatagt
901 acataccgtg tagtcagtgt tctcacagtg ctgcaccaag actggctcaa cggcaaagaa
961 taaaaatgca aagtgtccaa caaagcactc ccagccccta tcgagaagac tattagtaag
1021 gcaaaggggc agcctcgtga accacaggtg tacactctgc caccagtag agaggaaatg
1081 acaagaacc aagtctcatt gacctgcctg gtgaaaggct tctacccag cgacatcgcc
1141 gttgagtggt agagtaacgg tcagcctgag aacaattaca agacaacccc cccagtgtctg
1201 gatagtgcag ggtctttctt tctgtacagt aagctgactg tggacaagtc ccgctggcag
1261 cagggtaacg tcttcagctg ttcgtgatg cagaggcat tgcacaacca ctacaccag
1321 aagtcaactga gcctgagccc agggaag
```

Protein Sequence Defining the Full Length Sh14F11
IGHV2-70 Heavy Chain (Humanized Heavy Chain Variable
Region and Human IgG1 Constant Region) (SEQ ID
NO:202)

```
1  qvtlkesgpa lvkptqtltl tctfsgfsls tygmvgwir qpgkalewl adiwdddky
61  ynpslksrlt iskdtsknqv vltmtndpv dtavyycarr ghysamdyw qgtlvtvssa
121 stkgpsvfpl apsskctsgg taalgcclvd yfpepvtvsw nsgaltsgvh tfpavlqssg
181 lyslssvvtv pssslgtqty icnvnhkpsn tkvdkrvepk scdkthtcpp cpapellggp
241 svflfppkpk dtlmisrtpv vtcvvdvsh edpevkfnwy vdgvevhnak tkpreeqyns
301 tyrvvsvltv lhqdwlngke ykckvsnkai papiektisk akqgprepqv ytlppsreem
361 tknqvsltcl vkgyfysdia vewesngqpe nnykttppvl dsdgsfflys kltdksrwg
421 qgnvfscsvm healhnhytq kslslspgk
```

Nucleic Acid Sequence Encoding the Full Length Ch01G06
Chimeric Light Chain (Mouse Kappa Chain Variable Region
and Human Kappa Constant Region) (SEQ ID NO:203)

```
1  gacatccaaa tgaccagtc acccgcgagc ctttcggcgt cggtcggaga aacggtcacg
61  atcacgtgcc ggacatcaga gaatctccat aactacctcg cgtggtatca acagaagcag
121 gggaagtcgc cccagttgct tgtatcagat gcgaaaacgt tggcggatgg ggtgccgtcc
181 agattctcgg gatcgggctc ggggacgcag tactcgtca agatcaattc gctgcagccg
241 gaggactttg ggtcgtacta ttgtcagcat ttttggtcat caccgtatac atttgagggt
301 ggaacgaaac ttgagattaa gcgcacagtt gctgccccca gcgtgttcat ttccccacct
361 agcgtatgagc agctgaaaag cggctactgcc tctgtcgtat gcttgcctaa caacttttac
421 ccacgtgagg ctaagggtgca gtggaaagtg gataatgcac ttcaatctgg aaacagtcaa
481 gagtcctgta cagaacagga cagcaaagac tcaacttatt cactctcttc caccctgact
541 ctgtccaagg cagactatga aaaacacaag gtatacgctt gcgaggttac acaccagggt
601 ttgtctagtc ctgtcaccaa gtccttcaat agggggcgaat gt
```

Protein Sequence Defining the Full Length Ch01G06
Chimeric Light Chain (Mouse Kappa Chain Variable Region
and Human Kappa Constant Region) (SEQ ID NO:204)

```
1  diqmtqspas lsasvgetvt itcrtsenlh nylawyqqkq gkspqllvyd aktladgvps
61  rfsgsgsgtgq yslkinslqp edfgyycqh fwsspytfgg gtleikrtv aapsvfifpp
121 sdeqlksqta svvcllnnfy preakvqkv dnalqsgnsq esvteqdskd styslsstlt
181 lskadyekhk vyacevthqg lsspvtksfm rgec
```

Nucleic Acid Sequence Encoding the Full Length Hu01G06
IGKV1-39 Light Chain (Humanized Kappa Chain Variable
Region and Human Kappa Constant Region) (SEQ ID
NO:205)

```
1  gacatccaaa tgaccagtc gccgtcgtcg ctttcagcgt cggtagggga tcgggtcaca
61  attacgtgcc gaacgtcaga gaatttgcac aactacctcg cgtggtatca gcagaagccc
121 gggaagtcac cgaaactcct tgtctacgat gcgaaaacgc tggcggatgg agtgccgtcg
181 agattctcgg gaagcggatc cggtagcgac tatacgttta cgatctcacc gctccagccc
241 gaggactttg cgacgtacta ttgtcagcat ttttggtcgt cgcctacac atttgggcag
301 gggaccaagt tggaatcaa gcgcacagtt gctgccccca gcgtgttcat ttccccacct
361 agcgtatgagc agctgaaaag cggctactgcc tctgtcgtat gcttgcctaa caacttttac
421 ccacgtgagg ctaagggtgca gtggaaagtg gataatgcac ttcaatctgg aaacagtcaa
481 gagtcctgta cagaacagga cagcaaagac tcaacttatt cactctcttc caccctgact
541 ctgtccaagg cagactatga aaaacacaag gtatacgctt gcgaggttac acaccagggt
601 ttgtctagtc ctgtcaccaa gtccttcaat agggggcgaat gt
```

Protein Sequence Defining the Full Length Hu01G06
IGKV1-39 Light Chain (Humanized Kappa Chain Variable
Region and Human Kappa Constant Region) (SEQ ID
NO:206)

```

1 diqmtqspss lsasvgdrvt itcrtsenlh nylawyqqkp gkspkllvyd aktladgvps
61 rfsgsgsgtd ytltisslqp edfatyycqf fwsspytfqg gtleikrtv aapsvfifpp
121 sdeqlksgta svvcllnnfy preakvqwkf dnalqsgnsq esvteqdskd styslsstlt
181 lskadyekhk vyacevthqg lsspvtksfh rgec

```

Nucleic Acid Sequence Encoding the Full Length Hu01G06
IGKV1-39 S43A V48I Light Chain (Humanized Kappa
Chain Variable Region and Human Kappa Constant Region)
(Also Referred to Herein as the Full Length Hu01G06
IGFV1-39 F1 Light Chain; SEQ ID NO:207)

```

1 gacatccaaa tgaccagtc gccgtcgtcg ctttcagcgt cggtagggga tcgggtcaca
61 attacgtgcc gaacgtcaga gaatttgcac aactacctcg cgtggtatca gcagaagccc
121 gggaagggcc cgaaactcct tatctacgat gcgaaaacgc tggcggatgg agtgccgtcg
181 agattctcgg gaagcggatc cggtagcgac tatacgctta cgatctcacc gctccagccc
241 gaggaacttg cgacgtacta ttgtcagcat ttttggtcgt cgccctacac atttgggcag
301 gggaccaagt tggaaatcaa gcgcacagtt gctgccccca gcgtgttcat tttccacct
361 agcgtgagc agctgaaaag cggtagtcgc tctgtcgtat gcttgctcaa caacttttac
421 ccacgtgagg ctaagggtga gtggaaagtg gataatgcac ttcaatctgg aaacagtcaa
481 gagtcctga cagaacagga cagcaaagac tcaacttatt cactctcttc caccctgact
541 ctgtccaagg cagactatga aaaacacaag gtatacgctt gcgaggttac acaccagggg
601 ttgtctagtc ctgtcaccaa gtccttcaat agggggcgaat gt

```

Protein Sequence Defining the Full Length Hu01G06
IGKV1-39 S43A V48I Light Chain (Humanized Kappa
Chain Variable Region and Human Kappa Constant Region)
(also Referred to Herein as the Full Length Hu01G06
IGFV1-39 F1 Light Chain; SEQ ID NO:208)

```

1 diqmtqspss lsasvgdrvt itcrtsenlh nylawyqqkp gkapklliyd aktladgvps
61 rfsgsgsgtd ytltisslqp edfatyycqf fwsspytfqg gtleikrtv aapsvfifpp
121 sdeqlksgta svvcllnnfy preakvqwkf dnalqsgnsq esvteqdskd styslsstlt
181 lskadyekhk vyacevthqg lsspvtksfh rgec

```

Nucleic Acid Sequence Encoding the Full Length Hu01G06
IGKV1-39 V48I Light Chain (Humanized Kappa Chain
Variable Region and Human Kappa Constant Region) (SEQ
ID NO:209)

```

1 gacatccaaa tgaccagtc gccgtcgtcg ctttcagcgt cggtagggga tcgggtcaca
61 attacgtgcc gaacgtcaga gaatttgcac aactacctcg cgtggtatca gcagaagccc
121 gggaagtcac cgaaactcct tatctacgat gcgaaaacgc tggcggatgg agtgccgtcg
181 agattctcgg gaagcggatc cggtagcgac tatacgctta cgatctcacc gctccagccc

```

-continued

241 gaggactttg cgacgtacta ttgtcagcat ttttggtcgt cgcctacac atttgggcag
 301 gggaccaagt tggaaatcaa gcgcacagtt gctgccccca gcgtgttcat tttccacct
 361 agcgatgagc agctgaaaag cggactgcc tctgtcgtat gcttgetcaa caacttttac
 421 ccacgtgagg ctaaggtgca gtggaaagtg gataatgcac ttcaatctgg aaacagtcaa
 481 gagtccgtga cagaacagga cagcaaagac tcaacttatt cactctcttc caccctgact
 541 ctgtccaagg cagactatga aaaacacaag gtatacgctt gcgaggttac acaccagggt
 601 ttgtctagtc ctgtcaccaa gtccttcaat aggggcgaat gt

Protein Sequence Defining the Full Length Hu01G06
 IGKV1-39 V48I Light Chain (Humanized Kappa Chain
 Variable Region and Human Kappa Constant Region) (SEQ
 ID NO:210)

1 diqmtqspss lsasvgdrvt itcrtsenlh nylawyqqkp gkspklliyl aktladgvps
 61 rfsgsgsgtd ylttisslqp edfatyyqch fwsspytfqg gtleikrtv aapsvfifpp
 121 sdeqlksgta svvcllnnfy breakvqkv dnalqsgnsq esvteqdskd styslsstlt
 181 lskadyekhk vyacevthqg lsspvtksfm rgec

Nucleic Acid Sequence Encoding the Full Length Hu01G06
 IGKV1-39 F1 Light Chain (Humanized Kappa Chain Vari-
 able Region and Human Kappa Constant Region) (also
 referred to herein as the Full Length Hu01G06 IGKV1-39
 S43A V48I Light Chain; SEQ ID NO:207)

1 gacatccaaa tgaccagtc gccgtcgtcg ctttcagcgt cggtagggga tcgggtcaca
 61 attacgtgcc gaacgtcaga gaatttgcat aactacctcg cgtggtatca gcagaagccc
 121 gggaaagccc cgaactcct tatctacgat gcgaaaacgc tggcggtatg agtgccgtcg
 181 agattctcgg gaagcggatc cggtagcgac tatacgctta cgatctcatc gctccagccc
 241 gaggactttg cgacgtacta ttgtcagcat ttttggtcgt cgcctacac atttgggcag
 301 gggaccaagt tggaaatcaa gcgcacagtt gctgccccca gcgtgttcat tttccacct
 361 agcgatgagc agctgaaaag cggactgcc tctgtcgtat gcttgetcaa caacttttac
 421 ccacgtgagg ctaaggtgca gtggaaagtg gataatgcac ttcaatctgg aaacagtcaa
 481 gagtccgtga cagaacagga cagcaaagac tcaacttatt cactctcttc caccctgact
 541 ctgtccaagg cagactatga aaaacacaag gtatacgctt gcgaggttac acaccagggt
 601 ttgtctagtc ctgtcaccaa gtccttcaat aggggcgaat gt

Protein Sequence Defining the Full Length Hu01G06
 IGKV1-39 F1 Light Chain (Humanized Kappa Chain Vari-
 able Region and Human Kappa Constant Region) (also
 referred to herein as the Full Length Hu01G06 IGKV1-39
 S43A V48I Light Chain; SEQ ID NO:208)

1 diqmtqspss lsasvgdrvt itcrtsenlh nylawyqqkp gkapklliyl aktladgvps
 61 rfsgsgsgtd ylttisslqp edfatyyqch fwsspytfqg gtleikrtv aapsvfifpp
 121 sdeqlksgta svvcllnnfy breakvqkv dnalqsgnsq esvteqdskd styslsstlt
 181 lskadyekhk vyacevthqg lsspvtksfm rgec

Nucleic Acid Sequence Encoding the Full Length Hu01G06
IGKV1-39 F2 Light Chain (Humanized Kappa Chain
Variable Region and Human Kappa Constant Region) (SEQ
ID NO:263)

```
1  gacatccaaa tgaccagtc gccgctgctg ctttcagcgt cggtagggga tcgggtcaca
61  attacgtgcc gaacgtcaga gaatttgcac aactacctcg cgtggtatca gcagaagccc
121  gggaaagtcac cgaaactcct tatctacgat gcgaaaacgc tggcggatgg agtgccgctcg
181  agattctcgg gaagcgggac cggtagcgac tatacgctta cgatctcacc gctccagccc
241  gaggactttg cgacgtacta ttgtcagcat ttttggtcgg acccctacac atttgggcag
301  gggaccaagt tggaaatcaa gcgcacagtt gctgccccca gcgtgttcat tttccacct
361  agcgtgagc agctgaaaag cggtagctgcc tctgtcgtat gcttgetcaa caacttttac
421  ccacgtgagg ctaagtgca gtggaaagtg gataatgcac ttcaatctgg aaacagtcaa
481  gagtccgtga cagaacagga cagcaaagac tcaacttatt cactctcttc caccctgact
541  ctgtccaagg cagactatga aaaacacaag gtatacgctt gcgaggttac acaccagggt
601  ttgtctagtc ctgtcaccaa gtccttcaat aggggcgaat gt
```

Protein Sequence Defining the Full Length Hu01G06
IGKV1-39 F2 Light Chain (Humanized Kappa Chain
Variable Region and Human Kappa Constant Region) (SEQ
ID NO:264)

```
1  diqmtqspss lsasvdrvt itcrtsenlh nylawyqqkp gkspkllyd aktladgvps
61  rfsqsgsgtd ytltisslqp edfatyyqch fwsdpytfqg gtleikrtv aapsvfifpp
121  sdeqlksqta svvcllnfy breakvqwk dnalqsgnsq esvteqdskd styslsstlt
181  lskadyekhk vyacevthqg lsspvtksf n rgec
```

Nucleic Acid Sequence Encoding the Full Length Ch06C11
Chimeric Light Chain (Mouse Kappa Chain Variable Region
and Human Kappa Constant Region) (SEQ ID NO:211)

```
1  gatatcgta tgaccagtc ccagaagttc atgtcaactt cagtgggaga cagagtgtcc
61  gtcacatgta aagcctcgca aaatgtggga accaacgtag cgtggttcca gcagaaacct
121  ggccaatcac cgaaggcact gatctactcg gccagctata ggtactcggg agtaccagat
181  cggtttacgg ggtcggggag cgggacggac tttatctca ctatttcaa tgtccagtcg
241  gaggaccttg cggaatactt ctgccagcag tataacaact atcccctcac gtttggtgct
301  ggtacaaaat tggagttgaa gcgcacagtt gctgccccca gcgtgttcat tttccacct
361  agcgtgagc agctgaaaag cggtagctgcc tctgtcgtat gcttgetcaa caacttttac
421  ccacgtgagg ctaagtgca gtggaaagtg gataatgcac ttcaatctgg aaacagtcaa
481  gagtccgtga cagaacagga cagcaaagac tcaacttatt cactctcttc caccctgact
541  ctgtccaagg cagactatga aaaacacaag gtatacgctt gcgaggttac acaccagggt
601  ttgtctagtc ctgtcaccaa gtccttcaat aggggcgaat gt
```

Protein Sequence Defining the Full Length Ch06C11
Chimeric Light Chain (Mouse Kappa Chain Variable Region
and Human Kappa Constant Region) (SEQ ID NO:212)

```
1 divmtqsqkf mtsvsgdrvs vtckasqnvq tnvwfqqkp gqspkaliys asyrysgvdp
61 rftsgsgsgtd filtisnvqs edlaeyfcqq ynnyppltfga gtklelkrtv aapsvfifpp
121 sdeqlksgta svvcllnnfy preakvqwk v dnalqsgnsq esvteqdskd styslsstlt
181 lskadyekhk vyacevthqg lsspvtksfn rgec
```

Nucleic Acid Sequence Encoding the Full Length Sh06C11
IGKV1-16 Light Chain (Humanized Kappa Chain Variable
Region and Human Kappa Constant Region) (SEQ ID
NO:213)

```
1 gacatccaaa tgaccaate gccctcctcc ctctccgcat cagtagggga ccgcgtcaca
61 attacttgca aagcgtcgca gaacgtcgga acgaatgtgg cgtggtttca gcagaagccc
121 ggaaaagctc cgaagagctt gatctactcg gcctcatata ggtattcggg tgtgccgagc
181 cggtttagcg ggtcggggtc aggtactgat ttcacgctca caatttcato gttgcagcca
241 gaagatttcg ccacatatta ctgtcagcag tacaacaatt accctctgac gttcggccag
301 ggaaccaaac ttgagatcaa gcgcacagtt gctgccccca gcgtgttcat tttccacct
361 agcgatgagc agctgaaaag cgggtactgcc tctgtcgtat gcttgctcaa caacttttac
421 ccacgtgagg ctaaggtgca gtggaaagtg gataatgcac ttcaatctgg aaacagtcaa
481 gagtccgtga cagaacagga cagcaaagac tcaacttatt cactctcttc caccctgact
541 ctgtccaagg cagactatga aaaacacaag gtatacgctt gcgaggttac acaccagggt
601 ttgtctagtc ctgtcaccaa gtccttcaat aggggcgaat gt
```

Protein Sequence Defining the Full Length Sh06C11
IGKV1-16 Light Chain (Humanized Kappa Chain Variable
Region and Human Kappa Constant Region) (SEQ ID
NO:214)

```
1 diqmtqspss lsasvgdrvt itckasqnvq tnvwfqqkp gkapksliys asyrysgvps
61 rfsqsgsgtd fsltisslqp edfatyyccq ynnyppltfgq gtkleikrtv aapsvfifpp
121 sdeqlksgta svvcllnnfy preakvqwk v dnalqsgnsq esvteqdskd styslsstlt
181 lskadyekhk vyacevthqg lsspvtksfn rgec
```

Nucleic Acid Sequence Encoding the Full Length Ch14F11
Chimeric Light Chain (Mouse Kappa Chain Variable Region
and Human Kappa Constant Region) (SEQ ID NO:215)

```
1 gacatcgtga tgacacagtc acagaaattc atgtccacat ccgtcgggtga tagagtatcc
61 gtcacgtgta aggcctcgca aaacgtagga actaatgtgg cgtggatatca acagaagcca
121 ggacagtcac ccaaagcact catctacagc ccctcatatc ggtacagcgg ggtgccggac
181 aggttcacgg gatcggggag cgggaccgat tttacactga ccatttcgaa tgtccagtcg
241 gaggaccttg cggaatactt ctgccagcag tataactcgt accctcacac gtttggaggt
301 ggactaaagt tggagatgaa acgcacagtt gctgccccca gcgtgttcat tttccacct
361 agcgatgagc agctgaaaag cgggtactgcc tctgtcgtat gcttgctcaa caacttttac
```

-continued

421 ccacgtgagg ctaaggtgca gtggaaagtg gataatgcac ttcaatctgg aaacagtcaa
481 gagtccgtga cagaacagga cagcaaagac tcaacttatt cactctcttc caccctgact
541 ctgtccaagg cagactatga aaaacacaag gtatacgctt gcgaggttac acaccagggt
601 ttgtctagtc ctgtcaccaa gtccttcaat aggggcgaat gt

Protein Sequence Defining the Full Length Ch14F11
Chimeric Light Chain (Mouse Kappa Chain Variable Region
and Human Kappa Constant Region) (SEQ ID NO:216)

1 divmtqskf mtsvsgdrvs vtckasqnvq tnvaqqkpk gqspkaliys psyrysgvdp
61 rftsgsgstd ftltisnvqs edlaeyfcqq ynsyphtfgg gtklemkrtv aapsvfifpp
121 sdeqlksgta svvcllnnfy preakvqwkv dnalqsgnsq esvteqdskd styslsstlt
181 lskadyekhk vyacevthqg lsspvtksfm rgec

Nucleic Acid Sequence Encoding the Full Length Hu14F11
IGKV1-16 Light Chain (Humanized Kappa Chain Variable
Region and Human Kappa Constant Region) (SEQ ID
NO:217)

1 gatatccaga tgacacagtc accctcgtcg ctctcagctt ccgtaggcga cagggtcact
61 attacgtgta aagcatcaca gaacgtcggg acgaatgtgg cgtggtttca gcagaagccc
121 ggggaagagcc ccaaagcgct tatctactcc ccgtcgtatc ggtattccgg tgtgccaagg
181 agattttcgg ggtcaggttc gggaactgac ttaccctga ccattctctc cctccaaccg
241 gaagatttcg ccacgtactt ctgccagcag tacaacagct atcctcacac attcggacaa
301 gggacaaaagt tggagattaa acgcacagtt gctgccccca gcgtgttcat tttccacct
361 agcgatgagc agctgaaaag cggtagctgc tctgtctgat gcttgcctca caacttttac
421 ccacgtgagg ctaaggtgca gtggaaagtg gataatgcac ttcaatctgg aaacagtcaa
481 gagtccgtga cagaacagga cagcaaagac tcaacttatt cactctcttc caccctgact
541 ctgtccaagg cagactatga aaaacacaag gtatacgctt gcgaggttac acaccagggt
601 ttgtctagtc ctgtcaccaa gtccttcaat aggggcgaat gt

Protein Sequence Defining the Full Length Hu14F11
IGKV1-16 Light Chain (Humanized Kappa Chain Variable
Region and Human Kappa Constant Region) (SEQ ID
NO:218)

1 diqmtqspss lsasvsgdrvt itckasqnvq tnvaqqkpk gkspkaliys psyrysgvps
61 rfsqsgsgstd ftltisslqp edfatyfcqq ynsyphtfgg gtkleikrtv aapsvfifpp
121 sdeqlksgta svvcllnnfy preakvqwkv dnalqsgnsq esvteqdskd styslsstlt
181 lskadyekhk vyacevthqg lsspvtksfm rgec

[0311] Table 22 is a concordance chart showing the SEQ ID NO. of each sequence discussed in this Example.

TABLE 22

SEQ ID NO. Nucleic Acid or Protein	
171	Human IgG1 constant - nucleic acid
172	Human IgG1 constant - protein
173	Human Kappa constant - nucleic acid
174	Human Kappa constant - protein
175	Humanized Ch01G06 Chimeric Heavy Human Variable + Human IgG1 constant - nucleic acid
176	Humanized Ch01G06 Chimeric Heavy Human Variable + Human IgG1 constant - protein
177	Humanized Hu01G06 IGHV1-18 Heavy Human Variable + Human IgG1 constant - nucleic acid
178	Humanized Hu01G06 IGHV1-18 Heavy Human Variable + Human IgG1 constant - protein
179	Humanized Hu01G06 IGHV1-69 Heavy Human Variable + Human IgG1 constant - nucleic acid
180	Humanized Hu01G06 IGHV1-69 Heavy Human Variable + Human IgG1 constant - protein
181	Humanized Sh01G06 IGHV1-18 M69L Heavy Human Variable + Human IgG1 constant - nucleic acid
182	Humanized Sh01G06 IGHV1-18 M69L Heavy Human Variable + Human IgG1 constant - protein
183	Humanized Sh01G06 IGHV1-18 M69L K64Q G44S Heavy Human Variable + Human IgG1 constant - nucleic acid
184	Humanized Sh01G06 IGHV1-18 M69L K64Q G44S Heavy Human Variable + Human IgG1 constant - protein
185	Humanized Sh01G06 IGHV1-18 M69L K64Q Heavy Human Variable + Human IgG1 constant - nucleic acid
186	Humanized Sh01G06 IGHV1-18 M69L K64Q Heavy Human Variable + Human IgG1 constant - protein
187	Humanized Sh01G06 IGHV1-69 T30S I69L Heavy Human Variable + Human IgG1 constant - nucleic acid
188	Humanized Sh01G06 IGHV1-69 T30S I69L Heavy Human Variable + Human IgG1 constant - protein
189	Humanized Sh01G06 IGHV1-69 T30S K64Q I69L Heavy Human Variable + Human IgG1 constant - nucleic acid
190	Humanized Sh01G06 IGHV1-69 T30S K64Q I69L Heavy Human Variable + Human IgG1 constant - protein
255	Humanized Hu01G06 IGHV1-18 F1 Heavy Human Variable + Human IgG1 constant - nucleic acid
256	Humanized Hu01G06 IGHV1-18 F1 Heavy Human Variable + Human IgG1 constant - protein
257	Humanized Hu01G06 IGHV1-18 F2 Heavy Human Variable + Human IgG1 constant - nucleic acid
258	Humanized Hu01G06 IGHV1-18 F2 Heavy Human Variable + Human IgG1 constant - protein
259	Humanized Hu01G06 IGHV1-69 F1 Heavy Human Variable + Human IgG1 constant - nucleic acid
260	Humanized Hu01G06 IGHV1-69 F1 Heavy Human Variable + Human IgG1 constant - protein
261	Humanized Hu01G06 IGHV1-69 F2 Heavy Human Variable + Human IgG1 constant - nucleic acid
262	Humanized Hu01G06 IGHV1-69 F2 Heavy Human Variable + Human IgG1 constant - protein
191	Humanized Ch06C11 Chimeric Heavy Human Variable + Human IgG1 constant - nucleic acid
192	Humanized Ch06C11 Chimeric Heavy Human Variable + Human IgG1 constant - protein
193	Humanized HE LM 06C11 IGHV2-70 Heavy Human Variable + Human IgG1 constant - nucleic acid
194	Humanized HE LM 06C11 IGHV2-70 Heavy Human Variable + Human IgG1 constant - protein
195	Humanized Hu06C11 IGHV2-5 Heavy Human Variable + Human IgG1 constant - nucleic acid
196	Humanized Hu06C11 IGHV2-5 Heavy Human Variable + Human IgG1 constant - protein
197	Humanized Ch14F11 Chimeric Heavy Human Variable + Human IgG1 constant - nucleic acid
198	Humanized Ch14F11 Chimeric Heavy Human Variable + Human IgG1 constant - protein
199	Humanized Sh14F11 IGHV2-5 Heavy Human Variable + Human IgG1 constant - nucleic acid
200	Humanized Sh14F11 IGHV2-5 Heavy Human Variable + Human IgG1 constant - protein
201	Humanized Sh14F11-IGHV2-70 Heavy Human Variable + Human IgG1 constant - nucleic acid
202	Humanized Sh14F11-IGHV2-70 Heavy Human Variable + Human IgG1 constant - protein
203	Humanized Ch01G06 Chimeric Human Variable + Human Kappa constant - nucleic acid
204	Humanized Ch01G06 Chimeric Human Variable + Human Kappa constant - protein
205	Humanized Hu01G06 IGKV1-39 Human Variable + Human Kappa constant - nucleic acid
206	Humanized Hu01G06 IGKV1-39 Human Variable + Human Kappa constant - protein
207	Humanized Hu01G06 IGKV1-39 S43A V48I Human Variable + Human Kappa constant - nucleic acid
208	Humanized Hu01G06 IGKV1-39 S43A V48I Human Variable + Human Kappa constant - protein
209	Humanized Hu01G06 IGKV1-39 V48I Human Variable + Human Kappa constant - nucleic acid
210	Humanized Hu01G06 IGKV1-39 V48I Human Variable + Human Kappa constant - protein
207	Humanized Hu01G06 IGKV1-39 F1 Human Variable + Human Kappa constant - nucleic acid
208	Humanized Hu01G06 IGKV1-39 F1 Human Variable + Human Kappa constant - protein
263	Humanized Hu01G06 IGKV1-39 F2 Human Variable + Human Kappa constant - nucleic acid
264	Humanized Hu01G06 IGKV1-39 F2 Human Variable + Human Kappa constant - protein
211	Humanized Ch06C11 Chimeric Human Variable + Human Kappa constant - nucleic acid
212	Humanized Ch06C11 Chimeric Human Variable + Human Kappa constant - protein
213	Humanized Sh06C11 IGKV1-16 Human Variable + Human Kappa constant - nucleic acid
214	Humanized Sh06C11 IGKV1-16 Human Variable + Human Kappa constant - protein
215	Humanized Ch14F11 Chimeric Human Variable + Human Kappa constant - nucleic acid
216	Humanized Ch14F11 Chimeric Human Variable + Human Kappa constant - protein
217	Humanized Hu14F11 IGKV1-16 Human Variable + Human Kappa constant - nucleic acid
218	Humanized Hu14F11 IGKV1-16 Human Variable + Human Kappa constant - protein

[0312] Table 23 below shows antibodies containing chimeric immunoglobulin heavy and light chains and exem-

plary combinations of the full-length chimeric or humanized immunoglobulin heavy and light chains.

TABLE 23

Antibody Name	Light Chain	Heavy Chain
Hu01G06-1	Humanized Ch01G06 Chimeric Human Variable + Human Kappa constant (SEQ ID NO: 204)	Humanized Ch01G06 Chimeric Heavy Human Variable + Human IgG1 constant (SEQ ID NO: 176)
Hu01G06-46	Humanized Hu01G06 IGKV1-39 Human Variable + Human Kappa constant (SEQ ID NO: 206)	Humanized Hu01G06 IGHV1-18 Heavy Human Variable + Human IgG1 constant (SEQ ID NO: 178)
Hu01G06-52	Humanized Hu01G06 IGKV1-39 Human Variable + Human Kappa constant (SEQ ID NO: 206)	Humanized Hu01G06 IGHV1-69 Heavy Human Variable + Human IgG1 constant (SEQ ID NO: 180)
Hu01G06-100	Humanized Hu01G06 IGKV1-39 Human Variable + Human Kappa constant (SEQ ID NO: 206)	Humanized Sh01G06 IGHV1-18 M69L Heavy Human Variable + Human IgG1 constant (SEQ ID NO: 182)
Hu01G06-101	Humanized Hu01G06 IGKV1-39 Human Variable + Human Kappa constant (SEQ ID NO: 206)	Humanized Sh01G06 IGHV1-18 M69L K64Q Heavy Human Variable + Human IgG1 constant (SEQ ID NO: 186)
Hu01G06-102	Humanized Hu01G06 IGKV1-39 Human Variable + Human Kappa constant (SEQ ID NO: 206)	Humanized Sh01G06 IGHV1-18 M69L K64Q G44S Heavy Human Variable + Human IgG1 constant (SEQ ID NO: 184)
Hu01G06-103	Humanized Hu01G06 IGKV1-39 Human Variable + Human Kappa constant (SEQ ID NO: 206)	Humanized Sh01G06 IGHV1-69 T30S I69L Heavy Human Variable + Human IgG1 constant (SEQ ID NO: 188)
Hu01G06-104	Humanized Hu01G06 IGKV1-39 Human Variable + Human Kappa constant (SEQ ID NO: 206)	Humanized Sh01G06 IGHV1-69 T30S K64Q I69L Heavy Human Variable + Human IgG1 constant (SEQ ID NO: 190)
Hu01G06-105	Humanized Hu01G06 IGKV1-39 V48I Human Variable + Human Kappa constant (SEQ ID NO: 210)	Humanized Sh01G06 IGHV1-18 M69L Heavy Human Variable + Human IgG1 constant (SEQ ID NO: 182)
Hu01G06-106	Humanized Hu01G06 IGKV1-39 V48I Human Variable + Human Kappa constant (SEQ ID NO: 210)	Humanized Sh01G06 IGHV1-18 M69L K64Q Heavy Human Variable + Human IgG1 constant (SEQ ID NO: 186)
Hu01G06-107	Humanized Hu01G06 IGKV1-39 V48I Human Variable + Human Kappa constant (SEQ ID NO: 210)	Humanized Sh01G06 IGHV1-18 M69L K64Q G44S Heavy Human Variable + Human IgG1 constant (SEQ ID NO: 184)
Hu01G06-108	Humanized Hu01G06 IGKV1-39 V48I Human Variable + Human Kappa constant (SEQ ID NO: 210)	Humanized Sh01G06 IGHV1-69 T30S I69L Heavy Human Variable + Human IgG1 constant (SEQ ID NO: 188)
Hu01G06-109	Humanized Hu01G06 IGKV1-39 V48I Human Variable + Human Kappa constant (SEQ ID NO: 210)	Humanized Sh01G06 IGHV1-69 T30S K64Q I69L Heavy Human Variable + Human IgG1 constant (SEQ ID NO: 190)
Hu01G06-110	Humanized Hu01G06 IGKV1-39 S43A V48I Human Variable + Human Kappa constant (SEQ ID NO: 208)	Humanized Sh01G06 IGHV1-18 M69L Heavy Human Variable + Human IgG1 constant (SEQ ID NO: 182)
Hu01G06-111	Humanized Hu01G06 IGKV1-39 S43A V48I Human Variable + Human Kappa constant (SEQ ID NO: 208)	Humanized Sh01G06 IGHV1-18 M69L K64Q Heavy Human Variable + Human IgG1 constant (SEQ ID NO: 186)
Hu01G06-112	Humanized Hu01G06 IGKV1-39 S43A V48I Human Variable + Human Kappa constant (SEQ ID NO: 208)	Humanized Sh01G06 IGHV1-18 M69L K64Q G44S Heavy Human Variable + Human IgG1 constant (SEQ ID NO: 184)
Hu01G06-113	Humanized Hu01G06 IGKV1-39 S43A V48I Human Variable + Human Kappa constant (SEQ ID NO: 208)	Humanized Sh01G06 IGHV1-69 T30S I69L Heavy Human Variable + Human IgG1 constant (SEQ ID NO: 188)
Hu01G06-114	Humanized Hu01G06 IGKV1-39 S43A V48I Human Variable + Human Kappa constant (SEQ ID NO: 208)	Humanized Sh01G06 IGHV1-69 T30S K64Q I69L Heavy Human Variable + Human IgG1 constant (SEQ ID NO: 190)
Hu01G06-122	Humanized Hu01G06 IGKV1-39 F1 Human Variable + Human Kappa constant (SEQ ID NO: 208)	Humanized Hu01G06 IGHV1-18 F1 Heavy Human Variable + Human IgG1 constant (SEQ ID NO: 256)
Hu01G06-127	Humanized Hu01G06 IGKV1-39 F2 Human Variable + Human Kappa constant (SEQ ID NO: 264)	Humanized Hu01G06 IGHV1-18 F2 Heavy Human Variable + Human IgG1 constant (SEQ ID NO: 258)
Hu01G06-135	Humanized Hu01G06 IGKV1-39 F1 Human Variable + Human Kappa constant (SEQ ID NO: 208)	Humanized Hu01G06 IGHV1-69 F1 Heavy Human Variable + Human IgG1 constant (SEQ ID NO: 260)
Hu01G06-138	Humanized Hu01G06 IGKV1-39 F1 Human Variable + Human Kappa constant (SEQ ID NO: 208)	Humanized Hu01G06 IGHV1-69 F2 Heavy Human Variable + Human IgG1 constant (SEQ ID NO: 262)
Hu01G06-146	Humanized Hu01G06 IGKV1-39 F2 Human Variable + Human Kappa constant (SEQ ID NO: 264)	Humanized Hu01G06 IGHV1-69 F2 Heavy Human Variable + Human IgG1 constant (SEQ ID NO: 262)
Hu06C11-1	Humanized Ch06C11 Chimeric Human Variable + Human Kappa constant (SEQ ID NO: 212)	Humanized Ch06C11 Chimeric Heavy Human Variable + Human IgG1 constant (SEQ ID NO: 176)

TABLE 23-continued

Antibody Name	Light Chain	Heavy Chain
Hu06C11-27	Humanized Sh06C11 IGKV1-16 Human Variable + Human Kappa constant (SEQ ID NO: 214)	Humanized HE LM 06C11 IGHV2-70 Heavy Human Variable + Human IgG1 constant (SEQ ID NO: 194)
Hu06C11-30	Humanized Sh06C11 IGKV1-16 Human Variable + Human Kappa constant (SEQ ID NO: 214)	Humanized Hu06C11 IGHV2-5 Heavy Human Variable + Human IgG1 constant (SEQ ID NO: 196)
Hu14F11-1	Humanized Ch14F11 Chimeric Human Variable + Human Kappa constant (SEQ ID NO: 216)	Humanized Ch14F11 Chimeric Heavy Human Variable + Human IgG1 constant (SEQ ID NO: 198)
Hu14F11-23	Humanized Ch14F11 Chimeric Human Variable + Human Kappa constant (SEQ ID NO: 216)	Humanized Ch06C11 Chimeric Heavy Human Variable + Human IgG1 constant (SEQ ID NO: 192)
Hu14F11-24	Humanized Ch06C11 Chimeric Human Variable + Human Kappa constant (SEQ ID NO: 212)	Humanized Ch14F11 Chimeric Heavy Human Variable + Human IgG1 constant (SEQ ID NO: 198)
Hu14F11-39	Humanized Hu14F11 IGKV1-16 Human Variable + Human Kappa constant (SEQ ID NO: 218)	Humanized Sh14F11 IGHV2-5 Heavy Human Variable + Human IgG1 constant (SEQ ID NO: 200)
Hu14F11-47	Humanized Hu14F11 IGKV1-16 Human Variable + Human Kappa constant (SEQ ID NO: 218)	Humanized Sh14F11-IGHV2-70 Heavy Human Variable + Human IgG1 constant (SEQ ID NO: 202)

[0313] The antibody constructs containing the full length chimeric heavy and light chains are designated below:

[0314] Chimeric 01G06 (Hu01G06-1)=Full Length Ch01G06 Chimeric Heavy Chain (Mouse Heavy Chain Variable Region and Human IgG1 Constant Region) (SEQ ID NO:176) plus Full Length Ch01G06 Chimeric Light Chain (Mouse Kappa Chain Variable Region and Human Kappa Constant Region) (SEQ ID NO:204)

[0315] Chimeric 06C11 (Hu06C11-1)=Full Length Ch06C11 Chimeric Heavy Chain (Mouse Heavy Chain Variable Region and Human IgG1 Constant Region) (SEQ ID NO:192) plus Full Length Ch06C11 Chimeric Light Chain (Mouse Kappa Chain Variable Region and Human Kappa Constant Region) (SEQ ID NO:212)

[0316] Chimeric 14F11 (Hu14F11-1)=Full Length Ch14F11 Chimeric Heavy Chain (Mouse Heavy Chain Variable Region and Human IgG1 Constant Region) (SEQ ID NO:198) plus Full Length Ch14F11 Chimeric Light Chain (Mouse Kappa Chain Variable Region and Human Kappa Constant Region) (SEQ ID NO:216)

[0317] Fifteen of the possible antibody constructs containing the full length immunoglobulin heavy and light chains containing humanized variable regions are designated below:

[0318] Hu01G06-46=Full Length Hu01G06 IGHV1-18 Heavy Chain (Humanized Heavy Chain Variable Region and Human IgG1 Constant Region) (SEQ ID NO:178) plus Full Length Hu01G06 IGKV1-39 Light Chain (Humanized Kappa Chain Variable Region and Human Kappa Constant Region) (SEQ ID NO:206)

[0319] Hu01G06-52=Full Length Hu01G06 IGHV1-69 Heavy Chain (Humanized Heavy Chain Variable Region and Human IgG1 Constant Region) (SEQ ID NO:180) plus Full Length Hu01G06 IGKV1-39 Light Chain (Humanized Kappa Chain Variable Region and Human Kappa Constant Region) (SEQ ID NO:206)

[0320] Hu01G06-107=Full Length Sh01G06 IGHV1-18 M69L K64Q G44S Heavy Chain (Humanized Heavy Chain Variable Region and Human IgG1 Constant Region) (SEQ ID NO:184) plus Full Length Hu01G06 IGKV1-39 V48I Light Chain (Humanized

Kappa Chain Variable Region and Human Kappa Constant Region) (SEQ ID NO:210)

[0321] Hu01G06-108=Full Length Sh01G06 IGHV1-69 T30S I69L Heavy Chain (Humanized Heavy Chain Variable Region and Human IgG1 Constant Region) (SEQ ID NO:188) plus Full Length Hu01G06 IGKV1-39 V48I Light Chain (Humanized Kappa Chain Variable Region and Human Kappa Constant Region) (SEQ ID NO:210)

[0322] Hu01G06-112=Full Length Sh01G06 IGHV1-18 M69L K64Q G44S Heavy Chain (Humanized Heavy Chain Variable Region and Human IgG1 Constant Region) (SEQ ID NO:184) plus Full Length Hu01G06 IGKV1-39 S43A V48I Light Chain (Humanized Kappa Chain Variable Region and Human Kappa Constant Region) (SEQ ID NO:208)

[0323] Hu01G06-113=Full Length Sh01G06 IGHV1-69 T30S I69L Heavy Chain (Humanized Heavy Chain Variable Region and Human IgG1 Constant Region) (SEQ ID NO:188) plus Full Length Hu01G06 IGKV1-39 S43A V48I Light Chain (Humanized Kappa Chain Variable Region and Human Kappa Constant Region) (SEQ ID NO:208)

[0324] Hu01G06-122=Full Length Hu01G06 IGHV1-18 F1 Heavy Chain (Humanized Heavy Chain Variable Region and Human IgG1 Constant Region) (SEQ ID NO:256) plus Full Length Hu01G06 IGKV1-39 F1 Light Chain (Humanized Kappa Chain Variable Region and Human Kappa Constant Region) (SEQ ID NO:208)

[0325] Hu01G06-127=Full Length Hu01G06 IGHV1-18 F2 Heavy Chain (Humanized Heavy Chain Variable Region and Human IgG1 Constant Region) (SEQ ID NO:258) plus Full Length Hu01G06 IGKV1-39 F2 Light Chain (Humanized Kappa Chain Variable Region and Human Kappa Constant Region) (SEQ ID NO:264)

[0326] Hu01G06-135=Full Length Hu01G06 IGHV1-69 F1 Heavy Chain (Humanized Heavy Chain Variable Region and Human IgG1 Constant Region) (SEQ ID NO:260) plus Full Length Hu01G06 IGKV1-39 F1 Light Chain (Humanized Kappa Chain Variable Region and Human Kappa Constant Region) (SEQ ID NO:208)

[0327] Hu01G06-138=Full Length Hu01G06 IGHV1-69 F2 Heavy Chain (Humanized Heavy Chain Variable Region and Human IgG1 Constant Region) (SEQ ID NO:262) plus Full Length Hu01G06 IGKV1-39 F1 Light Chain (Humanized Kappa Chain Variable Region and Human Kappa Constant Region) (SEQ ID NO:208)

[0328] Hu01G06-146=Full Length Hu01G06 IGHV1-69 F2 Heavy Chain (Humanized Heavy Chain Variable Region and Human IgG1 Constant Region) (SEQ ID NO:262) plus Full Length Hu01G06 IGKV1-39 F2 Light Chain (Humanized Kappa Chain Variable Region and Human Kappa Constant Region) (SEQ ID NO:264)

[0329] Hu06C11-27=Full Length HE LM 06C11 IGHV2-70 Heavy Chain (Humanized Heavy Chain Variable Region and Human IgG1 Constant Region) (SEQ ID NO:194) plus Full Length Sh06C11 IGKV1-16 Light Chain (Humanized Kappa Chain Variable Region and Human Kappa Constant Region) (SEQ ID NO:214)

[0330] Hu06C11-30=Full Length Hu06C11 IGHV2-5 Heavy Chain (Humanized Heavy Chain Variable Region and Human IgG1 Constant Region) (SEQ ID NO:196) plus Full Length Sh06C11 IGKV1-16 Light Chain (Humanized Kappa Chain Variable Region and Human Kappa Constant Region) (SEQ ID NO:214)

[0331] Hu14F11-39=Full Length Sh14F11 IGHV2-5 Heavy Chain (Humanized Heavy Chain Variable Region and Human IgG1 Constant Region) (SEQ ID NO:200) plus Full Length Hu14F11 IGKV1-16 Light Chain (Humanized Kappa Chain Variable Region and Human Kappa Constant Region) (SEQ ID NO:218)

[0332] Hu14F11-47=Full Length Sh14F11-IGHV2-70 Heavy Chain (Humanized Heavy Chain Variable Region and Human IgG1 Constant Region) (SEQ ID NO:202) plus Full Length Hu14F11 IGKV1-16 Light Chain (Humanized Kappa Chain Variable Region and Human Kappa Constant Region) (SEQ ID NO:218)

Example 15: Binding Affinities of Humanized and Chimeric Anti-GDF15 Monoclonal Antibodies

[0333] The binding affinities and kinetics of binding of chimeric and humanized antibodies to mFc-rhGDF15 were measured by surface plasmon resonance, using a BIAcore® T100 instrument (GE Healthcare, Piscataway, NJ).

[0334] Goat anti-human IgGs (Fc fragment specific, Jackson ImmunoResearch, West Grove, PA) were immobilized on carboxymethylated dextran CM4 sensor chips by amine coupling, according to a standard protocol. Analyses were performed at 37° C. using PBS containing 0.05% surfactant P20 as running buffer. The antibodies were captured in individual flow cells at a flow rate of 10 L/minute. Injection time was varied for each antibody to yield an Rmax between 30 and 60 RU. Buffer or mFc-rhGDF15 diluted in running buffer was injected sequentially over a reference surface (no antibody captured) and the active surface (antibody to be tested) for 240 seconds at 60 L/minute. The dissociation phase was monitored for up to 1200 seconds. The surface was then regenerated with two 60-second injections of 10 mM Glycine-HCl, pH 2.25, at a flow rate of 30 L/minute. The GDF15 concentration range tested was 20 nM to 0.625 nM.

[0335] Kinetic parameters were determined using the kinetic function of the BIAevaluation software (GE Healthcare) with double reference subtraction. Kinetic parameters

for each antibody, k_a (association rate constant), k_d (dissociation rate constant), and K_D (equilibrium dissociation constant) were determined. Kinetic values of purified monoclonal antibodies on mFc-rhGDF15 are summarized in Table 24.

TABLE 24

Antibody	k_a (1/Ms)	k_d (1/s)	K_D (M)	n
Hu01G06-1	4.2E+06	6.4E-04	1.6E-10	8
Hu01G06-46	3.6E+06	3.8E-04	1.1E-10	11
Hu01G06-52	3.6E+06	3.6E-04	9.9E-11	10
Hu06C11-1	5.3E+06	8.4E-04	1.8E-10	2
Hu06C11-27	4.7E+06	8.2E-04	1.8E-10	2
Hu06C11-30	4.8E+06	8.7E-04	1.8E-10	2
Hu14F11-1	3.0E+06	4.6E-04	1.6E-10	2
Hu14F11-39	3.0E+06	1.9E-04	6.6E-11	2
Hu14F11-47	3.3E+06	1.8E-04	6.5E-11	2

[0336] The results in Table 24 demonstrate that the chimeric and each of the humanized antibodies, have fast association rates (k_a), very slow disassociation rates (k_d) and very high affinities (K_D). In particular, the antibodies have affinities ranging from about 65 pM to about 200 pM.

[0337] Kinetic values of chimeric 01G06 (Hu01G06-1), two initial lead humanized 01G06 monoclonal antibodies (Hu01G06-46 and -52), and sequence optimized humanized 01G06 monoclonal antibody variants Hu01G06-100 through -114 (in supernatant) on mFc-rhGDF15 are summarized in Table 25.

TABLE 25

Antibody	k_a (1/Ms)	k_d (1/s)	K_D (M)	n
Hu01G06-1	4.9E+06	7.1E-04	1.4E-10	3
Hu01G06-46	4.1E+06	4.3E-04	1.0E-10	3
Hu01G06-52	5.0E+06	4.4E-04	8.9E-11	3
Hu01G06-100	4.1E+06	6.2E-04	1.5E-10	3
Hu01G06-101	4.4E+06	6.3E-04	1.4E-10	3
Hu01G06-102	4.4E+06	4.6E-04	1.1E-10	3
Hu01G06-103	4.4E+06	4.7E-04	1.1E-10	3
Hu01G06-104	4.5E+06	5.2E-04	1.2E-10	3
Hu01G06-105	4.3E+06	5.6E-04	1.3E-10	3
Hu01G06-106	4.3E+06	7.0E-04	1.6E-10	3
Hu01G06-107	4.1E+06	4.7E-04	1.2E-10	3
Hu01G06-108	4.2E+06	4.6E-04	1.2E-10	3
Hu01G06-109	4.6E+06	5.6E-04	1.3E-10	4
Hu01G06-110	4.3E+06	5.8E-04	1.4E-10	4
Hu01G06-111	4.3E+06	6.6E-04	1.6E-10	3
Hu01G06-112	4.7E+06	5.3E-04	1.2E-10	3
Hu01G06-113	4.5E+06	4.8E-04	1.1E-10	3
Hu01G06-114	4.5E+06	5.4E-04	1.3E-10	3

[0338] The results in Table 25 demonstrate that the sequence optimized antibodies, Hu01G06-100 through -114, have binding affinities ranging from about 89 pM to about 160 pM.

[0339] Binding affinities and binding kinetics of mFc-rhGDF15 with chimeric 14F11 (Hu14F11-1), chimeric light 14F11 with chimeric heavy 06C11 (Hu14F11-23), and chimeric light 06C11 with chimeric heavy 14F11 (Hu14F11-24) heavy monoclonal antibody variants (in supernatant) were measured using biolayer interferometry (BLI) on an Octet™ QK instrument (ForteBio, Inc., Menlo Park, CA). The Octet analysis was performed at 30° C. using 1x Kinetics Buffer (ForteBio, Inc.) as assay buffer. Anti-human IgG Fc Capture (AHC) biosensors (ForteBio, Inc.) were used to capture human antibodies onto the sensors. Sensors

were saturated in assay buffer for at 300 seconds before the assay. Antibodies were loaded onto sensors by dipping the sensors into antibody supernatant solution for 220 seconds, which typically resulted in capture levels of 1.5-2 nm. Baseline was established by dipping the sensors into 1x assay buffer for 200 seconds. Next, association was monitored for 220 seconds in 400 nM mFc-rhGDF15 protein, and dissociation was followed for 600 seconds in buffer alone. **[0340]** Kinetic parameters for Hu14F11-1, Hu14F11-23, and Hu14F11-24 were determined using the kinetic function of the ForteBio Analysis Software Version 7.0. Kinetic parameters of the antibody, k_a , k_d , and K_D were determined. **[0341]** Kinetic values of Hu14F11-1, Hu14F11-23, and Hu14F11-24 heavy monoclonal antibody variants (in supernatant) on mFc-rhGDF15 are summarized in Table 26.

TABLE 26

Antibody	k_a (1/Ms)	k_d (1/s)	K_D (M)	n
Hu14F11-1	6.3E+05	1.9E-05	3.2E-11	3
Hu14F11-23	3.4E+05	6.2E-05	1.8E-10	1
Hu14F11-24	7.1E+05	2.2E-04	3.1E-10	1

[0342] The results in Table 26 demonstrate that Hu14F11-23 and Hu14F11-24, (i.e., antibodies that consist of one chimeric 06C11 chain (heavy or light) mixed with one chimeric 14F11 chain (heavy or light)), retain binding to GDF15. In particular, these antibodies have high affinities ranging from about 180 pM to about 310 pM.

Example 16: Binding Affinities of Affinity Matured Humanized Anti-GDF15 Monoclonal Antibodies

[0343] The binding affinities and kinetics of binding of chimeric and humanized antibodies to mFc-rhGDF15, cleaved-rhGDF15, rabbit Fc mature recombinant mouse GDF15 (rFc-rmGDF15), and mouse Fc mature recombinant cynomolgus monkey GDF15 (mFc-rcGDF15) were measured by surface plasmon resonance, using a BIAcore® T100 instrument (GE Healthcare, Piscataway, NJ).

[0344] Goat anti-human IgGs (Fc fragment specific, Jackson ImmunoResearch, West Grove, PA) were immobilized on carboxymethylated dextran CM4 sensor chips by amine coupling, according to a standard protocol. Analyses were performed at 37° C. using PBS containing 0.05% surfactant P20 as running buffer. The antibodies were captured in individual flow cells at a flow rate of 10 μ L/minute. Injection time was varied for each antibody to yield an Rmax between 30 and 60 RU. Buffer, mFc-rhGDF15, cleaved-rhGDF15, rFc-rmGDF15, or mFc-rcGDF15 diluted in running buffer was injected sequentially over a reference surface (no antibody captured) and the active surface (antibody to be tested) for 240 seconds at 60 μ L/minute. The dissociation phase was monitored for up to 1500 seconds. The surface was then regenerated with two 60-second injections of 10 mM Glycine-HCl, pH 2.25, at a flow rate of 30 μ L/minute. The GDF15 concentration range tested for each GDF15 protein was 5 nM to 0.3125 nM (two-fold dilutions).

[0345] Kinetic parameters were determined using the kinetic function of the BIAevaluation software (GE Healthcare) with double reference subtraction. Kinetic parameters for each antibody, k_a (association rate constant), k_d (dissociation rate constant), and K_D (equilibrium dissociation constant) were determined. Kinetic values of purified mono-

clonal antibodies on mFc-rhGDF15, mature human GDF15, rFc-rmGDF15, and mFc-rcGDF15 are summarized in Table 27.

TABLE 27

Protein	Antibody	k_a (1/Ms)	k_d (1/s)	K_D (M)	n
mFc-rhGDF15	Hu01G06-122	5.9E+06	2.1E-05	6.5E-12	5
	Hu01G06-127	4.6E+06	4.2E-05	1.8E-11	4
	Hu01G06-135	5.3E+06	4.4E-05	1.4E-11	5
	Hu01G06-138	5.9E+06	4.1E-05	1.1E-11	5
	Hu01G06-146	5.3E+06	2.6E-05	9.3E-12	5
Cleaved-rhGDF15	Hu01G06-122	7.9E+06	3.4E-05	7.9E-12	4
	Hu01G06-127	6.1E+06	3.6E-05	1.0E-11	4
	Hu01G06-135	7.3E+06	6.2E-05	1.0E-11	4
	Hu01G06-138	7.9E+06	2.5E-05	4.5E-12	4
	Hu01G06-146	6.5E+06	5.2E-05	1.1E-11	4
mFc-rcGDF15	Hu01G06-122	2.3E+06	2.4E-05	1.0E-11	4
	Hu01G06-127	1.8E+06	1.6E-05	9.5E-12	4
	Hu01G06-135	2.2E+06	7.9E-05	3.8E-11	4
	Hu01G06-138	2.3E+06	5.3E-05	2.5E-11	4
	Hu01G06-146	2.0E+06	1.5E-05	8.0E-12	4
rFc-rmGDF15	Hu01G06-122	2.2E+07	1.4E-03	6.3E-11	2
	Hu01G06-127	3.9E+07	2.1E-03	5.1E-11	2
	Hu01G06-135	3.7E+07	1.9E-03	5.5E-11	2
	Hu01G06-138	1.9E+07	8.0E-04	4.4E-11	2
	Hu01G06-146	1.1E+07	7.2E-04	6.3E-11	2

[0346] The results in Table 27 demonstrate that the chimeric and each of the humanized antibodies, have fast association rates (k_a), very slow disassociation rates (k_d) and very high affinities (K_D). In particular, the antibodies have affinities ranging from less than 5 pM (e.g., about 4.5 pM) to about 65 pM.

Example 17: Reversal of Cachexia in an HT-1080 Fibrosarcoma Xenograft Model

[0347] This Example demonstrates the reversal of cachexia (as indicated by body weight loss) by humanized 01G06, 06C11, 14F11 antibodies in an HT-1080 fibrosarcoma xenograft model. HT-1080 cells were grown in culture at 37° C. and inoculated subcutaneously into the flank of 8-week old female ICR-SCID mice as described above in Example 10. Body weight was measured daily. When body weight reached 93%, the mice were randomly divided into groups of ten mice each. Each group received one of the following treatments: murine IgG control, 01G06, 06C11, 14F11, and their respective humanized versions at 2 mg/kg. Treatment was administered once a day by intra-peritoneal injection. Antibody treatment with 01G06, Hu01G06-46 and Hu01G06-52 resulted in body weight increase to initial weight or 100% ($p < 0.001$) (FIG. 23). Statistical analysis was performed using ANOVA. Results for reversal of body weight on day in the HT-1080 model are shown in FIG. 23 and Table 28, respectively.

TABLE 28

Gr.	Treatment		% Body weight	ANOVA Analysis (compared to mIgG)
	Agent	mg/kg		
1	mIgG	2	79.5	NA
2	01G06	2	87.6	$p < 0.001$
3	Hu01G06-46	2	95.4	$p < 0.001$
4	Hu01G06-52	2	87.8	$p < 0.001$

[0348] The data in FIG. 23 and Table 28 indicated that antibodies 01G06, Hu01G06-46 and Hu01G06-52 can reverse cachexia in an HT-1080 fibrosarcoma xenograft model.

[0349] Antibody treatment with 06C11, Hu06C11-27, and Hu06C11-30 resulted in body weight increase relative to initial weight or about 100% ($p < 0.001$) (FIG. 24). Statistical analysis was performed using ANOVA. Results for reversal of body weight in the HT-1080 model are shown in FIG. 24 and Table 29.

TABLE 29

Treatment		% Body weight	ANOVA Analysis (compared to mIgG)
Gr.	Agent		
1	mIgG	2	87.7
2	01G06	2	93.6
3	Hu06C11-27	2	93.2
4	Hu06C11-30	2	89.8

[0350] The data in FIG. 24 and Table 29 indicate that antibodies 06C11, Hu06C11-27, and Hu06C11-30 can reverse cachexia in an HT-1080 fibrosarcoma xenograft model.

[0351] Antibody treatment with 14F11, Hu14F11-39, and Hu14F11-47 resulted in body weight increase relative to initial weight or about 100% ($p < 0.001$) (FIG. 25). Statistical analysis was performed using ANOVA. Results for reversal of body weight in the HT-1080 model are shown in FIG. 25 and Table 30.

TABLE 30

Treatment		% Body weight	ANOVA Analysis (compared to mIgG)
Gr.	Agent		
1	mIgG	2	87.7
2	14F11	2	96.6
3	Hu14F11-39	2	90.5
4	Hu14F11-47	2	90.7

[0352] The data in FIG. 25 and Table 30 indicated that antibodies 14F11, Hu14F11-39, and Hu14F11-47 can reverse cachexia in an HT-1080 fibrosarcoma xenograft model.

[0353] Antibody treatment with humanized 01G06 antibodies (i.e., antibodies Hu01G06-122, Hu01G06-127, Hu01G06-135, Hu01G06-138 and Hu01G06-146) resulted in body weight increase relative to initial weight or about 100% ($p < 0.001$) (FIG. 26). Statistical analysis was performed using ANOVA. Treatment with human IgG (hIgG) was used as a control. Results for reversal of body weights in the HT-1080 model are shown in FIG. 26 and Table 31.

TABLE 31

Treatment		% Body weight	ANOVA Analysis (compared to mIgG)
Gr.	Agent		
1	mIgG	2	84.2
2	Hu01G06-122	2	96.3
3	Hu01G06-127	2	96.1
4	Hu01G06-135	2	93.5

TABLE 31-continued

Treatment		% Body weight	ANOVA Analysis (compared to mIgG)
Gr.	Agent		
5	Hu01G06-138	2	91.9
6	Hu01G06-146	2	92.7

[0354] The data in FIG. 26 and Table 31 indicated that humanized anti-GDF15 antibodies Hu01G06-122, Hu01G06-127, Hu01G06-135, Hu01G06-138 and Hu01G06-146 can reverse cachexia in an HT-1080 fibrosarcoma xenograft model.

Example 18: Reversal of Cachexia in an mFc-rhGDF15-Induced Model

[0355] This Example demonstrates the reversal of cachexia (as indicated by body weight loss) by humanized 01G06 antibodies (i.e., antibody Hu01G06-122, Hu01G06-127, Hu01G06-135, Hu01G06-138, or Hu01G06-146) in an mFc-rhGDF15-induced cachexia model. mFc-rhGDF15 (1 μ g/g) was administered subcutaneously into the flank of 8-week old female ICR-SCID mice. Body weight was measured daily. When body weight reached 93%, the mice were randomly divided into six groups of ten mice each. Each group received one of the following treatments: human IgG control (hIgG), Hu01G06-122, Hu01G06-127, Hu01G06-135, Hu01G06-138 or Hu01G06-146 at 2 mg/kg. Treatment was administered once by intra-peritoneal injection. Treatment with antibody Hu01G06-122, Hu01G06-127, Hu01G06-135, Hu01G06-138 or Hu01G06-146 resulted in body weight increase relative to initial weight or about 100% ($p < 0.001$) (FIG. 27 and Table 32).

TABLE 32

Treatment		% Body weight	ANOVA Analysis (compared to mIgG)
Gr.	Agent		
1	mIgG	2	70.6
2	Hu01G06-122	2	101.7
3	Hu01G06-127	2	103.2
4	Hu01G06-135	2	102.5
5	Hu01G06-138	2	101.8
6	Hu01G06-146	2	102.5

[0356] The data in FIG. 27 and Table 32 indicate that the disclosed anti-GDF15 antibodies can reverse cachexia in an mFc-rhGDF15-induced mouse model (i.e., a non-tumor bearing mouse model).

[0357] These results indicate that humanized anti-GDF15 antibodies can reverse cachexia in an mFc-rhGDF15-induced cachexia model.

Example 19: Dose Response Reversal of Cachexia in an HT-1080 Fibrosarcoma Xenograft Model

[0358] This Example demonstrates the dose response reversal of cachexia (as indicated by body weight loss) by humanized Hu01G06-127 and Hu01G06-135 antibodies in an HT-1080 fibrosarcoma xenograft model. HT-1080 cells were grown in culture at 37° C. and inoculated subcutaneously into the flank of 8-week old female ICR-SCID mice as described above in Example 10. Body weight was measured daily. When body weight reached 93%, the mice were

randomly divided into groups of ten mice each. Each group received one of the following treatments: human IgG control (hIgG; 20 mg/kg), Hu01G06-127 (20 mg/kg, 2 mg/kg, or 0.2 mg/kg) and Hu01G06-135 (20 mg/kg, 2 mg/kg, or 0.2 mg/kg). Treatment was administered once a day by intravenous injection. Antibody treatment with Hu01G06-127 and Hu01G06-135 at 20 mg/kg resulted in body weight increase above the initial weight or 108% ($p<0.001$) (FIG. 28). Antibody treatment with Hu01G06-127 and Hu01G06-135 at 2 mg/kg resulted in limited body weight decrease compare to control (hIgG) from the initial weight or 88-85% ($p<0.001$) (FIG. 28). Statistical analysis was performed using ANOVA. Results for changes of body weight at the end of the study in the HT-1080 model are shown in FIG. 28 and Table 33.

TABLE 33

Treatment		% Body	ANOVA Analysis	
Gr.	Agent	mg/kg	weight	(compared to mIgG)
1	mIgG	20	75.2	NA
2	Hu01G06-127	20	108.9	$p < 0.001$
3	Hu01G06-127	2.0	88.1	$p < 0.001$
4	Hu01G06-127	0.2	80.0	NS
5	Hu01G06-135	20	108.6	$p < 0.001$
6	Hu01G06-135	2.0	85.2	$p < 0.001$
7	Hu01G06-135	0.2	77.3	NS

[0359] The data in FIG. 28 and Table 33 indicated that antibodies Hu01G06-127 and Hu01G06-135 can reverse cachexia in an HT-1080 fibrosarcoma xenograft model in a dose-dependent manner.

Example 20: Reversal of Muscle and Fat Loss in an HT-1080 Xenograft Tumor Model

[0360] This Example demonstrates the reversal of cachexia (as indicated by body weight loss, muscle mass loss and fat mass loss) by antibody 01G06 in an HT-1080 fibrosarcoma xenograft model. HT-1080 cells were grown in culture at 37° C. in an atmosphere containing 5% CO₂, using Eagle’s Minimum Essential Medium (ATCC, Catalog No. 30-2003) containing 10% FBS. Cells were inoculated subcutaneously into the flank of 8-week old female ICR SCID mice with 5×10⁶ cells per mouse in 50% matrigel. A cohort of ten 8-week old female ICR SCID mice with the same body weight was selected for subcutaneous inoculation into the flank with matrigel, as a non tumor (SHAM) control arm. Body weight was measured daily. When body weight reached 91% in the tumor bearing mice, the mice were randomly divided into two groups of ten mice each. Each group received one of the following treatments: human IgG control (hIgG) or Hu01G06-127 10 mg/kg on day 1, day 3 and day 6. Treatment was administered by intra-peritoneal injection. Treatment with antibody Hu01G06-127 resulted in body weight increase to 105% of initial weight compared to non-tumor bearing control mice (SHAM; $p<0.001$) (FIG. 29A and Table 34).

TABLE 34

Treatment		% Body	ANOVA Analysis	
Gr.	Agent	mg/kg	weight	(compared to hIgG)
1	hIgG	10	84.3	NA
2	Hu01G06-127	10	105.4	$p < 0.001$
2	SHAM no tumor control	none	101.9	$p < 0.001$

[0361] The data in FIG. 29A and Table 34 indicate that the disclosed anti-GDF15 antibody can completely reverse cachexia in an HT-1080 fibrosarcoma xenograft model.

[0362] In this experiment, a group of ten mice were sacrificed at the time of dosing (baseline or 91% body weight loss, without treatment) and at the end of the study (eight days post dose, either hIgG or Hu01G06-127 as well SHAM non tumor control mice). Gonadal fat and the gastrocnemius muscles were removed surgically and weighed. As shown in FIG. 29B, significant gonadal fat mass loss was observed seven days post dose with hIgG, but not in the group treated with antibody Hu01G06-127 compared to baseline control (91% body weight loss). Moreover, treatment with Hu01G06-127 not only prevented further fat loss (compared to baseline group), but also, was able to restore the normal levels of gonadal fat (compared to SHAM non-tumor control) (FIG. 29B). In addition, significant gastrocnemius muscle mass loss was observed seven days post dose with hIgG, but not in the group treated with antibody Hu01G06-127 compared to baseline control (91% body weight loss) (FIG. 29C). Treatment with Hu01G06-127 not only prevented further muscle loss (compare to baseline group), but also was able to restore the normal levels of gastrocnemius muscle (compared to SHAM non tumor control) (FIG. 29C).

[0363] These results indicate that the disclosed anti-GDF15 antibodies can completely reverse cachexia measured by the loss of muscle mass, loss of fat and involuntary weight loss in an HT-1080 xenograft tumor model.

INCORPORATION BY REFERENCE

[0364] The entire disclosure of each of the patent documents and scientific articles referred to herein is incorporated by reference for all purposes.

EQUIVALENTS

[0365] The invention may be embodied in other specific forms without departing from the spirit or essential characteristics thereof. The foregoing embodiments are therefore to be considered in all respects illustrative rather than limiting on the invention described herein. Scope of the invention is thus indicated by the appended claims rather than by the foregoing description, and all changes that come within the meaning and the range of equivalency of the claims are intended to be embraced therein.

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SEQ ID NO: 11 FEATURE REGION	moltype = AA length = 16 Location/Qualifiers 1..16 note = Description of Artificial Sequence: Synthetic peptide	
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SEQ ID NO: 18	moltype = AA length = 9	
FEATURE	Location/Qualifiers	
REGION	1..9	
	note = Description of Artificial Sequence: Synthetic peptide	
source	1..9	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 18		
RGYDDYWG Y		9
SEQ ID NO: 19	moltype = AA length = 9	
FEATURE	Location/Qualifiers	
REGION	1..9	
	note = Description of Artificial Sequence: Synthetic peptide	
source	1..9	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 19		
RGHYSAMDY		9
SEQ ID NO: 20	moltype = AA length = 11	
FEATURE	Location/Qualifiers	
REGION	1..11	
	note = Description of Artificial Sequence: Synthetic peptide	
source	1..11	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 20		
RVGGLLEG YFD Y		11
SEQ ID NO: 21	moltype = AA length = 11	
FEATURE	Location/Qualifiers	
REGION	1..11	
	note = Description of Artificial Sequence: Synthetic peptide	
source	1..11	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 21		
RTSENHNYL A		11
SEQ ID NO: 22	moltype = AA length = 15	
FEATURE	Location/Qualifiers	
REGION	1..15	
	note = Description of Artificial Sequence: Synthetic peptide	
source	1..15	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 22		
RASESVDNYG ISFMN		15
SEQ ID NO: 23	moltype = AA length = 11	
FEATURE	Location/Qualifiers	
REGION	1..11	
	note = Description of Artificial Sequence: Synthetic peptide	
source	1..11	
	mol_type = protein	

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SEQUENCE: 23	organism = synthetic construct	
KASQNVGTNV A		11
SEQ ID NO: 24	moltype = AA length = 11	
FEATURE	Location/Qualifiers	
REGION	1..11	
	note = Description of Artificial Sequence: Synthetic peptide	
source	1..11	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 24		
RASGNIHNYL A		11
SEQ ID NO: 25	moltype = AA length = 15	
FEATURE	Location/Qualifiers	
REGION	1..15	
	note = Description of Artificial Sequence: Synthetic peptide	
source	1..15	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 25		
RASQSVTSR FSYMH		15
SEQ ID NO: 26	moltype = AA length = 7	
FEATURE	Location/Qualifiers	
REGION	1..7	
	note = Description of Artificial Sequence: Synthetic peptide	
source	1..7	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 26		
DAKTLAD		7
SEQ ID NO: 27	moltype = AA length = 7	
FEATURE	Location/Qualifiers	
REGION	1..7	
	note = Description of Artificial Sequence: Synthetic peptide	
source	1..7	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 27		
AASNQGS		7
SEQ ID NO: 28	moltype = AA length = 7	
FEATURE	Location/Qualifiers	
REGION	1..7	
	note = Description of Artificial Sequence: Synthetic peptide	
source	1..7	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 28		
SASYRYS		7
SEQ ID NO: 29	moltype = AA length = 7	
FEATURE	Location/Qualifiers	
REGION	1..7	
	note = Description of Artificial Sequence: Synthetic peptide	
source	1..7	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 29		
NAKTLAD		7
SEQ ID NO: 30	moltype = AA length = 7	
FEATURE	Location/Qualifiers	
REGION	1..7	
	note = Description of Artificial Sequence: Synthetic peptide	
source	1..7	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 30		
SPSYRYS		7
SEQ ID NO: 31	moltype = AA length = 7	
FEATURE	Location/Qualifiers	

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REGION	1..7	
	note = Description of Artificial Sequence: Synthetic peptide	
source	1..7	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 31		
YASNLES		7
SEQ ID NO: 32	moltype = AA length = 9	
FEATURE	Location/Qualifiers	
REGION	1..9	
	note = Description of Artificial Sequence: Synthetic peptide	
source	1..9	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 32		
QHFWSPPYT		9
SEQ ID NO: 33	moltype = AA length = 9	
FEATURE	Location/Qualifiers	
REGION	1..9	
	note = Description of Artificial Sequence: Synthetic peptide	
source	1..9	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 33		
QQSKEVPWT		9
SEQ ID NO: 34	moltype = AA length = 9	
FEATURE	Location/Qualifiers	
REGION	1..9	
	note = Description of Artificial Sequence: Synthetic peptide	
source	1..9	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 34		
QQYNSYPYT		9
SEQ ID NO: 35	moltype = AA length = 9	
FEATURE	Location/Qualifiers	
REGION	1..9	
	note = Description of Artificial Sequence: Synthetic peptide	
source	1..9	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 35		
QQYNNYPLT		9
SEQ ID NO: 36	moltype = AA length = 9	
FEATURE	Location/Qualifiers	
REGION	1..9	
	note = Description of Artificial Sequence: Synthetic peptide	
source	1..9	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 36		
QQYNSYPHT		9
SEQ ID NO: 37	moltype = AA length = 9	
FEATURE	Location/Qualifiers	
REGION	1..9	
	note = Description of Artificial Sequence: Synthetic peptide	
source	1..9	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 37		
QHSWEIPYT		9
SEQ ID NO: 38	moltype = AA length = 7	
FEATURE	Location/Qualifiers	
REGION	1..7	
	note = Description of Artificial Sequence: Synthetic peptide	
source	1..7	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 38		

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GYTFTDY 7

SEQ ID NO: 39 moltype = DNA length = 360
 FEATURE Location/Qualifiers
 misc_feature 1..360
 note = Description of Artificial Sequence: Synthetic polynucleotide
 source 1..360
 mol_type = other DNA
 organism = synthetic construct

SEQUENCE: 39
 gaggtcctgc tgcaacagtc tggacctgag ctggtgaagc ctggggcttc agtgaagata 60
 ccctgcaagg cttctggata cacattcact gactacaaca tggactgggt gaagcagagc 120
 catggaaaga gccttgagtg gattggacaa attaatccta acaatgggtg tattttcttc 180
 aaccagaagt tcaagggcaa ggccacattg actgtagaca agtcctccaa tacagccttc 240
 atggaggtcc gcagcctgac atctgaggac actgcagtct attactgtgc aagagaggca 300
 attactacgg taggcgctat ggactactgg ggtcaaggaa cctcagtcac cgtctcctca 360

SEQ ID NO: 40 moltype = AA length = 120
 FEATURE Location/Qualifiers
 REGION 1..120
 note = Description of Artificial Sequence: Synthetic polypeptide
 source 1..120
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 40
 EVLLQQSGPE LVKPGASVKI PCKASGYTFT DYNMDWVKQS HGKSLEWIGQ INPNNGGIF 60
 NQKFKGKATL TVDKSSNTAF MEVRSLTSED TAVYYCAREA ITTVGAMDYV GQGTSTVTVSS 120

SEQ ID NO: 41 moltype = DNA length = 354
 FEATURE Location/Qualifiers
 misc_feature 1..354
 note = Description of Artificial Sequence: Synthetic polynucleotide
 source 1..354
 mol_type = other DNA
 organism = synthetic construct

SEQUENCE: 41
 cagggtccaac tgcagcagcc tggggctgaa ctggtgaagc ctggggcttc agtgaagctg 60
 tcctgcaagg cttctggcta caccttcacc agctactgga ttcactgggt gaaccagagg 120
 cctggacaag gccttgagtg gattggagac attaatccta gcaacggccg tagtaagtat 180
 aatgagaagt tcaagaacaa ggccacaatg actgcagaca aatcctccaa cacagcctac 240
 atgcaactca gcagcctgac atctgaggac tctgcggtct attactgtgc aagagagggt 300
 ctggatggtg ctatggacta ctgggggtcaa ggaacctcag tcaccgtctc ctca 354

SEQ ID NO: 42 moltype = AA length = 118
 FEATURE Location/Qualifiers
 REGION 1..118
 note = Description of Artificial Sequence: Synthetic polypeptide
 source 1..118
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 42
 QVQLQQPGAE LVKPGASVKL SCKASGYTFT SYWIHWVNQR PGQGLEWIGD INPSNGRSKY 60
 NEKFKNKATM TADKSSNTAY MQLSSLTSED SAVYYCAREV LDGAMDYWGQ GTSVTVSS 118

SEQ ID NO: 43 moltype = DNA length = 357
 FEATURE Location/Qualifiers
 misc_feature 1..357
 note = Description of Artificial Sequence: Synthetic polynucleotide
 source 1..357
 mol_type = other DNA
 organism = synthetic construct

SEQUENCE: 43
 caggttactc tgaagagtc tggccctggg atattgcagc cctcccagac cctcagtcgt 60
 acttggtctt tctctgggtt ttcactgagc acttatggta tgggtgtgac ctggattcgt 120
 cagccttcag gaaagggctc ggagtggtcg gcacacattt actgggatga tgacaagcgc 180
 tataacccat cccctgaagag ccggctcaca atctccaagg atacctccaa caaccaggta 240
 ttctcaaga tcaccagtgt ggacactgca gatactgcca catactactg tgctcaaagc 300
 gggatatagta acttggttgc ttactggggc caagggactc tggtcactgt ctctgca 357

SEQ ID NO: 44 moltype = AA length = 119
 FEATURE Location/Qualifiers

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REGION 1..119
note = Description of Artificial Sequence: Synthetic polypeptide

source 1..119
mol_type = protein
organism = synthetic construct

SEQUENCE: 44
QVTLKESGPG ILQPSQTLSTL TCSFSGFSLT TYGMGVTWIR QPSGKGLEWL AHYWDDDKR 60
YNPSLKSRLT ISKDTSNQV FLKITSVDTA DTATYYCAQT GYSNLFAYWG QGTLVTVSA 119

SEQ ID NO: 45 moltype = DNA length = 357
FEATURE Location/Qualifiers
misc_feature 1..357
note = Description of Artificial Sequence: Synthetic polynucleotide

source 1..357
mol_type = other DNA
organism = synthetic construct

SEQUENCE: 45
caggttactc tgaagagtc tggccctggg atattgcagc cctcccagac cctcagtcgt 60
acttggtctt tctctgggtt ttactgaac acttatggta tgggtgtgag ctggattcgt 120
cagccttcag gaaagggtct ggagtggctg gcacacattt actgggatga tgacaagcgc 180
tataacccat ccctgaagag cgggtcaca atctccaagg atgcctcaa caaccgggtc 240
ttctcaaga tcaccagtggt ggacactgca gatactgcca catactactg tgctcaaaga 300
ggttatgatg attactgggg ttactggggc caagggactc tggctactat ctctgca 357

SEQ ID NO: 46 moltype = AA length = 119
FEATURE Location/Qualifiers
REGION 1..119
note = Description of Artificial Sequence: Synthetic polypeptide

source 1..119
mol_type = protein
organism = synthetic construct

SEQUENCE: 46
QVTLKESGPG ILQPSQTLSTL TCSFSGFSLN TYGMGVSWIR QPSGKGLEWL AHYWDDDKR 60
YNPSLKSRLT ISKDASNVR FLKITSVDTA DTATYYCAQR GYDDYWG YWG QGTLVTISA 119

SEQ ID NO: 47 moltype = DNA length = 360
FEATURE Location/Qualifiers
misc_feature 1..360
note = Description of Artificial Sequence: Synthetic polynucleotide

source 1..360
mol_type = other DNA
organism = synthetic construct

SEQUENCE: 47
gaggtcctgc tgcaacagtc tggacctgag gtggtgaagc ctggggcttc agtgaagata 60
ccctgcaagg cttctggata cacattcact gactacaaca tggactgggt gaagcagagc 120
catggaaaga gccttgagtg gattggagag attaatccta acaatggtgg tactttctac 180
aaccagaagt tcaagggcaa ggccacattg actgtagaca agtcctccag cacagcctac 240
atggagctcc gcagcctgac atctgaggac actgcagtct attactgtgc aagagaggca 300
attactacgg taggcgctat ggactactgg ggtcaaggaa cctcagtcac cgtctctca 360

SEQ ID NO: 48 moltype = AA length = 120
FEATURE Location/Qualifiers
REGION 1..120
note = Description of Artificial Sequence: Synthetic polypeptide

source 1..120
mol_type = protein
organism = synthetic construct

SEQUENCE: 48
EVLLQQSGPE VVPGASVKI PCKASGYTFT DYNMDWVKQS HGKSLEWIGE INPNNGGTFY 60
NQFKGKATL TVDKSSSTAY MELRSLTSED TAVYYCAREA ITTVGAMDYV GQGTSTVTVSS 120

SEQ ID NO: 49 moltype = DNA length = 357
FEATURE Location/Qualifiers
misc_feature 1..357
note = Description of Artificial Sequence: Synthetic polynucleotide

source 1..357
mol_type = other DNA
organism = synthetic construct

SEQUENCE: 49
caggttactc tgaagagtc tggccctgga atattgcagc cctcccagac cctcagtcgt 60

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acttggtctt tctctgggtt ttcactgagc acttatggta tgggtgtagg ctggattcgt 120
cagccttcag gaaaggggtc agagtggctg gcagacattt ggtgggatga cgataagtac 180
tataacccat cccctgaagag ccggctcaca atctccaagg atacctccag caatgaggta 240
ttctcaaga tcgccattgt ggacactgca gatactgcca cttactactg tgctcgaaga 300
ggtcactact ctgctatgga ctactgggtt caaggaacct cagtcaacgt ctctca 357

SEQ ID NO: 50      moltype = AA length = 119
FEATURE           Location/Qualifiers
REGION           1..119
                  note = Description of Artificial Sequence: Synthetic
                  polypeptide
source           1..119
                  mol_type = protein
                  organism = synthetic construct

SEQUENCE: 50
QVTLKESGPG ILQPSQTLSTL TCSFSGFSLT TYGMGVGWIR QPSGKGLEWL ADIWDDDKY 60
YNPSLKSRLT ISKDTSSNEV FLKIAIVDTA DTATYYCARR GHYSAMDYWG QGTSVTVSS 119

SEQ ID NO: 51      moltype = DNA length = 363
FEATURE           Location/Qualifiers
misc_feature      1..363
                  note = Description of Artificial Sequence: Synthetic
                  polynucleotide
source           1..363
                  mol_type = other DNA
                  organism = synthetic construct

SEQUENCE: 51
caggttactc tgaagagatc tggccctggg atattgcagc cctcccagac cctcagtcgt 60
acttggtctt tctctgggtt ttcactgagc acttctggta tgggtgtgag ttggattcgt 120
cagccttcag gaaaggggtc ggagtggctg gcacacaatg actgggatga tgacaagcgc 180
tataagtcac cccctgaagag ccggctcaca atatccaagg atacctccag aaaccaggta 240
ttctcaaga tcaccagtggt ggacactgca gatactgcca catactactg tgctcgaaga 300
gttgggggat tagagggcta ttttgattac tggggccaag gcaccactct cacagtctcc 360
tca 363

SEQ ID NO: 52      moltype = AA length = 121
FEATURE           Location/Qualifiers
REGION           1..121
                  note = Description of Artificial Sequence: Synthetic
                  polypeptide
source           1..121
                  mol_type = protein
                  organism = synthetic construct

SEQUENCE: 52
QVTLKESGPG ILQPSQTLSTL TCSFSGFSLT TSGMGVSWIR QPSGKGLEWL AHNDWDDDKR 60
YKSSLKSRLT ISKDTSRNQV FLKITSVDTA DTATYYCARR VGGLEGYFDY WQGTTLTVS 120
S 121

SEQ ID NO: 53      moltype = DNA length = 360
FEATURE           Location/Qualifiers
misc_feature      1..360
                  note = Description of Artificial Sequence: Synthetic
                  polynucleotide
source           1..360
                  mol_type = other DNA
                  organism = synthetic construct

SEQUENCE: 53
caagtgcacac ttgtgcagtc ggggtgcggaa gtcaaaaagc cgggagcgtc ggtgaaagta 60
tcgtgtaaag cgtcgggata tacgtttacg gactataaca tggactgggt acgacaggca 120
ccggggaaat cgttggaatg gatcgacagc attaatccga acaatggggg aattttcttt 180
aatcagaaat tcaaaggacg ggcgacgttg acggtcgata catcgacgaa tacggcgtat 240
atggaattga ggtcgcttcg ctccggacgat acggcgggtc attactgcgc caggaggcgc 300
atcacgacgg tagggcgcat ggattattgg ggacagggga cgcttgtagc ggtatcgctc 360

SEQ ID NO: 54      moltype = AA length = 120
FEATURE           Location/Qualifiers
REGION           1..120
                  note = Description of Artificial Sequence: Synthetic
                  polypeptide
source           1..120
                  mol_type = protein
                  organism = synthetic construct

SEQUENCE: 54
QVQLVQSGAE VKKPGASVKV SCKASGYTFT DYNMDWVRQA PGKSLEWIGQ INPNNGGIF 60
NQKFKGRATL TVDTSTNTAY MELRSLRSDD TAVYYCAREA ITTVGAMDYWG GQGLTVTVSS 120

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SEQ ID NO: 55 moltype = DNA length = 360
 FEATURE Location/Qualifiers
 misc_feature 1..360
 note = Description of Artificial Sequence: Synthetic
 polynucleotide
 source 1..360
 mol_type = other DNA
 organism = synthetic construct

SEQUENCE: 55
 caagtccagc ttgtccagtc gggagcggaa gtgaagaaac cggggcgctc ggtcaaagta 60
 tcgtgtaaaag cgtcgggata tacgtttacg gactataaca tggattgggt acgacaggct 120
 ccgggaaaat cattggaatg gattggacag attaatccga ataatggggg tatcttcttt 180
 aatcaaaagt ttaaagggag ggcgacgttg acggtggaca aatcgacaaa tacggcgctat 240
 atggaattgt cgtcgcttcg gtcggaggac acggcggtgt attactcgcg gagggaggcg 300
 atcacgacgg tcggggcgat ggattattgg ggacagggaa cgcttgtagc ggtatcgctc 360

SEQ ID NO: 56 moltype = AA length = 120
 FEATURE Location/Qualifiers
 REGION 1..120
 note = Description of Artificial Sequence: Synthetic
 polypeptide
 source 1..120
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 56
 QVQLVQSGAE VKKPGSSVKV SCKASGYTFT DYNMDWVRQA PGKSLEWIGQ INPNNGGIFP 60
 NQKFGRATL TVDKSTNTAY MELSSLRSED TAVYYCAREA ITTVGAMDYV GQGLVTVSS 120

SEQ ID NO: 57 moltype = DNA length = 360
 FEATURE Location/Qualifiers
 misc_feature 1..360
 note = Description of Artificial Sequence: Synthetic
 polynucleotide
 source 1..360
 mol_type = other DNA
 organism = synthetic construct

SEQUENCE: 57
 cagggtccagc ttgtgcaatc gggagcggaa gtgaagaaac cgggagcgctc ggtaaaagtc 60
 tcgtgcaaaag cgtcggggta tacgtttacg gactataaca tggactgggt gcgccaagcg 120
 cctggacagg gtcttgaatg gatggggcag attaatccga ataatggagg gatcttcttt 180
 aatcagaaat tcaaaggaag ggtaacgctg acgacagaca cgtcaacatc gacggcctat 240
 atggaattgc ggtcgttgcg atcagatgat acggcggtct actattgtgc gagggaggcg 300
 attacgacgg tgggagcgat ggattattgg ggacagggga cgttggtaac ggtatcgctc 360

SEQ ID NO: 58 moltype = AA length = 120
 FEATURE Location/Qualifiers
 REGION 1..120
 note = Description of Artificial Sequence: Synthetic
 polypeptide
 source 1..120
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 58
 QVQLVQSGAE VKKPGASVKV SCKASGYTFT DYNMDWVRQA PGQGLEWMGQ INPNNGGIFP 60
 NQKFGRVTL TDTSTSTAY MELRSLRSD TAVYYCAREA ITTVGAMDYV GQGLVTVSS 120

SEQ ID NO: 59 moltype = DNA length = 360
 FEATURE Location/Qualifiers
 misc_feature 1..360
 note = Description of Artificial Sequence: Synthetic
 polynucleotide
 source 1..360
 mol_type = other DNA
 organism = synthetic construct

SEQUENCE: 59
 cagggtccagc ttgtgcaatc gggagcggaa gtgaagaaac cgggagcgctc ggtaaaagtc 60
 tcgtgcaaaag cgtcggggta tacgtttacg gactataaca tggactgggt gcgccaagcg 120
 cctggacaga gccttgaatg gatggggcag attaatccga ataatggagg gatcttcttt 180
 aatcagaaat tccaggggaag ggtaacgctg acgacagaca cgtcaacatc gacggcctat 240
 atggaattgc ggtcgttgcg atcagatgat acggcggtct actattgtgc gagggaggcg 300
 attacgacgg tgggagcgat ggattattgg ggacagggga cgttggtaac ggtatcgctc 360

SEQ ID NO: 60 moltype = AA length = 120
 FEATURE Location/Qualifiers
 REGION 1..120
 note = Description of Artificial Sequence: Synthetic

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source                polypeptide
                      1..120
                      mol_type = protein
                      organism = synthetic construct

SEQUENCE: 60
QVQLVQSGAE VKKPGASVKV SCKASGYTFT DYNMDWVRQA PGQSLEWMGQ INPNNGGIF 60
NQKFKGRVTL TTDSTSTAY MELRSLRSDD TAVYYCAREA ITTVGAMDYV GQGLTVTVSS 120

SEQ ID NO: 61         moltype = DNA length = 360
FEATURE              Location/Qualifiers
misc_feature          1..360
                      note = Description of Artificial Sequence: Synthetic
                      polynucleotide
source                1..360
                      mol_type = other DNA
                      organism = synthetic construct

SEQUENCE: 61
cagggtccagc ttgtgcaatc gggagcggaa gtgaagaaac cgggagcgtc ggtaaaagt 60
tcgtgcaaaag cgctcggggta tacgtttacg gactataaca tggactgggt gcgccaagcg 120
cctggacagg gtcttgaatg gatggggcag attaatccga ataatggagg gatcttcttt 180
aatcagaaat tccaggggaag ggtaacgctg acgacagaca cgtaacatc gacggcctat 240
atggaattgc ggtcggtgcg atcagatgat acggcggtct actattgtgc gagggaggcg 300
attacgacgg tgggagcgat ggattattgg ggacagggga cgttggtaac ggtatcgctc 360

SEQ ID NO: 62         moltype = AA length = 120
FEATURE              Location/Qualifiers
REGION               1..120
                      note = Description of Artificial Sequence: Synthetic
                      polypeptide
source                1..120
                      mol_type = protein
                      organism = synthetic construct

SEQUENCE: 62
QVQLVQSGAE VKKPGASVKV SCKASGYTFT DYNMDWVRQA PGQGLEWMGQ INPNNGGIF 60
NQKFKGRVTL TTDSTSTAY MELRSLRSDD TAVYYCAREA ITTVGAMDYV GQGLTVTVSS 120

SEQ ID NO: 63         moltype = DNA length = 360
FEATURE              Location/Qualifiers
misc_feature          1..360
                      note = Description of Artificial Sequence: Synthetic
                      polynucleotide
source                1..360
                      mol_type = other DNA
                      organism = synthetic construct

SEQUENCE: 63
caagtacagc ttgtacagtc gggagcggaa gtcaagaaac cgggatcgtc ggtcaaagt 60
tcgtgtaaaag cgctcgggata tacgtttacg gactataaca tggattgggt gcgacaagcg 120
cctgggcagg gacttgaatg gatgggtcag atcaatccga ataatggggg aatctttttc 180
aatcagaaat ttaaaggggag ggtaacgctg acggcgggata aaagcacgtc aacggcgat 240
atggagttgt cgctcggtgcg gtcggaggac acggcggtct attactgcgc gagggagcg 300
attacgacgg tgggagcgat ggattattgg gggcagggaa cgcttgtaac ggtgtcatcg 360

SEQ ID NO: 64         moltype = AA length = 120
FEATURE              Location/Qualifiers
REGION               1..120
                      note = Description of Artificial Sequence: Synthetic
                      polypeptide
source                1..120
                      mol_type = protein
                      organism = synthetic construct

SEQUENCE: 64
QVQLVQSGAE VKKPGSSVKV SCKASGYTFS DYNMDWVRQA PGQGLEWMGQ INPNNGGIF 60
NQKFKGRVTL TADKSTSTAY MELSSLRSDD TAVYYCAREA ITTVGAMDYV GQGLTVTVSS 120

SEQ ID NO: 65         moltype = DNA length = 360
FEATURE              Location/Qualifiers
misc_feature          1..360
                      note = Description of Artificial Sequence: Synthetic
                      polynucleotide
source                1..360
                      mol_type = other DNA
                      organism = synthetic construct

SEQUENCE: 65
caagtacagc ttgtacagtc gggagcggaa gtcaagaaac cgggatcgtc ggtcaaagt 60
tcgtgtaaaag cgctcgggata tacgtttacg gactataaca tggattgggt gcgacaagcg 120
cctgggcagg gacttgaatg gatgggtcag atcaatccga ataatggggg aatctttttc 180

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aatcagaagt ttcaggggag ggtaacgctg acggcggata aaagcacgtc aacggcgtat 240
atggagttgt cgtcggttgcg gtcggaggac acggcgggtct attactgcgc gaggggaagcg 300
attacgacgg tgggagcgat ggattattgg gggcagggaa cgcttgtaac ggtgtcatcg 360

SEQ ID NO: 66      moltype = AA  length = 120
FEATURE           Location/Qualifiers
REGION           1..120
                  note = Description of Artificial Sequence: Synthetic
                  polypeptide
source           1..120
                  mol_type = protein
                  organism = synthetic construct

SEQUENCE: 66
QVQLVQSGAE VKKPGSSVKV SCKASGYTFS DYNMDWVRQA PGQGLEWMGQ INPNNGGIF 60
NQKFPQGRVTL TADKSTSTAY MELSSLRSED TAVYYCAREA ITTVGAMDYV QGTLVTVSS 120

SEQ ID NO: 67      moltype = DNA  length = 357
FEATURE           Location/Qualifiers
misc_feature      1..357
                  note = Description of Artificial Sequence: Synthetic
                  polynucleotide
source           1..357
                  mol_type = other DNA
                  organism = synthetic construct

SEQUENCE: 67
caggtgactt tgaagaatc cggtcgccga ttggtaaagc caaccagac acttacgctc 60
acatgtacat ttccggatt cagcttgaac acttacggga tgggagtgtc gtggattcgg 120
caacatccgg ggaaggctct ggagtggctg gcgcacatct actgggatga tgacaaaagg 180
tataaccctt cacttaaaac gagactgacg atctcgaagg acacaagcaa gaatcaggtc 240
gtcctcacga ttacgaatgt agaccgggtg gatactgccg tctattactg cgcgcaacgc 300
gggatgatg actactgggg atattggggg cagggcaccc tcgtgaccat ctcgtca 357

SEQ ID NO: 68      moltype = AA  length = 119
FEATURE           Location/Qualifiers
REGION           1..119
                  note = Description of Artificial Sequence: Synthetic
                  polypeptide
source           1..119
                  mol_type = protein
                  organism = synthetic construct

SEQUENCE: 68
QVTLKESGPA LVKPTQTLLT TCTFSGFSLN TYGMGVSWIR QPPGKALEWL AHYWDDEDKR 60
YNPSLKRRLT ISKDTSKNQV VLITINVDPV DTAIVYCAQR GYDDYWGYYG QGTLVTVSS 119

SEQ ID NO: 69      moltype = DNA  length = 357
FEATURE           Location/Qualifiers
misc_feature      1..357
                  note = Description of Artificial Sequence: Synthetic
                  polynucleotide
source           1..357
                  mol_type = other DNA
                  organism = synthetic construct

SEQUENCE: 69
caagtaacgc tcaaggagtc cggaccaccc ttggtgaagc cgacgcagac cttgactctt 60
acgtgcactt tctcgggggt ttactgaat acgtacggga tgggtgtctc atggatcagg 120
caacatccgg ggaaggattt ggaatggctg gcgcacatct actgggatga cgataagaga 180
tataacccaa gcctcaagtc gcgggtcacc attacaaaag atacatcgaa aaatcaggtc 240
gtacttacta tcacgaacat ggaccccggt gacacagcaa catattactg tgcccagcgc 300
ggctatgacg attattgggg ttactgggga cagggaacac tggtcacggt gtccagc 357

SEQ ID NO: 70      moltype = AA  length = 119
FEATURE           Location/Qualifiers
REGION           1..119
                  note = Description of Artificial Sequence: Synthetic
                  polypeptide
source           1..119
                  mol_type = protein
                  organism = synthetic construct

SEQUENCE: 70
QVTLKESGPT LVKPTQTLLT TCTFSGFSLN TYGMGVSWIR QPPGKLEWL AHYWDDEDKR 60
YNPSLKSRLT ITKDTSKNQV VLITINMDPV DTATYYCAQR GYDDYWGYYG QGTLVTVSS 119

SEQ ID NO: 71      moltype = DNA  length = 357
FEATURE           Location/Qualifiers
misc_feature      1..357
                  note = Description of Artificial Sequence: Synthetic

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source                polynucleotide
                      1..357
                      mol_type = other DNA
                      organism = synthetic construct

SEQUENCE: 71
cagatcactt tgaagaaag cggaccgacc ttggtcaagc ccacacaaac cctcacgctc 60
acgtgtacat tttcgggggt ctcgctttca acttacggga tgggagtagg gtggattcgc 120
cagccgcctg gtaaagcgtt ggagtgggct gcagacatct ggtgggacga cgataagtac 180
tataatccct cgctcaagtc cagactgacc atcacgaaag atacgagcaa gaaccaggtc 240
gtgctgacaa tgactaacat ggaccagtg gatacggcta catattactg cgccaggcgg 300
ggtcactact cagcgatgga ttattggggc cagggaaacac tggtaacggg gtcgtcc 357

SEQ ID NO: 72         moltype = AA length = 119
FEATURE              Location/Qualifiers
REGION               1..119
                    note = Description of Artificial Sequence: Synthetic
                    polypeptide
source               1..119
                    mol_type = protein
                    organism = synthetic construct

SEQUENCE: 72
QITLKESGPT LVKPTQTLLT TCTFSGFSLT TYGMGVGWIR QPPGKALEWL ADIWDDDKY 60
YNPSLKSRLT ITKDTSKNQV VLTMTNMDPV DTATYYCARR GHYSAMDYWG QGTLLTVSS 119

SEQ ID NO: 73         moltype = DNA length = 357
FEATURE              Location/Qualifiers
misc_feature         1..357
                    note = Description of Artificial Sequence: Synthetic
                    polynucleotide
source               1..357
                    mol_type = other DNA
                    organism = synthetic construct

SEQUENCE: 73
caagtgactc tcaaggagtc cggacccgcc ctggtcaaac caacgcagac actgacgctc 60
acatgcacct tcagcggatt ttctgtgtca acgtacggca tgggtgtggg gtggattcgc 120
cagcctcccg ggaaagccct tgaatgggtg gcggacatct ggtgggatga tgacaagtac 180
tataatccct cacttaagtc acggttgacg atctcgaaag acaccagcaa gaaccaggta 240
gtgctgacaa tgactaacat ggacccggtc gatacagcgg tctactattg tgctagaagg 300
ggacactact ccgcaatgga ttattggggg caggggacgc tcgtaaccgt gtcgtcg 357

SEQ ID NO: 74         moltype = AA length = 119
FEATURE              Location/Qualifiers
REGION               1..119
                    note = Description of Artificial Sequence: Synthetic
                    polypeptide
source               1..119
                    mol_type = protein
                    organism = synthetic construct

SEQUENCE: 74
QVTLKESGPA LVKPTQTLLT TCTFSGFSLT TYGMGVGWIR QPPGKALEWL ADIWDDDKY 60
YNPSLKSRLT ISKDTSKNQV VLTMTNMDPV DTAVYYCARR GHYSAMDYWG QGTLLTVSS 119

SEQ ID NO: 75         moltype = DNA length = 321
FEATURE              Location/Qualifiers
misc_feature         1..321
                    note = Description of Artificial Sequence: Synthetic
                    polynucleotide
source               1..321
                    mol_type = other DNA
                    organism = synthetic construct

SEQUENCE: 75
gacatccaga tgactcagtc tccagcctcc ctatctgcat ctgtgggaga aactgtcacc 60
atcacatgtc gaacaagtga gaatcttcac aattatttag catggatca gcagaaacag 120
ggaaaatctc ctcagctcct ggtctatgat gcaaaaacct tagcagatgg tgtgccatca 180
aggttcagtg gcagtggatc aggaacacaa tattctctca agatcaacag cctgcagcct 240
gaagattttg ggagttatta ctgtcaacat ttttggagta gtccttacac gttcggaggg 300
gggaccaagc tggaaataaa a 321

SEQ ID NO: 76         moltype = AA length = 107
FEATURE              Location/Qualifiers
REGION               1..107
                    note = Description of Artificial Sequence: Synthetic
                    polypeptide
source               1..107
                    mol_type = protein
                    organism = synthetic construct

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SEQUENCE: 76
 DIQMTQSPAS LSASVGETVT ITCRTSENLH NYLAWYQQKQ GKSPQLLVYD AKTLADGVPS 60
 RFGSGSGGTQ YSLKINSLQP EDFGSYYCQH FWSSPYTFGG GTKLEIK 107

SEQ ID NO: 77 moltype = DNA length = 333
 FEATURE Location/Qualifiers
 misc_feature 1..333
 note = Description of Artificial Sequence: Synthetic
 polynucleotide
 source 1..333
 mol_type = other DNA
 organism = synthetic construct

SEQUENCE: 77
 gacattgtgt tgacccaatc tccagcttct ttggctgtgt ctctagggca gagggccacc 60
 atctcctgca gagccagcga aagtgttgat aattatggca ttagttttat gaactgggtc 120
 caacagaaac caggacagcc acccaaactc ctcatctatg ctgcatccaa ccaaggctcc 180
 ggggtcccctg ccagggtttag ttggcagtggg tctgggacag acttcagcct caacatccat 240
 cctatggagg aggatgatac tgcaatgtat ttctgtcagc aaagtaagga ggttccgtgg 300
 acgttcgggt gaggtccaa gctggaatc aaa 333

SEQ ID NO: 78 moltype = AA length = 111
 FEATURE Location/Qualifiers
 REGION 1..111
 note = Description of Artificial Sequence: Synthetic
 polypeptide
 source 1..111
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 78
 DIVLTQSPAS LAVSLGQRAT ISCRASESVD NYGISFMNWF QQKPGQPPKL LIYAASNQGS 60
 GVPARFSGSG SGTDFSLNIH PMEEDDTAMY FCQQSKEVPW TFGGSKLEI K 111

SEQ ID NO: 79 moltype = DNA length = 321
 FEATURE Location/Qualifiers
 misc_feature 1..321
 note = Description of Artificial Sequence: Synthetic
 polynucleotide
 source 1..321
 mol_type = other DNA
 organism = synthetic construct

SEQUENCE: 79
 gacattgtga tgacccagtc tcaaaaattc atgtccacat cagtaggaga cagggtcagc 60
 gtcacctgca aggccagtcga gaatgtgggt actaatgtag cctgggtatca acagaaatta 120
 ggacaatctc ctaaaacact gatttactcg gcattcctacc ggtacagtgg agtccctgat 180
 cgcttcacag gcagtggatc tgggacagat ttcactctca ccacagcaa tgtgcagtct 240
 gaagacttgg cagagtattt ctgtcagcaa tataacagct atccgtacac gttcggaggg 300
 gggaccaagc tggaaataaa a 321

SEQ ID NO: 80 moltype = AA length = 107
 FEATURE Location/Qualifiers
 REGION 1..107
 note = Description of Artificial Sequence: Synthetic
 polypeptide
 source 1..107
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 80
 DIVMTQSQKF MSTSVGDRVS VTCKASQNVG TNVAWYQQKL QSPKTLIYS ASYRSGVDP 60
 RFTGSGSGTD FTLTISNVQS EDLAEYFCQQ YNSYPYTFGG GTKLEIK 107

SEQ ID NO: 81 moltype = DNA length = 321
 FEATURE Location/Qualifiers
 misc_feature 1..321
 note = Description of Artificial Sequence: Synthetic
 polynucleotide
 source 1..321
 mol_type = other DNA
 organism = synthetic construct

SEQUENCE: 81
 gacattgtga tgacccagtc tcaaaaattc atgtccacat cagtaggaga cagggtcagc 60
 gtcacctgca aggccagtcga gaatgtgggt actaatgtag cctgggttca acagaaacca 120
 ggtaaatctc ctaaaacact gatttactcg gcattcctacc ggtacagtgg agtccctgat 180
 cgcttcacag gcagtggatc tgggacagat ttcattctca ccacagcaa tgtgcagtct 240
 gaagacctgg cagagtattt ctgtcagcaa tataacaact atcctctcac gttcgggtgt 300
 gggaccaagc tggagctgaa a 321

SEQ ID NO: 82	moltype = AA length = 107	
FEATURE	Location/Qualifiers	
REGION	1..107	
	note = Description of Artificial Sequence: Synthetic polypeptide	
source	1..107	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 82		
DIVMTQSQKF MSTSVGDRVS	VTCKASQNVG TNVAWFQQKP	QQSPKALIYS ASYRYSGVPD 60
RFTGSGSGTD FILTISNVQS	EDLAEYFCQQ YNNYPLTFGA	GTKLELK 107
SEQ ID NO: 83	moltype = DNA length = 321	
FEATURE	Location/Qualifiers	
misc_feature	1..321	
	note = Description of Artificial Sequence: Synthetic polynucleotide	
source	1..321	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 83		
gacatccaga tgactcagtc	tccagcctcc ctatctgcat	ctgtggggaga aactgtcacc 60
atcacatgtc gagcaagtgg	gaatattcac aattattttag	catgggtatca gcagaaacag 120
ggaaaatctc ctcagctctc	ggctctataat gcaaaaaacct	tagcagatgg tgtgccatca 180
aggttcagtg gcagctggatc	aggaacacaa tattctctca	agatcaacag cctgcagcct 240
gaagattttg ggagttatta	ctgtcaacat ttttgaggat	ctccttacac gttcggaggg 300
gggaccaagc tggaaataaa	a	321
SEQ ID NO: 84	moltype = AA length = 107	
FEATURE	Location/Qualifiers	
REGION	1..107	
	note = Description of Artificial Sequence: Synthetic polypeptide	
source	1..107	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 84		
DIQMTQSPAS LSASVGETVT	ITCRASGNIH NYLAWYQQQK	GKSPQLLVYN AKTLADGVPS 60
RFSGSGSGTQ YSLKINSIQP	EDFGSYFCQH FWSSPYTFGG	GTKLEIK 107
SEQ ID NO: 85	moltype = DNA length = 321	
FEATURE	Location/Qualifiers	
misc_feature	1..321	
	note = Description of Artificial Sequence: Synthetic polynucleotide	
source	1..321	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 85		
gcattgtgaa tgaccagtc	tcaaaaattc atgtccacat	cagtaggaga cagggtcagc 60
gtcacctgca aggccagtc	gaatgtgggt actaatgtag	cctgggtatca acagaaacca 120
gggcaatctc ctaaagcact	gatttactcg ccatcctacc	ggtacagtgg agtccctgat 180
cgcttcacag gcagctggatc	tgggacagat ttcactctca	ccatcagcaa tgtgcagctc 240
gaagacttgg cagaatat	ctgtcagcaa tataacagct	atcctcacac gttcggaggg 300
gggaccaagc tggaaatgaa	a	321
SEQ ID NO: 86	moltype = AA length = 107	
FEATURE	Location/Qualifiers	
REGION	1..107	
	note = Description of Artificial Sequence: Synthetic polypeptide	
source	1..107	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 86		
DIVMTQSQKF MSTSVGDRVS	VTCKASQNVG TNVAWFQQKP	QQSPKALIYS PSYRYSGVPD 60
RFTGSGSGTD FTLTISNVQS	EDLAEYFCQQ YNSYPHTFGG	GTKLEMK 107
SEQ ID NO: 87	moltype = DNA length = 333	
FEATURE	Location/Qualifiers	
misc_feature	1..333	
	note = Description of Artificial Sequence: Synthetic polynucleotide	
source	1..333	
	mol_type = other DNA	
	organism = synthetic construct	

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SEQUENCE: 87
gacattgtgc tgacacagtc tectgcttcc ttagctgtat ctctggggca gagggccacc 60
atctcatgca gggccagcca aagtgtcagt acatctaggt ttagttatat gcactgggtc 120
caacagaaac caggacaggc acccaaacct ctcataagat atgcatccaa cctagaatct 180
gggggtccctg ccagggttcag tggcagtggtg tctggggacag acttcaccct caacatccat 240
cctgtggagg gggaggatag tgcaacatat tactgtcagc acagttggga gattccgtac 300
acgttcggag gggggaccaa gctggaata aaa 333

SEQ ID NO: 88 moltype = AA length = 111
FEATURE Location/Qualifiers
REGION 1..111
 note = Description of Artificial Sequence: Synthetic
 polypeptide
source 1..111
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 88
DIVLTQSPAS LAVSLGQRAT ISCRASQSVS TSRFSYMHWF QQKPGQAPKL LIKYASNLES 60
GVPARFSGSG SGTDFTLNIH PVEGEDTATY YCQHSWEIPY TFGGGTKLEI K 111

SEQ ID NO: 89 moltype = DNA length = 321
FEATURE Location/Qualifiers
misc_feature 1..321
 note = Description of Artificial Sequence: Synthetic
 polynucleotide
source 1..321
 mol_type = other DNA
 organism = synthetic construct

SEQUENCE: 89
gacatccaaa tgacccagtc gccgtcgctcg ctttcagcgt cggtagggga tcgggtcaca 60
attacgtgcc gaacgtcaga gaatttgcac aactacctcg cgtgggtatca gcagaagccc 120
gggaagtcac cgaaactcct tgtctacgat gcgaaaacgc tggcggatgg agtgccgctcg 180
agattctcgg gaagcggatc cggtagcgac tatacgctta cgatctcatc gctccagccc 240
gaggactttg cgacgtacta ttgtcagcat ttttggtcgt cgccctacac atttgggcag 300
gggaccaagt tggaaatcaa g 321

SEQ ID NO: 90 moltype = AA length = 107
FEATURE Location/Qualifiers
REGION 1..107
 note = Description of Artificial Sequence: Synthetic
 polypeptide
source 1..107
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 90
DIQMTQSPSS LSASVGDRVT ITCRTSENLH NYLAWYQKP GKSPKLLVYD AKTLADGVPS 60
RFSGSGSGTD YTLTISSLQP EDFATYYCQH FWSSPYTFGQ GTKLEIK 107

SEQ ID NO: 91 moltype = DNA length = 321
FEATURE Location/Qualifiers
misc_feature 1..321
 note = Description of Artificial Sequence: Synthetic
 polynucleotide
source 1..321
 mol_type = other DNA
 organism = synthetic construct

SEQUENCE: 91
gacatccaaa tgacccagtc gccgtcgctcg ctttcagcgt cggtagggga tcgggtcaca 60
attacgtgcc gaacgtcaga gaatttgcac aactacctcg cgtgggtatca gcagaagccc 120
gggaagggcc cgaaactcct tatctacgat gcgaaaacgc tggcggatgg agtgccgctcg 180
agattctcgg gaagcggatc cggtagcgac tatacgctta cgatctcatc gctccagccc 240
gaggactttg cgacgtacta ttgtcagcat ttttggtcgt cgccctacac atttgggcag 300
gggaccaagt tggaaatcaa g 321

SEQ ID NO: 92 moltype = AA length = 107
FEATURE Location/Qualifiers
REGION 1..107
 note = Description of Artificial Sequence: Synthetic
 polypeptide
source 1..107
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 92
DIQMTQSPSS LSASVGDRVT ITCRTSENLH NYLAWYQKP GKAPKLLIYD AKTLADGVPS 60
RFSGSGSGTD YTLTISSLQP EDFATYYCQH FWSSPYTFGQ GTKLEIK 107

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SEQ ID NO: 93 moltype = DNA length = 321
 FEATURE Location/Qualifiers
 misc_feature 1..321
 note = Description of Artificial Sequence: Synthetic
 polynucleotide
 source 1..321
 mol_type = other DNA
 organism = synthetic construct

SEQUENCE: 93
 gacatccaaa tgacccagtc gccgtcgctc ctttcagcgt cggtagggga tcgggtcaca 60
 attacgtgcc gaacgtcaga gaatttgc at aactacctcg cgtgggtatca gcagaagccc 120
 gggaagtcac cgaaactcct tatctacgat gcgaaaacgc tggcggatgg agtgccgctcg 180
 agattctcgg gaagcggatc cggtacggac tatacgctta cgatctcatc gctccagccc 240
 gaggactttg cgacgtacta ttgtcagcat ttttggctcg cgcctacac atttgggcag 300
 gggaccaagt tggaaatcaa g 321

SEQ ID NO: 94 moltype = AA length = 107
 FEATURE Location/Qualifiers
 REGION 1..107
 note = Description of Artificial Sequence: Synthetic
 polypeptide
 source 1..107
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 94
 DIQMTQSPSS LSASVGDVRT ITCRTSENLH NYLAWYQQKP GKSPKLLIYD AKTLADGVPS 60
 RFSGSGSGTD YLTISLQP EDFATYYCQH FWSSPYTFGQ GTKLEIK 107

SEQ ID NO: 95 moltype = DNA length = 321
 FEATURE Location/Qualifiers
 misc_feature 1..321
 note = Description of Artificial Sequence: Synthetic
 polynucleotide
 source 1..321
 mol_type = other DNA
 organism = synthetic construct

SEQUENCE: 95
 gacatccaaa tgacccaatc gccctcctcc ctctccgcat cagtagggga ccgcgtcaca 60
 attacttgca aagcgtcgca gaacgtcgga acgaatgtgg cgtgggttca gcagaagccc 120
 ggaaaagctc cgaagagcct gatctactcg gcctcatata ggtattcggg tgtgccgagc 180
 cggtttagcg ggtcggggtc aggtactgat ttacagctca caatttcate gttgcagcca 240
 gaagatttcg ccacatatta ctgtcagcag tacaacaatt accctctgac gttcggccag 300
 ggaaccaaac ttgagatcaa g 321

SEQ ID NO: 96 moltype = AA length = 107
 FEATURE Location/Qualifiers
 REGION 1..107
 note = Description of Artificial Sequence: Synthetic
 polypeptide
 source 1..107
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 96
 DIQMTQSPSS LSASVGDVRT ITCKASQNVG TNVAFQQKP GKAPKSLIYS ASYRYSGVPS 60
 RFSGSGSGTD FTLTISLQP EDFATYYCQ YNNYPLTFGQ GTKLEIK 107

SEQ ID NO: 97 moltype = DNA length = 321
 FEATURE Location/Qualifiers
 misc_feature 1..321
 note = Description of Artificial Sequence: Synthetic
 polynucleotide
 source 1..321
 mol_type = other DNA
 organism = synthetic construct

SEQUENCE: 97
 gatatccaga tgacacagtc accctcgctc ctctcagctt ccgtaggcga cagggtcact 60
 attacgtgta aagcatcaca gaacgtcgga acgaatgtgg cgtgggttca gcagaagccc 120
 gggaagagcc ccaaagcgct tatctactcc ccgtcgatc ggtattccgg tgtgccaaagc 180
 agattttcgg ggtcagggtc gggaactgac tttaccctga ccatctcgtc cctccaaccg 240
 gaagatttcg ccacgtactt ctgccagcag tacaacagct atcctcacac attcggaaca 300
 gggacaaagt tggagattaa a 321

SEQ ID NO: 98 moltype = AA length = 107
 FEATURE Location/Qualifiers
 REGION 1..107
 note = Description of Artificial Sequence: Synthetic

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polypeptide
source          1..107
                mol_type = protein
                organism = synthetic construct

SEQUENCE: 98
DIQMTQSPSS LSASVGDVRT ITCKASQNVG TNVAVFQQKP GKSPKALIYS PSYRYSQVPS 60
RFSGSGSGTDT FTLTISSLQP EDFATYFCQQ YNSYPHTFGQ GTKLEIK 107

SEQ ID NO: 99      moltype = DNA length = 1332
FEATURE           Location/Qualifiers
misc_feature       1..1332
                  note = Description of Artificial Sequence: Synthetic
                  polynucleotide
source            1..1332
                  mol_type = other DNA
                  organism = synthetic construct

SEQUENCE: 99
gaggctcctgc tgcaacagtc tggacctgag ctggtgaagc ctggggcttc agtgaagata 60
ccctgcaagg cttctggata cacattcact gactacaaca tggactgggt gaagcagagc 120
catggaaaga gccttgatg gattggacaa attaatccta acaatgggtg tattttcttc 180
aaccagaagt tcaagggcaa ggccacattg actgtagaca agtcctccaa tacagccttc 240
atggaggtcc gcagcctgac atctgaggac actgcagtct attactgtgc aagagaggca 300
attactacgg taggcctgat ggactactgg ggtcaaggaa cctcagtcac cgtctcctca 360
gccaaaacga ccccccatc tgtctatcca ctggcccctg gatctgctgc ccaaaactaac 420
tccatggtga ccctgggatg cctggtcaag ggctatttcc ctgagccagt gacagtgacc 480
tggaaactctg gatccctgac cagcgggtgt cacaccttcc cagctgtcct gcagtctgac 540
ctctacactc tgagcagctc agtgactgtc cctccagcga cctggcccag cgagaccgtc 600
acctgcaacg ttgcccaccc ggccagcagc accaagggtg acaagaaaat tgtgccccag 660
gattgtggtt gtaagccttg catatgtaca gtcccagaag tatcatctgt cttcatcttc 720
cccccaagc ccaaggatgt gctcaccatt actctgactc ctaaggctac gtgtgtttgt 780
gtagacatca gcaaggatga tcccaggtc cagttcagct ggtttgtaga tgatgtggag 840
gtgcacacag ctacagcagc accccgggag gagcagttca acagcacttt ccgctcagtc 900
agtgaacttc ccatcatgca ccaggactgg ctcaatggca aggagttcaa atgcagggtc 960
aacagtgacg ctttccctgc ccccatcgag aaaaccatct ccaaaaccaa aggcagaccg 1020
aaggctccac aggtgtacac cattccacct cccaaggagc agatggccaa ggataaaagtc 1080
agtctgacct gcatgataac agactcttcc cctgaagaca ttactgtgga gtggcagtg 1140
aatgggcagc acgaggagaa ctacaagaac actcagccca tcatggacac agatggctct 1200
tactctgctc acagcaagct caatgtgcag aagagcaact gggaggcagg aaatacttcc 1260
acctgctctg tgttatcatga gggcctgcac aaccaccata ctgagaagag cctctccacc 1320
tctcctggta aa 1332

SEQ ID NO: 100      moltype = AA length = 444
FEATURE           Location/Qualifiers
REGION           1..444
                  note = Description of Artificial Sequence: Synthetic
                  polypeptide
source            1..444
                  mol_type = protein
                  organism = synthetic construct

SEQUENCE: 100
EVLQQSGPE LVKPGASVKI PCKASGYTFT DYNMDWVKQS HGKSLEWIGQ INPNNGGIF 60
NQKFKGKATL TVDKSSNTAF MEVRLTSED TAVYYCAREA ITTVGAMDYV GQGTSTVTVSS 120
AKTTPPSVYP LAPGSAQTN SMVTLGCLVK GYFPEPVTVT WNSGSLSSGV HTFPAVLQSD 180
LYTLSSSVTV PSSTWPSSETV TCNVAHPASS TKVDKKIVPR DCGCKPCICT VPEVSSVFIF 240
PKPKDVLTI TLTPKVTCVV VDISKDDPEV QFSWFVDDVE VHTAQTPRE EQFNSTFRSV 300
SELPIMHQDW LNGKEFKCRV NSAAFPAPIE KTISKTKGRP KAPQVYTIPP PKEQMAKDKV 360
SLTCMITDFP PEDITVEWQW NGQPAENYKN TQPIMDTDGS YFVYSKLNQV KSNWEAGNTF 420
TCSVLHEGLH NHHTEKSLSH SPGK 444

SEQ ID NO: 101      moltype = DNA length = 642
FEATURE           Location/Qualifiers
misc_feature       1..642
                  note = Description of Artificial Sequence: Synthetic
                  polynucleotide
source            1..642
                  mol_type = other DNA
                  organism = synthetic construct

SEQUENCE: 101
gacatccaga tgactcagtc tccagcctcc ctatctgcat ctgtgggaga aactgtcacc 60
atcacatgtc gaacaagtga gaatcttcac aattatttag catgggatca gcagaaacag 120
ggaaaatctc ctacgctcct ggtctatgat gcaaaaacct tagcagatgg tgtgccatca 180
aggttcagtg gcagtggtac aggaacacaa tattctctca agatcaacag cctgcagcct 240
gaagattttg ggagttatta ctgtcaacat ttttgagta gtccttacac gttcggaggg 300
gggaccaagc tggaaataaa acgggctgat gctgcacca ctgtatccat cttcccacca 360
tccagtgagc agttaacatc tggaggtgcc tcagtcgtgt gcttcttgaa caactctac 420
cccaagaca tcaatgtcaa gtggaagatt gatggcagtg aacgacaaaa tggcgtcctg 480

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aacagtgtgga ctgatcagga cagcaaaagac agcacctaca gcatgagcag caccctcacg 540
ttgaccaagg acgagtatga acgacataac agctatacct gtgaggccac tcacaagaca 600
tcaacttcac ccattgtcaa gagcttcaac aggaatgagt gt 642

SEQ ID NO: 102      moltype = AA  length = 214
FEATURE            Location/Qualifiers
REGION            1..214
                  note = Description of Artificial Sequence: Synthetic
                  polypeptide
source            1..214
                  mol_type = protein
                  organism = synthetic construct

SEQUENCE: 102
DIQMTQSPAS LSASVGETVT ITCRTSENLH NYLAWYQQKQ GKSPQLLVYD AKTLADGVPS 60
RFSGSGSGTQ YSLKINSLQP EDFGSYYCQH FWSSPYTFGG GTKLEIKRAD AAPTVISIFPP 120
SSEQLTSGGA SVVCFLLNFF PKDINVKKKI DGSERQNGVL NSWTDQDSKD STYSMSSTLT 180
LTKDEYERHN SYTCEATHKT STSPIVKSPN RNEC 214

SEQ ID NO: 103      moltype = DNA  length = 1326
FEATURE            Location/Qualifiers
misc_feature       1..1326
                  note = Description of Artificial Sequence: Synthetic
                  polynucleotide
source            1..1326
                  mol_type = other DNA
                  organism = synthetic construct

SEQUENCE: 103
caggtccaac tgcagcagcc tggggctgaa ctggtgaagc ctggggcttc agtgaagctg 60
tcctgcaagg cttctggcta caccctcacc agctactgga ttcactgggt gaaccagagg 120
cctggacaag gccttgagtg gattggagac attaatccta gcaacggccg tagtaagtat 180
aatgagaagt tcaagaacaa ggccacaatg actgcagaca aatcctccaa cacagcctac 240
atgcaactca gcagcctgac atctgaggac tctgcggtct attactgtgc aagagaggtt 300
ctggatggtg ctatggacta ctgggggtcaa ggaacctcag tcaccgtctc ctcagccaaa 360
acgacacccc catctgtcta tccactggcc cctggatctg ctgcccaaac taactccatg 420
gtgaccctgg gatgcctggt caagggctat ttccctgagc cagtgcagtg gacctggaac 480
tctggatccc tgtccagcgg tgtgcacacc ttcccagctg tctgcagtc tgacctctac 540
actctgagca gctcagtgac tgtcccctcc agcacctggc ccagcgagac cgtcacctgc 600
aacgttgccc acccgccagc cagcaccaag gtggacaaga aaattgtgcc cagggattgt 660
ggttgtaagc cttgcatatg tacagtccca gaagtatcat ctgtcttcat cttcccccca 720
aagcccaagg atgtgctcac cattactctg actcctaagg tcacgtgtgt tgtggtagac 780
atcagcaagg atgatccga ggtccagttc agctggtttg tagatgatgt ggaggtgcac 840
acagctcaga cgcaaccocg ggaggagcag ttcaacagca ctttcgctc agtcagtgaa 900
cttcccatca tgcaccagga ctggctcaat ggcaaggagt tcaaatgcag ggtcaacagt 960
gcagctttcc tgcgccccat cgagaaaaacc atctccaaaa ccaaaggcag accgaaggct 1020
ccacaggtgt acaccattcc acctcccaag gagcagatgg ccaaggataa agtcagtctg 1080
acctgcagta taacagacct cttccctgaa gacattactg tggagtggca gtggaatggg 1140
cagccagcgg agaactacaa gaacactcag cccatcatgg acacagatgg ctcttacttc 1200
gtctacagca agctcaatgt gcagaagagc aactgggagg caggaaatac ttccacttgc 1260
tctgtgttac atgagggcct gcacaaccac catactgaga agagcctctc ccactctcct 1320
ggtaaa 1326

SEQ ID NO: 104      moltype = AA  length = 442
FEATURE            Location/Qualifiers
REGION            1..442
                  note = Description of Artificial Sequence: Synthetic
                  polypeptide
source            1..442
                  mol_type = protein
                  organism = synthetic construct

SEQUENCE: 104
QVQLQQPGAE LVKPGASVKL SCKASGYTFT SYWIHWNQR PGQGLEWIGD INPSNGRSKY 60
NEKFKNKATM TADKSSNTAY MQLSSLTSED SAVYYCAREV LDGAMDYWGQ GTSVTVSSAK 120
TTPPSVYPLA PGSAAQNTSM VTLGCLVKGY FPEPVTVTNW SGSLSSGVHT FPAVLQSDLY 180
TLSSSVTVPS STWPSSETVC NVAHPASSTK VDKKIVPRDC GCKPCICTVP EVSSVFIFPP 240
KPKDVLITIL TPKVTCVVVD ISKDDPEVQF SWFVDDVEVH TAQTQPREEQ FNSTFRSVSE 300
LPIMHQDWLN GKEFKCRVNS AAFPAPIEKT ISKTKGRPKA PQVYTIPPPK EQMAKDKVSL 360
TCMITDFPPE DITVEWQWNG QPAENYKNTQ PIMDTDGSYF VYSKLNQVKS NWEAGNTFTC 420
SVLHEGLHNN HTEKSLSHSP GK 442

SEQ ID NO: 105      moltype = DNA  length = 654
FEATURE            Location/Qualifiers
misc_feature       1..654
                  note = Description of Artificial Sequence: Synthetic
                  polynucleotide
source            1..654
                  mol_type = other DNA

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organism = synthetic construct

SEQUENCE: 105
gacattgtgt tgacccaatc tccagcttct ttggctgtgt ctctagggca gagggccacc 60
atctcctgca gagccagcga aagtgttgat aattatggca ttagttttat gaactgggtc 120
caacagaaac caggacagcc acccaactc ctcatctatg ctgcatccaa ccaaggctcc 180
ggggctccctg ccaggtttag tggcagtggt tctgggacag acttcagcct caacatccat 240
cctatggagg aggatgatac tgcaatgtat ttctgtcagc aaagtaagga ggttccgtgg 300
acgttcgggtg gaggtcccaa gctggaaatc aaacgggctg atgctgcacc aactgtatcc 360
atcttccac catccagtga gcagttaaca tctggagggtg cctcagtcgt gtgcttcttg 420
aacaacttct accccaaga catcaatgtc aagtgggaag ttgatggcag tgaacgacaa 480
aatggcgctc tgaacagttg gactgatcag gacagcaaa acagcaccta cagcatgagc 540
agcaccctca cggtgaccaa ggacgagtat gaacgacata acagctatac ctgtgaggcc 600
actcacaaga catcaacttc acccattgtc aagagcttca acaggaatga gtgt 654

SEQ ID NO: 106      moltype = AA length = 218
FEATURE            Location/Qualifiers
REGION              1..218
                    note = Description of Artificial Sequence: Synthetic
                    polypeptide
source              1..218
                    mol_type = protein
                    organism = synthetic construct

SEQUENCE: 106
DIVLTQSPAS LAVSLGQRAT ISCRASESVD NYGISFMNWF QQKPGQPPKL LIYAASNQGS 60
GVPARFSGSG SGTDFSLNIH PMEEDDTAMY FCQQSKEVPW TFGGSKLEI KRADAAPTVS 120
IFPPSSSEQLT SGGASVVCFL NNFYPKDINV KWKIDGSEKQ NGVLNSWTDQ DSKDSTYSMS 180
STLTLTKDEY ERHNSYTCEA THKTSTSPIV KSFNRNEC 218

SEQ ID NO: 107      moltype = DNA length = 1329
FEATURE            Location/Qualifiers
misc_feature        1..1329
                    note = Description of Artificial Sequence: Synthetic
                    polynucleotide
source              1..1329
                    mol_type = other DNA
                    organism = synthetic construct

SEQUENCE: 107
caggttactc tgaagagatc tggccctggg atattgcagc cctcccagac cctcagtcgt 60
acttgttctt tctctgggtt ttactgagc acttatggta tgggtgtgac ctggattcgt 120
cagccttcag gaaagggtct ggagtggctg gcacacattt actgggatga tgacaagcgc 180
tataacccat cctgaagag ccggctcaca atctccaagg atacctccaa caaccaggta 240
ttctcaaga tcaccagtggt ggacactgca gatactgcca catactactg tgctcaaacg 300
gggtatagta acttgtttgc ttactggggc caagggactc tggtcactgt cctgcagcc 360
aaaacgacac ccccatctgt ctatccactg gcccttggat ctgtgcacca aactaactcc 420
atggtgaccc tgggatgcct ggtcaagggc tatttccctg agccagtgac agtgacctgg 480
aaactctggat cctgttcag cgtgtgtcac accttcccag ctgtcctgca gtctgacctc 540
tacactctga gcagctcagt gactgtcccc tccagcacct ggcccagcga gaccgtcacc 600
tgcaacgttg cccaccggc cagcagcacc aaggtggaca agaaaattgt gccacgggat 660
tgtgtgtgta agccttgcat atgtacagtc ccagaagtat catctgtctt catcttcccc 720
ccaaagccca aggatgtgct caccattact ctgactccta aggtcacgtg tgttgttgta 780
gacatcagca aggatgatcc cgaggtccag ttcagctggt ttgtagatga tgtggagggtg 840
cacacagctc agacgcaccc ccgggaggag cagttcaaca gcactttccg ctcagtcagt 900
gaacttccca tcatgcacca ggactgggtc aatggcaagg agttcaaatg cagggtcaac 960
agtgcagctt tccctgcccc catcgagaaa accatctcca aaaccaaagg cagaccgaag 1020
gctccacagg ttacacccat tccactccc aaggagcaga tggccaagga taaagtcagt 1080
ctgacctgca tgataacaga cttcttcctt gaagacatta ctgtggagtg gcagtggagt 1140
gggcagccag cggagaacta caagaacact cagcccatca tggacacaga tggctcttac 1200
ttcgtctaca gcaagctcaa tgtgcagaag agcaactggg aggcaggaaa tactttcacc 1260
tgctctgtgt tacatgaggg cctgcacaac caccatactg agaagagcct ctcccactct 1320
cctggtaaa 1329

SEQ ID NO: 108      moltype = AA length = 443
FEATURE            Location/Qualifiers
REGION              1..443
                    note = Description of Artificial Sequence: Synthetic
                    polypeptide
source              1..443
                    mol_type = protein
                    organism = synthetic construct

SEQUENCE: 108
QVTLKESGPG ILQPSQTLSL TCSFSGFSLs TYGMGVTWIR QPSGKGLEWL AHYWDDEK 60
YNPSLKSRLL ISKDTSNQV FLKITSVDTA DTATYYCAQT GYSNLFAYWG QGTLVTVSAA 120
KTPPSVYPL APGSAQNS MVTLGCLVKG YFPEPVTWV NSGSLSSGVH TFPVAVLQSDL 180
YTLSSSVTVP SSTWPSETVT CNVAHPASST KVDKKIVPRD CGCKPCICTV PEVSSVFIFP 240
PKPKDVLTTT LTPKVTVCVV DISKDDPEVQ FSWFVDDDEV HTAQTPREE QFNSTFRSVS 300
ELPIMHQDWL NGKEFKCRVN SAAFPAPIEK TISKTGRPK APQVYTIIPP KEQMAKDKVS 360

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LTCMITDFFP EDITVEWQWN GQPAENYKNT QPIMDTDGSY FVYSKLVNQK SNWEAGNTFT 420
 CSVLHEGLHN HTEKSLSHS PGK 443

SEQ ID NO: 109 moltype = DNA length = 642
 FEATURE Location/Qualifiers
 misc_feature 1..642
 note = Description of Artificial Sequence: Synthetic
 polynucleotide
 source 1..642
 mol_type = other DNA
 organism = synthetic construct

SEQUENCE: 109

gacattgtga	tgacccagtc	tcaaaaattc	atgtccacat	cagtaggaga	cagggtcagc	60
gtcacctgca	aggccagtc	gaatgtgggt	actaatgtag	cctgggtatca	acagaaatta	120
ggacaatctc	ctaaaacact	gatttactcg	gcacccctacc	ggtagcagtg	agtcacctgat	180
cgcttcacag	gcagtggtcg	tggtgacagat	ttcactctca	ccatcagcaa	tgtgcagtc	240
gaagacttgg	cagagtattt	ctgtcagcaa	tataacagct	atccgtacac	gttcggagg	300
gggaccaagc	tggaaataaa	acgggctgat	gctgcaccaa	ctgtatccat	cttcccacca	360
tccagtgcgc	agttaacatc	tggaggtgcc	tcagtcgtgt	gcttcctgaa	caacttctac	420
cccaaagaca	tcaatgtcaa	gtggaagatt	gatggcagtg	aacgacaaaa	tggcgctcctg	480
aacagttgga	ctgatcagga	cagcaaagac	agcacctaca	gcatgagcag	caccctcacg	540
ttgaccaagg	acgagtatga	acgacataac	agctatacct	gtgaggccac	tcacaagaca	600
tcaacttcac	ccattgtcaa	gagcttcaac	aggaatgagt	gt		642

SEQ ID NO: 110 moltype = AA length = 214
 FEATURE Location/Qualifiers
 REGION 1..214
 note = Description of Artificial Sequence: Synthetic
 polypeptide
 source 1..214
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 110

DIVMTQSQKF	MSTSVGDRVS	VTCKASQNVG	TNVANYQQL	QSPKTLIYS	ASRYSGVDP	60
RFTGSGSGTD	FTLTISNVQS	EDLAIFYCQQ	YNSYPYTFGG	GTKLEIKRAD	AAPTVISIFPP	120
SSEQLTSGGA	SVVCFLNNFY	PKDINVKWKI	DGSEKQNGVL	NSWTDQDSKD	STYSMSSTLT	180
LTKDEYERHN	SYTCEATHKT	STSPIVKSFN	RNEC			214

SEQ ID NO: 111 moltype = DNA length = 1329
 FEATURE Location/Qualifiers
 misc_feature 1..1329
 note = Description of Artificial Sequence: Synthetic
 polynucleotide
 source 1..1329
 mol_type = other DNA
 organism = synthetic construct

SEQUENCE: 111

caggttactc	tgaagagtc	tggccctggg	atattgcagc	cctcccagac	cctcagtcgt	60
acttgttctt	tctctgggtt	ttcactgaac	acttatggta	tgggtgtgag	ctggattcgt	120
cagccttcag	gaaagggtct	ggagtggctg	gcacacattt	actgggatga	tgacaagcgc	180
tataacccat	ccctgaagag	ccggctcaca	atctccaagg	atgcctccaa	caaccgggtc	240
ttctcaaga	tcaccagtgt	ggacactgca	gatactgcca	catactactg	tgctcaaaga	300
ggttatgatg	attactgggg	ttactggggc	caaggggactc	tggtcactat	ctctgcagcc	360
aaaacgcac	ccccatctgt	ctatccactg	gcccctggat	ctgctgcccc	aactaactcc	420
atggtgaccc	tgggatgcct	ggtcaagggc	tatttccctg	agccagtgc	agtgcactgg	480
aactctggat	ccctgtccag	cgggtgtgc	accttcccag	ctgtcctgca	gtctgacctc	540
tacactctga	gcagctcagt	gactgtcccc	tccagcacct	ggcccagcga	gaccgtcacc	600
tgcaacgttg	cccacccggc	cagcagcacc	aaggtggaca	agaaaattgt	gcccagggat	660
tgtggttgta	agccttgcat	atgtacagtc	ccagaagtat	catctgtctt	catcttcccc	720
ccaaagccca	aggatgtgct	caccattact	ctgactccta	aggtcacgtg	tgttgtggta	780
gacatcagca	aggatgatcc	cgagggtccag	ttcagctggg	ttgtagatga	tgtggagggtg	840
cacacagctc	agacgcaacc	ccgggaggag	cagttcaaca	gcactttccg	ctcagtcagt	900
gaacttccca	tcatgcacca	ggactggctc	aatggcaagg	agttcaaatg	cagggtcaac	960
agtgcagctt	tcctgtcccc	catcgagaaa	accatctcca	aaaccaaagg	cagaccgaag	1020
gctccacagg	tgtacacccat	tccactccc	aaggagcaga	tggccaagga	taaagtcagt	1080
ctgacctgca	tgataacaga	cttcttccct	gaagacatta	ctgtggagtg	gcagtgggaat	1140
gggcagccag	cggagaacta	caagaacact	cagcccatca	tggacacaga	tggctcttac	1200
ttcgtctaca	gcaagctcaa	tgtgcagaag	agcaactggg	aggcaggaaa	tactttcacc	1260
tgctctgtgt	tacatgaggg	cctgcacaa	caccatactg	agaagagcct	ctcccactct	1320
cctggtaaa						1329

SEQ ID NO: 112 moltype = AA length = 443
 FEATURE Location/Qualifiers
 REGION 1..443
 note = Description of Artificial Sequence: Synthetic
 polypeptide

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source                1..443
                      mol_type = protein
                      organism = synthetic construct

SEQUENCE: 112
QVTLKESGPG ILQPSQTLSL TCSFSGFSLN TYGMGVSWIR QPSGKGLEWL AHYWDDDKR 60
YNPSLKSRLT ISKDASNNRV FLKITSVDTA DTATYYCAQR GYDDYWGYWG QGTLVTISAA 120
KTPPSVYPL APGSAAQTNM MVTLGCLVKG YFPEPVTWVW NSGSLSSGVH TFPVLQSDL 180
YTLSSSVTVP SSTWPSETVT CNVAHPASST KVDKKIVPRD CGCKPCICTV PEVSSVFIFP 240
PKPKDVLIT LTPKVTVCVV DISKDDPEVQ FSWFVDDVEV HTAQTPREE QFNSTFRSVS 300
ELPIMHQDWL NGKEFKCRVN SAAFPAPIEK TISKTGRPK APQVYTIPPP KEQMAKD KVS 360
LTCMITDFFP EDITVEWQWN GQPAENYKNT QPIMDTDGSY FVYSKLVNQK SNWEAGNTFT 420
CSVLHEGLHN HHTKSLSHS PGK 443

SEQ ID NO: 113        moltype = DNA length = 642
FEATURE              Location/Qualifiers
misc_feature          1..642
                      note = Description of Artificial Sequence: Synthetic
                      polynucleotide
source                1..642
                      mol_type = other DNA
                      organism = synthetic construct

SEQUENCE: 113
gacattgtga tgaccacagtc tcaaaaattc atgtccacat cagtaggaga cagggtcagc 60
gtcacctgca aggcacagtc gaattgtggg actaatgtag cctggtttca acagaaacca 120
ggtaaatctc ctaaaagcact gatttactcg gcatcttacc ggtacagtgg agtccctgat 180
cgcttcacag gcagtggtatc tgggacagat ttcattctca ccatcagcaa tgtgcagtct 240
gaagacctgg cagagtattt ctgtcagcaa tataacaact atcctctcac gttcgggtgt 300
gggaccaagc tggagctgaa acgggctgat gctgcaccaa ctgtatccat ctccaccaca 360
tccagtgcgc agttaacatc tggaggtgcc tcagtcgtgt gcttcttgaa caactctctc 420
cccaagaca tcaatgtcaa gtggaagatt gatggcagtg aacgacaaaa tggcgtcctg 480
aacagttgga ctgatcagga cagcaaaagc agcacctaca gcatgagcag caccctcacg 540
ttgaccaagg acgagtatga acgacataac agctatacct gtgaggccac tcacaagaca 600
tcaacttcac ccattgtcaa gagcttcaac aggaatgagt gt 642

SEQ ID NO: 114        moltype = AA length = 214
FEATURE              Location/Qualifiers
REGION              1..214
                      note = Description of Artificial Sequence: Synthetic
                      polypeptide
source                1..214
                      mol_type = protein
                      organism = synthetic construct

SEQUENCE: 114
DIVMTQSQKF MSTSVGDRVS VTCKASQNVG TNVAFQQKP GQSPKALIYS ASYRSGVDP 60
RFTSGSGSTD FILTISNVQS EDLAEIFCQQ YNNYPLTFGA GTKLELKRAD AAPTVSIFPP 120
SSEQLTSGGA SVVCFLNNFY PKDINVKKWI DGSEKQNGVL NSWTDQDSKD STYSMSSTLT 180
LTKDEYERHN SYTCEATHKT STSPIVKSFN RNEC 214

SEQ ID NO: 115        moltype = DNA length = 1368
FEATURE              Location/Qualifiers
misc_feature          1..1368
                      note = Description of Artificial Sequence: Synthetic
                      polynucleotide
source                1..1368
                      mol_type = other DNA
                      organism = synthetic construct

SEQUENCE: 115
gaggtcctgc tgcaacagtc tggacctgag gtgggtgaagc ctggggcttc agtgaagata 60
ccctgcaagg cttctggata cacattcact gactacaaca tggactgggt gaagcagagc 120
catggaaaga gccttgagtg gattggagag attaatccta acaatggtgg tactttctac 180
aaaccagaagt tcaagggcga ggccacattg actgtagaca agtcctccag cacagcctac 240
atggagctcc gcagcctgac atctgaggac actgcagctc attactgtgc aagagaggca 300
attactacgg taggcgctat ggactactgg ggtcaaggaa cctcagtcac cgtctcctca 360
gccccaaaca ccccccatc agtctatcca ctggcccctg ggtgtggaga tacaactggt 420
tcctccgtga cctgggatg ctgggtcaag ggctacttcc ctgagtcagt gactgtgact 480
tgggaactctg gatccctgac cagcagtggt cacaccttcc cagctctcct gcagcttgga 540
ctctacacta tgagcagctc agtgactgtc cctccagcag cctggccaag tcagaccgtc 600
acctgcagcg ttgctcacc agccagcagc accacgggtg acaaaaaact tgagccagc 660
gggcccattt caacaatcaa cccctgtcct ccatgcaagg agtgtcaca atgccagct 720
cctaacctcg aggggtggac atccgtcttc atcttccctc caaatatcaa ggatgtactc 780
atgatctccc tgacacccaa ggtcacgtgt gtgggtgggt atgtgagcga ggatgaccca 840
gacgtccaga tcagctgggt tgtgaacaac gtggaagtac acacagctca gacacaaacc 900
catagagagg attacaacag tactatcccg gtggtcagca cctcccccat ccagcaccag 960
gactggatga gtggcaagga gttcaaatgc aaggtaaca acaagacct cccatcacc 1020
atcgagagaa ccatctcaaa aattaaagg ctagtccagag ctcacaagat atacatcttg 1080
ccgccaccag cagagcagtt gtccaggaaa gatgtcagtc tcaacttgct ggtcgtgggc 1140

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ttcaaccctg gagacatcag tgtggagtgg accagcaatg ggcatacaga ggagaactac 1200
aaggacaccc caccagtcct agactctgac ggttcttact tcatatatag caagctcaat 1260
atgaaaaaaa gcaagtggga gaaaacagat tccttctcat gcaacgtgag acacgagggt 1320
ctgaaaaaatt actacctgaa gaagaccatc tcccggcttc cgggtaaa 1368

SEQ ID NO: 116      moltype = AA length = 456
FEATURE
REGION              Location/Qualifiers
                    1..456
                    note = Description of Artificial Sequence: Synthetic
                        polypeptide
source              1..456
                    mol_type = protein
                    organism = synthetic construct

SEQUENCE: 116
EVLLQQSGPE VVPGASVKI PCKASGYTFT DYNMDWVKQS HGKSLEWIGE INPNNGGTFY 60
NQKFKGKATL TVDKSSSTAY MELRSLTSED TAVYYCAREA ITTVGAMDYW GQGTSTVTSS 120
AKTTPPSVYP LAPGCGDTTG SSVTLGCLVK GYFPESVTVT WNSGSLSSSV HTFPALLQSG 180
LYTMSSSVTV PSSTWPSQTV TCSVAHPASS TTVDKKLEPS GPISTINPCP PCKECKKCPA 240
PNLEGGPSVF IFPPNIKDVL MISLTPKVT C VVDVSEDDP DVQISWVFN VEVHTAQQT 300
HREDYNSTIR VVSTLPIQH Q DWMSGKEFKC KVNNDLPLSP IERTISKIKG LVRAPQVYIL 360
PPPAEQLSRK DVSLTCLVVG FNPGDISVEW TSNHTEENY KDTAPVLDS GSYFIYSKLN 420
MKTSKWEKTD SFSCNVRHEG LKNYYLKKTI SRSPGK 456

SEQ ID NO: 117      moltype = DNA length = 642
FEATURE
misc_feature         Location/Qualifiers
                    1..642
                    note = Description of Artificial Sequence: Synthetic
                        polynucleotide
source              1..642
                    mol_type = other DNA
                    organism = synthetic construct

SEQUENCE: 117
gacatccaga tgactcagtc tccagcctcc ctatctgcat ctgtgggaga aactgtcacc 60
atcacatgtc gagcaagtgg gaatatccac aattatttag catggatca gcagaaacag 120
ggaaaatctc ctacagtcct ggtctataat gcaaaaacct tagcagatgg tgtgccatca 180
agggttcagtg gcagtggtatc aggaacacaa tattctctca agatcaacag cctgcagcct 240
gaagattttg ggagttatta ctgtcaacat ttttggagtt ctccttacac gttcggaggg 300
gggaccaagc tggaaataaa acgggctgat gctgcaccaa ctgtatccat ctccccacca 360
tccagtgagc agttaaacat tggagtggtc tcagtcgtgt gcttcttgaa caactctac 420
cccaaagaca tcaatgtcaa gtggaagatt gatggcagtg aacgacaaaa tggcgtcctg 480
aacagttgga ctgatcagga cagcaaaagc agcacctaca gcatgagcag caccctcacg 540
tgaccaagg acgagtatga acgacataac agctatacct gtgaggccac tcacaagaca 600
tcaacttcac ccattgtcaa gagcttcaac aggaatgagt gt 642

SEQ ID NO: 118      moltype = AA length = 214
FEATURE
REGION              Location/Qualifiers
                    1..214
                    note = Description of Artificial Sequence: Synthetic
                        polypeptide
source              1..214
                    mol_type = protein
                    organism = synthetic construct

SEQUENCE: 118
DIQMTQSPAS LSASVGETVT ITCRASGNIH NYLAWYQQKQ GKSPQLLVYN AKTLADGVPS 60
RFGSGSGGTQ YSLKINSLQP EDFGSYYCQH FWSSPYTFGG GTKLEIKRAD AAPTVISIFPP 120
SSEQLTSGGA SVVCFLNIFY PKDINVWKWI DGSEKQNGVL NSWDQDSKD STYSMSSTLT 180
LTKDEYERHN SYTCEATHKT STSPIVKSPN RNEC 214

SEQ ID NO: 119      moltype = DNA length = 1329
FEATURE
misc_feature         Location/Qualifiers
                    1..1329
                    note = Description of Artificial Sequence: Synthetic
                        polynucleotide
source              1..1329
                    mol_type = other DNA
                    organism = synthetic construct

SEQUENCE: 119
caggttactc tgaagagtc tggccctgga atattgcagc cctcccagac cctcagtcgt 60
acttgttctt tctctgggtt ttcactgagc acttatggta tgggtgtagg ctggattcgt 120
cagccttcag gaaagggctc agagtggctg gcagacattt ggtgggatga cgataagtac 180
tataacccat cctgaagag cgggtccaca atctccaagg atacctccag caatgaggta 240
ttcctcaaga tcgccattgt ggacactgca gatactgcca cttactactg tgcctgaaga 300
ggtcactact ctgctatgga ctactgggtt caaggaacct cagtcaccgt ctctcagcc 360
aaaacgacac ccccatctgt ctatccactg gccctctggat ctgctgcccc aactaactcc 420
atggtgaccc tgggatgcct ggtcaagggc tatttcctg agccagtgac agtgacctgg 480
aactctggat ccctgtccag cgggtgtcac accttcccag ctgtcctgca gtctgacctc 540

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tacactctga	gcagctcagt	gactgtcccc	tcacgacact	ggccccagca	gaccgtcacc	600
tgcaacgctt	cccacccggc	cagcagcacc	aaggtggaca	agaaaattgt	gcccagggat	660
tgtggttgta	agccttgcat	atgtacagtc	ccagaagtat	catctgtctt	catcttcccc	720
ccaaagccca	aggatgtgct	caccattact	ctgcactcta	aggtcacgtg	tgtgtgtgta	780
gacatcagca	aggatgatcc	cgagggtccag	ttcagctggt	ttgtagatga	tgtggaggtg	840
cacacagctc	agacgcaacc	ccgggaggag	cagttcaaca	gcactttccg	ctcagtcagt	900
gaacttccca	tgatgcacca	ggactggctc	aatggcaagg	agttcaaatg	caggggtcaac	960
agtgcatcgt	ttccctgccc	catcgagaaa	accatctcca	aaaccaaagg	cagaccgaag	1020
gctccacagg	tgtacaccat	tccacctccc	aaggagcaga	tggcccaagg	taaaagtcagt	1080
ctgacctgca	tgataacaga	cttcttccct	gaagacatta	ctgtggagtg	gcagtggaat	1140
gggacgcagc	cggagaacta	caagaacact	cagcccatca	tggacacaga	tggtctttac	1200
ttcgtctaca	gcaagctcaa	tgtgcgaagt	agcaactggg	aggcaggaaa	tactttcaac	1260
tgctctgtgt	tacatgaggg	cctgcacaac	caccatactg	agaagagcct	ctcccactct	1320
cctggtaaa						1329
SEQ ID NO: 120	moltype = AA length = 443					
FEATURE	Location/Qualifiers					
REGION	1..443					
	note = Description of Artificial Sequence: Synthetic polypeptide					
source	1..443					
	mol_type = protein					
	organism = synthetic construct					
SEQUENCE: 120						
QVTLKESGPG	ILQPSQTLSL	TCSFSGFSLS	TYGMGVGWIR	QPSGKGLEWL	ADIWDDDKY	60
YNPSLKSRLT	ISKDTSSENV	FLKIAIVDTA	DTATYYCARR	GHYSAMDYWG	QGTSTVTVSSA	120
KTPPSVYPL	APGSAAQTNS	MVTLGCLVKG	YFPEPVTVTW	NSGSLSSGVH	TFPAVLQSDL	180
YTLSSSVTVP	SSTWPSSETVT	CNVAHPASST	KVDKKIVPRD	CGCKPCICTV	PEVSSVFIFP	240
PKPKDVLIT	LTPKVTCTVV	DISKDDPEVQ	FSWFVDVVEV	HTAQTPQREE	QPNSTFRSVS	300
ELPIMHQDWL	NGKEPKCRVN	SAAFPAPIEK	TISKTKGRPK	APQVYITPPP	KEQMAKDKVS	360
LTCMITDFFP	EDITVEWQWN	GQPAENYKNT	QPIMDTDGSY	PVYSKLNQVK	SNWEAGNTFT	420
CSVLHEGLHN	HHTEKLSLHS	PGK				443
SEQ ID NO: 121	moltype = DNA length = 642					
FEATURE	Location/Qualifiers					
misc_feature	1..642					
	note = Description of Artificial Sequence: Synthetic polynucleotide					
source	1..642					
	mol_type = other DNA					
	organism = synthetic construct					
SEQUENCE: 121						
gacattgttaa	tgaccacagtc	tcaaaaattc	atgtccacat	cagtaggaga	cagggtcagc	60
gtcacctgca	aggccagtc	gaatgtgggt	actaatgtag	cctgggtatca	acagaaacca	120
ggccaatctc	ctaaagcact	gatttactcg	ccatctctacc	ggtagcagtgg	agtcacctgat	180
cgcttcacag	gcagtggtatc	tgggacagat	ttcactctca	ccatcagcaa	tgtgcagtc	240
gaagacttgg	cagaatattt	ctgtcagcaa	tataacagct	atcctcacac	gttcggagg	300
gggaccaaag	tggaaatgaa	acgggctgat	gctgcacca	ctgtatccat	cttccacca	360
tccagtgaag	agttaacatc	tggagggtgc	tcagtcgtgt	gctctctgaa	caactctac	420
cccaaaagca	tcaatgtcaa	gtggaaagatt	gatggcagtg	aacgacaaa	tggcgctctg	480
acagcttgga	ctgatcagga	cagcaaaagc	agcacctaca	gcattgagcag	caccctcacg	540
ttgaccaagg	acgagtatga	acgacataac	agctatacct	gtgaggccac	tcacaagaca	600
tcaacttcac	ccattgtcaa	gagcttcaac	aggaatgagt	gt		642
SEQ ID NO: 122	moltype = AA length = 214					
FEATURE	Location/Qualifiers					
REGION	1..214					
	note = Description of Artificial Sequence: Synthetic polypeptide					
source	1..214					
	mol_type = protein					
	organism = synthetic construct					
SEQUENCE: 122						
DIVMTSQSKF	MSTSVGDRVS	VTCKASQNVG	TNVAWYQQKP	QSPKALIYS	PSYRYSGVDP	60
FTFTSGSGST	FVLTTISNVQS	EDLAEYFCQI	YNSYPHTFGG	GTKLEMKRAD	AAPTVSIFPP	120
SSEQLTSGGA	STVCLNNFY	PKDINVKWKI	DGSEQRQNGV	NSWTDQDSKD	STYSMSSTLT	180
LTKDEYERHN	SYTCEATHKT	STSPIVKSFN	RNEC			214
SEQ ID NO: 123	moltype = DNA length = 1335					
FEATURE	Location/Qualifiers					
misc_feature	1..1335					
	note = Description of Artificial Sequence: Synthetic polynucleotide					
source	1..1335					
	mol_type = other DNA					
	organism = synthetic construct					

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SEQUENCE: 123

cagggttactc	tgaagagatc	tggccctggg	atattgcagc	cctcccagac	cctcagtcgt	60
acttgttctt	tctctgggtt	ttcactgagc	acttcttgta	tgggtgtgag	ttggattcgt	120
cagccttcag	gaaagggtct	ggagtggctg	gcacacaatg	actgggatga	tgacaagcgc	180
tataagtcac	ccctgaagag	ccggctcaca	atatccaagg	atacctccag	aaaccaggta	240
ttcctcaaga	tcaccagtg	ggacactgca	gatactgcca	catactactg	tgctcgaa	300
gttgggggat	tagagggtct	ttttgattac	tggggccaag	gcaccactct	cacagtctcc	360
tcagccaaaa	cgacaccccc	atctgtctat	ccactggccc	ctggatctgc	tgcccaact	420
aactccatgg	tgaccctggg	atgctctgtc	aagggtatt	tccttgagcc	agtgcagtg	480
acctggaact	ctggatccct	gtccagcgg	gtgcacacct	tcctcagctg	cctgcagctc	540
gacctctaca	ctctgagcag	ctcagtgact	gtccctccca	gcacctggcc	cagcgagacc	600
gtcacctgca	acgttgccca	cccggccagc	agcaccaagg	tggacaagaa	aattgtgccc	660
agggattgtg	gttgtaagcc	tgtcatatgt	acagtcccag	aagtatcatc	tgtcttcac	720
ttcccccaca	agcccaagga	ttgtctcacc	attactctga	ctcctaaggt	cacgtgtgtt	780
gtggtagaca	tcagcaagga	tgatcccag	gtccagtcca	gctggtttgt	agatgatgtg	840
gaggtgcaca	cagctcagac	gcaaccccg	gaggagcagt	tcaacagcac	tttccgtca	900
gtcagtgaa	ttcccatcat	gcaccaggac	tggtccaatg	gcaaggaggt	caaatgcagg	960
gtcaacagtg	cagctttccc	tgcccccatc	gagaaaacca	tctccaaaac	caaaggcaga	1020
ccgaaggctc	cacaggtgta	caccattcca	cctcccaagg	agcagatggc	caaggataaa	1080
gtcagtgctg	ctctcatgat	aacagacttc	ttccctgaag	acattactgt	ggagtggcag	1140
tggatgggg	agccagcgga	gaactacaag	aacactcagc	ccatcatgga	cacagatggc	1200
tcttaactgc	ctacagcaaa	gctcaatgtg	cagaagagca	actgggaggg	aggaaatact	1260
ttcacctgct	ctgtgttaca	tgagggacct	cacaaccacc	atactgagaa	gagcctctcc	1320
cactctcctg	gtaaaa					1335

SEQ ID NO: 124 moltype = AA length = 445

FEATURE Location/Qualifiers

REGION 1..445

 note = Description of Artificial Sequence: Synthetic polypeptide

source 1..445

 mol_type = protein

 organism = synthetic construct

SEQUENCE: 124

QVTLKESGPG	ILQPSQTLSL	TCSFSGFSL	TSGMGVSWIR	QPSGKGLEWL	AHNDWDDDKR	60
YKSSLSKRLT	ISKDTSRNV	FLKITSVDTA	DTATYYCARR	VGGLEGYFDY	WGQGTTLTVS	120
SAKTTPPSVY	PLAPGSAQT	NSMVTLGCLV	KGYFPEPVT	TWNSGSLSSG	VHTFPAVLQS	180
DLYTLSSSVT	VPSSTWPS	VTCNVHPAS	STKVDKKIVP	RDCGCKPCIC	TVPEVSSVFI	240
FPPKPKDVL	ITLTPKVT	VVDISKDDPE	VQFSWFVDDV	EVHTAQTPR	EEQFNSTFRS	300
VSELPIMHQD	WLNKKEFKR	VNSAAPPAPI	EKTISKTKGR	PKAPQVYTIP	PPKEQMAKDK	360
VSLTCMITDF	FPEDITVEWQ	WNGQPAENYK	NTQPIIMDTG	SYFVYSKLVN	QKSNWEAGNT	420
FTCSVLHEGL	HNHHTKSL	HSPGK				445

SEQ ID NO: 125 moltype = DNA length = 654

FEATURE Location/Qualifiers

misc_feature 1..654

 note = Description of Artificial Sequence: Synthetic polynucleotide

source 1..654

 mol_type = other DNA

 organism = synthetic construct

SEQUENCE: 125

gacattgtgc	tgacacagtc	tctctgttcc	ttagctgtat	ctctggggca	gagggccacc	60
atctcatgca	gggccagcca	aagtgtcagt	acatctaggt	ttagttatat	gcactgggtc	120
caacagaaa	caggacagcc	acccaaactc	ctcatcaagt	atgcattcaa	cctagaatct	180
gggggtccctg	ccagggttcag	tggcagtggt	tctgggacag	acttcacctc	caacatccat	240
cctgtggagg	gggaggatac	tgcaacatat	tactgtcagc	acagttggga	gattccgtac	300
acgttcggag	gggggaccaa	gctggaaata	aaacgggctg	atgctgcacc	aactgtatcc	360
atcttcccac	catccagtga	gcagttaaca	tctggaggtg	cctcagtcgt	gtgcttcttg	420
aacaacttct	accccaaaga	catcaatgtc	aagtgggaag	ttgatggcag	tgaacgacaa	480
aatggcgctc	tgaacagttg	gactgatcag	gacagcaaa	acagcaccta	cagcatgagc	540
agcaccctca	cgttgaccaa	ggacgagtat	gaacgacata	acagctatac	ctgtgaggcc	600
actcacaaga	catcaacttc	acccattgtc	aagagcttca	acaggaatga	gtgt	654

SEQ ID NO: 126 moltype = AA length = 218

FEATURE Location/Qualifiers

REGION 1..218

 note = Description of Artificial Sequence: Synthetic polypeptide

source 1..218

 mol_type = protein

 organism = synthetic construct

SEQUENCE: 126

DIVLTQSPAS	LAVSLGQRAT	ISCRASQSVS	TSRFSYMHWF	QKPGQAPKL	LIKYASNLES	60
GVPARFSGSG	SGTDFTLNIH	PVEGEDTATY	YCQHSWEIPY	TFGGKTLEI	KRADAAPT	120
IPPPSSEQLT	SGGASVVCFL	NNFYPKDINV	KWKIDGSE	NGVLNSWTDQ	DSKDSTYSMS	180

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STLTLTKDEY ERHNSYTCEA THKTSTSPIV KSFNRNEC 218

SEQ ID NO: 127 moltype = DNA length = 360
 FEATURE Location/Qualifiers
 misc_feature 1..360
 note = Description of Artificial Sequence: Synthetic polynucleotide
 source 1..360
 mol_type = other DNA
 organism = synthetic construct

SEQUENCE: 127
 gaagtgttgt tgcagcagtc agggccggag ttggtaaaac cgggagcgtc ggtgaaaatc 60
 ccgtgcaaaag cgctcggggtg tacgtttacg gactataaca tggattgggt gaaacagtcg 120
 catgggaaat cgcttgaatg gattggtcag atcaatccga ataattggagg aatcttcttt 180
 aatcagaagt ttaaaggaaa agcgacgctt acagtcgata agtcgtcgaa cacggcggtc 240
 atggaagtac ggtcgcttac gtcggaagat acggcggtct attactgtgc gagggaggcg 300
 attacgacgg tgggagcgat ggactattgg ggacaaggga cgtcggtcac ggtatcgctc 360

SEQ ID NO: 128 moltype = AA length = 7
 FEATURE Location/Qualifiers
 REGION 1..7
 note = Description of Artificial Sequence: Synthetic peptide
 source 1..7
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 128
 GYTFTSY 7

SEQ ID NO: 129 moltype = DNA length = 357
 FEATURE Location/Qualifiers
 misc_feature 1..357
 note = Description of Artificial Sequence: Synthetic polynucleotide
 source 1..357
 mol_type = other DNA
 organism = synthetic construct

SEQUENCE: 129
 cagggtgacac tcaaaagaatc agggaccgga atccttcagc ccagccagac ctgtcgctg 60
 acttggtcgt tctccgggtt cagcctgaat acttatggga tgggtgtgtc atggatcagg 120
 caaccgtccg ggaagggatt ggaagtggctc gcgcacatct actgggacga tgacaaacgc 180
 tacaatcctt cgctgaagag cggattgacg atttccaagg atgcctcgaa caaccgggta 240
 tttcttaaga tcacgtcggt cgatacggca gacacggcga cctattactg cgcccaaaga 300
 gggtagcatg actattgggg atattggggc caggggacac tcgtcacaat ttcagct 357

SEQ ID NO: 130 moltype = AA length = 9
 FEATURE Location/Qualifiers
 REGION 1..9
 note = Description of Artificial Sequence: Synthetic peptide
 source 1..9
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 130
 GFSLSTYGM 9

SEQ ID NO: 131 moltype = DNA length = 357
 FEATURE Location/Qualifiers
 misc_feature 1..357
 note = Description of Artificial Sequence: Synthetic polynucleotide
 source 1..357
 mol_type = other DNA
 organism = synthetic construct

SEQUENCE: 131
 cagggtcagc tgaaagagtc aggtcccgga atccttcaac cttcgagac attgtcactc 60
 acatgttctt tctcggggtt ctgcgtctcg acttatggca tgggtgtagg atggattcgg 120
 cagcccagcg ggaaggggct tgagtgggtt gcggatatct ggtgggacga cgacaaatc 180
 tacaatccga gcctgaagtc ccgcctcacc atttcgaaa atacgtcatc aaacgaagtc 240
 tttttgaaga tcgccatcgt ggacacggcg gatacagcga cgtattactg cgccagaagg 300
 ggacactaca gcgcaatgga ttattgggga caggggacct cggtgactgt gtcgtcc 357

SEQ ID NO: 132 moltype = AA length = 9
 FEATURE Location/Qualifiers
 REGION 1..9
 note = Description of Artificial Sequence: Synthetic peptide
 source 1..9
 mol_type = protein

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organism = synthetic construct

SEQUENCE: 132
GFSLNTYGM 9

SEQ ID NO: 133      moltype = DNA  length = 321
FEATURE            Location/Qualifiers
misc_feature        1..321
                    note = Description of Artificial Sequence: Synthetic
                    polynucleotide
source              1..321
                    mol_type = other DNA
                    organism = synthetic construct

SEQUENCE: 133
gacatccaaa tgaccagtc acccgcgagc ctttcggcgt cggtcggaga aacggtcacg 60
atcacgtgcc ggacatcaga gaatctccat aactacctcg cgtgggatca acagaagcag 120
gggaagtcgc cccagttgct tgtatacgat gcgaaaacgt tggcggatgg ggtgccgtcc 180
agattctcgg gatcgggctc ggggacgcag tactcgctca agatcaattc gctgcagccg 240
gaggactttg ggtcgtacta ttgtcagcat ttttggtcat caccgtatac atttgagggt 300
ggaacgaaac ttgagattaa g 321

SEQ ID NO: 134      moltype = AA  length = 9
FEATURE            Location/Qualifiers
REGION             1..9
                    note = Description of Artificial Sequence: Synthetic peptide
source              1..9
                    mol_type = protein
                    organism = synthetic construct

SEQUENCE: 134
GFSLSTSGM 9

SEQ ID NO: 135      moltype = DNA  length = 321
FEATURE            Location/Qualifiers
misc_feature        1..321
                    note = Description of Artificial Sequence: Synthetic
                    polynucleotide
source              1..321
                    mol_type = other DNA
                    organism = synthetic construct

SEQUENCE: 135
gatatcgta tgaccagtc ccagaagttc atgtcaactt cagtgggaga cagagtgtcc 60
gtcacatgta aagcctcgca aaatgtggga accaacgtag cgtgggtcca gcagaaacct 120
ggccaatcac cgaaggcact gatctactcg gccagctata ggtactcggg agtaccagat 180
cgggtttacgg ggtcggggag cgggacggac tttatcctca ctatttccaa tgtccagtcg 240
gaggaccttg cggaataact ctgccagcag tataacaact atccctcac gtttggtgct 300
ggtacaaaat tggagttgaa g 321

SEQ ID NO: 136      moltype = AA  length = 8
FEATURE            Location/Qualifiers
REGION             1..8
                    note = Description of Artificial Sequence: Synthetic peptide
source              1..8
                    mol_type = protein
                    organism = synthetic construct

SEQUENCE: 136
GYTFDDYN 8

SEQ ID NO: 137      moltype = DNA  length = 321
FEATURE            Location/Qualifiers
misc_feature        1..321
                    note = Description of Artificial Sequence: Synthetic
                    polynucleotide
source              1..321
                    mol_type = other DNA
                    organism = synthetic construct

SEQUENCE: 137
gacatcgtga tgacacagtc acagaaattc atgtccacat ccgtcggatga tagagtatcc 60
gtcacgtgta aggccctcgca aaacgtagga actaatgtgg cgtgggatca acagaagcca 120
ggacagtcac ccaaagcact catctacagc cctcatatc ggtacagcgg ggtgccggac 180
aggttcacgg gatcggggag cgggaccgat tttactga ccatttcgaa tgtccagtcg 240
gaggaccttg cggaataact ctgccagcag tataactcgt accctcacac gtttgagggt 300
ggcactaagt tggagatgaa a 321

SEQ ID NO: 138      moltype = AA  length = 8
FEATURE            Location/Qualifiers
REGION             1..8
                    note = Description of Artificial Sequence: Synthetic peptide

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source	1..8 mol_type = protein organism = synthetic construct	
SEQUENCE: 138 GYTFTSYW		8
SEQ ID NO: 139 FEATURE REGION	moltype = AA length = 5 Location/Qualifiers 1..5 note = Description of Artificial Sequence: Synthetic peptide	
source	1..5 mol_type = protein organism = synthetic construct	
SEQUENCE: 139 GGGGS		5
SEQ ID NO: 140 FEATURE REGION	moltype = AA length = 10 Location/Qualifiers 1..10 note = Description of Artificial Sequence: Synthetic peptide	
source	1..10 mol_type = protein organism = synthetic construct	
SEQUENCE: 140 GFSLSTYGMG		10
SEQ ID NO: 141 FEATURE REGION	moltype = AA length = 10 Location/Qualifiers 1..10 note = Description of Artificial Sequence: Synthetic peptide	
source	1..10 mol_type = protein organism = synthetic construct	
SEQUENCE: 141 GFSLNTYGMG		10
SEQ ID NO: 142 FEATURE REGION	moltype = AA length = 10 Location/Qualifiers 1..10 note = Description of Artificial Sequence: Synthetic peptide	
source	1..10 mol_type = protein organism = synthetic construct	
SEQUENCE: 142 GFSLSTSGMG		10
SEQ ID NO: 143 FEATURE REGION	moltype = AA length = 6 Location/Qualifiers 1..6 note = Description of Artificial Sequence: Synthetic peptide	
source	1..6 mol_type = protein organism = synthetic construct	
SEQUENCE: 143 NPNNGG		6
SEQ ID NO: 144 FEATURE REGION	moltype = AA length = 6 Location/Qualifiers 1..6 note = Description of Artificial Sequence: Synthetic peptide	
source	1..6 mol_type = protein organism = synthetic construct	
SEQUENCE: 144 NPSNGR		6
SEQ ID NO: 145 FEATURE REGION	moltype = AA length = 5 Location/Qualifiers 1..5 note = Description of Artificial Sequence: Synthetic peptide	
source	1..5 mol_type = protein organism = synthetic construct	
SEQUENCE: 145 YWDDD		5

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SEQ ID NO: 146	moltype = AA length = 5	
FEATURE	Location/Qualifiers	
REGION	1..5	
	note = Description of Artificial Sequence: Synthetic peptide	
source	1..5	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 146		
WWDDD		5
SEQ ID NO: 147	moltype = AA length = 5	
FEATURE	Location/Qualifiers	
REGION	1..5	
	note = Description of Artificial Sequence: Synthetic peptide	
source	1..5	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 147		
DWDDD		5
SEQ ID NO: 148	moltype = AA length = 8	
FEATURE	Location/Qualifiers	
REGION	1..8	
	note = Description of Artificial Sequence: Synthetic peptide	
source	1..8	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 148		
INPNNGGI		8
SEQ ID NO: 149	moltype = AA length = 8	
FEATURE	Location/Qualifiers	
REGION	1..8	
	note = Description of Artificial Sequence: Synthetic peptide	
source	1..8	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 149		
INPSNGRS		8
SEQ ID NO: 150	moltype = AA length = 7	
FEATURE	Location/Qualifiers	
REGION	1..7	
	note = Description of Artificial Sequence: Synthetic peptide	
source	1..7	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 150		
IYWDDDK		7
SEQ ID NO: 151	moltype = AA length = 8	
FEATURE	Location/Qualifiers	
REGION	1..8	
	note = Description of Artificial Sequence: Synthetic peptide	
source	1..8	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 151		
INPNNGGT		8
SEQ ID NO: 152	moltype = AA length = 7	
FEATURE	Location/Qualifiers	
REGION	1..7	
	note = Description of Artificial Sequence: Synthetic peptide	
source	1..7	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 152		
IWWDDDK		7
SEQ ID NO: 153	moltype = AA length = 7	
FEATURE	Location/Qualifiers	
REGION	1..7	
	note = Description of Artificial Sequence: Synthetic peptide	
source	1..7	
	mol_type = protein	

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SEQUENCE: 153	organism = synthetic construct	
NDWDDDK		7
SEQ ID NO: 154	moltype = AA length = 13	
FEATURE	Location/Qualifiers	
REGION	1..13	
	note = Description of Artificial Sequence: Synthetic peptide	
source	1..13	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 154		
AREAITVGA MDY		13
SEQ ID NO: 155	moltype = AA length = 11	
FEATURE	Location/Qualifiers	
REGION	1..11	
	note = Description of Artificial Sequence: Synthetic peptide	
source	1..11	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 155		
AREVLGDGAMD Y		11
SEQ ID NO: 156	moltype = AA length = 11	
FEATURE	Location/Qualifiers	
REGION	1..11	
	note = Description of Artificial Sequence: Synthetic peptide	
source	1..11	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 156		
AQTGYSNLFA Y		11
SEQ ID NO: 157	moltype = AA length = 11	
FEATURE	Location/Qualifiers	
REGION	1..11	
	note = Description of Artificial Sequence: Synthetic peptide	
source	1..11	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 157		
AQRGYDDYWG Y		11
SEQ ID NO: 158	moltype = AA length = 11	
FEATURE	Location/Qualifiers	
REGION	1..11	
	note = Description of Artificial Sequence: Synthetic peptide	
source	1..11	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 158		
ARRGHYSAMD Y		11
SEQ ID NO: 159	moltype = AA length = 13	
FEATURE	Location/Qualifiers	
REGION	1..13	
	note = Description of Artificial Sequence: Synthetic peptide	
source	1..13	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 159		
ARRVGLEGY FDY		13
SEQ ID NO: 160	moltype = AA length = 6	
FEATURE	Location/Qualifiers	
REGION	1..6	
	note = Description of Artificial Sequence: Synthetic peptide	
source	1..6	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 160		
ENLHNY		6
SEQ ID NO: 161	moltype = AA length = 10	
FEATURE	Location/Qualifiers	

-continued

REGION	1..10	
	note = Description of Artificial Sequence: Synthetic peptide	
source	1..10	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 161		
ESVDNYGISF		10
SEQ ID NO: 162	moltype = AA length = 6	
FEATURE	Location/Qualifiers	
REGION	1..6	
	note = Description of Artificial Sequence: Synthetic peptide	
source	1..6	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 162		
QNVGTN		6
SEQ ID NO: 163	moltype = AA length = 6	
FEATURE	Location/Qualifiers	
REGION	1..6	
	note = Description of Artificial Sequence: Synthetic peptide	
source	1..6	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 163		
GNIHNY		6
SEQ ID NO: 164	moltype = AA length = 10	
FEATURE	Location/Qualifiers	
REGION	1..10	
	note = Description of Artificial Sequence: Synthetic peptide	
source	1..10	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 164		
QSVTSRFSY		10
SEQ ID NO: 165	moltype = DNA length = 972	
FEATURE	Location/Qualifiers	
source	1..972	
	mol_type = genomic DNA	
	organism = Mus sp.	
SEQUENCE: 165		
gccaaaacga caccceccatc tgtctatcca ctggcccctg gatctgctgc ccaaactaac	60	
tccatggtga cctgggatg cctgggtcaag ggctatttcc ctgagccagt gacagtgacc	120	
tggaaactctg gatecctgtc cagcgggtgtg cacaccttcc cagctgtcct gcagtctgac	180	
ctctacactc tgagcagctc agtgactgtc cctccagca cctggcccag cgagaccgtc	240	
acctgcaacg ttgccaccc ggccagcagc accaagggtgg acaagaaaat tgtgccagg	300	
gattgtggtt gtaagccttg catatgtaca gtcccagaag tatcatctgt cttcatcttc	360	
ccccaaagc ccaaggatgt gctcaccatt actctgactc ctaaggtcac gtgtgttggtg	420	
gtagacatca gcaaggatga tcccgaggtc cagttcagct ggtttgtaga tgatgtggag	480	
gtgcacacag ctacagacgca accccgggag gagcagttca acagcacttt ccgctcagtc	540	
agtgaacttc ccatcatgca ccaggactgg ctcaatggca aggagttcaa atgcagggtc	600	
aacagtgcaag ctttccctgc ccccatcgag aaaaccatct ccaaaaccaa aggcagaccg	660	
aaggctccac aggtgtacac cattccacct cccaaggagc agatggccaa ggataaagtc	720	
agtctgacct gcatgataac agactcttc cctgaagaca ttactgtgga gtggcagtg	780	
aatgggcagc cagcgagaaa ctacaagaac actcagccca tcatggacac agatggctct	840	
tacttcgtct acagcaagct caatgtgcag aagagcaact gggagggcagg aaatactttc	900	
acctgctctg tgttacatga gggcctgcac aaccaccata ctgagaagag cctctccac	960	
tctcctggta aa	972	
SEQ ID NO: 166	moltype = AA length = 324	
FEATURE	Location/Qualifiers	
source	1..324	
	mol_type = protein	
	organism = Mus sp.	
SEQUENCE: 166		
AKTTPPSVYP LAPGSAAQTN SMVTLGCLVK GYFPEPVTVT WNSGSLSSGV HTFPAVLQSD	60	
LYTLSSSVTV PSSTWPSETV TCNVVHPASS TKVDKKIVPR DCGCKPCICT VPEVSSVFIF	120	
PKPKDVLTI TLTPKVTCTV VDISKDDPEV QFSWFVDDVE VHTAQTPRE EQFNSTFRSV	180	
SELPIMHQDW LNGKEFKCRV NSAAFPAPIE KTISKTKGRP KAPQVYTIPP PKEQMAKDKV	240	
SLTCMITDFD PEDITVEQW NGQPAENYKN TQPIMDTGS YFVYSKLVNQ KSNWEAGNTF	300	
TCSVLHEGLH NHHTEKSLSH SPGK	324	
SEQ ID NO: 167	moltype = DNA length = 1008	

-continued

FEATURE
source

Location/Qualifiers
1..1008
mol_type = genomic DNA
organism = Mus sp.

SEQUENCE: 167

```

gccaaaacaa caccceccatc agtctatcca ctggcccctg ggtgtggaga tacaactggt 60
tcctccgtga ctctgggatg cctgggtcaag ggctacttcc ctgagtcagt gactgtgact 120
tggaactctg gatccctgtc cagcagtggt cacaccttcc cagctctcct gcagtctgga 180
ctctacacta tgagcagctc agtgactgtc cctccagca cctggccaag tcagaccgtc 240
acctgcagcg ttgtccacc agccagcagc accacgggtg acaaaaaact tgagcccagc 300
gggcccattt caacaatcaa cccctgtcct ccatgcaagg agtgtcacia atgccagct 360
cctaacctcg aggggtggacc atccgtcttc atcttccctc caaatatcaa ggatgtactc 420
atgatctccc tgacacccaa ggtcacgtgt gtggtggtgg atgtgagcga ggatgacca 480
gacgtccaga tcagctgggt tgtgaacaac gtggaagta acacagctca gacacaaacc 540
catagagagg attacaacag tactatccgg gtggtcagca cctcccccac ccagcaccag 600
gactggatga gtggcaagga gttcaaatgc aagggtcaaca acaagacct cccatcacc 660
atcgagagaa ccatctcaaa aattaaagg ctagtccagag ctccacaagt atacatcttg 720
cgccaccagg cagagcagtt gtcaggaaa gatgtcagtc tcaacttgctt ggtcgtgggc 780
ttcaaccctg gagacatcag tgtggagtgg accagcaatg ggcatacaga ggagaactac 840
aaggacaccg cacagctcct agactctgac ggttcttact tcatatatag caagctcaat 900
atgaaaacaa gcaagtgagg gaaaacagat tcttctcat gcaacgtgag acacgagggt 960
ctgaaaaatt actacctgaa gaagaccatc tcccggcttc cgggtaaa 1008

```

SEQ ID NO: 168

FEATURE
source

moltype = AA length = 336
Location/Qualifiers
1..336
mol_type = protein
organism = Mus sp.

SEQUENCE: 168

```

AKTTPPSVYP LAPGCGDTTG SSVTLGCLVK GYFPESVTVT WNSGSLSSSV HTFPALLQSG 60
LYTMSSSVTV PSSTWPSQTV TCSVAHPASS TTVDKKLEPS GPISTINPCP PCKECHKCPA 120
PNLEGGPSVF IFPPNIKDLV MISLTPKVT C VVVDVSEDDP DVQISWFVNN VEVHTAQQT 180
HREDYNSTR VVSTLPIQH Q DWMSGKEFKC KVNNDLPS IERTISKIKG LVRAPQVYIL 240
PPPAEQLSRK DVSLTCLVVG FNPGLISVEV TSNHGTEENY KDTAPVLDSG GSYFIYSKLN 300
MKTSKWEKTD SFSCNVRHEG LKNYYLKKTI SRSPGK 336

```

SEQ ID NO: 169

FEATURE
source

moltype = DNA length = 321
Location/Qualifiers
1..321
mol_type = genomic DNA
organism = Mus sp.

SEQUENCE: 169

```

cgggctgatg ctgccaacac tgtatccatc ttcccacat ccagtgagca gttaacatct 60
ggaggtgcct cagctgtgtg cttcttgaac aacttctacc ccaagacat caatgtcaag 120
tggaagattg atggcagtg aagacaaaat ggcgtcctga acagttggac tgatcaggac 180
agcaagaca gccactacag catgagcagc accctcacgt tgaccaagga cgagtatgaa 240
cgacataaca gctatacctg tgaggccact cacaagacat caacttcacc cattgtcaag 300
agcttcaaca ggaatgagtg t 321

```

SEQ ID NO: 170

FEATURE
source

moltype = AA length = 107
Location/Qualifiers
1..107
mol_type = protein
organism = Mus sp.

SEQUENCE: 170

```

RADAAPT VSI FPPSSEQLTS GGASVVCFLN NFYPKDINVK WKIDGSE RQN GVLNSWTDQD 60
SKDSTYS MSS TLTLTKDEYE RHNSYTCEAT HKTSTSPIVK SFNRNEC 107

```

SEQ ID NO: 171

FEATURE
source

moltype = DNA length = 990
Location/Qualifiers
1..990
mol_type = genomic DNA
organism = Homo sapiens

SEQUENCE: 171

```

gcctcaacaa aaggaccaag tgtgttccca ctgccccta gcagcaagag tacatccggg 60
ggcactgcag cactcggctg cctcgtcaag gattattttc cagagccagt aaccgtgagc 120
tggaacagtg gagcactcac ttctggtgtc catacttttc ctgctgtcct gcaaaactct 180
ggcctgtact cactcagctc cgtcgtgacc gtgccatctt catctctggg cactcagacc 240
tacatctgta atgtaaacca caagcctagc aatactaagg tgcataagcg ggtggaaccc 300
aagagctcgc acaagactca cacttgctcc ccatgccctg cccctgaact tctggggcgt 360
cccagcgtct ttttggtccc accaaagcct aaagatactc tgatgataag tagaacaccc 420
gaggtgacat gtgttgttgt agacgtttcc cagcaggacc cagaggttaa gttcaactgg 480
tacgttgatg gagtcgaagt acataatgct aagaccaagc ctagagagga gcagtataat 540
agtacatacc gtgtatgcag tgttctcaca gtgctgcacc aagactggct caacggcaaa 600
gaatacaaat gcaaaagtgc caacaaagca ctcccagccc ctatcgagaa gactattagt 660
aaggcaagg ggcagcctcg tgaaccacag gtgtacactc tgccacccag tagagaggaa 720

```

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atgacaaaga accaagtctc attgacctgc ctggtgaaag gcttctaccc cagcgacatc 780
gccgttgagt gggagagtaa cggtcagcct gagaacaatt acaagacaac cccccagtg 840
ctggatagtg acgggtcttt cttctgttac agtaagctga ctgtggacaa gtcccgtcg 900
cagcagggta acgtcttcag ctgtccgtg atgcacgagg cattgcacaa ccactacacc 960
cagaagtcac tgagcctgag ccaggggaag 990

```

```

SEQ ID NO: 172      moltype = AA length = 330
FEATURE            Location/Qualifiers
source             1..330
                  mol_type = protein
                  organism = Homo sapiens

```

```

SEQUENCE: 172
ASTKGPSVFP LAPSSKSTSG GTAALGCLVK DYFPEPVTVS WNSGALTSGV HTFPAVLQSS 60
GLYSLSSVVT VPSSSLGTQT YICNVNHKPS NTKVDKRVEP KSCDKTHTCP PCPAPELLGG 120
PSVFLFPPKP KDTLMISRTPEVTCVVVDVS HEDPEVKFNW YVDGVEVHNA KTKPREEQYN 180
STYRVVSVLT VLNQDNLNGK EYKCKVSNKA LPAPIEKTIK KAKGQPREPQ VYTLPPSREE 240
MTKNQVSLTC LVKGFYPSDI AVEWESNGQP ENNYKTTTPV LDSDGSFFLY SKLTVDKSRW 300
QQGNVFSCSV MHEALHNHYT QKSLSLSPGK 330

```

```

SEQ ID NO: 173      moltype = DNA length = 321
FEATURE            Location/Qualifiers
source             1..321
                  mol_type = genomic DNA
                  organism = Homo sapiens

```

```

SEQUENCE: 173
cgcacagttg ctgccccag cgtgttcatt ttcccaccta gcgatgagca gctgaaaagc 60
ggtaactgct ctgtcgtatg cttgctcaac aacttttacc cactgtgagg taagggtcag 120
tggaagtggg ataagtgcac tcaatctgga aacagtcagg agtcctgac agaacaggac 180
agcaagactc caactttatc actctcttcc accctgactc tgtccaagcg agactatgaa 240
aaacacaagg tatacgctcg cgagggttaca caccagggtt tgtctagtcc tgtcaccaag 300
tccttcaata ggggcgaagt t 321

```

```

SEQ ID NO: 174      moltype = AA length = 107
FEATURE            Location/Qualifiers
source             1..107
                  mol_type = protein
                  organism = Homo sapiens

```

```

SEQUENCE: 174
RTVAAPSVFI FPPSDEQLKS GTASVVCLLN NFYPREAKVQ WKVDNALQSG NSQESVTEQD 60
SKDSTYLSLS TLTLKADYE KHKVYACEVT HQGLSSPVTK SFNRGEC 107

```

```

SEQ ID NO: 175      moltype = DNA length = 1350
FEATURE            Location/Qualifiers
misc_feature       1..1350
                  note = Description of Artificial Sequence: Synthetic
                  polynucleotide
source             1..1350
                  mol_type = other DNA
                  organism = synthetic construct

```

```

SEQUENCE: 175
gaagtgttgt tgcagcagtc agggccggag ttggtaaaac cgggagcgtc ggtgaaaatc 60
ccgtgcaaaag cgtcgggggta tacgttttac gactataaca tggattgggt gaaacagtcg 120
catgggaaat cgcttgaaat gatttggtcag atcaatccga ataattggag aatctctctt 180
aatcagaagt ttaaggaaaa agcgacgctt acagtcgata agtcgtcgaa caccggcgctc 240
atggaaagtac ggtcgcttac gtcggaagat acggcggtct attactgtgc gagggaggcg 300
attacgacgg tgggagcgat ggactattgg ggacaaggga cgtcggtcac ggtatcgctg 360
gcctcaacaa aaggaccaag tctgttccca ctgcgcccta gcagcaagag tacatccggg 420
ggcactgcag cactcggctg cctcgtcaag gattattttc cagagccagt aaccgtgagc 480
tggaacagtg gagcactcac ttctggtgtc catacttttc ctgctgtcct gcaaagctct 540
ggcctgtact cactcagctc cgtcgtgacc gtgccatctt catctctggg cactcagacc 600
tacatctgta atgtaaaaca caagcctagc aatactaagg tcgataagcg ggtggaacct 660
aagagctgcg acaagactca cacttgtccc ccatgccctg cccctgaact tctgggcggt 720
cccagcgtct ttttgttccc accaaagcct aaagatactc tgatgataag tagaacacct 780
gaggtgacat gtgttgttgt agacgtttcc cagcaggacc cagagggtta gttcaactgg 840
tacgttgatg gagtcgaagt acataatgct aagaccaagc ctagagagga gcagtataat 900
agtacatacc gtgtagttag tgttctcaca gtgctgcacc aagactggct caacgcgcaa 960
gaatacaaat gcaaaagtgc caacaaagca ctcccagccc ctatcgagaa gactattagt 1020
aaggcaaaag ggcagcctcg tgaaccacag gtgtacactc tgccaccagc tagagaggaa 1080
atgacaaaga accaagtctc attgacctgc ctggtgaaag gcttctaccc cagcgacatc 1140
gccgttgagt gggagagtaa cggtcagcct gagaacaatt acaagacaac cccccagtg 1200
ctggatagtg acgggtcttt cttctgttac agtaagctga ctgtggacaa gtcccgtcg 1260
cagcagggta acgtcttcag ctgttccgtg atgcacgagg cattgcacaa ccactacacc 1320
cagaagtcac tgagcctgag ccaggggaag 1350

```

```

SEQ ID NO: 176      moltype = AA length = 450
FEATURE            Location/Qualifiers

```

-continued

REGION 1..450
note = Description of Artificial Sequence: Synthetic polypeptide

source 1..450
mol_type = protein
organism = synthetic construct

SEQUENCE: 176

EVLLQQSGPE	LVKPGASVKI	PCKASGYTFT	DYNMDWVKQS	HGKSLEWIGQ	INPNNGGIF	60
NQKFKGKATL	TVDKSSNTAF	MEVRSLTSED	TAVYYCAREA	ITTVGAMDYD	GQGTSTVTSS	120
ASTKGPSVFP	LAPSSKSTSG	GTAALGCLVK	DYFPEPVTVS	WNSGALTSGV	HTFPAVLQSS	180
GLYSLSSVVT	VPSSSLGTQT	YICNVNHKPS	NTKVDKRVEP	KSCDKTHTCP	PCPAPELLGG	240
PSVFLFPPKP	KDTLMISRTF	EVTGVVVDVS	HEDPEVKFNW	YVDGVEVHNA	KTKPREEQYN	300
STYRVVSVLT	VLHQDWLNGK	EYKCKVSNKA	LPAPIEKTIS	KAKGQPREPQ	VYTLPPSREE	360
MTKNQVSLTC	LVKGFYPSDI	AVEWESNGQP	ENNYKTTTPV	LDSDGSFFLY	SKLTVDKSRW	420
QQGNVFSCSV	MHEALHNHYT	QKSLSLSPGK				450

SEQ ID NO: 177 moltype = DNA length = 1350
FEATURE Location/Qualifiers
misc_feature 1..1350
note = Description of Artificial Sequence: Synthetic polynucleotide

source 1..1350
mol_type = other DNA
organism = synthetic construct

SEQUENCE: 177

caagtgcac	ttgtgcagtc	gggtgcggaa	gtcaaaaagc	cgggagcgtc	ggtgaaagta	60
tcgtgtaaag	cgctcgggata	tacgttttacg	gactataaca	tggactgggt	acgacaggca	120
ccggggaaat	cggtggaatg	gatcggacag	attaatccga	acaatggggg	aattttcttt	180
aatcagaaat	tcaaaggacg	ggcgacgttg	acggtcgata	catcgacgaa	tacggcggtat	240
atggaattga	ggctcgcttcg	ctcggacgat	acggcgggtct	attactgcgc	cagggaggcg	300
atcacgacgg	taggggcatg	ggattattgg	ggacaggggg	cgcttgtagc	ggtatcgctc	360
gcctcaacaa	aaggaccaag	tgtgttccca	ctcgcccca	gcagcaagag	tacatccggg	420
ggcactgcag	cactcggctg	cctcgtaaac	gattattttc	cagagccagt	aaccgtgagc	480
tggaacagtg	gagcactcac	ttctgggtgc	catacttttc	ctgctgtcct	gcaaagctct	540
ggcctgtact	cactcagctc	cgctcgtgac	gtgccatctt	catctctggg	cactcagacc	600
tacatctgta	atgtaaaaca	caagcctagc	aatactaagg	tcgataagcg	ggtggaaccc	660
aagagctgcg	acaagactca	cacttgttcc	ccatgccctg	cccctgaact	tctggggcgt	720
cccagcgtct	ttttgttccc	accaaagcct	aaagatactc	tgatgataag	tagaacaccc	780
gaggtgacat	gtgttgttgt	agacgtttcc	cacgaggacc	cagaggttaa	gttcaactgg	840
tacgttgatg	gagtcgaagt	acataatgct	aagaccaagc	ctagagagga	gcagtataat	900
agtagacacc	gtgtagtcag	tgttttcaca	gtgctgcacc	aagactgggt	caacggcaaa	960
gaatacaaat	gcaaaagtgc	caacaaagca	ctcccagccc	ctatcgagaa	gactattagt	1020
aaggcaaaag	ggcagcctcg	tgaaccacag	gtgtacactc	tgccacccag	tagagaggaa	1080
atgacaaaag	accaaagctc	attgacctgc	ctgggtgaaag	gcttctaccc	cagcgacatc	1140
gccgttgagt	gggagagtaa	cggtcagcct	gagaacaatt	acaagacaac	ccccccagtg	1200
ctggatagtg	acgggtcttt	ctttctgtac	agtaagctga	ctgtggacaa	gtcccgtctg	1260
cagcagggta	acgtcttcag	ctgttccgtg	atgcacgagg	cattgcacaa	ccactacacc	1320
cagaagtcac	tgagcctgag	cccagggaag				1350

SEQ ID NO: 178 moltype = AA length = 450
FEATURE Location/Qualifiers
REGION 1..450
note = Description of Artificial Sequence: Synthetic polypeptide

source 1..450
mol_type = protein
organism = synthetic construct

SEQUENCE: 178

QVQLVQSGAE	VKKPGASVKV	SCKASGYTFT	DYNMDWVRQA	PGKSLEWIGQ	INPNNGGIF	60
NQKFKGRATL	TVDSTNTAY	MELRSLRSD	TAVYYCAREA	ITTVGAMDYD	GQGTSLVTSS	120
ASTKGPSVFP	LAPSSKSTSG	GTAALGCLVK	DYFPEPVTVS	WNSGALTSGV	HTFPAVLQSS	180
GLYSLSSVVT	VPSSSLGTQT	YICNVNHKPS	NTKVDKRVEP	KSCDKTHTCP	PCPAPELLGG	240
PSVFLFPPKP	KDTLMISRTF	EVTGVVVDVS	HEDPEVKFNW	YVDGVEVHNA	KTKPREEQYN	300
STYRVVSVLT	VLHQDWLNGK	EYKCKVSNKA	LPAPIEKTIS	KAKGQPREPQ	VYTLPPSREE	360
MTKNQVSLTC	LVKGFYPSDI	AVEWESNGQP	ENNYKTTTPV	LDSDGSFFLY	SKLTVDKSRW	420
QQGNVFSCSV	MHEALHNHYT	QKSLSLSPGK				450

SEQ ID NO: 179 moltype = DNA length = 1350
FEATURE Location/Qualifiers
misc_feature 1..1350
note = Description of Artificial Sequence: Synthetic polynucleotide

source 1..1350
mol_type = other DNA
organism = synthetic construct

SEQUENCE: 179

-continued

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caagtcacgc ttgtccagtc gggagcggaa gtgaagaaac cggggctcgc ggtcaaagta 60
tcgtgtaaaag cgtcgggata tacgtttacg gactataaca tggattgggt acgacaggct 120
ccgggaaaaat cattggaatg gattggacag attaatccga ataattgggg tatcttcttt 180
aatcaaaagt ttaaagggag ggcgacgttg acgggtggaca aatcgacaaa tacggcgtat 240
atgggaattgt cgtcgcttcg gtcggaggac acggcgggtg attactgcgc gagggaggcg 300
atcacgcagcg tcggggcgat ggattattgg ggacagggaa cgcttctgac ggtatcgctc 360
gcctcaacaa aaggaccaag tgtgttccca ctgcgcccta gcagcaagag tacatccggg 420
ggcactgcag cactcggtcg cctcgtaag gattatttct cagagccagt aaccgtgagc 480
tggaaacagt gagcactcac ttctgtgtgc catacttttc ctgctgtcct gcaaaactct 540
ggcctgtact cactcagctc cgtcgtgacc gtgccatctt catctctggg cactcagacc 600
tacatctgta atgtaaaacca caagcctagc aataactaagg tcgataagcg ggtggaaacc 660
aagagctgcg acaagactca cacttgtccc ccatgccctg cccctgaact tctggggcgt 720
cccagcgtct ttttgttccc accaaagcct aaagatactc tgatgataag tagaacaccc 780
gaggtgacat gtgtgtgtgt agacgttttc acgaggagcc cagaggttaa gttcaactgg 840
tacgttgatg gagtcgaagt acataatgct aagaccaagc ctagagaggaa gcagtataat 900
agtacatacc gtgtatgcag tgttctcaca gtgctgcacc aagactggct caacggcaaa 960
gaatacaaat gcaaaagtgc caacaaagca ctcccagccc ctatcgagaa gactattagt 1020
aaggcaaaag ggcagcctcg tgaaccacag gtgtacactc tgccaccag tagagaggaa 1080
atgacaaaaga accaagtctc attgacctgc ctggtgaaag gcttctaccc cagcgacatc 1140
gccgttgagt gggagagtaa cggtcagcct gagaacaatt acaagacaac cccccagtg 1200
ctggatagtg acgggtcttt ctttctgtac agtaagctga ctgtggacaa gtcccgtctg 1260
cagcagggta acgtcttcag ctgttccgtg atgcacgagg cattgcacaa ccactacacc 1320
cagaagtcac tgagcctgag cccagggaag                                     1350

SEQ ID NO: 180      moltype = AA length = 450
FEATURE            Location/Qualifiers
REGION            1..450
                    note = Description of Artificial Sequence: Synthetic
                    polypeptide
source            1..450
                    mol_type = protein
                    organism = synthetic construct

SEQUENCE: 180
QVQLVQSGAE VKKPGSSVKV SCKASGYTFT DYNMDWVRQA PGKSLEWIGQ INPNNGGIFV 60
NQKFKGRATL TVDKSTNTAY MELSSLRSED TAVYYCAREA ITTVGAMDYV GQGLTLTVSS 120
ASTKGPSVFP LAPSSKSTSG GTAALGCLVK DYFPEPTVS WNSGALTSGV HTPFAVLQSS 180
GLYSLSSVVT VPSSSLGTQT YICNVNHKPS NTKVDKRVPE KSCDKHTTCP PCPAPELLGG 240
PSVFLFPPPK KDTLMISRTPEVTCVVVDVS HEDPEVKFNW YVDGVEVHNA KTKPREEQYN 300
STYRVVSVLT VLNQDNLNGK EYCKVSNKA LPAPIEKTIS KAKGQPREPQ VYTLPPSREE 360
MTKNQVSLTC LVKGFYPSDI AVEWESNGQP ENNYKTTTPV LDSDGSFFLY SKLTVDKSRW 420
QQGNVFCSCV MHEALHNHYT QKSLSLSPGK                                     450

SEQ ID NO: 181      moltype = DNA length = 1350
FEATURE            Location/Qualifiers
misc_feature       1..1350
                    note = Description of Artificial Sequence: Synthetic
                    polynucleotide
source            1..1350
                    mol_type = other DNA
                    organism = synthetic construct

SEQUENCE: 181
caggtccagc ttgtgcaatc gggagcggaa gtgaagaaac cgggagcgc ggtaaaagtc 60
tcgtgcaaaag cgtcggggta tacgtttacg gactataaca tggactgggt gcgccaagcg 120
cctggacagg gtcttgaatg gatggggcag attaatccga ataattggag gatcttcttt 180
aatcagaaat tcaaagggaag ggtaacgctg acgacagaca cgtcaacatc gacggcctat 240
atgggaattgc ggtcgttgcg atcagatgat acggcgggtc actattgtgc gagggaggcg 300
attacgacgg tgggagcgat ggattattgg ggacagggga cgttggtaac ggtatcgctc 360
gcctcaacaa aaggaccaag tgtgttccca ctgcgcccta gcagcaagag tacatccggg 420
ggcactgcag cactcggtcg cctcgtaag gattatttct cagagccagt aaccgtgagc 480
tggaaacagt gagcactcac ttctgtgtgc catacttttc ctgctgtcct gcaaaactct 540
ggcctgtact cactcagctc cgtcgtgacc gtgccatctt catctctggg cactcagacc 600
tacatctgta atgtaaaacca caagcctagc aataactaagg tcgataagcg ggtggaaacc 660
aagagctgcg acaagactca cacttgtccc ccatgccctg cccctgaact tctggggcgt 720
cccagcgtct ttttgttccc accaaagcct aaagatactc tgatgataag tagaacaccc 780
gaggtgacat gtgtgtgtgt agacgttttc acgaggagcc cagaggttaa gttcaactgg 840
tacgttgatg gagtcgaagt acataatgct aagaccaagc ctagagaggaa gcagtataat 900
agtacatacc gtgtatgcag tgttctcaca gtgctgcacc aagactggct caacggcaaa 960
gaatacaaat gcaaaagtgc caacaaagca ctcccagccc ctatcgagaa gactattagt 1020
aaggcaaaag ggcagcctcg tgaaccacag gtgtacactc tgccaccag tagagaggaa 1080
atgacaaaaga accaagtctc attgacctgc ctggtgaaag gcttctaccc cagcgacatc 1140
gccgttgagt gggagagtaa cggtcagcct gagaacaatt acaagacaac cccccagtg 1200
ctggatagtg acgggtcttt ctttctgtac agtaagctga ctgtggacaa gtcccgtctg 1260
cagcagggta acgtcttcag ctgttccgtg atgcacgagg cattgcacaa ccactacacc 1320
cagaagtcac tgagcctgag cccagggaag                                     1350

SEQ ID NO: 182      moltype = AA length = 450

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FEATURE Location/Qualifiers
 REGION 1..450
 note = Description of Artificial Sequence: Synthetic polypeptide
 source 1..450
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 182

QVQLVQSGAE	VKKPGASVKV	SCKASGYTFT	DYNMDWVRQA	PGQGLEWMGQ	INPNNGGIF	60
NQKFQGRVTL	TTDTSTSTAY	MELRSLRSD	TAVYYCAREA	ITTVGAMDY	WGQGLVTVSS	120
ASTKGPSVFP	LAPSSKSTSG	GTAALGCLVK	DYFPEPVTVS	WNSGALTSGV	HTFPAVLQSS	180
GLYSLSVV	VPSSSLGTQT	YICNVNHKPS	NTKVDKRV	KSCDKTHTCP	PCPAPELLGG	240
PSVFLFPPKP	KDTLMISRTP	EVTCTVVDVS	HEDPEVKFNW	YVDGVEVHNA	KTKPREEQYN	300
STYRVVSVLT	VLHQDWLNGK	EYCKKVSNA	LPAPIEKTIS	KAKGQPREPQ	VYTLPPSREE	360
MTKNQVSLTC	LVKGFYPSDI	AVEWESNGQP	ENNYKTTTPV	LDSGGSFFLY	SKLTVDKSRW	420
QQGNVFSCSV	MHEALHNHYT	QKSLSLSPGK				450

SEQ ID NO: 183 moltype = DNA length = 1350
 FEATURE Location/Qualifiers
 misc_feature 1..1350
 note = Description of Artificial Sequence: Synthetic polynucleotide
 source 1..1350
 mol_type = other DNA
 organism = synthetic construct

SEQUENCE: 183

cagggtccagc	ttgtgcaatc	gggagcggaa	gtgaagaac	cgggagcgtc	ggtaaaagtc	60
tcgtgcaaac	cgctggggta	tacgtttacg	gactataaca	tggactgggt	gcgccaagcg	120
cctggacaga	gccttgaatg	gatggggcag	attaatccga	ataatggagg	gatcttcttt	180
aatcagaaat	tccaggggaag	ggtaacgctg	acgacagaca	cgtaacatc	gacggcctat	240
atggaattgc	ggtcgttgcg	atcagatgat	acggcgggtc	actattgtgc	gagggaggcg	300
attacgacgg	tgggagcgat	ggattattgg	ggacagggga	cgttggtaac	ggtatcgctg	360
gcctcaacaa	aaggaccaag	tgtgttccca	ctcgcccca	gcagcaagag	tacatccggg	420
ggcactgcag	cactcggctg	cctcgtcaag	gattattttc	cagagcgagt	aaccgtgagc	480
tggaaacagt	gagcactcac	ttctggtgtc	catacttttc	ctgctgtcct	gcaaagctct	540
ggcctgtact	cactcagctc	cgtcgtgacc	gtgccatctt	catctctggg	cactcagacc	600
tacatctgta	atgtaaaaca	caagcctagc	aatactaagg	tcgataagcg	ggtggaaccc	660
aagagctgcg	acaagactca	caactgtccc	ccatgccctg	ccctgaact	tctgggcggt	720
cccagcgtct	ttttgttccc	accaaagcct	aaagatactc	tgatgataag	tagaacaacc	780
gaggtgacat	gtgttgttgt	agacgtttcc	cacgaggacc	cagagggtta	gttcaactgg	840
tacgttgatg	gagtcgaagt	acataatgct	aagaccaagc	ctagagagga	gcagataaat	900
agtaacatac	gtgtagtcag	tggttctaca	gtgctgcacc	aagactggct	caacggcaaa	960
gaatacaaat	gcaagatgtc	caacaagaga	ctcccagccc	ctatcgagaa	gactattagt	1020
aaggcaaaag	ggcagcctcg	tgaaccacag	gtgtacactc	tgccaccag	tagagaggaa	1080
atgacaaaag	accaagtctc	attgacctgc	ctgggtgaaag	gcttctaccc	cagcgacatc	1140
gccgttgagt	gggagagtaa	cggtcagcct	gagaacaatt	acaagacaac	ccccccagtg	1200
ctggatagtg	acgggtcttt	ctttctgtac	agtaagctga	ctgtgggaca	gtcccgcctg	1260
cagcagggta	acgtcttcag	ctgttccgtg	atgcacgagg	cattgcacaa	ccactacacc	1320
cagaagtcac	tgagcctgag	cccagggaag				1350

SEQ ID NO: 184 moltype = AA length = 450
 FEATURE Location/Qualifiers
 REGION 1..450
 note = Description of Artificial Sequence: Synthetic polypeptide
 source 1..450
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 184

QVQLVQSGAE	VKKPGASVKV	SCKASGYTFT	DYNMDWVRQA	PGQSLEWMGQ	INPNNGGIF	60
NQKFQGRVTL	TTDTSTSTAY	MELRSLRSD	TAVYYCAREA	ITTVGAMDY	WGQGLVTVSS	120
ASTKGPSVFP	LAPSSKSTSG	GTAALGCLVK	DYFPEPVTVS	WNSGALTSGV	HTFPAVLQSS	180
GLYSLSVV	VPSSSLGTQT	YICNVNHKPS	NTKVDKRV	KSCDKTHTCP	PCPAPELLGG	240
PSVFLFPPKP	KDTLMISRTP	EVTCTVVDVS	HEDPEVKFNW	YVDGVEVHNA	KTKPREEQYN	300
STYRVVSVLT	VLHQDWLNGK	EYCKKVSNA	LPAPIEKTIS	KAKGQPREPQ	VYTLPPSREE	360
MTKNQVSLTC	LVKGFYPSDI	AVEWESNGQP	ENNYKTTTPV	LDSGGSFFLY	SKLTVDKSRW	420
QQGNVFSCSV	MHEALHNHYT	QKSLSLSPGK				450

SEQ ID NO: 185 moltype = DNA length = 1350
 FEATURE Location/Qualifiers
 misc_feature 1..1350
 note = Description of Artificial Sequence: Synthetic polynucleotide
 source 1..1350
 mol_type = other DNA
 organism = synthetic construct

-continued

SEQUENCE: 185

cagggtccagc	ttgtgcaatc	gggagcggaa	gtgaagaaac	cgggagcgtc	ggtaaaagtc	60
tcgtgcacaa	cgctcgggta	tacgtttacg	gactataaca	tggactgggt	gcgccaagcg	120
cctggacagg	gtcttgaatg	gatggggcag	attaatccga	ataatggagg	gatcttcttt	180
aatcagaaat	tccagggaa	ggtaacgctg	acgacagaca	cgtaacatc	gacggcctat	240
atgggaattgc	ggtcgttgcg	atcagatgat	acggcgggtct	actattgtgc	gagggaggcg	300
attacgacgg	tgggagcgat	ggattattgg	ggacagggga	cgttggtaac	ggtatcgctg	360
gcctcaacaa	aaggaccaag	tgtgttccca	ctcgcccta	gcagcaagag	tacatccggg	420
ggcactgcag	cactcggctg	cctcgtaag	gattattttc	cagagccagt	aaccgtgagc	480
tggacacagt	gagcactcac	ttctgggtgc	catacttttc	ctgctgtcct	gcaaagctct	540
ggcctgtact	cactcagctc	cgctcgtgac	gtgccatctt	catctctggg	cactcagacc	600
tacatctgta	atgtaaacca	caagcctagc	aatactaagg	tcgataagcg	ggtggaaccc	660
aagagctcgc	acaagactca	cacttgtccc	ccatgccctg	ccctgaact	tctggggcgt	720
cccagcgtct	tttgtttccc	accaaagcct	aaagatactc	tgatgataag	tagaacaccc	780
gaggtgacat	gtgttgttgt	agacgtttcc	cacgaggacc	cagaggttaa	gttcaactgg	840
tacgttgatg	gagtcgaagt	acataatgct	aagaccaagc	ctagagagga	gcagtataat	900
agtacatacc	gtgtagtcag	tgttttcaca	gtgctgcacc	aagactggct	caacggcaaa	960
gaatacaaat	gcaaagtgtc	caacaaagca	ctcccagccc	ctatcgagaa	gactattagt	1020
aaggcaaaag	ggcagcctcg	tgaaccacag	gtgtacactc	tgccacccag	tagagaggaa	1080
atgacaaaag	accaaagtctc	attgacctgc	ctgggtgaaag	gcttctaccc	cagcgacatc	1140
gccgtttagt	gggagagtaa	cggtcagcct	gagaacaatt	acaagacaac	ccccccagt	1200
ctggatagtg	acgggtcttt	ctttctgtac	agtaagctga	ctgtggacaa	gtcccgtctg	1260
cagcagggta	acgtcttcag	ctgttcctgt	atgcacgagg	cattgcacaa	ccactacacc	1320
cagaagtcac	tgagcctgag	cccagggaag				1350

SEQ ID NO: 186 moltype = AA length = 450

FEATURE Location/Qualifiers

REGION 1..450

 note = Description of Artificial Sequence: Synthetic

 polypeptide

source 1..450

 mol_type = protein

 organism = synthetic construct

SEQUENCE: 186

QVQLVQSGAE	VKKPGASVKV	SKKASGYTFT	DYNMDWVRQA	PGQGLEWMGQ	INPNNGGIFV	60
NQKFQGRVTL	TTDSTSTAY	MELRSLRSD	TAVYYCAREA	ITTVGAMDYW	GQGTLVTVSS	120
ASTKGFSVFP	LAPSSKSTSG	GTAALGCLVK	DYFPEPVTVS	WNSGALTSGV	HTFPAVLQSS	180
GLYSLSSVVT	VPSSSLGTQT	YICNVNHKPS	NTKVDKRVPE	KSCDKTHTCP	PCPAPELLGG	240
PSVFLFPPPK	KDTLMI SRTP	EVTCTVVDVS	HEDPEVKFNW	YVDGVEVHNA	KTKPREEQYN	300
STYRVVSVLT	VLHQDWLNGK	EYCKKVSNKA	LPAPIEKTIS	KAKGQPREPQ	VYTLPPSREE	360
MTKNQVSLTC	LVKGFYPSDI	AVEWESNGQP	ENNYKTPPV	LDSDGSFFLY	SKLTVDKSRW	420
QQQNVFSCSV	MHEALHNHYT	QKSLSLSPGK				450

SEQ ID NO: 187 moltype = DNA length = 1350

FEATURE Location/Qualifiers

misc_feature 1..1350

 note = Description of Artificial Sequence: Synthetic

 polynucleotide

source 1..1350

 mol_type = other DNA

 organism = synthetic construct

SEQUENCE: 187

caagtacagc	ttgtacagtc	gggagcggaa	gtcaagaaac	cgggatcgtc	ggtcaaaagt	60
tcgtgtaaa	cgctcgggata	tacgtttagc	gactataaca	tggattgggt	gcgacaagcg	120
cctgggcagg	gacttgaatg	gatgggtcag	atcaatccga	ataatggggg	aatctttttc	180
aatcagaagt	ttaaagggag	ggtaacgctg	acggcgggata	aaagcacgtc	aacggcgctat	240
atggagttgt	cgctcgttgcg	gtcggaggac	acggcgggtct	attactgcgc	gaggggaagcg	300
attacgacgg	tgggagcgat	ggattattgg	gggcagggaa	cgcttgtaac	ggtgtcatcg	360
gcctcaacaa	aaggaccaag	tgtgttccca	ctcgcccta	gcagcaagag	tacatccggg	420
ggcactgcag	cactcggctg	cctcgtaag	gattattttc	cagagccagt	aaccgtgagc	480
tggacacagt	gagcactcac	ttctgggtgc	catacttttc	ctgctgtcct	gcaaagctct	540
ggcctgtact	cactcagctc	cgctcgtgac	gtgccatctt	catctctggg	cactcagacc	600
tacatctgta	atgtaaacca	caagcctagc	aatactaagg	tcgataagcg	ggtggaaccc	660
aagagctcgc	acaagactca	cacttgtccc	ccatgccctg	ccctgaact	tctggggcgt	720
cccagcgtct	tttgtttccc	accaaagcct	aaagatactc	tgatgataag	tagaacaccc	780
gaggtgacat	gtgttgttgt	agacgtttcc	cacgaggacc	cagaggttaa	gttcaactgg	840
tacgttgatg	gagtcgaagt	acataatgct	aagaccaagc	ctagagagga	gcagtataat	900
agtacatacc	gtgtagtcag	tgttttcaca	gtgctgcacc	aagactggct	caacggcaaa	960
gaatacaaat	gcaaagtgtc	caacaaagca	ctcccagccc	ctatcgagaa	gactattagt	1020
aaggcaaaag	ggcagcctcg	tgaaccacag	gtgtacactc	tgccacccag	tagagaggaa	1080
atgacaaaag	accaaagtctc	attgacctgc	ctgggtgaaag	gcttctaccc	cagcgacatc	1140
gccgtttagt	gggagagtaa	cggtcagcct	gagaacaatt	acaagacaac	ccccccagt	1200
ctggatagtg	acgggtcttt	ctttctgtac	agtaagctga	ctgtggacaa	gtcccgtctg	1260
cagcagggta	acgtcttcag	ctgttcctgt	atgcacgagg	cattgcacaa	ccactacacc	1320
cagaagtcac	tgagcctgag	cccagggaag				1350

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SEQ ID NO: 188 moltype = AA length = 450
 FEATURE Location/Qualifiers
 REGION 1..450
 note = Description of Artificial Sequence: Synthetic
 polypeptide
 source 1..450
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 188
 QVQLVQSGAE VKKPGSSVKV SCKASGYTFS DYNMDWVRQA PGQGLEWMGQ INPNNGGIF 60
 NQKFKGRVTL TADKSTSTAY MELSSLRSED TAVYYCAREA ITTVGAMDYV GQGT LVTVSS 120
 ASTKGPSVFP LAPSSKSTSG GTAALGCLVK DYFPEPTVS WNSGALTSGV HTPFAVLQSS 180
 GLYSLSSVVT VPSSSLGTQT YICNVNHHKPS NTKVDKRVEP KSCDKTHTCP PCPAPELLGG 240
 PSVFLFPPPKP KDTLMISRTPEVTCVVVDVSHEDPEVKFNW YVDGVEVHNA KTKPREEQYN 300
 STYRVVSVLT VLNQDNLNGK EYKCKVSNKA LPAPIEKTIS KAKGQPREPQ VYTLPPSREE 360
 MTKNQVSLTC LVKGFYPSDI AVEWESNGQP ENNYKTTPPV LDSDGSFPLY SKLTVDKSRW 420
 QQGNVFSCSV MHEALHNHYT QKSLSLSPGK 450

SEQ ID NO: 189 moltype = DNA length = 1350
 FEATURE Location/Qualifiers
 misc_feature 1..1350
 note = Description of Artificial Sequence: Synthetic
 polynucleotide
 source 1..1350
 mol_type = other DNA
 organism = synthetic construct

SEQUENCE: 189
 caagtacagc tctgtacagtc gggagcggaa gtcaagaaac cgggatcgtc ggtcaaagtg 60
 tcgtgtaaag cgtcgggata tacgtttagc gactataaca tggattgggt gcgacaagcg 120
 cctgggcagg gacttgaatg gatgggtcag atcaatccga ataattgggg aatctttttc 180
 aatcagaagt ttcaggggag ggtaacgctg acggcggata aaagcacgct aacggcgat 240
 atggagttgt cgtcgttgcg gtcggaggac acggcggctc attactgcgc gagggagcg 300
 attacgacgg tgggagcgat ggattattgg gggcaggga cgttctgaac ggtgtcatcg 360
 gcctcaacaa agggaccaag tgtgttccca ctgcgcccta gcagcaagag tacatccggg 420
 ggcactgcag cactcggctg cctcgtcaag gattattttc cagagccagt aaccgtgagc 480
 tggaaacagt gagcactcac ttctgtgtgc catacttttc ctgctgtcct gcaaagctct 540
 ggctgtgact cactcagctc cgtcgtgacc gtgccatctt catctctggg cactcagacc 600
 tacatctgta atgtaaacca caagcctagc aataactaag tcgataagcg ggtggaaacc 660
 aagagctgcg acaagactca cacttgctcc ccatgccctg cccctgaact tctgggcggt 720
 ccagcgtctc tttgtttccc accaaagcct aaagatactc tgatgataag tagaacaccc 780
 gaggtgacat gtgtgtgtgt agacgtttcc cagcaggacc cagaggttaa gttcaactgg 840
 tacgttgatg gagtcgaagt acataatgct aagaccaagc ctagagagga gcagtataat 900
 agtacatacc gtgtagtcac gtgtctcaca gtgctgcacc aagactggct caacggcaaa 960
 gaatacaaat gcaaaagtgc caacaagca ctcacagccc ctatcgagaa gactattagt 1020
 aaggcaaaag ggcagcctcg tgaaccacag gtgtacactc tgccaccagg tagagaggaa 1080
 atgacaaaag accaagctctc attgacctgc ctggtgaaaag gcttctaccc cagcgacatc 1140
 gccgttgagt gggagagtaa cggtcagcct gagaacaatt acaagacaac cccccagtg 1200
 ctggatagtg acgggtcttt ctttctgtac agtaagctga ctgtggacaa gtcccgctgg 1260
 cagcagggta acgtcttcag ctgttccgtg atgcacgagg cattgcacaa ccactacacc 1320
 cagaagtcac tgagcctgag cccagggaag 1350

SEQ ID NO: 190 moltype = AA length = 450
 FEATURE Location/Qualifiers
 REGION 1..450
 note = Description of Artificial Sequence: Synthetic
 polypeptide
 source 1..450
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 190
 QVQLVQSGAE VKKPGSSVKV SCKASGYTFS DYNMDWVRQA PGQGLEWMGQ INPNNGGIF 60
 NQKFKGRVTL TADKSTSTAY MELSSLRSED TAVYYCAREA ITTVGAMDYV GQGT LVTVSS 120
 ASTKGPSVFP LAPSSKSTSG GTAALGCLVK DYFPEPTVS WNSGALTSGV HTPFAVLQSS 180
 GLYSLSSVVT VPSSSLGTQT YICNVNHHKPS NTKVDKRVEP KSCDKTHTCP PCPAPELLGG 240
 PSVFLFPPPKP KDTLMISRTPEVTCVVVDVSHEDPEVKFNW YVDGVEVHNA KTKPREEQYN 300
 STYRVVSVLT VLNQDNLNGK EYKCKVSNKA LPAPIEKTIS KAKGQPREPQ VYTLPPSREE 360
 MTKNQVSLTC LVKGFYPSDI AVEWESNGQP ENNYKTTPPV LDSDGSFPLY SKLTVDKSRW 420
 QQGNVFSCSV MHEALHNHYT QKSLSLSPGK 450

SEQ ID NO: 191 moltype = DNA length = 1347
 FEATURE Location/Qualifiers
 misc_feature 1..1347
 note = Description of Artificial Sequence: Synthetic
 polynucleotide
 source 1..1347
 mol_type = other DNA

-continued

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organism = synthetic construct

SEQUENCE: 191
caggtgacac tcaagaatc aggaccgga atccttcagc ccagccagac cttgtcgctg 60
acttggtcgt tctccggttt cagcctgaat acttatggga tgggtgtgtc atggatcagg 120
caaccgtccg ggaaaggatt ggagtggtc gcgcacatct actgggacga tgacaaacgc 180
tacaatcctt cgctgaagag ccgattgacg atttccaagg atgcctcgaa caaccgggta 240
tttcttaaga tcacgtcggt cgatacggca gacacggcga cctattactg cgcccaaaga 300
gggtacgatg actattgggg atattggggc caggggacac tcgtcacaaat ttcagctgcc 360
tcaacaaaag gaccaagtgt gttccactc gccctagca gcaagagtag atccgggggc 420
actgcagcac tcggctgctt cgtcaaggat tattttccag agccagtaac cgtgagctgg 480
aacagtggag cactcacttc tgggtgccat acttttccctg ctgtcctgca aagctctggc 540
ctgtactcac tcagctccgt cgtgaccgtg ccatcttcat ctctggggcac tcagacctac 600
atctgtaatg taaccacaa gccctagcaat actaaggctg ataagcgggt ggaacccaag 660
agctgcgaca agactcacac ttgtccccc tgccctgccc ctgaacttct gggcggtccc 720
agcgtctttt tgttcccacc aaagcctaaa gatactctga tgataagtag aacacccgag 780
gtgacatgtg ttgtttaga cgtttccccc gaggaccag aggttaagtt caactggtag 840
gttgatggag tcgaagtaca taatgctaag accaagccta gagaggagca gtataatagt 900
acataccgtg tagtcagtgt tctcacagt ctcgaccaag actggctcaa cggcaaaaga 960
tacaatatga aagtgtccaa caaagcactc ccagccccta tcgagaagac tattagtaag 1020
gcaaaggggc agcctcgtga accacaggtg tacactctgc caccagtag agaggaaatg 1080
acaaagaacc aagtctcatt gacctgcctg gtgaaaggct tctaccccag cgacatcgcc 1140
gttgagtggg agagtaacgg tcagcctgag aacaattaca agacaacccc cccagtgtg 1200
gatagtgcg ggtctttctt tctgtacagt aagctgactg tggacaagtc ccgctggcag 1260
cagggtaacg tcttcagctg ttccgtgatg cagcaggcat tgcacaacca ctacacccag 1320
aagtcactga gcctgagccc aggggaag 1347

SEQ ID NO: 192      moltype = AA length = 449
FEATURE
REGION              Location/Qualifiers
                    1..449
                    note = Description of Artificial Sequence: Synthetic
                        polypeptide
source              1..449
                    mol_type = protein
                    organism = synthetic construct

SEQUENCE: 192
QVTLKESGPG ILQPSQTLSL TCSFSGFSLN TYGMGVSWIR QPSGKGLEWL AHYWDDEDKR 60
YNPSLKSRLT ISKDSNNRVR FLKITSVDTA DTATYYCAQR GYDDYWGWWG QGTLVTISAA 120
STKGPSVFPPL APSSKSTSGG TAALGCLVKD YFPEPVTVSW NSGALTSGVH TFPVAVLQSSG 180
LYSLSSVVTV PSSSLGQTY ICNVNHKPSN TKVDRKVEPK SCDKTHTCPP CPAPELLGGP 240
SVFLFPPKPK DTLMISRPE VTCVVVDVSH EDPEVKFNWY VDGVEVHNAK TKPREEQYNS 300
TYRVVSVLTV LHQDNLNGKE YKCKVSNKAL PAPIEKTISK AKGQPREPQV YTLPPSREEM 360
TKNQVSLTCL VKGFYPSDIA VEWESNGQPE NNYKTPPVL DSDGSFFLYS KLTVDKSRWQ 420
QGNVFSQSVN HEALHNNHTQ KSLSLSPGK 449

SEQ ID NO: 193      moltype = DNA length = 1347
FEATURE
misc_feature         Location/Qualifiers
                    1..1347
                    note = Description of Artificial Sequence: Synthetic
                        polynucleotide
source              1..1347
                    mol_type = other DNA
                    organism = synthetic construct

SEQUENCE: 193
caggtgactt tgaaagaatc cggtcccgca ttggtaaagc caaccagac acttacgctc 60
acatgtacat tttccggttt cagcctgaat acttacggga tgggagtgtc gtggattcgg 120
caacctccgg ggaaggctct ggagtggtc gcgcacatct actgggatga tgacaaaagg 180
tataaccctt cacttaaaac gagactgacg atctcgagg acacaagcaa gaatcaggtc 240
gtcctcagca ttacgaatgt agaccgggtg gatactgccg tctattactg cgcgcaacgc 300
gggtatgatg actactgggg atattggggc cagggcaccc tcgtgacctt ctctgagacc 360
tcaacaaaag gaccaagtgt gttccactc gccctagca gcaagagtag atccgggggc 420
actgcagcac tcgctgcctt cgtcaaggat tattttccag agccagtaac cgtgagctgg 480
aacagtggag cactcacttc tgggtgccat acttttccctg ctgtcctgca aagctctggc 540
ctgtactcac tcagctccgt cgtgaccgtg ccatcttcat ctctggggcac tcagacctac 600
atctgtaatg taaccacaa gccctagcaat actaaggctg ataagcgggt ggaacccaag 660
agctgcgaca agactcacac ttgtccccc tgccctgccc ctgaacttct gggcggtccc 720
agcgtctttt tgttcccacc aaagcctaaa gatactctga tgataagtag aacacccgag 780
gtgacatgtg ttgtttaga cgtttccccc gaggaccag aggttaagtt caactggtag 840
gttgatggag tcgaagtaca taatgctaag accaagccta gagaggagca gtataatagt 900
acataccgtg tagtcagtgt tctcacagt ctcgaccaag actggctcaa cggcaaaaga 960
tacaatatga aagtgtccaa caaagcactc ccagccccta tcgagaagac tattagtaag 1020
gcaaaggggc agcctcgtga accacaggtg tacactctgc caccagtag agaggaaatg 1080
acaaagaacc aagtctcatt gacctgcctg gtgaaaggct tctaccccag cgacatcgcc 1140
gttgagtggg agagtaacgg tcagcctgag aacaattaca agacaacccc cccagtgtg 1200
gatagtgcg ggtctttctt tctgtacagt aagctgactg tggacaagtc ccgctggcag 1260
cagggtaacg tcttcagctg ttccgtgatg cagcaggcat tgcacaacca ctacacccag 1320
aagtcactga gcctgagccc aggggaag 1347

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SEQ ID NO: 194      moltype = AA  length = 449
FEATURE            Location/Qualifiers
REGION            1..449
                  note = Description of Artificial Sequence: Synthetic
                  polypeptide
source            1..449
                  mol_type = protein
                  organism = synthetic construct

```

```

SEQUENCE: 194
QVTLKESGPA LVKPTQTLTL TCTFSGFSLN TYGMGVSWIR QPPGKALEWL AHYWDDDKR 60
YNPSLKRRLT ISKDTSKNQV VLTITNVDPV DTAVYYCAQR GYDDYWGYWG QGTLVTVSSA 120
STKGPSVFPL APSSKSTSGG TAALGCLVKD YFPEPVTVSW NSGALTSGVH TFPVQLQSSG 180
LYSLSSVVTV PSSSLGTQTY ICNVNHKPSN TKVDRKVEPK SCDKTHTCPP CPAPELLGGP 240
SVFLFPPKPK DTLMISRTPV VTCVVVDVSH EDPEVKFNWY VDGVEVHNAK TKPREEQYNS 300
TYRVVSVLTV LHQDWLNGKE YKCKVSNKAL PAPIEKTISK AKGQPREPQV YTLPPSREEM 360
TKNQVSLTCL VKGFYPSDIA VEWESNGQPE NNYKTPPVVL DSDGSFFLYS KLTVDKSRWQ 420
QGNVFSCSVM HEALHNHYTQ KSLSLSPGK 449

```

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SEQ ID NO: 195      moltype = DNA  length = 1347
FEATURE            Location/Qualifiers
misc_feature       1..1347
                  note = Description of Artificial Sequence: Synthetic
                  polynucleotide
source            1..1347
                  mol_type = other DNA
                  organism = synthetic construct

```

```

SEQUENCE: 195
caagtaacgc tcaaggagtc cggaccacc ttggtgaagc cgacgcagac ctgactctt 60
acgtgcactt tctcgggggt ttactgaat acgtacggga tgggtgtctc atggatcagg 120
caactctcgg ggaaggattt ggaatggctg gcgcacatct actgggatga cgataagaga 180
tataacccaa gcctcaagtc gcggctcacc attacaaaag atacatcgaa aaatcaggtc 240
gtacttacta tcacgaacat ggaccccggt gacacagcaa catattactg tgcacagcgc 300
ggctatgacg attattgggg ttactgggga cagggaacac tggtcacggt gtccagcgcc 360
tcaacaaaag gaccaagtgt gttcccaact gccctagca gcaagagtac atccgggggc 420
actgcagcac tcggtcgctt cgtcaaggat tattttccag agccagtaac cgtgagctgg 480
aacagtggag cactcacttc tgggtgtccat acttttctct ctgtcctgca aagctctggc 540
ctgtactcac tcagctcctg cgtgacgctg ccatcttcat ctctgggcac tcagacctac 600
atctgtaatg taaaccacaa gccctagcaat actaaggctg ataagcgggt ggaacccaag 660
agctgcgaca agactcacac ttgtccccc tgccctgccc ctgaacttct gggcggtccc 720
agcgtctttt tgttcccacc aaagcctaaa gatactctga tgataagtag aacaccgcag 780
gtgacatgtg ttgtgttaga cgtttccccc gaggaccagc aggttaagtt caactggtag 840
gttgatggag tcgaagtaca accaagccta gagaggagca gtataatagt 900
acataccgtg tagtcagtgt tctcacagtg ctgcaccaag actggctcaa cggcaaaaga 960
tacaaatgca aagtgtccaa caaagcactc ccagccccta tcgagaagac tattagtaga 1020
gcaaaagggc agcctcgtga accacaggtg tacactctgc caccacagtag agaggaaatg 1080
acaaagaacc aagttctcatt gacctgcctg gtgaaaggct tctacccagc cgacatcgcc 1140
gttgagtggg agagtaacgg tcagcctgag aacaattaca agacaacccc cccagtgtgt 1200
gatagtgcag ggtctttctt tctgtacagt aagctgactg tggacaagtc ccgctggcag 1260
cagggtaacg tcttcagctg ttccgtgatg cactgagcat tgcacaacca ctacaccag 1320
aagtcactga gcctgagccc aggggaag 1347

```

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SEQ ID NO: 196      moltype = AA  length = 449
FEATURE            Location/Qualifiers
REGION            1..449
                  note = Description of Artificial Sequence: Synthetic
                  polypeptide
source            1..449
                  mol_type = protein
                  organism = synthetic construct

```

```

SEQUENCE: 196
QVTLKESGPT LVKPTQTLTL TCTFSGFSLN TYGMGVSWIR QPPGKGLEWL AHYWDDDKR 60
YNPSLKSRLT ITKDTSKNQV VLTITNMDPV DTATYYCAQR GYDDYWGYWG QGTLVTVSSA 120
STKGPSVFPL APSSKSTSGG TAALGCLVKD YFPEPVTVSW NSGALTSGVH TFPVQLQSSG 180
LYSLSSVVTV PSSSLGTQTY ICNVNHKPSN TKVDRKVEPK SCDKTHTCPP CPAPELLGGP 240
SVFLFPPKPK DTLMISRTPV VTCVVVDVSH EDPEVKFNWY VDGVEVHNAK TKPREEQYNS 300
TYRVVSVLTV LHQDWLNGKE YKCKVSNKAL PAPIEKTISK AKGQPREPQV YTLPPSREEM 360
TKNQVSLTCL VKGFYPSDIA VEWESNGQPE NNYKTPPVVL DSDGSFFLYS KLTVDKSRWQ 420
QGNVFSCSVM HEALHNHYTQ KSLSLSPGK 449

```

```

SEQ ID NO: 197      moltype = DNA  length = 1347
FEATURE            Location/Qualifiers
misc_feature       1..1347
                  note = Description of Artificial Sequence: Synthetic
                  polynucleotide
source            1..1347

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mol_type = other DNA
organism = synthetic construct

SEQUENCE: 197
cagggtcacgc tgaagagagtc aggtccccga atccttcaac ctctgcagac attgtcactc 60
acatgttcct tctccgggtt ctgcgtctcg acttatggca tgggtgtagg atggattcgg 120
cagccccagcg ggaaggggct tgagtggttg gcggatatct ggtgggacga cgacaaatac 180
tacaatccga gcctgaagtc ccgcctcacc atttcgaaag atacgtcatc aaacgaagtc 240
tttttgaaga tcgccaatcg ggacacggcg gatacagcga cgtattactg cgccagaagg 300
ggacactaca gcgcaatgga ttattgggga caggggacct cggtgactgt gtcgtccgcc 360
tcaacaaaag gaccaagtgt gttcccactc gccctagca gcaagagtac atccgggggc 420
actgcagcac tcggctgect cgtcaaggat tattttccag agccagtaac cgtgagctgg 480
aacagtggag cactcacttc tgggtgccat acttttctcg ctgtcctgca aagctctggc 540
ctgtactcac tcagctccgt cgtgaccgtg ccatcttcat ctctggggcac tcagacctac 600
atctgtaatg taaaccacaa gcctagcaat actaaggctg ataagcgggt ggaacccaag 660
agctgcgaca agactcacac ttgtccccc tgccttgccc ctgaacttct gggcgggtccc 720
agcgtctttt tgttcccacc aaagcctaaa gatactctga tgataagtag aacacccgag 780
gtgacatgtg ttgtttaga cgtttccccc gaggaaccag aggttaagtt caactggtag 840
gttgatggag tcgaagtaca taatgctaag accaagccta gagaggagca gtataatagt 900
acataccgtg tagtcagtgt tctcacagtg ctgcaccaag actgggtcaa cggcaaaaga 960
tacaatatga aagtggtcaa caaagcactc ccagccccta tcgagaagac tattagtaag 1020
gcaaaggggc agcctcgtga accacaggtg tacactctgc caccagtag agaggaaatg 1080
acaaagaacc aagtctcatt gacctgctg gtgaaaggct tctaccccag cgacatcgcc 1140
gttgagtggg agagtaacgg tcagctgtag aacaattaca agacaacccc cccagtgtctg 1200
gatagtgcag ggtctttctt tctgtacagt aagctgactg tggacaagtc ccgctggcag 1260
cagggtaacg tcttcagctg ttcogtgatg cagcaggcat tgcacaacca ctacacccag 1320
aagtcactga gcctgagccc aggggaag                                     1347

SEQ ID NO: 198      moltype = AA length = 449
FEATURE            Location/Qualifiers
REGION              1..449
                    note = Description of Artificial Sequence: Synthetic
                    polypeptide
source              1..449
                    mol_type = protein
                    organism = synthetic construct

SEQUENCE: 198
QVTLKESGPG ILQPSQTLSL TCSFSGFSLs TYGMGVGWIR QPSGKGLEWL ADIWWDDDKY 60
YNPSLKSRLL ISKDTSSNEV FLKIAIVDTA DTATYYCARR GHYSAMDYWG QGTSVTVSSA 120
STKGPSVFPL APSSKSTSGG TAALGLCLVD YFPEPVTWSW NSGALTSGVH TFPVAVLQSSG 180
LYSLSSVVTV PSSSLGTQTY ICNVNHHKPSN TKVDKRVEPK SCDKHTHTCP CPAPELLGGP 240
SVFLFPPKPK DTLMSIRIPE VTCVVVDVSH EDPEVKFNWY VDGVEVHNAK TKPREEQYNS 300
TYRVVSVLTV LHQDWLNGKE YKCKVSNKAL PAPIEKTISK AKGQPREPQV YTLPPSREEM 360
TKNQVSLTCL VKGFYPSDIA VEWESNGQPE NNYKTTTPVL DSDGSFFLYS KLTVDKSRWQ 420
QGNVFSCSVM HEALHNNHTQ KSLSLSPGK                                     449

SEQ ID NO: 199      moltype = DNA length = 1347
FEATURE            Location/Qualifiers
misc_feature        1..1347
                    note = Description of Artificial Sequence: Synthetic
                    polynucleotide
source              1..1347
                    mol_type = other DNA
                    organism = synthetic construct

SEQUENCE: 199
cagatcactt tgaagaaaag cggaccgacc ttggtcaagc ccacacaaac cctcacgctc 60
acgtgtacat tttcgggggt ctgcgtttca acttacggga tgggagtagg gtggattcgc 120
cagccgcctg gtaaacggtt ggagtggtct gcagacatct ggtgggacga cgataagtag 180
tataatccct cgctcaagtc cagactgacc atcacgaaag atacgagcaa gaaccaggtc 240
gtgctgacaa tgactaacat ggacccagtg gatacggcta catattactg cgccaggcgg 300
ggctactact cagcgatgga ttattggggc cagggaaacac tggtaacggt gtcgtccgcc 360
tcaacaaaag gaccaagtgt gttcccactc gccctagca gcaagagtac atccgggggc 420
actgcagcac tcggctgect cgtcaaggat tattttccag agccagtaac cgtgagctgg 480
aacagtggag cactcacttc tgggtgccat acttttctcg ctgtcctgca aagctctggc 540
ctgtactcac tcagctccgt cgtgaccgtg ccatcttcat ctctggggcac tcagacctac 600
atctgtaatg taaaccacaa gcctagcaat actaaggctg ataagcgggt ggaacccaag 660
agctgcgaca agactcacac ttgtccccc tgccttgccc ctgaacttct gggcgggtccc 720
agcgtctttt tgttcccacc aaagcctaaa gatactctga tgataagtag aacacccgag 780
gtgacatgtg ttgtttaga cgtttccccc gaggaaccag aggttaagtt caactggtag 840
gttgatggag tcgaagtaca taatgctaag accaagccta gagaggagca gtataatagt 900
acataccgtg tagtcagtgt tctcacagtg ctgcaccaag actgggtcaa cggcaaaaga 960
tacaatatga aagtggtcaa caaagcactc ccagccccta tcgagaagac tattagtaag 1020
gcaaaggggc agcctcgtga accacaggtg tacactctgc caccagtag agaggaaatg 1080
acaaagaacc aagtctcatt gacctgctg gtgaaaggct tctaccccag cgacatcgcc 1140
gttgagtggg agagtaacgg tcagctgtag aacaattaca agacaacccc cccagtgtctg 1200
gatagtgcag ggtctttctt tctgtacagt aagctgactg tggacaagtc ccgctggcag 1260
cagggtaacg tcttcagctg ttcogtgatg cagcaggcat tgcacaacca ctacacccag 1320

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aagtcaactga gcctgagccc aggggaag 1347

SEQ ID NO: 200      moltype = AA length = 449
FEATURE            Location/Qualifiers
REGION             1..449
                   note = Description of Artificial Sequence: Synthetic
                   polypeptide
source             1..449
                   mol_type = protein
                   organism = synthetic construct

SEQUENCE: 200
QITLKESGPT LVKPTQTLTL TCTFSGFSLT TYGMGVGWIR QPPGKALEWL ADIWWDDEKY 60
YNPSLKSRLT ITKDTSKNQV VLTMTNMDPV DTATYYCARR GHYSAMDYWG QGTLVTVSSA 120
STKGPSVFPPL APSSKSTSGG TAALGCLVKD YFPEPVTVSW NSGALTSGVH TFPVAVLQSSG 180
LYSLSSVVTV PSSSLGTQTY ICNVNHKPSN TKVDRKVEPK SCDKTHTCPP CPAPELLGGP 240
SVFLFPPKPK DTLMISRTP VTCVVVDVSH EDPEVKFNWY VDGVEVHNAK TKPREEQYNS 300
TYRVVSVLTV LHQDWLNGKE YKCKVSNKAL PAPIEKTISK AKGQPREPQV YTLPPSREEM 360
TKNQVSLTCL VKGFYPSDIA VEWESNGQPE NNYKTTTPVL DSDGSFFLYS KLTVDKSRWQ 420
QGNVFSCSVM HEALHNHYTQ KSLSLSPGK 449

SEQ ID NO: 201      moltype = DNA length = 1347
FEATURE            Location/Qualifiers
misc_feature        1..1347
                   note = Description of Artificial Sequence: Synthetic
                   polynucleotide
source             1..1347
                   mol_type = other DNA
                   organism = synthetic construct

SEQUENCE: 201
caagtgaactc tcaaggagtc cggaccgcc ctggtcaaac caacgcagac actgacgctc 60
acatgcacct tcagcggatt ttctgtgtca acgtacggca tgggtgtggg gtggattcgc 120
cagcctccgg ggaaagccct tgaatgggtg gcggacatct ggtgggatga tgacaagtac 180
tataatccct cacttaagtc acggttgacg atctcgaaag acaccagcaa gaaccaggta 240
gtgctgacaa tgactaacat ggaccgggtc gatacagcgg tctactattg tgctagaagg 300
ggacactact ccgcaatgga ttattggggt caggggacgc tcgtaaccgt gtcgtcgccc 360
tcaacaaaag gaccaagtgt gttccactc gccctagaca gcaagagtag atccgggggc 420
actgcagcac tcggtgcgct cgtcaaggat tattttccag agccagtaac cgtgagctgg 480
aacagtggag cactcacttc tgggtgtccat acttttctcg ctgtcctgca aagctctggc 540
ctgtactcac tcagctccgt cgtgaccgtg ccatcttcat ctctgggcac tcagacctac 600
atctgtaatg taaaccacaa gcctagcaat actaaggctc ataagcgggt ggaacccaag 660
agctgcgaca agactcacac ttgtccccc tgccctgccc ctgaactctt gggcgggtccc 720
agcgtctttt tgttcccacc aaagcctaaa gatactctga tgataagtag aacacccgag 780
gtgacatgtg ttgtgttaga cgtttcccac gaggaccacg aggttaagtt caactggtag 840
gttgatggag tcgaagtaca taatgctaag accaagccta gagaggagca gtataatagt 900
acataccgtg tagtcagtg tctcacagtg ctgcaccaag actgggtcaa cggcaaaaga 960
tacaaatgca aagtgtccaa caaagcactc ccagccccta tcgagaagac tattagtaag 1020
gcaaaggggc agcctcgtga accacaggtg tacactctgc caccagtag agaggaaatg 1080
acaaagaacc aagtctcatt gacctgcctg gtgaaaggct tctacccag cgacatcgcc 1140
gttgatggag agagtacagg tcagcctgag aacaattaca agacaacccc cccagtgctg 1200
gatagtgaag ggtctttctt tctgtacagt aagctgactg tggacaagtc ccgctggcag 1260
cagggtaacg tcttcagctg ttccgtgatg cagcaggcat tgcacaacca ctacacccag 1320
aagtcaactga gcctgagccc aggggaag 1347

SEQ ID NO: 202      moltype = AA length = 449
FEATURE            Location/Qualifiers
REGION             1..449
                   note = Description of Artificial Sequence: Synthetic
                   polypeptide
source             1..449
                   mol_type = protein
                   organism = synthetic construct

SEQUENCE: 202
QVTLKESGPA LVKPTQTLTL TCTFSGFSLT TYGMGVGWIR QPPGKALEWL ADIWWDDEKY 60
YNPSLKSRLT ISKDTSKNQV VLTMTNMDPV DTAVYYCARR GHYSAMDYWG QGTLVTVSSA 120
STKGPSVFPPL APSSKSTSGG TAALGCLVKD YFPEPVTVSW NSGALTSGVH TFPVAVLQSSG 180
LYSLSSVVTV PSSSLGTQTY ICNVNHKPSN TKVDRKVEPK SCDKTHTCPP CPAPELLGGP 240
SVFLFPPKPK DTLMISRTP VTCVVVDVSH EDPEVKFNWY VDGVEVHNAK TKPREEQYNS 300
TYRVVSVLTV LHQDWLNGKE YKCKVSNKAL PAPIEKTISK AKGQPREPQV YTLPPSREEM 360
TKNQVSLTCL VKGFYPSDIA VEWESNGQPE NNYKTTTPVL DSDGSFFLYS KLTVDKSRWQ 420
QGNVFSCSVM HEALHNHYTQ KSLSLSPGK 449

SEQ ID NO: 203      moltype = DNA length = 642
FEATURE            Location/Qualifiers
misc_feature        1..642
                   note = Description of Artificial Sequence: Synthetic
                   polynucleotide

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source                1..642
                      mol_type = other DNA
                      organism = synthetic construct

SEQUENCE: 203
gacatccaaa tgaccagtc acccgcgagc ctttcggcgt cggtcggaga aacggtcacg 60
atcacgtgcc ggacatcaga gaattctccat aactacctcg cgtgggtatca acagaagcag 120
gggaagtcgc cccagttgct tgtatcagat gcgaaaacgt tggcggatgg ggtgccgtcc 180
agattctcgg gatcgggctc ggggacgcag tactcgctca agatcaattc gctgcagccg 240
gaggactttg ggtcgtagta ttgtcagcat ttttggtcat caccgtatac atttggagg 300
ggaacgaaac ttgagattaa gcgcacagtt gctgccccca gcgtgttcat ttcccacct 360
agcgatgagc agctgaaaag cggtagtgcc tctgtcgat gcttgctcaa caacttttac 420
ccacgtgagg ctaagggtgca gtggaaagt gataatgcac ttcaatctgg aaacagtcaa 480
gagtcctgta cagaacagga cagcaaagac tcaacttatt cactctcttc caccctgact 540
ctgtccaagg cagactatga aaaacacaag gtatacgctc gcgagggtac acaccagggt 600
ttgtctagtc ctgtcaccaa gtccttcaat aggggcgaat gt 642

SEQ ID NO: 204        moltype = AA length = 214
FEATURE               Location/Qualifiers
REGION               1..214
                      note = Description of Artificial Sequence: Synthetic
                      polypeptide
source               1..214
                      mol_type = protein
                      organism = synthetic construct

SEQUENCE: 204
DIQMTQSPAS LSASVGETVT ITCRTSENLH NYLAWYQQKQ GKSPQLLVYD AKTLADGVPS 60
RFGSGSGGTQ YSLKINSLQP EDFGSYYCQH FWSSPYTFGG GTKLEIKRTV AAPSVFIFPP 120
SDEQLKSGTA SVVCLLNIFY PREAKVQWKV DNALQSGNSQ ESVTEQDSKD STYLSSTLT 180
LSKADYEKHK VYACEVTHQG LSSPVTKSFN RGEK 214

SEQ ID NO: 205        moltype = DNA length = 642
FEATURE               Location/Qualifiers
misc_feature         1..642
                      note = Description of Artificial Sequence: Synthetic
                      polynucleotide
source               1..642
                      mol_type = other DNA
                      organism = synthetic construct

SEQUENCE: 205
gacatccaaa tgaccagtc gccgtcgctc ctttcagcgt cggtagggga tcgggtcaca 60
attacgtgcc gaacgtcaga gaatttgcac aactacctcg cgtgggtatca gcagaagccc 120
gggaagtcac cgaaactcct tgtctacgat gcgaaaacgc tggcggatgg agtgccgtcg 180
agattctcgg gaagcggatc cggtagcgac tatacgctta cgatctcatc gctccagccc 240
gaggactttg cgacgtacta ttgtcagcat ttttggtcgt cgccctacac atttggggcag 300
gggaccaaag tggaaatcaa gcgcacagtt gctgccccca gcgtgttcat ttcccacct 360
agcgatgagc agctgaaaag cggtagtgcc tctgtcgat gcttgctcaa caacttttac 420
ccacgtgagg ctaagggtgca gtggaaagt gataatgcac ttcaatctgg aaacagtcaa 480
gagtcctgta cagaacagga cagcaaagac tcaacttatt cactctcttc caccctgact 540
ctgtccaagg cagactatga aaaacacaag gtatacgctc gcgagggtac acaccagggt 600
ttgtctagtc ctgtcaccaa gtccttcaat aggggcgaat gt 642

SEQ ID NO: 206        moltype = AA length = 214
FEATURE               Location/Qualifiers
REGION               1..214
                      note = Description of Artificial Sequence: Synthetic
                      polypeptide
source               1..214
                      mol_type = protein
                      organism = synthetic construct

SEQUENCE: 206
DIQMTQSPSS LSASVGDRV TITCRTSENLH NYLAWYQQK GKSPKLLVYD AKTLADGVPS 60
RFGSGSGGTD YTLTISSLQP EDFATYYCQH FWSSPYTFGQ GTKLEIKRTV AAPSVFIFPP 120
SDEQLKSGTA SVVCLLNIFY PREAKVQWKV DNALQSGNSQ ESVTEQDSKD STYLSSTLT 180
LSKADYEKHK VYACEVTHQG LSSPVTKSFN RGEK 214

SEQ ID NO: 207        moltype = DNA length = 642
FEATURE               Location/Qualifiers
misc_feature         1..642
                      note = Description of Artificial Sequence: Synthetic
                      polynucleotide
source               1..642
                      mol_type = other DNA
                      organism = synthetic construct

SEQUENCE: 207
gacatccaaa tgaccagtc gccgtcgctc ctttcagcgt cggtagggga tcgggtcaca 60
attacgtgcc gaacgtcaga gaatttgcac aactacctcg cgtgggtatca gcagaagccc 120

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gggaaggccc cgaaactcct tatctacgat gcgaaaacgc tggcggatgg agtgccgctcg 180
agattctcgg gaagcggatc cggtagcgac tatacgctta cgatctcatc gctccagccc 240
gaggactttg cgacgtacta ttgtcagcat ttttggtcgt cgcctacac atttgggcag 300
gggaccaagt tggaaatcaa gcgcacagt gctgccccca gcgtgttcat ttcccacct 360
agcgatgagc agctgaaaag cggtagctgc tctgtcgtat gcttgctcaa caacttttac 420
ccacgtgagg ctaaggtgca gtggaaagt gataatgcac ttcaatctgg aaacagtcaa 480
gagtcctgta cagaacagga cagcaaagac tcaacttatt cactctcttc caccctgact 540
ctgtccaagg cagactatga aaaacacaag gtatacgctt gcgaggttac acaccagggt 600
ttgtctagtc ctgtcaccaa gtccttcaat aggggcgaat gt 642

SEQ ID NO: 208      moltype = AA  length = 214
FEATURE            Location/Qualifiers
REGION             1..214
                   note = Description of Artificial Sequence: Synthetic
                   polypeptide
source             1..214
                   mol_type = protein
                   organism = synthetic construct

SEQUENCE: 208
DIQMTQSPSS LSASVGRVT ITCRTSEN LH NYLAWYQQK GKAPKLLIYD AKTLADGVPS 60
RFSGSGSGTD YTLTISSLQ EDFAITYCQH FWSSPYTFGQ GTKLEIKRTV AAPSVFI FPP 120
SDEQLKSGTA SVVCLLNIFY PREAKVQWKV DNALQSGNSQ ESVTEQDSKD STYLSSTLT 180
LSKADYEKHK VYACEVTHQG LSSPVTKSFN RGE C 214

SEQ ID NO: 209      moltype = DNA  length = 642
FEATURE            Location/Qualifiers
misc_feature       1..642
                   note = Description of Artificial Sequence: Synthetic
                   polynucleotide
source             1..642
                   mol_type = other DNA
                   organism = synthetic construct

SEQUENCE: 209
gacatccaaa tgaccagtc gccgtcgtcg ctttcagcgt cggtagggga tcgggtcaca 60
attacgtgcc gaacgtcaga gaatttgcac aactacctcg cgtgggtatca gcagaagccc 120
gggaagtcac cgaaactcct tatctacgat gcgaaaacgc tggcggatgg agtgccgctcg 180
agattctcgg gaagcggatc cggtagcgac tatacgctta cgatctcatc gctccagccc 240
gaggactttg cgacgtacta ttgtcagcat ttttggtcgt cgcctacac atttgggcag 300
gggaccaagt tggaaatcaa gcgcacagt gctgccccca gcgtgttcat ttcccacct 360
agcgatgagc agctgaaaag cggtagctgc tctgtcgtat gcttgctcaa caacttttac 420
ccacgtgagg ctaaggtgca gtggaaagt gataatgcac ttcaatctgg aaacagtcaa 480
gagtcctgta cagaacagga cagcaaagac tcaacttatt cactctcttc caccctgact 540
ctgtccaagg cagactatga aaaacacaag gtatacgctt gcgaggttac acaccagggt 600
ttgtctagtc ctgtcaccaa gtccttcaat aggggcgaat gt 642

SEQ ID NO: 210      moltype = AA  length = 214
FEATURE            Location/Qualifiers
REGION             1..214
                   note = Description of Artificial Sequence: Synthetic
                   polypeptide
source             1..214
                   mol_type = protein
                   organism = synthetic construct

SEQUENCE: 210
DIQMTQSPSS LSASVGRVT ITCRTSEN LH NYLAWYQQK GKSPKLLIYD AKTLADGVPS 60
RFSGSGSGTD YTLTISSLQ EDFAITYCQH FWSSPYTFGQ GTKLEIKRTV AAPSVFI FPP 120
SDEQLKSGTA SVVCLLNIFY PREAKVQWKV DNALQSGNSQ ESVTEQDSKD STYLSSTLT 180
LSKADYEKHK VYACEVTHQG LSSPVTKSFN RGE C 214

SEQ ID NO: 211      moltype = DNA  length = 642
FEATURE            Location/Qualifiers
misc_feature       1..642
                   note = Description of Artificial Sequence: Synthetic
                   polynucleotide
source             1..642
                   mol_type = other DNA
                   organism = synthetic construct

SEQUENCE: 211
gatatcgta tgaccagtc ccagaagttc atgtcaactt cagtgaggaga cagagtgtcc 60
gtcacatgta aagcctcgca aaatgtggga accaacgtag cgtgggtcca gcagaacct 120
ggccaatcac cgaaggcact gcttactcg gccagctata ggtactcggg agtaccagat 180
cggtttacgg ggtcggggag cgggacggac tttatctca ctatttcaa tgtccagtcg 240
gaggaccttg cggaatactt ctgccagcag tataacaact atccctcac gtttgggtgt 300
ggtacaaaat tggagttgaa gcgcacagt gctgccccca gcgtgttcat ttcccacct 360
agcgatgagc agctgaaaag cggtagctgc tctgtcgtat gcttgctcaa caacttttac 420
ccacgtgagg ctaaggtgca gtggaaagt gataatgcac ttcaatctgg aaacagtcaa 480

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gagtcctgta cagaacagga cagcaaaagac tcaacttatt cactctcttc caccctgact 540
ctgtccaagg cagactatga aaaacacaag gtatacgctt gcgagggtac acaccagggt 600
ttgtctagtc ctgtcaccaa gtccttcaat aggggcgaat gt 642

SEQ ID NO: 212      moltype = AA length = 214
FEATURE            Location/Qualifiers
REGION            1..214
                  note = Description of Artificial Sequence: Synthetic
                  polypeptide
source            1..214
                  mol_type = protein
                  organism = synthetic construct

SEQUENCE: 212
DIVMTQSQKF MSTSVGDRVS VTCKASQNVG TNVAVFQQKP GQSPKALIYS ASYRYSGVDP 60
RFTGSGSGTD FILTISNVQS EDLAIFYCQQ YNNYPLTFGA GTKLELKRIV AAPSVFIFPP 120
SDEQLKSGTA SVVCLLNIFY PREAKVQWKV DNALQSGNSQ ESVTEQDSKD STYLSSTLT 180
LSKADYEKHK VYACEVTHQG LSSPVTKSFN RGEC 214

SEQ ID NO: 213      moltype = DNA length = 642
FEATURE            Location/Qualifiers
misc_feature       1..642
                  note = Description of Artificial Sequence: Synthetic
                  polynucleotide
source            1..642
                  mol_type = other DNA
                  organism = synthetic construct

SEQUENCE: 213
gacatccaaa tgacccaatc gccctcctcc ctctccgcat cagtagggga ccgcgtcaca 60
attacttgca aagcgtcgca gaacgtcgga acgaatgtgg cgtgggttca gcagaagccc 120
ggaaaagctc cgaagagctt gatctactcg gccctcatata ggtattcggg tgtgccgagc 180
cgggtttagcg ggtcggggcg aggtactgat ttcacgctca caatttcac gttgcagcca 240
gaagatttcg ccacatatta cgttcagcag tacaacaatt accctctgac gttcggccag 300
ggaacccaac ttgagatcaa gcgcacagtt gctgccccca gcgtgttcat ttcccacct 360
agcgatgagc agctgaaaag cggtagctgc tctgtcgtat gcttgctcaa caacttttac 420
ccacgtgagg ctaaggtgca gtggaaagtg gataatgcac ttcaatctgg aaacagtcaa 480
gagtcctgta cagaacagga cagcaaaagac tcaacttatt cactctcttc caccctgact 540
ctgtccaagg cagactatga aaaacacaag gtatacgctt gcgagggtac acaccagggt 600
ttgtctagtc ctgtcaccaa gtccttcaat aggggcgaat gt 642

SEQ ID NO: 214      moltype = AA length = 214
FEATURE            Location/Qualifiers
REGION            1..214
                  note = Description of Artificial Sequence: Synthetic
                  polypeptide
source            1..214
                  mol_type = protein
                  organism = synthetic construct

SEQUENCE: 214
DIQMTQSPSS LSASVGRVIT ITCKASQNVG TNVAVFQQKP GKAPKSLIYS ASYRYSGVPS 60
RFTGSGSGTD FTLTISLQP EDFATYYCQQ YNNYPLTFGA GTKLEIKRIV AAPSVFIFPP 120
SDEQLKSGTA SVVCLLNIFY PREAKVQWKV DNALQSGNSQ ESVTEQDSKD STYLSSTLT 180
LSKADYEKHK VYACEVTHQG LSSPVTKSFN RGEC 214

SEQ ID NO: 215      moltype = DNA length = 642
FEATURE            Location/Qualifiers
misc_feature       1..642
                  note = Description of Artificial Sequence: Synthetic
                  polynucleotide
source            1..642
                  mol_type = other DNA
                  organism = synthetic construct

SEQUENCE: 215
gacatcgtga tgacacagtc acagaaattc atgtccacat ccgtcgggtga tagagtatcc 60
gtcacgtgta aggcctcgca aaacgttaga actaatgtgg cgtgggtatca acagaagcca 120
ggacagtcac ccaagcactc catctacagc ccctcatatc ggtacacggg ggtgccggac 180
agggttcacgg gatcgggggag cgggaccgat ttacactga ccatttcgaa tgtccagctc 240
gaggaccttg cggaataactt ctgccagcag tataactcgt accctcacac gtttgagggt 300
ggcactaagt tggagatgaa acgcacagtt gctgccccca gcgtgttcat ttcccacct 360
agcgatgagc agctgaaaag cggtagctgc tctgtcgtat gcttgctcaa caacttttac 420
ccacgtgagg ctaaggtgca gtggaaagtg gataatgcac ttcaatctgg aaacagtcaa 480
gagtcctgta cagaacagga cagcaaaagac tcaacttatt cactctcttc caccctgact 540
ctgtccaagg cagactatga aaaacacaag gtatacgctt gcgagggtac acaccagggt 600
ttgtctagtc ctgtcaccaa gtccttcaat aggggcgaat gt 642

SEQ ID NO: 216      moltype = AA length = 214
FEATURE            Location/Qualifiers

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REGION 1..214
note = Description of Artificial Sequence: Synthetic polypeptide

source 1..214
mol_type = protein
organism = synthetic construct

SEQUENCE: 216

DIVMTQSQKF	MSTSVGDRVS	VTCKASQNVG	TNVAWYQQKP	GQSPKALIYS	PSYRYSGVDP	60
RFTGSGSGTD	FTLTISNVQS	EDLAIFYCQQ	YNSYPHTFGG	GTKLEMKRTV	AAPSVFIFPP	120
SDEQLKSGTA	SVVCLLNIFY	PREAKVQWKV	DNALQSGNSQ	ESVTEQDSKD	STYLSSTLT	180
LSKADYEKHK	VYACEVTHQG	LSSPVTKSFN	RGEC			214

SEQ ID NO: 217 moltype = DNA length = 642
FEATURE Location/Qualifiers
misc_feature 1..642
note = Description of Artificial Sequence: Synthetic polynucleotide

source 1..642
mol_type = other DNA
organism = synthetic construct

SEQUENCE: 217

gatatccaga	tgacacagtc	accctcgtcg	ctctcagctt	ccgtaggcga	cagggtcact	60
attacgtgta	aagcatcaca	gaacgtcggg	acgaatgtgg	cgtgggttca	gcagaagccc	120
gggaagagcc	ccaaagcgct	tatctactcc	ccgtcgtatc	ggatttcggg	tgtgccaaagc	180
agattttcgg	ggtcaggttc	gggaactgac	tttaccctga	ccatctcgtc	cctccaaccg	240
gaagatttcg	ccacgtactt	ctgccagcag	tacaacagct	atcctcacac	attcggacaa	300
gggacaaagt	tggagattaa	acgcacagtt	gctgccccca	gcgtgttcat	tttcccacct	360
agcgtatgag	agctgaaaag	cggtaactgc	tctgtcgtat	gcttgctcaa	caacttttac	420
ccacgtgagg	ctaagggtgca	gtggaaaagt	gataatgcac	ttcaatctgg	aaacagtcaa	480
gagtcctgta	cagaacagga	cagcaaaagc	tcaacttatt	cactctcttc	cacctgact	540
ctgtccaagg	cagactatga	aaaacacaa	gtatacgctc	gcgaggttac	acaccagggt	600
ttgtctagtc	ctgtcaccaa	gtccttcaat	aggggcgaat	gt		642

SEQ ID NO: 218 moltype = AA length = 214
FEATURE Location/Qualifiers
REGION 1..214
note = Description of Artificial Sequence: Synthetic polypeptide

source 1..214
mol_type = protein
organism = synthetic construct

SEQUENCE: 218

DIQMTQSPSS	LSASVGDVRT	ITCKASQNVG	TNVAWFQKP	GKSPKALIYS	PSYRYSGVPS	60
RFTGSGSGTD	FTLTISLQP	EDFATYFCQQ	YNSYPHTFGG	GTKLEIKRTV	AAPSVFIFPP	120
SDEQLKSGTA	SVVCLLNIFY	PREAKVQWKV	DNALQSGNSQ	ESVTEQDSKD	STYLSSTLT	180
LSKADYEKHK	VYACEVTHQG	LSSPVTKSFN	RGEC			214

SEQ ID NO: 219 moltype = DNA length = 1035
FEATURE Location/Qualifiers
misc_feature 1..1035
note = Description of Artificial Sequence: Synthetic polynucleotide

source 1..1035
mol_type = other DNA
organism = synthetic construct

SEQUENCE: 219

gggtgtaaac	cctgcactcg	cacgggtgcg	gaggtgtcct	ccgtctttat	cttcctctcc	60
aaacccaagg	atgtgctgac	aatcactttg	actccaaaag	tcacatgcgt	agtcgtggac	120
atctcgaaag	acgacccgga	agtgacgttc	tcgtgggttg	ttgatgatgt	agaagtgcac	180
accgtctaaa	cccagccgag	ggaagaacag	tttaacagca	cgtttaggag	tgtgtcggaa	240
ctgcccatta	tgacaccagg	ttgggttaat	gggaaggagt	tcaaatgtcg	cgtgaatagt	300
gcggtcgttc	cagcccctat	tgaaaagact	atttccaaaa	cgaagggtcg	gccccaaagt	360
ccccaaagt	atcacaaatccc	tccgcccga	gaacaaatgg	caaaagacaa	agtgaagtgt	420
acgtgcactg	tcacggactt	tttcccggag	gatatcaccg	tcgaatggca	atggaatggg	480
caactcgccg	aaaactacaa	gaatacacaa	cccattatgg	ataccgatgg	atcgtatttc	540
gtctactcaa	agttgaacgt	acagaagtca	aattgggagg	cagggaatac	gttcacttgc	600
agtgttttgc	acgaaggcct	ccataaccac	catacggaaa	agtcactgtc	gcactccccg	660
ggaaaaatcg	agggcagaat	ggatggtgga	ggagggtcgg	cgcgcaacgg	ggaccactgt	720
ccgtcggggc	ccgggctgtg	ctgccgtctg	cacacggtcc	gcgcgtcgct	ggaagacctg	780
ggctggggcg	attgggtgct	gtcgccacgg	gaggtgcaag	tgaccatgtg	catcggcgcg	840
tgcccagacc	agttccgggc	ggcaaacatg	cacgcgcaga	tcaagacgag	cctgcaccgc	900
ctgaagcccg	acacgggtgc	agcgcctgc	tgcgtgcccc	ccagctacaa	tccatgggtg	960
ctcattcaaa	agaccgacac	cggggtgtcg	ctccagacct	atgatgactt	gttagccaaa	1020
gactgccact	gcata					1035

SEQ ID NO: 220 moltype = AA length = 345

-continued

FEATURE Location/Qualifiers
 REGION 1..345
 note = Description of Artificial Sequence: Synthetic polypeptide
 source 1..345
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 220

GCKPCICTVP	EVSSVFIFPP	KPKDVLITL	TPKVTCVVVD	ISKDDPEVQF	SWFVDDVEVH	60
TAQTQPREEQ	FNSTFRSVSE	LPIMHQDWLN	GKEFKCRVNS	AAFPAPIEKT	ISKTGRPKA	120
PQVYTIPPPK	EQMAKDKVS	TCMITDFFPE	DITVEWQWNG	QPAENYKNTQ	PIMDTDGSYF	180
VYSKLNQVQS	NWEAGNTFTC	SVLHEGLHNH	HTEKSLSHSP	GKIEGRMDGG	GG SARNGDHC	240
PLGPGRCCLL	HTVRASLEDL	GWADWVLSR	EVQVTMCIGA	CPSQFRAANM	HAQIKTSLHR	300
LKPDTPVAPC	CVPASYNPMV	LIQKTDGVS	LQTYDILLAK	DCHCI		345

SEQ ID NO: 221 moltype = DNA length = 1026
 FEATURE Location/Qualifiers
 misc_feature 1..1026
 note = Description of Artificial Sequence: Synthetic polynucleotide
 source 1..1026
 mol_type = other DNA
 organism = synthetic construct

SEQUENCE: 221

tcgaaaccca	cttgccctcc	tccggagctg	ttggggcggac	cctccgtgtt	tatctttccc	60
ccgaagccga	aagataccct	tatgatctca	cggacgcggg	aggtcacttg	cgtagtagtg	120
gatgtgtcgg	aggatgaccc	cgaagtccag	ttcacctggt	atatcaataa	cgagcaagtg	180
aggacagcga	ggcccccact	tagggagcag	cagttcaact	ccacaattcg	ggtcgtcagc	240
actttgccca	tcgctcatga	ggactggctc	cgcgaaaaag	agttcaagtg	taagggtgcat	300
aacaaggcat	tgccagcgcc	tattgaaaag	acaatctcga	aggcgcgagg	gcagccgctc	360
gagcccaaa	gtatatacga	aggacccccg	agggagaaga	tgtcgtcgcg	ctcagtaagc	420
cttacgtgca	tgtattaacg	tttctaccct	agcgacatca	gcgtagagtg	ggaaaagaat	480
ggaaaggcgg	aggataacta	caagacgact	cccgcgggtg	tggattcgga	tggttcgtac	540
tttctgtata	gcaaattgtc	agtcgccgac	tcagaatggc	agaggggtga	cgtgttcacg	600
tgctccgtga	tgcacgaagc	acttcacaat	cactacaccc	agaaatcaat	ctcgcgggtc	660
ccaggcaaa	gtggaggagg	gtcgggtcac	gcccccctc	gcgattcggt	tccgctgggg	720
cctggtgat	gtgtcatctc	cgagacagtc	caggccacgc	tggaggacct	cgggtggtca	780
gactgggttc	tgtccccacg	acaactgcag	ctttcgatgt	gcgtggggga	atgtccgcac	840
ttgtacagat	cggcgaatac	ccacgctcag	attaaggcac	gactccatgg	tttgacagcca	900
gataaagtcc	ccgcaccttg	ctgtgtcccc	agctcatata	ctcctgtcgt	actcatgcat	960
cggacagaca	gcggcgtgtc	gcttcaaacg	tatgacgacc	tcgtagcgag	aggatgtcat	1020
tgcgcc						1026

SEQ ID NO: 222 moltype = AA length = 342
 FEATURE Location/Qualifiers
 REGION 1..342
 note = Description of Artificial Sequence: Synthetic polypeptide
 source 1..342
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 222

SKPTCPPPEL	LGGPSVFIFP	KPKDITLMIS	RTPEVTCVVV	DVSEDDPEVQ	FTWYINNEQV	60
RTARPPPLREQ	QFNSTIRVVS	TLPIAHEDWL	RGKEFKCKVH	NKALPAPIEK	TISKARGQPL	120
EPKVYTMGPP	REELSSRSVS	LTCMINGFYP	SDISVEWEKN	GKAEDNYKTT	PAVLDSGGSY	180
FLYSKLSVPT	SEWQRGDVFT	CSVMHEALHN	HYTQKSISRS	PGKGGGGSAA	AHPRDSCPLG	240
PGRCCHLETV	QATLEDLGWS	DWVLSRQLQ	LSMVCVGECPH	LYRSANTHAQ	IKARLHGLQP	300
DKVPAPCCVP	SSYTPVVLHM	RTDSGVSLQT	YDDLVARGCH	CA		342

SEQ ID NO: 223 moltype = AA length = 350
 FEATURE Location/Qualifiers
 REGION 1..350
 note = Description of Artificial Sequence: Synthetic polypeptide
 source 1..350
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 223

DKTHTCPPCP	APELLGGPSV	FLFPPKPKDT	LMISRTPEVT	CVVVDVSHED	PEVKFNWYVD	60
GVEVHNATK	PREEQYNSTY	RVVSVLTVLH	QDWLNGKEYK	CKVSNKALPA	PIEKTISKAK	120
GQPREPQVYT	LPPSREEMTK	NQVSLTCLVK	GFYPSDIAVE	WESNGQPENN	YKTPPVLDL	180
DGSFFLYSKL	TVDKSRWQQG	NVFSQCSVMHE	ALHNHYTQKS	LSLSPGKIEG	RMDGGGGSAR	240
NGDHCPLGPG	RCCRLHTVRA	SLEDLGWADW	VLSPREVQVT	MCIGACPSQF	RAANMHAQIK	300
TSLHRLKPD	VPAPCCVPAS	YNPMVLIQKT	DTGVSLQTYD	DLLAKDCHCI		350

SEQ ID NO: 224 moltype = AA length = 344

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FEATURE Location/Qualifiers
 REGION 1..344
 note = Description of Artificial Sequence: Synthetic polypeptide
 source 1..344
 mol_type = protein
 organism = synthetic construct
 SEQUENCE: 224
 DKHTTCPPCP APELLGGPSV FLFPPKPKDT LMISRTPEVT CVVVDVSHED PEVKFNWYVD 60
 GVEVHNAKTK PREEQYNSTY RVVSVLTVLH QDWLNGKEYK CKVSNKALPA PIEKTISKAK 120
 GQPREPQVYT LPPSREEMTK NQVSLTCLVK GFYPSDIAVE WESNGQPENN YKTTTPPVLDS 180
 DGSFFLYSKL TVDKSRWQQG NVFSCSVME ALHNHYTQKS LSLSPGKGGG GSARNGDHCP 240
 LGPGRCCRLH TVRASLEDLG WADWVLSPRE VQVTCIGAC PSQFRAANMH AQIKTSLHRL 300
 KPDTVPAPCC VPASYNPMVL IQKTDGTGVS LQTYDDLLAKD CHCI 344

SEQ ID NO: 225 moltype = DNA length = 22
 FEATURE Location/Qualifiers
 misc_feature 1..22
 note = Description of Artificial Sequence: Synthetic primer
 source 1..22
 mol_type = other DNA
 organism = synthetic construct
 SEQUENCE: 225
 ctaatacgac tcactatagg gc 22

SEQ ID NO: 226 moltype = DNA length = 21
 FEATURE Location/Qualifiers
 misc_feature 1..21
 note = Description of Artificial Sequence: Synthetic primer
 source 1..21
 mol_type = other DNA
 organism = synthetic construct
 SEQUENCE: 226
 tatgcaaggc ttacaaccac a 21

SEQ ID NO: 227 moltype = DNA length = 21
 FEATURE Location/Qualifiers
 misc_feature 1..21
 note = Description of Artificial Sequence: Synthetic primer
 source 1..21
 mol_type = other DNA
 organism = synthetic construct
 SEQUENCE: 227
 aggacagggg ttgattgttg a 21

SEQ ID NO: 228 moltype = DNA length = 27
 FEATURE Location/Qualifiers
 misc_feature 1..27
 note = Description of Artificial Sequence: Synthetic primer
 source 1..27
 mol_type = other DNA
 organism = synthetic construct
 SEQUENCE: 228
 ctcattcctg ttgaagctct tgacaat 27

SEQ ID NO: 229 moltype = DNA length = 23
 FEATURE Location/Qualifiers
 misc_feature 1..23
 note = Description of Artificial Sequence: Synthetic primer
 source 1..23
 mol_type = other DNA
 organism = synthetic construct
 SEQUENCE: 229
 aagcagtggg atcaacgcag agt 23

SEQ ID NO: 230 moltype = DNA length = 23
 FEATURE Location/Qualifiers
 misc_feature 1..23
 note = Description of Artificial Sequence: Synthetic primer
 source 1..23
 mol_type = other DNA
 organism = synthetic construct
 SEQUENCE: 230
 cgactgaggg acctccagat gtt 23

SEQ ID NO: 231 moltype = DNA length = 17

-continued

FEATURE	Location/Qualifiers	
misc_feature	1..17	
	note = Description of Artificial Sequence: Synthetic primer	
source	1..17	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 231		
gtaaacgac ggccagt		17
SEQ ID NO: 232	moltype = DNA length = 18	
FEATURE	Location/Qualifiers	
misc_feature	1..18	
	note = Description of Artificial Sequence: Synthetic primer	
source	1..18	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 232		
caggaaacag ctatgacc		18
SEQ ID NO: 233	moltype = DNA length = 45	
FEATURE	Location/Qualifiers	
misc_feature	1..45	
	note = Description of Artificial Sequence: Synthetic primer	
source	1..45	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 233		
ctaatacgac tcactatagg gcaagcagtg gtatcaacgc agagt		45
SEQ ID NO: 234	moltype = AA length = 7	
FEATURE	Location/Qualifiers	
REGION	1..7	
	note = Description of Artificial Sequence: Synthetic peptide	
source	1..7	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 234		
GYTFSDY		7
SEQ ID NO: 235	moltype = AA length = 8	
FEATURE	Location/Qualifiers	
REGION	1..8	
	note = Description of Artificial Sequence: Synthetic peptide	
source	1..8	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 235		
GYTFSDYN		8
SEQ ID NO: 236	moltype = AA length = 17	
FEATURE	Location/Qualifiers	
REGION	1..17	
	note = Description of Artificial Sequence: Synthetic peptide	
source	1..17	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 236		
QINPYNHLIF FNQKFQG		17
SEQ ID NO: 237	moltype = AA length = 17	
FEATURE	Location/Qualifiers	
REGION	1..17	
	note = Description of Artificial Sequence: Synthetic peptide	
source	1..17	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 237		
QINPNNGLIF FNQKFQG		17
SEQ ID NO: 238	moltype = AA length = 17	
FEATURE	Location/Qualifiers	
REGION	1..17	
	note = Description of Artificial Sequence: Synthetic peptide	
source	1..17	
	mol_type = protein	
	organism = synthetic construct	

-continued

SEQUENCE: 238
QINPNNGLIF FNQKFKG 17

SEQ ID NO: 239 moltype = AA length = 17
FEATURE Location/Qualifiers
REGION 1..17
note = Description of Artificial Sequence: Synthetic peptide
source 1..17
mol_type = protein
organism = synthetic construct

SEQUENCE: 239
QINPYNHLIF FNQKFKG 17

SEQ ID NO: 240 moltype = AA length = 6
FEATURE Location/Qualifiers
REGION 1..6
note = Description of Artificial Sequence: Synthetic peptide
source 1..6
mol_type = protein
organism = synthetic construct

SEQUENCE: 240
NPYNHL 6

SEQ ID NO: 241 moltype = AA length = 6
FEATURE Location/Qualifiers
REGION 1..6
note = Description of Artificial Sequence: Synthetic peptide
source 1..6
mol_type = protein
organism = synthetic construct

SEQUENCE: 241
NPNNGL 6

SEQ ID NO: 242 moltype = AA length = 8
FEATURE Location/Qualifiers
REGION 1..8
note = Description of Artificial Sequence: Synthetic peptide
source 1..8
mol_type = protein
organism = synthetic construct

SEQUENCE: 242
INPYNHLI 8

SEQ ID NO: 243 moltype = AA length = 8
FEATURE Location/Qualifiers
REGION 1..8
note = Description of Artificial Sequence: Synthetic peptide
source 1..8
mol_type = protein
organism = synthetic construct

SEQUENCE: 243
INPNNGLI 8

SEQ ID NO: 244 moltype = AA length = 9
FEATURE Location/Qualifiers
REGION 1..9
note = Description of Artificial Sequence: Synthetic peptide
source 1..9
mol_type = protein
organism = synthetic construct

SEQUENCE: 244
QHFWSDPYT 9

SEQ ID NO: 245 moltype = DNA length = 360
FEATURE Location/Qualifiers
misc_feature 1..360
note = Description of Artificial Sequence: Synthetic polynucleotide
source 1..360
mol_type = other DNA
organism = synthetic construct

SEQUENCE: 245
cagggtccagc ttgtgcaatc gggagcggaa gtgaagaaac cgggagcgctc ggtaaaagtc 60
tcgtgcaaaag cgctcggggta tacgtttacg gactataaca tggactgggt ggcccaagcg 120
cctggacaga gccttgaatg gatggggcag attaatccgt acaatcacct gatcttcttt 180
aatcagaaat tccaggaag ggtaacgctg acgacagaca cgtaacatc gacggcctat 240

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atggaattgc ggtcgttgcg atcagatgat acggcgggtct actattgtgc gagggaggcg 300
attacgacgg tgggagcgat ggattattgg ggacagggga cgttggtaac ggtatcgtcg 360

SEQ ID NO: 246      moltype = AA length = 120
FEATURE            Location/Qualifiers
REGION            1..120
                  note = Description of Artificial Sequence: Synthetic
                  polypeptide
source            1..120
                  mol_type = protein
                  organism = synthetic construct

SEQUENCE: 246
QVQLVQSGAE VKKPGASVKV SCKASGYTFT DYNMDWVRQA PGQSLEWMGQ INPYNHLIFF 60
NQKFQGRVTL TTDSTSTAY MELRSLRSDD TAVYYCAREA ITTVGAMDYV GQGLVTVSS 120

SEQ ID NO: 247      moltype = DNA length = 360
FEATURE            Location/Qualifiers
misc_feature       1..360
                  note = Description of Artificial Sequence: Synthetic
                  polynucleotide
source            1..360
                  mol_type = other DNA
                  organism = synthetic construct

SEQUENCE: 247
caggtcacgc ttgtgcaatc gggagcggaa gtgaagaaac cgggagcgctc ggtaaaagtc 60
tcgtgcaaaag cgtcggggta tacgtttacg gactataaca tggactgggt ggcgcaagcg 120
cctggacaga gccttgaatg gatggggcag attaatccga ataattggact gatcttcttt 180
aatcagaaat tccagggaag ggtaacgctg acgacagaca cgtcaacatc gacggcctat 240
atggaattgc ggtcgttgcg atcagatgat acggcgggtct actattgtgc gagggaggcg 300
attacgacgg tgggagcgat ggattattgg ggacagggga cgttggtaac ggtatcgtcg 360

SEQ ID NO: 248      moltype = AA length = 120
FEATURE            Location/Qualifiers
REGION            1..120
                  note = Description of Artificial Sequence: Synthetic
                  polypeptide
source            1..120
                  mol_type = protein
                  organism = synthetic construct

SEQUENCE: 248
QVQLVQSGAE VKKPGASVKV SCKASGYTFT DYNMDWVRQA PGQSLEWMGQ INPNNGLIFF 60
NQKFQGRVTL TTDSTSTAY MELRSLRSDD TAVYYCAREA ITTVGAMDYV GQGLVTVSS 120

SEQ ID NO: 249      moltype = DNA length = 360
FEATURE            Location/Qualifiers
misc_feature       1..360
                  note = Description of Artificial Sequence: Synthetic
                  polynucleotide
source            1..360
                  mol_type = other DNA
                  organism = synthetic construct

SEQUENCE: 249
caagtacagc ttgtacagtc gggagcggaa gtcaagaaac cgggatcgctc ggtcaaagtc 60
tcgtgtaaaag cgtcgggata tacgtttacg gactataaca tggattgggt ggcgacaagcg 120
cctgggcagg gacttgaatg gatgggtcag atcaatccga ataattgggt gatctttttc 180
aatcagaagt ttaaagggag ggtaacgctg acggcggata aaagcacgct aacggcgctat 240
atggagttgt cgtcgttgcg gtcggaggac acggcgggtct attactgcgc gaggggaagcg 300
attacgacgg tgggagcgat ggattattgg gggcagggaa cgcttgtaac ggtgtcatcg 360

SEQ ID NO: 250      moltype = AA length = 120
FEATURE            Location/Qualifiers
REGION            1..120
                  note = Description of Artificial Sequence: Synthetic
                  polypeptide
source            1..120
                  mol_type = protein
                  organism = synthetic construct

SEQUENCE: 250
QVQLVQSGAE VKKPGSSVKV SCKASGYTFS DYNMDWVRQA PGQGLEWMGQ INPNNGLIFF 60
NQKFQGRVTL TADKSTSTAY MELSSLRSED TAVYYCAREA ITTVGAMDYV GQGLVTVSS 120

SEQ ID NO: 251      moltype = DNA length = 360
FEATURE            Location/Qualifiers
misc_feature       1..360
                  note = Description of Artificial Sequence: Synthetic
                  polynucleotide

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source                1..360
                      mol_type = other DNA
                      organism = synthetic construct

SEQUENCE: 251
caagtacagc ttgtacagtc gggagcggaa gtcaagaaac cgggatcgtc ggtcaaagtg 60
tcgtgtaaaag cgtcggggata tacgttttagc gactataaca tggattgggt gcgacaagcg 120
cctgggcagg gacttgaatg gatgggtcag atcaatccgt acaatcacct gatctttttc 180
aatcagaagt ttaaaggagg ggtaacgctg acggcgggata aaagcacgtc aacggcgat 240
atggagttgt cgtcgttgcg gtcggaggac acggcgggtct attactgcgc gaggggaagcg 300
attacgacgg tgggagcgat ggattattgg gggcagggaa cgcttgaac ggtgtcatcg 360

SEQ ID NO: 252        moltype = AA length = 120
FEATURE              Location/Qualifiers
REGION               1..120
                      note = Description of Artificial Sequence: Synthetic
                      polypeptide

source                1..120
                      mol_type = protein
                      organism = synthetic construct

SEQUENCE: 252
QVQLVQSGAE VKKPGSSVKV SCKASGYTFS DYNMDWVRQA PGQGLEWMGQ INPYNHLIF 60
NQKFKGRVTL TADKSTSTAY MELSSLRSED TAVYYCAREA ITTVGAMDYW GQGTLTVTSS 120

SEQ ID NO: 253        moltype = DNA length = 321
FEATURE              Location/Qualifiers
misc_feature          1..321
                      note = Description of Artificial Sequence: Synthetic
                      polynucleotide

source                1..321
                      mol_type = other DNA
                      organism = synthetic construct

SEQUENCE: 253
gacatccaaa tgaccagtc gccgtcgctc ctttcagcgt cggtagggga tcgggtcaca 60
attacgtgcc gaacgtcaga gaatttgcac aactacctcg cgtggatca gcagaagccc 120
gggaagtcac cgaaactcct tatctacgat gcgaaaacgc tggcggatgg agtgccgctc 180
agattctcgg gaagcggatc cggtagcgac tatacgctta cgatctcatc gctccagccc 240
gaggactttg cgacgtacta ttgtcagcat ttttggtcgg acccctacac atttgggcag 300
gggaccaagt tggaaatcaa g                                     321

SEQ ID NO: 254        moltype = AA length = 107
FEATURE              Location/Qualifiers
REGION               1..107
                      note = Description of Artificial Sequence: Synthetic
                      polypeptide

source                1..107
                      mol_type = protein
                      organism = synthetic construct

SEQUENCE: 254
DIQMTQSPSS LSASVGRVIT ITCRTSENH NYLAWYQKP GKSPKLLIYD AKTLADGVPS 60
RFSGSGSGTD YTLTISSLQP EDFATYYCQH FWSDPYTFGQ GTKLEIK 107

SEQ ID NO: 255        moltype = DNA length = 1350
FEATURE              Location/Qualifiers
misc_feature          1..1350
                      note = Description of Artificial Sequence: Synthetic
                      polynucleotide

source                1..1350
                      mol_type = other DNA
                      organism = synthetic construct

SEQUENCE: 255
caggtcacgc ttgtgcaatc gggagcggaa gtgaagaaac cgggagcgtc ggtaaaagt 60
tcgtgcgaag cgtcggggta tacgtttacg gactataaca tggactgggt gcgccaagcg 120
cctggacaga gccttgaatg gatggggcag attaatccgt acaatcacct gatcttttt 180
aatcagaagt tccaggggaag ggtaacgctg acgacagaca cgtcaacatc gacggcctat 240
atggaaattgc ggtcgttgcg atcagatgat acggcgggtct actattgtgc gagggaggcg 300
attacgacgg tgggagcgat ggattattgg ggacagggga cgttggtaac ggtatcgctc 360
gcctcaacaa aaggaccaag tgtgttccca ctgcgcccta gcagcaagag tacatccggg 420
ggcactgcag cactcgctcg cctcgtaag gattattttc cagagccagt aaccgtgagc 480
tggaacagtg gagcactcac ttctggtgtc catacttttc ctgctgtcct gcaaagctct 540
ggcctgtact cactcagctc cgtcgtgacc gtgccattct catctctggg cactcagacc 600
tacatctgta atgtaaaaca caagcctagc aatactaagg tcgataagcg ggtggaaccc 660
aagagctgcg acaagactca cacttgtccc ccatgccctg cccctgaact tctgggcggt 720
cccagcgtct ttttggtccc accaaagcct aaagatactc tgatgataag tagaacaccc 780
gaggtgacat gtgttgttgt agacgtttcc cagcaggacc cagaggttaa gttcaactgg 840
tacgttgatg gagtcgaagt acataatgct aagaccaagc ctagagagga gcagtataat 900
agtacatacc gtgtagtcag tgttctcaca gtgctgcacc aagactggct caacggcaaa 960

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gaatacaaat gcaaagtgtc caacaaagca ctcccagccc ctatcgagaa gactattagt 1020
aaggcaaaagg ggcagcctcg tgaaccacag gtgtacactc tgccacccag tagagaggaa 1080
atgacaaaaga accaagtctc attgacctgc ctggtgaaag gcttctaccc cagcgacatc 1140
gccgttgagt gggagagtaa cggtcagcct gagaacaatt acaagacaac cccccagtg 1200
ctggatagtg acgggtcttt ctttctgtac agtaagctga ctgtggacaa gtcccgtgg 1260
cagcagggta acgtcttcag ctgttccgtg atgcacgagg cattgcacaa ccactacacc 1320
cagaagtcac tgagcctgag cccagggaag 1350

SEQ ID NO: 256      moltype = AA length = 450
FEATURE
REGION            1..450
                  note = Description of Artificial Sequence: Synthetic
                  polypeptide
source            1..450
                  mol_type = protein
                  organism = synthetic construct

SEQUENCE: 256
QVQLVQSGAE VKKPGASVKV SCKASGYTFT DYNMDWVRQA PGQSLEWMGQ INPNHLIFF 60
NQKFQGRVTL TTDSTSTAY MELRSLRSD D TAVYYCAREA ITTVGAMDYW GQGT LVTVSS 120
ASTKGPSVFP LAPSSKSTSG GTAALGCLVK DYFPEPVTVS WNSGALTSGV HTFPAVLQSS 180
GLYSLSSVVT VPSSSLGTQT YICNVNHKPS NTKVDKRVPE KSCDKTHTCP PCPAPELLGG 240
PSVFLFPPKP KDTLMISRTP EVTCVVVDVS HEDPEVKFNW YVDGVEVHNA KTKPREEQYN 300
STYRVSVLT VLHQDWLNGK EYCKKVS NKA LPAPIEKTIS KAKGQPREPQ VYTLPPSREE 360
MTKNQVSLTC LVKGFYPSDI AVEWESNGQP ENNYKTPPV LDSDGSFFLY SKLTVDKSRW 420
QQGNVFCSCV MHEALHNHYT QKSLSLSPGK 450

SEQ ID NO: 257      moltype = DNA length = 1350
FEATURE
misc_feature      1..1350
                  note = Description of Artificial Sequence: Synthetic
                  polynucleotide
source            1..1350
                  mol_type = other DNA
                  organism = synthetic construct

SEQUENCE: 257
cagggtccagc ttgtgcaatc gggagcggaa gtgaagaaac cgggagcgtc ggtaaaagt 60
tcgtgcaaaag cgctcggggtat tacgtttacg gactataaca tggactgggt gcgccaagcg 120
cctggacaga gccttgaaatg gatggggcag attaatccga ataattggact gatctctctt 180
aatcagaaat tccagggaag ggtaacgctg acgacagaca cgtaacatc gacggcctat 240
atggaattgc ggtcgttgctg atcagatgat acggcgggtc actattgtgc gagggaggcg 300
attacgacgg tgggagcgat ggattattgg ggacagggga cgttggtaac ggtatcgctg 360
gcctcaacaa aaggaccaag tgtgttccca ctgcgcccta gcagcaagag tacatccggg 420
ggcactcgag cactcggctg cctcgtcaag gattatttct cagagccagt aaccgtgagc 480
tggaacagtg gagcactcac ttctggtgtc catacttttc ctgctgtcct gcaaaagctc 540
ggcctgtact cactcagctc cgtcgtgacc gtgccatctt catctctggg cactcagacc 600
tacatctgta atgtaaaacca caagcctagc aatactaagg tcgataagcg ggtggaaccc 660
aagagctcgc acaagactca caettgtccc ccatgccctg cccctgaact tctgggagg 720
cccagcgtct ttttgttccc accaaagcct aaagatactc tgatgataag tagaacaccc 780
gaggtgacat gtgtgttttc acaggtttcc cagcaggacc cagagggtta gttcaactgg 840
tacgttgatg gagtcgaagt acataatgct aagaccaagc ctagagagga gcagtataat 900
agtacatacc gtgtagtcag tgttctcaca gtgctgcacc aagactggct caacggcaaa 960
gaatacaaat gcaaaagtgc caacaaagca ctcccagccc ctatcgagaa gactattagt 1020
aaggcaaaagg ggcagcctcg tgaaccacag gtgtacactc tgccacccag tagagaggaa 1080
atgacaaaaga accaagtctc attgacctgc ctggtgaaag gcttctaccc cagcgacatc 1140
gccgttgagt gggagagtaa cggtcagcct gagaacaatt acaagacaac cccccagtg 1200
ctggatagtg acgggtcttt ctttctgtac agtaagctga ctgtggacaa gtcccgtgg 1260
cagcagggta acgtcttcag ctgttccgtg atgcacgagg cattgcacaa ccactacacc 1320
cagaagtcac tgagcctgag cccagggaag 1350

SEQ ID NO: 258      moltype = AA length = 450
FEATURE
REGION            1..450
                  note = Description of Artificial Sequence: Synthetic
                  polypeptide
source            1..450
                  mol_type = protein
                  organism = synthetic construct

SEQUENCE: 258
QVQLVQSGAE VKKPGASVKV SCKASGYTFT DYNMDWVRQA PGQSLEWMGQ INPNNGLIFF 60
NQKFQGRVTL TTDSTSTAY MELRSLRSD D TAVYYCAREA ITTVGAMDYW GQGT LVTVSS 120
ASTKGPSVFP LAPSSKSTSG GTAALGCLVK DYFPEPVTVS WNSGALTSGV HTFPAVLQSS 180
GLYSLSSVVT VPSSSLGTQT YICNVNHKPS NTKVDKRVPE KSCDKTHTCP PCPAPELLGG 240
PSVFLFPPKP KDTLMISRTP EVTCVVVDVS HEDPEVKFNW YVDGVEVHNA KTKPREEQYN 300
STYRVSVLT VLHQDWLNGK EYCKKVS NKA LPAPIEKTIS KAKGQPREPQ VYTLPPSREE 360
MTKNQVSLTC LVKGFYPSDI AVEWESNGQP ENNYKTPPV LDSDGSFFLY SKLTVDKSRW 420
QQGNVFCSCV MHEALHNHYT QKSLSLSPGK 450

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SEQ ID NO: 259      moltype = DNA length = 1350
FEATURE            Location/Qualifiers
misc_feature        1..1350
                    note = Description of Artificial Sequence: Synthetic
                    polynucleotide
source              1..1350
                    mol_type = other DNA
                    organism = synthetic construct

```

```

SEQUENCE: 259
caagtacagc ttgtacagtc gggagcggaa gtcaagaaac cgggatcgtc ggtcaaaagt 60
tcgtgtaaag cgtcgggata tacgttttagc gactataaca tggattgggt gcgacaagcg 120
cctgggcagg gacttgaatg gatgggtcag atcaatccga ataattgggt gatctttttc 180
aatcagaagt ttaaagggag ggtaacgctg acggcggata aaagcacgtc aacggcgat 240
atggagttgt cgtcgttgct gtcggaggac acggcgggtct attactgcgc gagggaagcg 300
attacgacgg tgggagcgat ggattattgg gggcagggaa cgcttgtaac ggtgtcatcg 360
gcctcaacaa aaggaccaag tgtgttccca ctgcgcccta gcagcaagag tacatccggg 420
ggcactgcag cactcggctg cctcgtcaag gattattttc cagagccagt aaccgtgagc 480
tggaaacagt gagcactcac ttctggtgtc catacttttc ctgctgtcct gcaaagctct 540
ggcctgtact cactcagctc cgtcgtgacc gtgccatctt catctctggg cactcagacc 600
tacatctgta atgtaaacca caagcctagc aatactaagg tcgataagcg ggtggaaccc 660
aagagctcgc acaagactca cacttgtccc ccatgccctg cccctgaact tctggggcgt 720
cccagcgtct tttgttccc accaaagcct aaagatactc tgatgataag tagaacaccc 780
gaggtgacat gtgttgttgt agacgtttcc cagcaggacc cagaggttaa gttcaactgg 840
tacgttgatg gactcgaagt acataatgct aagaccaagc cttagagagga gcagtataat 900
agtacatacc gtgtagtcag tgtttccaca gtgctgcacc aagactggct caacggcaaa 960
gaatacaaat gcaaaagtgc caacaaagca ctcccagccc ctatcgagaa gactattagt 1020
aaggcaaaag ggcagcctcg tgaaccacag gtgtacactc tgccacccag tagagaggaa 1080
atgacaaaag accaagtctc attgacctgc ctgggtgaaag gcttctaccc cagcgacatc 1140
gccgttgagt gggagagtaa cggtcagcct gagaacaatt acaagacaac cccccagtg 1200
ctggatagtg acgggtcttt ctttctgtac agtaagctga ctgtggacaa gtcccgttg 1260
cagcagggta acgtcttcag ctgttcctgt atgcacgagg cattgcacaa ccactacacc 1320
cagaagtcac tgagcctgag cccagggaag 1350

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SEQ ID NO: 260      moltype = AA length = 450
FEATURE            Location/Qualifiers
REGION             1..450
                    note = Description of Artificial Sequence: Synthetic
                    polypeptide
source              1..450
                    mol_type = protein
                    organism = synthetic construct

```

```

SEQUENCE: 260
QVQLVQSGAE VKKPGSSVKV SCKASGYTFS DYNMDWVRQA PGQGLEWMGQ INPNNGLIFF 60
NQKFKGRVTL TADKSTSTAY MELSSLSRSED TAVYYCAREA ITTVGAMDYV GQGTLLVTVSS 120
ASTKGPSVFP LAPSSKSTSG GTAALGCLVK DYFPEPTVS WNSGALTSKV HTFPAVLQSS 180
GLYSLSSVVT VPSSSLGTQT YICNVNHKPS NTKVDKRVEP KSCDKTHTCP PCPAPELLGG 240
PSVFLFPPPK KDTLMI SRTP EVTCVVVDVS HEDPEVKFNW YVDGVEVHNA KTKPREEQYN 300
STYRVVSVLT VLHQDWLNGK EYKCKVSNKA LPAPIEKTIS KAKGQPREPQ VYTLPPSREE 360
MTKNQVSLTC LVKGFYPSDI AVEWESNGQP ENNYKTTTPV LDSGGSFPLY SKLTVDKSRW 420
QQGNVFSCSV MHEALHNHYT QKSLSLSPGK 450

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SEQ ID NO: 261      moltype = DNA length = 1350
FEATURE            Location/Qualifiers
misc_feature        1..1350
                    note = Description of Artificial Sequence: Synthetic
                    polynucleotide
source              1..1350
                    mol_type = other DNA
                    organism = synthetic construct

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```

SEQUENCE: 261
caagtacagc ttgtacagtc gggagcggaa gtcaagaaac cgggatcgtc ggtcaaaagt 60
tcgtgtaaag cgtcgggata tacgttttagc gactataaca tggattgggt gcgacaagcg 120
cctgggcagg gacttgaatg gatgggtcag atcaatccgt acaatcacct gatctttttc 180
aatcagaagt ttaaagggag ggtaacgctg acggcggata aaagcacgtc aacggcgat 240
atggagttgt cgtcgttgct gtcggaggac acggcgggtct attactgcgc gagggaagcg 300
attacgacgg tgggagcgat ggattattgg gggcagggaa cgcttgtaac ggtgtcatcg 360
gcctcaacaa aaggaccaag tgtgttccca ctgcgcccta gcagcaagag tacatccggg 420
ggcactgcag cactcggctg cctcgtcaag gattattttc cagagccagt aaccgtgagc 480
tggaaacagt gagcactcac ttctggtgtc catacttttc ctgctgtcct gcaaagctct 540
ggcctgtact cactcagctc cgtcgtgacc gtgccatctt catctctggg cactcagacc 600
tacatctgta atgtaaacca caagcctagc aatactaagg tcgataagcg ggtggaaccc 660
aagagctcgc acaagactca cacttgtccc ccatgccctg cccctgaact tctggggcgt 720
cccagcgtct tttgttccc accaaagcct aaagatactc tgatgataag tagaacaccc 780
gaggtgacat gtgttgttgt agacgtttcc cagcaggacc cagaggttaa gttcaactgg 840
tacgttgatg gactcgaagt acataatgct aagaccaagc cttagagagga gcagtataat 900

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agtcataacc gtgtagtcag tgttttcaca gtgctgcacc aagactggct caacggcaaa 960
gaatacaaat gcaaagtgtc caacaaagca ctcccagccc ctatcgagaa gactattagt 1020
aaggcaaaag ggcagcctcg tgaaccacag gtgtacacct tgccaccacag tagagaggaa 1080
atgacaaaaga accaagtctc attgacctgc ctggtgaaaag gcttctaccc cagcgacatc 1140
gccgttgagt gggagagtta cggtcagcct gagaacaatt acaagacaac cccccagtg 1200
ctggatagtg acgggtcttt ctttctgtac agtaagctga ctgtggacaa gtcccgtgg 1260
cagcagggta acgtcttcag ctgttccgtg atgcacgagg cattgcacaa ccactacacc 1320
cagaagtcac tgagcctgag cccagggaag                                     1350

SEQ ID NO: 262          moltype = AA length = 450
FEATURE                Location/Qualifiers
REGION                 1..450
                        note = Description of Artificial Sequence: Synthetic
                        polypeptide
source                 1..450
                        mol_type = protein
                        organism = synthetic construct

SEQUENCE: 262
QVQLVQSGAE VKKPGSSVKV SCKASGYTFS DYNMDWVRQA PGQGLEWMGQ INPYNHLIFF 60
NQKFKGRVTL TADKSTSTAY MELSSLRSED TAVYYCAREA ITTVGAMDYW GQGTLLTVSS 120
ASTKGPSVFP LAPSSKSTSG GTAALGCLVK DYFPEPVTVS WNSGALTSGV HTPFAVLQSS 180
GLYSLSSVVT VPSSSLGTQT YICNVNHNKPS NTKVDKRVEP KSCDKHTTCP PCPAPELLGG 240
PSVFLFPPPK KDTLMISRTPEVTCVVVDVS HEDPEVKFNW YVDGVEVHNA KTKPREEQYN 300
STYRVSVLT VLHQDWLNGK EYKCKVSNKA LPAPIEKTIS KAKGQPREPQ VYTLPPSREE 360
MTKNQVSLTC LVKGFYPSDI AVEWESNGQP ENNYKTTTPV LDSDGSFFLY SKLTVDKSRW 420
QQGNVFCSCV MHEALHNHYT QKSLSLSPGK                                     450

SEQ ID NO: 263          moltype = DNA length = 642
FEATURE                Location/Qualifiers
misc_feature           1..642
                        note = Description of Artificial Sequence: Synthetic
                        polynucleotide
source                 1..642
                        mol_type = other DNA
                        organism = synthetic construct

SEQUENCE: 263
gacatccaaa tgcccagtc gccgtcgtcg ctttcagcgt cggtagggga tcgggtcaca 60
attacgtgcc gaacgtcaga gaatttgcac aactacctcg cgtgggtatca gcagaagccc 120
gggaagtcac cgaaactcct tatctacgat gcgaaaacgc tggcggatgg agtgccgtcg 180
agattctcgg gaagcggatc cggtagcgac tatacgctta cgatctcatc gctccagccc 240
gaggactttg cgacgtacta ttgtcagcat ttttggtcgg acccctacac atttggggcag 300
gggaccaagt tggaaatcaa gcgcacagtt gctgccccca gcgtgttcat ttccaccct 360
agcgatgagc agctgaaaag cggtagctgcc tctgtcgtat gcttgctcaa caacttttac 420
ccacgtgagg ctaaggtgca gtggaaagtg gataatgcac ttcaatctgg aaacagtcaa 480
gagtcctgta cagaacagga cagcaaaagac tcaacttatt cactctcttc caccctgact 540
ctgtccaagg cagactatga aaacacaag gtatacgctc gcgaggttac acaccagggt 600
ttgtctagtc ctgtcaccaa gtccttcaat agggggcgaat gt                                     642

SEQ ID NO: 264          moltype = AA length = 214
FEATURE                Location/Qualifiers
REGION                 1..214
                        note = Description of Artificial Sequence: Synthetic
                        polypeptide
source                 1..214
                        mol_type = protein
                        organism = synthetic construct

SEQUENCE: 264
DIQMTQSPSS LSASVGRVIT ITCRTSENH NYLAWYQQKP GKSPKLLIYD AKTLADGVPS 60
RFSGSGSGTD YTLTISSLQP EDFATYYCQH FWSDPYTFGQ GTKLEIKRTV AAPSVFIFPP 120
SDEQLKSGTA SVVCLLNIFY PREAKVQWKV DNALQSGNSQ ESVTEQDSKD STYLSSTLT 180
LSKADYEKHK VYACEVTHQG LSSPVTKSFN RGEK                                     214

SEQ ID NO: 265          moltype = AA length = 4
FEATURE                Location/Qualifiers
REGION                 1..4
                        note = Description of Artificial Sequence: Synthetic peptide
source                 1..4
                        mol_type = protein
                        organism = synthetic construct

SEQUENCE: 265
GGGG                                                                4

SEQ ID NO: 266          moltype = AA length = 6
FEATURE                Location/Qualifiers
REGION                 1..6
                        note = Description of Artificial Sequence: Synthetic 6xHis

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-continued

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tag
source      1..6
            mol_type = protein
            organism = synthetic construct
SEQUENCE: 266
HHHHHH

```

6

What is claimed is:

1. A method of treating cachexia in a mammal comprising administering an effective amount of an anti-GDF15 antibody or antigen-binding fragment thereof.

2. The method of claim 1, wherein the cachexia is associated with an underlying disease selected from the group consisting of cancer, chronic heart failure, chronic kidney disease, COPD, AIDS, multiple sclerosis, rheumatoid arthritis, sepsis, and tuberculosis.

3. The method of claim 1, further comprising administering a second agent to the mammal in need thereof, wherein the second agent is selected from the group consisting of an inhibitor of Activin-A, an inhibitor of ActRIIB, an inhibitor of IL-6, an inhibitor of IL-6R, a melanocortin peptide inhibitor, a melanocortin receptor inhibitor, a ghrelin, a ghrelin mimetic, a GHS-R1a agonist, a SARM, a TNF α inhibitor, an IL-1 α inhibitor, a myostatin inhibitor, a beta-blocker and an anti-cancer agent.

4. The method of claim 1, wherein the mammal is a human.

5. A method of inhibiting loss of muscle mass associated with cachexia comprising administering an effective amount of an anti-GDF15 antibody or antigen-binding fragment thereof.

6. The method of claim 5, wherein the cachexia is associated with an underlying disease selected from the group consisting of cancer, chronic heart failure, chronic kidney disease, COPD, AIDS, multiple sclerosis, rheumatoid arthritis, sepsis, and tuberculosis.

7. The method of claim 5, wherein the loss of muscle mass is accompanied by a loss of fat mass.

8. The method of claim 5, further comprising administering a second agent to the mammal in need thereof, wherein the second agent is selected from the group consisting of an inhibitor of Activin-A, an inhibitor of ActRIIB, an inhibitor of IL-6, an inhibitor of IL-6R, a melanocortin peptide inhibitor, a melanocortin receptor inhibitor, a ghrelin, a ghrelin mimetic, a GHS-R1a agonist, a SARM, a TNF α inhibitor, an IL-1 α inhibitor, a myostatin inhibitor, a beta-blocker and an anti-cancer agent.

9. A method of inhibiting loss of organ mass associated with cachexia comprising administering an effective amount of an anti-GDF15 antibody or antigen-binding fragment thereof.

10. The method of claim 9, wherein the cachexia is associated with an underlying disease selected from the

group consisting of cancer, chronic heart failure, chronic kidney disease, COPD, AIDS, multiple sclerosis, rheumatoid arthritis, sepsis, and tuberculosis.

11. The method of claim 9, wherein the organ is kidney, liver, heart or spleen.

12. The method of claim 9, wherein the loss of organ mass is accompanied by a loss of muscle mass, a loss of fat mass or involuntary weight loss.

13. The method of claim 9 further comprising administering a second agent to the mammal in need thereof, wherein the second agent is selected from the group consisting of an inhibitor of Activin-A, an inhibitor of ActRIIB, an inhibitor of IL-6, an inhibitor of IL-6R, a melanocortin peptide inhibitor, a melanocortin receptor inhibitor, a ghrelin, a ghrelin mimetic, a GHS-R1a agonist, a SARM, a TNF α inhibitor, an IL-1 α inhibitor, a myostatin inhibitor, a beta-blocker and an anti-cancer agent.

14. A method of treating sarcopenia associated with cachexia in a mammal comprising administering an effective amount of an anti-GDF15 antibody or antigen-binding fragment thereof.

15. A method of decreasing the incidence and/or severity of cachexia in a mammal, thereby increasing the maximum tolerated dose of an anti-cancer agent capable of causing cachexia in the mammal, the method comprising administering an effective amount of an anti-GDF15 antibody or antigen-binding fragment thereof.

16. The method of claim 14, wherein the antibody or antigen-binding fragment thereof is administered in combination with the anti-cancer agent.

17. The method of claim 15, wherein the mammal is a human.

18. A method of increasing appetite in a mammal suffering from cachexia comprising administering an effective amount of an anti-GDF15 antibody or antigen-binding fragment thereof.

19. The method of claim 18, wherein the cachexia is associated with an underlying disease selected from the group consisting of cancer, chronic heart failure, chronic kidney disease, COPD, AIDS, multiple sclerosis, rheumatoid arthritis, sepsis, and tuberculosis.

20. The method of claim 16, wherein the mammal is a human.

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